

# CONTENTS

## 1. Number System

- Important Facts and Formulae QUEA-9
- Model Questions QUEA-14
  - Level-I QUEA-14
  - Level-II QUEA-17
  - Level-III QUEA-20
- Quantity based questions QUEA-22
- Data sufficiency questions QUEA-22

## 2. Simplification

- Important Facts and Formulae QUEA-38
- Model Questions QUEA-44
  - Level-I

## 6. Profit and Loss (Including Discounts)

- Important Facts and Formulae QUEA-140
- Model Questions QUEA-141
  - Level-I QUEA-141
  - Level-II QUEA-144
  - Level-III QUEA-147
- Quantity based questions QUEA-150
- Data sufficiency questions QUEA-150

## 7. Ratio and Proportion

- Important Facts and Fo
- Model Q

### 11. Time and Distance

- Important Facts and Formulae
- Model Questions

Level-I

Level-II

Level-III

Quantity based questions

Data sufficiency questions

QUEA-279  
QUEA-280  
QUEA-280  
QUEA-284  
QUEA-287  
QUEA-290  
QUEA-291

### 12. Simple Interest

- Important Facts and Formulae
- Model Questions

QUEA-311  
QUEA-

### 16. Probability

- Important Facts and Formulae
- Model Questions

Level-I

Level-II

Level-III

Quantity based questions

QUEA-387  
QUEA-388  
QUEA-388  
QUEA-392  
QUEA-395  
QUEA-398

### 17. Equations and Inequalities

- Important Facts and Formulae
- Model Questions

QUEA-416  
QUEA-419  
QUEA-419

# 1

# Number System

## IMPORTANT FACTS AND FORMULAE

Numbers are collection of certain symbols or figures called digits. The most common Number System in use is Decimal System which has ten symbols, each representing a digit. These are 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9. A combination of these figures representing a number is called a numeral.

We also have Binary Number System. It uses only 0 and 1. There are other Number Systems

It may be noted that zero is a to this even-odd classification.

**Cyclic Number** : It is such an integer which on being multiplied by any 1 through  $n$ , gives a product containing  $n$  digits as in the original number and its are in the same cyclic order in the

For example, 142857

$$2 \times 142857 = 285714; \quad 3 \times 1428$$

$$4 \times 142857 = 571428; \quad 5 \times 1428$$

**Fraction** : A fraction is a number represents a ratio or division of two integers

expressed in the form  $\frac{p}{q}$  where 'p' and 'q' are integers.

'p' is called the numerator and 'q' is called the denominator.

A fraction cannot have zero as denominator since division by zero is not defined.

**Zero divided by any integer is zero.**

A fraction with 1 as the denominator is called a whole number which is the same as the whole number which is the numerator.

For example,  $\frac{8}{1}$  is 8;  $\frac{0}{5}$  is 0.

**Real Number** : Any measurement in the physical world gives some magnitude or value or number. This number is called a Real Number. It consists of two groups: (i) Rational Number and (ii) Irrational Number.

**(i) Rational Number** : A rational number can always be represented by a fraction  $\frac{p}{q}$ , where  $p$  and  $q$  are integers and  $q \neq 0$ .

where  $p$  and  $q$  are integers and  $q \neq 0$ . All integers and fractions are rational numbers.

For example,  $2\left(\frac{2}{1}\right), \frac{2}{3}, \frac{4}{5}, \frac{6}{13}$

**(ii) Irrational Number** : An irrational number is a real number which cannot be expressed in the form of a fraction  $\frac{p}{q}$ , where  $p$  and  $q$  are integers and  $q \neq 0$ .

For example,  $\sqrt{3}, \sqrt{2}$ . It gives only an approximate answer in the form of a fraction or decimal number. The digits after the decimal point are non-ending. The same holds true for  $\pi$ , which again is irrational. Alternately, it can be expressed as a decimal number.

For example,  $\sqrt{3}, \sqrt{2}$ . It gives only an approximate answer in the form of a fraction or decimal number. The digits after the decimal point are non-ending. The same holds true for  $\pi$ , which again is irrational. Alternately, it can be expressed as a decimal number.



the remainder in the same manner. We continue this till division has been performed by all the divisors.

The last quotient obtained is the quotient we would have obtained by performing actual division by all the divisors.

The actual remainder will be given by different relations depending on the number of divisors.

Let us denote divisor by  $D$  and remainder

When division is by 2-divisors :  $R = R_1 + D_1$

When division is by 3-divisors :

$$R = R_1 + (R_2 \times D_1) + (R_3 \times D_1 \times D_2)$$

When division is by 4-divisors :

$$R = R_1 + (R_2 \times D_1) + (R_3 \times D_1 \times D_2) + (R_4 \times D_1 \times D_2 \times D_3)$$

In a similar way, we can write the remainder for division by any number of divisors.

Now let us try some examples using the method.

**Examples :**

$$(a) \frac{3567}{2 \times 9}$$

| Divisors            | 3567 | Remainders         |
|---------------------|------|--------------------|
| $D_1 \rightarrow 2$ | 1783 | $1 \leftarrow R_1$ |
| $D_2 \rightarrow 9$ | 198  | $1 \leftarrow R_2$ |

**Answer :**

Quotient,  $Q = 198$

Remainder,  $R = R_1 + (R_2 \times D_1) = 1 + (1 \times 2)$

$$(b) \frac{35367}{2 \times 3 \times 6}$$

| Divisors            | 35367 | Remainders         |
|---------------------|-------|--------------------|
| $D_1 \rightarrow 2$ | 17683 | $1 \leftarrow R_1$ |
| $D_2 \rightarrow 3$ | 5894  | $1 \leftarrow R_2$ |
| $D_3 \rightarrow 6$ | 982   | $2 \leftarrow R_3$ |

Quotient,  $Q = 982$

Remainder,  $R = R_1 + (R_2 \times D_1) + (R_3 \times D_1 \times D_2)$   
 $= 1 + (1 \times 2) + (2 \times 2 \times 3) = 15$

$$(c) \frac{7325}{2 \times 4 \times 9}$$

| Divisors            | 7325 | Remainders         |
|---------------------|------|--------------------|
| $D_1 \rightarrow 2$ | 3662 | $1 \leftarrow R_1$ |
| $D_2 \rightarrow 4$ | 915  | $2 \leftarrow R_2$ |
| $D_3 \rightarrow 9$ | 101  | $6 \leftarrow R_3$ |

$Q = 101$

## LEVEL-I

- Unit digit in  $13^{2003}$  is  
 (1) 1 (2) 3 (3) 7  
 (5) None of these
- If  $(2^a - 1) = 68$  a 19476735, w  
 digit, then the value of a is  
 (1) 7 (2) 3  
 (3) 5 (4) 8  
 (5) None of these
- Given the numbers :  $2^{5555}, 3^{3333}$   
 can be written in ascending o  
 (1)  $2^{5555}, 3^{3333}, 6^{2222}$  (2)  $3^{3333}, 2^{5555}, 6^{2222}$   
 (3)  $2^{5555}, 6^{2222}, 3^{3333}$  (4)  $6^{2222}, 2^{5555}, 3^{3333}$   
 (5) None of these
- The prime number 1999 can be v  
 $- b^2$ , where  $a$  and  $b$  are natura  
 Then the value of  $a^2 + b^2$  is  
 (1) 1998000 (2) 19980  
 (3) 1999000 (4) 19990  
 (5) None of these
- The number  $25^{64} \times 64^{25}$  is the  
 natural number  $n$ . The sum of d  
 (1) 7 (2) 14 (3) 21  
 (5) None of these
- The number  $(2^{48} - 1)$  is exactly  
 two numbers between 60 and 7  
 bers are :  
 (1) 63 and 65 (2) 63 an  
 (3) 61 and 65 (4) 65 an  
 (5) None of these
- How many 9's are the  
 $99999899999^2$ ?  
 (1) 7 (2) 9 (3) 11  
 (5) None of these
- In a farm there are cows and b  
 are counted they are 180, if leg  
 they are 420. The number of cov  
 is  
 (1) 130 (2) 150 (3) 50  
 (5) 35
- For numbers  $a$  and  $b$ , define \* :  
 $b) \div (a - b)$ . Then the value of (  
 (1)  $-\frac{2}{3}$  (2)  $-\frac{1}{5}$  (3) 0  
 (5) None of these

# MOD

## LEVEL-I

- Unit digit in  $13^{2003}$  is  
(1) 1 (2) 3 (3) 7 (4) 9  
(5) None of these
- If  $(2^{30} - 1) = 68n19470735$ , where  $n$  is a natural number, then the value of  $n$  is  
(1) 7 (2) 3  
(3) 5 (4) 8  
(5) None of these
- Given the numbers :  $2^{5555}, 3^{3333}, 6^{2222}$ . They can be written in ascending order as  
(1)  $2^{5555}, 3^{3333}, 6^{2222}$  (2)  $3^{3333}, 2^{5555}, 6^{2222}$   
(3)  $2^{5555}, 6^{2222}, 3^{3333}$  (4)  $6^{2222}, 2^{5555}, 3^{3333}$   
(5) None of these
- The prime number 1999 can be written as  $a^2 - b^2$ , where  $a$  and  $b$  are natural numbers. Then the value of  $a^2 + b^2$  is  
(1) 1998000 (2) 1998001  
(3) 1999000 (4) 1999001  
(5) None of these
- The number  $25^{64} \times 64^{25}$  is the square of a natural number  $n$ . The sum of digits of  $n$  is  
(1) 7 (2) 14 (3) 21 (4) 28  
(5) None of these
- The number  $(2^{48} - 1)$  is exactly divisible by two numbers between 60 and 70. The numbers are :  
(1) 63 and 65 (2) 63 and 67  
(3) 61 and 65 (4) 65 and 67  
(5) None of these
- How many 9's are there in  $999998999999^2$ ?  
(1) 7 (2) 9 (3) 11 (4) 13  
(5) None of these
- In a farm there are cows and hens. If heads are counted they are 180, if legs are counted they are 420. The number of cows in the farm is  
(1) 130 (2) 150 (3) 50 (4) 30  
(5) 35
- For numbers  $a$  and  $b$ , define  $\star$  as  $a \star b = (a + b) \div (a - b)$ . Then the value of  $(1 \star 2) \star 3$  is  
(1)  $-\frac{2}{3}$  (2)  $-\frac{1}{5}$  (3) 0 (4)  $\frac{1}{2}$   
(5) None of these

31. The difference between the sum of consecutive odd numbers and the sum of consecutive even numbers together is 5. The smallest even number is 5 more than the smallest odd number. What is the smallest odd number and the smallest even number?  
(1) 75 (2) 77 (3) 79  
(4) Cannot be determined
32. Sum of square of first  $n$  natural numbers and square of second number together is 10. The value of the second number is 8. What is the value of the first number? (assuming  $n$  and the second number are positive)  
(1) 18 (2) 8 (3) 10  
(5) None of these
33. The unit digit in the product of 123456789 is  
(1) 7 (2) 3 (3) 9  
(5) 2
34. There are four consecutive powers of 2 and four consecutive powers of 3. The sum of the highest power of 2 and highest odd number is 10. What is the sum of all the four consecutive powers?  
(1) 94 (2) 108 (3) 88  
(5) 120
35. Two-third of the first number is equal to the cube of the second number. The sum of the first and second number is equal to twelve percent of the sum of the first and second number.  
(1) 330 (2) 360 (3) 390  
(5) None of these
36. How many natural numbers between 134 and 467 which are divisible by 7?  
(1) 46 (2) 47 (3) 48  
(5) None of these
37. In a three digit number the digit in the tens place is twice the digit in the units place and the digit in the hundreds place is 1.5 times the digit in the tens place. The sum of all the three digits is 13, what is the number?  
(1) 364 (2) 436 (3) 234  
(5) None of these
38. If a number of two digits is  $k$  times the sum of its digits, then the number obtained by interchanging the digits is the original number multiplied by:  
(1)  $9 + k$  (2)  $10 + k$  (3)  $11 + k$   
(5) None of these

the square of the average of  
en even numbers, what is t  
number?

- (1) 11      (2) 13      (3) 17  
(5) 9

8. The ten's digit of a three digit  
If the digits of X are inter  
number thus formed is 396  
previous one. The sum of un  
dred digit is 14, then what is  
(1) 480      (2) 539      (3) 59  
(5) Other than those given a

9. The product of digits of a 2-  
is 16. When the digits of X a  
resultant number is 54 more  
the value of 75% of X?  
(1) 15      (2) 20      (3) 21  
(5) 61.5

10. The average of a series of fo  
even numbers ( $S_1$ ) is 31. If t  
ber of another series of five  
numbers ( $S_2$ ) is 13 less than  
number of  $S_1$ , what is the dif  
the highest number of  $S_1$  an  
(1) 7      (2) 13      (3) 5  
(5) 11

11. Father's age is 30 years mor  
age. Ten years hence, the f  
become three times the son  
What is son's present age (t  
(1) 8      (2) 7      (3) 5  
(4) Cannot be determined  
(5) None of these

12. When positive numbers a, b, c  
vided by 13, the remainders ar  
respectively. What will be the re  
(a + 2b + 5c) is divided by 13 ?  
(1) 8      (2) 9      (3) 5  
(5) 7

13. The numerator of a fraction i  
its denominator. If the num  
creased by 1 and the denon  
creased by 5, then the denomi  
4 times the numerator. Find t

- (1)  $\frac{5}{11}$       (2)  $\frac{3}{11}$       (3)  $\frac{4}{11}$   
(5)  $\frac{6}{11}$

**LEVEL-**

1. There are three perfect secutive even numbers whose sum is 2360. What is the difference between the smallest and the largest numbers in the series?  
(1) 258 (2) 296 (3) 272  
(4) 280
2. The digit at unit's place of a number is increased by 100% and the digit at ten's place of the same number is increased by 50%. The new number thus formed is 1625 more than the original number. What is the original number?  
(1) 22 (2) 63 (3) 123  
(4) All of these (5) None of these
3. There are three positive integers. The sum of the first and the second number is 100, the sum of the second and the third number is 150, and the product of the second and the third number is 4320, then what is the sum of the first and the third numbers?  
(1) 120 (2) 177 (3) 135  
(4) 150
4. Five rectangular boxes of capacity of containing 75 switches each. Three square boxes of containing 45 switches each. Two screws fixed with it. How many screws are there in square boxes and rectangular boxes?  
(1) 4000 (2) 3800 (3) 3600  
(4) 4700
5. Which of the following is true?  
A.  $(99/101) < (97/99) < (95/97)$   
B.  $(95/97) < (97/99) < (99/101)$   
C.  $(95/97)^2 > (97/99)^2 > (99/101)^2$   
D.  $(99/101)^2 > (97/99)^2 > (95/97)^2$   
(1) Only A (2) Only B  
(3) B and C (4) B and D  
(5) None of these
6. A tin of oil was  $4/5$  full. When 4 bottles of oil were taken out and four more bottles of oil were poured into it, it was  $3/4$  full. How many bottles of oil can it contain?  
(1) 20 (2) 10 (3) 15  
(4) 30 (5) None of these
7. At the first stop on his round trip, a bus driver loaded  $2/5$  of the packages.



28. A child was asked to add numbers (that is  $1 + 2 + 3$ ) with patience permitted. As he added the sum as 575. When the child checked the result wrong, the child discovered that he missed one number in the addition. The no. he missed is :  
 (1) less than 10 (2) 10  
 (3) 15 (4) more than 15
29. Each family in a locality has at least one adult, and no family has more than two children. Considering all the families together there are more adults than boys and more girls than boys. Then the minimum possible number of families in the locality is :  
 (1) 4 (2) 5 (3) 2  
 (5) 6
30. The no. of employees in Obaidullah's firm is a Prime number and is less than 100. The ratio of the no. of employees in the firm to that of employees in the firm above, to that of employees in the firm below, is not, can possibly be—  
 (1) 101 : 88 (2) 88 : 101  
 (3) 110 : 111 (4) 88 : 110  
 (5) 97 : 84

### QUANTITY BASED QUESTIONS

**Directions (1-5) :** In each of the following questions, two quantities are given, solve each and choose the correct option as given below :

- (1) Quantity I > Quantity II  
 (2) Quantity I  $\geq$  Quantity II  
 (3) Quantity I < Quantity II  
 (4) Quantity II  $\geq$  Quantity I  
 (5) Quantity I = Quantity II or no relation can be established

1. **Quantity I :** Find the unit digit of  $12^{12}$ .  
**Quantity II :** Find the unit digit of  $12^{12}$ .

2. **Quantity I :** Find the unit digit of  $646 \times 252 \times 119 \times 218$ .

**Quantity II :** Find the unit digit of the product of all prime numbers upto 100.

3. **Quantity I :** If  $5^{2x-1} = \frac{1}{125^{(x-3)}}$

**Quantity II :** If  $2^{x-1} + 2^{x+1} = 4$ , find the value of  $x$ .

4. **Quantity I :** Find the remainder when  $10^{10}$  is divided by 15.

**Quantity II :** Find the remainder when  $10^{10}$  is divided by 14.

10. (4) Let quotient =  $Q$  and remainder =  $R$

$$\therefore \text{Divisor} = 10Q = 5R$$

$$\text{Now, } R = 46 \Rightarrow 10Q = 5 \times 46$$

$$\Rightarrow Q = \frac{5 \times 46}{10} = 23$$

$$\text{Now, } Q = 23, R = 46$$

$$\text{and divisor} = 5 \times 46 = 230$$

$$\text{Dividend} = \text{Divisor} \times \text{quotient} + \text{Remainder}$$

$$= 230 \times 23 + 46$$

$$= 5290 + 46 = 5336$$

11. (2) **Tricky approach**

A number is divisible by 33 if it is divisible by 3 and 11 both

For divisibility by 11,

$$\therefore 5 + B + 7 = A + B + A$$

$$\therefore 2A = 12 \Rightarrow A = 6$$

$\therefore$  For divisibility of 5 6 B B 7 6 by 3,  $B = 3$

$$\therefore \text{Number} = 563376$$

$$\therefore A + B = 6 + 3 = 9$$

12. (2) If the numbers be  $x$  and  $y$ , then  $x + y = 42$   
 $xy = 437$

$$(x - y)^2 = (x + y)^2 - 4xy$$

$$= (42)^2 - 4 \times 437$$

$$= 1764 - 1748 = 16$$

$$\Rightarrow x - y = \sqrt{16} = 4$$

13. (1) Number =  $x$

$$\therefore \frac{x}{2} + \frac{x}{3} + \frac{x}{4} = x + 12$$

$$\Rightarrow \frac{6x + 4x + 3x}{12} = x + 12$$

$$\Rightarrow \frac{13x}{12} = x + 12$$

$$\Rightarrow x + \frac{x}{12} = x + 12 \Rightarrow \frac{x}{12} = 12$$

$$\Rightarrow x = 12 \times 12 = 144$$

14. (3) Let the numbers be  $a$  and  $b$  where  $a > b$ .

According to the question,

$$a^3 + b^3 = 793$$

$$\text{and } a + b = 13$$

$$\therefore (a + b)^3 = a^3 + b^3 + 3ab(a + b)$$

$$\therefore (13)^3 = 793 + 3ab \times 13$$

$$2197 = 793 + 39ab$$

$$39ab = 2197 - 793 = 1404$$

$$ab = \frac{1404}{39} = 36$$

$$30. (2) \frac{3}{5} + \frac{4}{7} = \frac{21+20}{35} = \frac{41}{35} \text{ and}$$

$$\frac{4}{5} + \frac{3}{7} = \frac{28+15}{35} = \frac{43}{35}$$

31. (2) Let the consecutive odd numbers

$x, x+2, x+4$  and  $x+6$

$\therefore$  Largest even number

$$= x+11$$

$\therefore$  Other even numbers

$$= x+7 \text{ and } x+9$$

$$\therefore x + x + 2 + x + 4 + x$$

$$[x+7+x+9+x+11] = 20$$

$$\Rightarrow x - 15 = 20$$

$$\Rightarrow x = 15 + 20 = 35$$

$\therefore$  Required sum

$$= x + x + 7$$

$$= 2x + 7 = 2 \times 35 + 7 = 77$$

32. (3) Let the first number be  $x$  and the number be  $y$ .

$$\therefore y^2 = 8^2 - 15 = 64 - 15 = 49$$

$$\therefore y = 7$$

$$\therefore x^2 + 7^2 = 568$$

$$\Rightarrow x^2 + 343 = 568$$

$$\Rightarrow x^2 = 568 - 343 = 225$$

$$\therefore x = \sqrt{225} = 15$$

$$\therefore 15 \times \frac{3}{5} = 9$$

33. (1)  $7^1 = 7, 7^2 = 49, 7^3 = 343,$

$$7^4 = 2401, 7^5 = 16807$$

i.e. after index 4, the unit's digit is repeated

$\therefore$  On dividing 153 by 4, remainder = 1

$\therefore$  Unit's digit in the expansion of  $(2467)$

$7^1 = 7$  and unit's digit in the expansion

$$(341)^{72} = 1$$

$\therefore$  Required unit's digit

$$= 7 \times 1 = 7$$

34. (5) Suppose,

Consecutive odd numbers

$$\Rightarrow x, x+1, x+2, x+3$$

Consecutive even numbers

$$\Rightarrow y, y+1, y+2, y+3$$

According to the question,

$$x+3+y+3=33$$

$$\Rightarrow x+y=33-6=27 \dots (i)$$

$$\therefore x+x+1+x+2+x+3+y+y+1+y+2+y+3$$

$$= 4x+4y+12 = 4(x+y)+12$$

$$= 4 \times 27 + 12$$

$$= 108 + 12 = 120$$

No. of students above 10 years

$$= x - \frac{5x}{24} - \frac{x}{4}$$

$$= \frac{24x - 5x - 6x}{24} = \frac{13x}{24}$$

$\Rightarrow$  No. of students above 10 years = 260

$$\Rightarrow \frac{13x}{24} = 260$$

$$\Rightarrow x = \frac{24 \times 260}{13}$$

$$\Rightarrow x = 480$$

$$\therefore \text{No. of boys} = \frac{5 \times 480}{8} = 5 \times 60 = 300$$

3. (1) Sum of the present age of old family = 94 + 20 = 114 years

Father + Mother + Son + Daughter-in-law = 92 years

Required difference between daughter & daughter in law

$$= 114 - 92 = 22 \text{ years}$$

$$4. (4) (x^{2a})^b = (x)^{\frac{4b}{2c}}$$

$$\Rightarrow 2ab = \frac{2b}{c} \Rightarrow ac = 1$$

$a = c = 1$   $\therefore$  Both  $a$  and  $c$  are natural numbers] .....(i)

$$\frac{x^{4b}}{x^{3a}} = x^{2(a-b)} \cdot x^b$$

$$\Rightarrow x^{4b-3a} = x^{3a-2b}$$

$$\Rightarrow 4b - 3a = 3a - 2b$$

$$\Rightarrow 6b = 6a$$

$$\Rightarrow a = b$$

..... (ii)

From (i) and (ii) we have,

$$a = b = c$$

5. (3) Let the number of sweets be  $x$ .

According to question,

$$\frac{x}{495} - \frac{x}{891} = 4$$

$$\Rightarrow \frac{891x - 495x}{891 \times 495} = 4$$

$$\Rightarrow 396x = 4 \times 891 \times 495$$

$$\Rightarrow x = \frac{4 \times 891 \times 495}{396} = 4455$$

(3) Pratibha's present age =  $x$  years

$\therefore$  Amit's present age =  $3x$  years

$\therefore$  Father's present age

$$= \frac{3x \times 13}{9} = \frac{13x}{3} \text{ years}$$

7. (1) L

$$2x +$$

$$\therefore T$$

$$2x +$$

Acco

$$\left( \frac{2x}{\dots} \right)$$

$$- \left( \frac{2x}{\dots} \right)$$

$$= 50$$

$$\Rightarrow (2$$

$$\Rightarrow (2$$

$$\Rightarrow (4$$

$$\Rightarrow 4$$

$$\Rightarrow 4$$

$$\Rightarrow x$$

$$\therefore S$$

$$= 2x$$

8. (2) U

$$\therefore H$$

$$\therefore O$$

$$= (14$$

$$= 14$$

$$= 14$$

New

$$= 10$$

$$= 99$$

$$\therefore 99$$

$$\Rightarrow 19$$

$$\Rightarrow 19$$

$$\Rightarrow x$$

$$\therefore H$$

$$= 14$$

$$\therefore R$$

$$\therefore bc = 77$$

$$\therefore bcd = 1001$$

$$\therefore d = \frac{bcd}{bc} = \frac{1001}{77} = 13$$

$$10. (4) \quad \begin{array}{r} 5 \quad A \quad 7 \\ 3 \quad 3 \quad 5 \\ \hline 8 \quad B \quad 2 \end{array}$$

8B12 is exactly divisible by 3.

$$\therefore 8 + B + 2 = \text{multiple of 3}$$

$$\therefore B = 2 \text{ or } 5 \text{ or } 8$$

$$\therefore A + 1 + 3 = 8$$

$$\therefore A = 4$$

17. (1) Let  $x$  tickets of each value be bought.

$$\therefore 45x + 75x + 95x = 3870$$

$$\Rightarrow 215x = 3870$$

$$\Rightarrow x = \frac{3870}{215} = 18$$

18. (3)  $1\frac{2}{7} = \frac{9}{7} \rightarrow$  Present age of Farah  
 $\frac{2}{7} \rightarrow$  Age of the time of marriage of Farah

$$\text{Difference} = 9 - 7 = 2$$

$$\therefore 2 \text{ unit} = 8 \text{ years}$$

$$\therefore 1 \text{ unit} = 4 \text{ years}$$

$$\therefore \text{Present age of Farah}$$

$$= 9 \text{ unit} = 9 \times 4 = 36 \text{ years}$$

Now, present age of her daughter

$$= \frac{1}{6} \times 36 = 6 \text{ years}$$

$$\therefore \text{Daughter's age 3 years ago}$$

$$= 6 - 3 = 3 \text{ years.}$$

$\Rightarrow$  Option (3) is right.

19. (4) Number of seats in each bus = 10 (let)

$$\text{Total passengers} = \frac{30 \times 4}{5} = 24$$

$$\frac{1}{4} \text{ of the passengers leave the bus.}$$

Remaining passengers

$$= 24 \times \frac{3}{4} = 18$$

$$\therefore \text{Required answer} = \frac{18}{20} = \frac{9}{10}$$

20. (2) Let the three fractions be  $p$ ,  $q$  and  $r$ , where  $p < q < r$ .

According to the question,

$$\frac{r}{p} = \frac{7}{6} \Rightarrow r = \frac{7}{6}p$$

Again, middle fraction

21. (

22. (1

23. (1) C

be x.

$\therefore N$

Acco

$$5 \frac{9}{12}$$

$$= 5 \frac{30}{48}$$

$$= 5 \frac{60}{96}$$

$$\Rightarrow \frac{270}{12}$$

$$= 48$$

$$\Rightarrow 69$$

$$= 837$$

$$\Rightarrow x =$$

24. (2) Let

Numb

Case

$$x +$$

$$\begin{array}{r|l} 29. (4) & 1. \ 2 \mid 72 \\ & 2 \mid 36 \\ & 2 \mid 18 \\ & 3 \mid 9 \\ & 3 \end{array}$$

$$\therefore 72 = 2^3 \times 3^2$$

$$\therefore \text{Number of factors} = (3 + 1)(2 + 1) = 12$$

$$\text{II. Sum of first } n \text{ odd numbers} \\ = n^2$$

$$\therefore \text{Sum of first 20 odd numbers} = 20^2 = 400$$

**Look :**

$$1 + 3 = 4 = 2^2;$$

$$1 + 3 + 5 = 9 = 3^2;$$

$$1 + 3 + 5 + 7 = 16 = 4^2.$$

III. 97 is the largest 2-digit prime number.

$$30. (3) A = 2^{32} = (2^8)^4 = (65536)^2$$

$$B = 2^{31} + 2^{30} + 2^{29} + \dots + 2^0$$

$$= 1 + 2 + 2^2 + \dots + 2^{31}$$

It is a geometric progression.

First term  $a = 1$ ; Common ratio  $r = 2$

Number of terms = 32

$$= \frac{1(2^{32} - 1)}{2 - 1}$$

$$\left[ S_n = \frac{a(r^n - 1)}{r - 1} \right]$$

$$= 2^{32} - 1$$

$$C = 3^0 + 3^1 + 3^2 + \dots + 3^{15}$$

It is a geometric series.

$$a = 1; r = 3, n = 16$$

$$\therefore C = \frac{1(3^{16} - 1)}{2} = \frac{(3^4)^4 - 1}{2}$$

$$= \frac{(81)^4 - 1}{2} = \frac{(6561)^2 - 1}{2}$$

### LEVEL-III

$$1. (4) x^2 + (x + 2)^2 + (x + 4)^2$$

$$= 2360$$

$$\Rightarrow x^2 + x^2 + 4 + 4x + x^2 + 16 + 8x = 2360$$

$$\Rightarrow 3x^2 + 12x + 20 = 2360$$

$$\Rightarrow 3x^2 + 12x - 2340 = 0$$

$$\Rightarrow x^2 + 4x - 780 = 0$$

$$\Rightarrow x^2 + 30x - 26x - 780 = 0$$

$$\Rightarrow x(x + 30) - 26(x + 30) = 0$$

$$\Rightarrow (x + 30)(x - 26) = 0$$

$$\Rightarrow x = -30 \text{ or } 26$$



17. (3) The numbers are

$x, x + 6, x + 12, x + 18, x + 24$  and  $(x + 30)$ .

Average

$$= \frac{\{x + (x + 6) + (x + 12) + (x + 18) + (x + 24) + (x + 30)\}}{6}$$

$$= 48$$

$$\Rightarrow \frac{6x + 90}{6} = 48$$

$$\Rightarrow x + 15 = 48$$

$$\Rightarrow x = 48 - 15 = 33$$

$$\text{Sixth number} = 33 + 30 = 63$$

So next three consecutive numbers = 69, 75 and 81

$$\text{Required average} = \frac{(69 + 75 + 81)}{3} = \frac{225}{3} = 75$$

$$18. (1) \text{ Remainder of } \frac{2^{256}}{17} = \frac{(2^4)^{64}}{17} = \frac{16^{64}}{17} = \frac{16^{64}}{16 + 1}$$

$\frac{a^n}{a+1}$  gives remainder 1, when  $n$  is even.

Hence remainder of  $\frac{16^{64}}{16+1}$  will be 1, since  $n = 64$  (even).

**Method-2 :** Remainder of  $\frac{2}{17} \rightarrow 2$

$$\text{Remainder of } \frac{2^2}{17} \rightarrow 4$$

$$\text{Remainder of } \frac{2^3}{17} \rightarrow 8$$

$$\text{Remainder of } \frac{2^4}{17} \rightarrow 16$$

$$\text{Remainder of } \frac{2^5}{17} \rightarrow 15$$

$$\text{Remainder of } \frac{2^6}{17} \rightarrow 13$$

$$\text{Remainder of } \frac{2^7}{17} \rightarrow 9$$

$$\text{Remainder of } \frac{2^8}{17} \rightarrow 1$$

Cyclic

hence

as rem

19. (2) The

Let no.

To 1st

Diamor

give  $\frac{18}{2}$

To thir

hence

Option

1st will

2nd will

3rd will

Thief is

also giv

correct

**Method**

before h

After gi

the thie

$$x - \left[ \frac{1}{2}x \right]$$

i.e. After

is left w

monds b

$$y - \left[ \frac{1}{2}y + 2 \right]$$

to 1st wat

monds, te.

The thief c

20. (3) Rem. of

$$\text{Rem. of } \frac{14}{1}$$

27. (4)

|   |                   |
|---|-------------------|
| 5 | N                 |
| 7 | $A \rightarrow 2$ |
| 9 | $B \rightarrow 3$ |
| 1 | $\rightarrow 4$   |

$$\therefore B = 9 \times 1 + 4 = 13$$

$$A = 7 \times 13 + 3 = 94$$

$$N = 5 \times 94 + 2 = 472$$

$$\therefore \text{Sum of the digits} = 4 + 7 + 2$$

28. (4) We know the sum of first 'n'

bers is equal to  $\frac{n(n+1)}{2}$ .

Let us try some numbers,

If  $n = 25$

$$\rightarrow \text{Sum would be } \frac{(25)(25+1)}{2} = 2$$

But  $n$  should be near 575; let's try a higher value.

$$\text{If } n = 35 \rightarrow \text{sum} = \frac{35 \times 36}{2} = 35$$

Try for a value less than 35. If  $n =$

$$\rightarrow \text{Sum} = \frac{30 \times 31}{2} = 15 \times 31 = 465$$

If  $n = 31$  sum would be  $465 + 31 =$

If  $n = 32$  sum would be  $496 + 32 =$

If  $n = 33$  sum would be  $528 + 33 =$

If  $n = 34$  sum would be  $461 + 34 =$

Sum must be greater than 575 and

has missed one number. Sum of

no. greater than 575 is 595, hence

can be  $595 - 575 = 20$  (this is the

29. (4) Starting from the least number, consider option (3) first, i.e. number 2.

We are given that no. of adults > no. of girls > no. of families.

If no. of families = 2  $\Rightarrow$  no. of girls

then no. of boys is at least 4 and

is at least 5, but in two families the

max. 4 adults, hence option (3) is

Now let no. of families = 3, then

at least 4, no. of boys is at least 5

adults is at least 6.

In 3 families we can have maximum

hence minimum no. of families is

(5) Here, employees who are greater

above plus employees who are not

all the employees of the company

n to us to be a prime no. less

sum of ratios should be a prime

# 2

## IMPO

The simplification of  
ing a number of fractions  
one fraction by performing  
operations namely, addition  
plication and division etc.  
enlarge our horizon of study  
the discussion of various re  
and worked out problems

**Fraction** : A fraction

number written in the form  
are integers and  $n \neq 0$ . The  
known as **denominator** the  
and the upper number 'm'  
**tor**.

**Improper Fraction** :  
numerator is greater than o  
nator, is called improper fr

For example,  $\frac{5}{4}, \frac{8}{7}, \frac{17}{7}$  etc

**Proper Fraction** : The  
merator is less than the de  
proper fraction.

For example,  $\frac{1}{2}, \frac{1}{3}, \frac{3}{7}$  etc

**Decimal Fraction** : Th  
nominator is 10 and its high  
decimal fraction.

For example,  $\frac{2}{10}, \frac{7}{1000}, \frac{1}{10}$

They can be expressed  
0.007, 0.0017

**Complex Fraction** : A com  
fraction in which numerator &  
both are fractions.

For example,  $\frac{\frac{2}{3} + \frac{1}{5}}{\frac{3}{4} + \frac{6}{10}}$

**Continued Fraction** : Frac  
tain addition fraction or a series

**Note :** You should know the common square roots. ( Approximate places)

|                     |                     |                   |
|---------------------|---------------------|-------------------|
| $\sqrt{0} = 0$      | $\sqrt{1} = 1$      | $\sqrt{2} = 1.4$  |
| $\sqrt{4} = 2$      | $\sqrt{5} = 2.236$  | $\sqrt{6} = 2.4$  |
| $\sqrt{9} = 3$      | $\sqrt{10} = 3.162$ | $\sqrt{11} = 3.3$ |
| $\sqrt{14} = 3.742$ | $\sqrt{15} = 3.873$ | $\sqrt{17} = 4.1$ |
| $\sqrt{21} = 4.583$ | $\sqrt{22} = 4.690$ | $\sqrt{23} = 4.7$ |
| $\sqrt{36} = 6$     | $\sqrt{49} = 7$     | $\sqrt{64} = 8$   |
| $\sqrt{100} = 10$   | $\sqrt{121} = 11$   | $\sqrt{144} = 12$ |

\* : Important

Square root of a number greater than 1 but less than 100 consists of even number of digits.

Square root of a number greater than 100 but less than 10000 consists of odd number of digits.

In general, if the given number has  $n$  digits, its square root will have  $\frac{n}{2}$  digits when  $n$  is even.

and  $\frac{n+1}{2}$  digits when  $n$  is odd. This holds true for the case of pure decimal fractions too.

**Some Properties of Exact Square Roots (When Square Roots are Whole Numbers)**

I. A pure square number ending in 1 or 9 as the last digit in its square root.

For example,  $\sqrt{81} = 9$ ;  $\sqrt{121} = 11$

II. If a square ends in 4, its square root ends in 2 or 8 as the last digit.

For example,  $\sqrt{64} = 8$ ;  $\sqrt{144} = 12$

III. If a square ends in 5 or 00, its square root ends in 5 or 0 respectively as the last digit.

For example,  $\sqrt{625} = 25$ ;  $\sqrt{225} = 15$

$\sqrt{100} = 10$ ;  $\sqrt{400} = 20$

IV. A square ending in 6 has 4 or 6 as the last digit in its square root.

For example,  $\sqrt{256} = 16$ ;  $\sqrt{196} = 14$

V. A square ending in 9 has 3 or 7 as the last digit in its square root.

For example,  $\sqrt{169} = 13$ ;  $\sqrt{729} = 27$

#### Method B : By Division

It is the most general method of finding square roots and is applicable to all cases.

I. Mark off groups of two-digits, starting from the right. The extreme group may be either a single digit or a pair.

$$= 1 = a^0$$

$$= a^m \cdot a = a^{m+1}$$

**Case III:** When  $m < n$

$$\text{In this case } \frac{a^m}{a^n} = \frac{a \times a \times a \dots \text{to } m \text{ factors}}{a \times a \times a \dots \text{to } n \text{ factors}}$$

$$= \frac{1}{a \times a \times a \dots \text{to } (n-m) \text{ factors}}$$

$$= \frac{1}{a^{n-m}} = a^{-(n-m)} = a^{m-n}$$

$$\text{Therefore } \frac{a^m}{a^n} = a^{m-n} \text{ whether } m > n \text{ or } m = n \text{ or } m < n$$

For example, (i)  $3^3 \div 3^2 = 3^{3-2} = 3$

$$(ii) \left(\frac{2}{5}\right)^5 \div \left(\frac{2}{5}\right)^2 = \left(\frac{2}{5}\right)^{5-2} = \left(\frac{2}{5}\right)^3$$

**3rd Law:** If  $a$  is any real number and  $m, n$  are positive integers, then

$$(a^m)^n = a^{mn} = (a^n)^m$$

**Proof:**  $(a^m)^n = a^m \times a^m \times a^m \dots$  to 'n' factors

$$= (a \times a \times a \dots \text{to } m \text{ factors}) \times (a \times a \times a \dots \text{to } m \text{ factors}) \times$$

$$(a \times a \times a \dots \text{to } m \text{ factors}) \dots \text{to } n \text{ factors}$$

$$= a \times a \times a \dots \text{to } (mn) \text{ factors}$$

$$= a^{mn}$$

$$\text{Similarly, } (a^n)^m = a^{nm}$$

**Corollary:**  $((a^m)^n)^p = (a^{mn})^p = a^{mnp}$  and so on.

**Conversely,**  $a^{mnp} = ((a^m)^n)^p$

$$\text{For example, (i) } (2^3)^5 = 2^{15}$$

$$(ii) ((3^4)^5)^2 = 3^{4 \times 5 \times 2} = 3^{40}$$

**4th Law:** If  $a, b$  are real numbers and  $m, n$  are positive integers, then

$$(i) (ab)^n = a^n b^n \quad (ii) \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}, b \neq 0$$

**Proof:** (i)  $(ab)^n = (ab) \times (ab) \times \dots$  to  $n$  factors

$$= (a \times a \times a \dots \text{to } n \text{ factors}) \times (b \times b \times b \times \dots \text{to } n \text{ factors})$$

$$= a^n \times b^n = a^n b^n$$

$$(ii) \left(\frac{a}{b}\right)^n = \left(\frac{a}{b}\right) \times \left(\frac{a}{b}\right) \times \left(\frac{a}{b}\right) \times \dots \text{to } n \text{ factors}$$

$$= \frac{a \times a \times a \dots \text{to } n \text{ factors}}{b \times b \times b \times \dots \text{to } n \text{ factors}} = \frac{a^n}{b^n}$$

For example, (i)  $6^5 = (2 \times 3)^5 = 2^5 \times 3^5$

$$(ii) \left(\frac{2}{3}\right)^4 = \frac{2^4}{3^4} = \frac{16}{81}$$



## MODEL QUESTIONS

## LEVEL-1

**Directions (1-140) :** What will come in place of the question mark (?) in the following equations?

1.  $(\sqrt{7\%} \text{ of } \sqrt{1764} \times 5) = 149.8 - 112$

- (1)  $\sqrt{18}$  (2) 18 (3) 324 (4) 24

(5) None of these

2.  $\sqrt{53124} \times \sqrt{5601} - 832^2 = (?)^2 + (37)^2$

- (1) 37 (2) 33 (3) 34 (4) 28

(5) None of these

3.  $(\sqrt{3} - 2)^2 = 9 - \sqrt{12} - \sqrt{36}$

- (1)  $13 - 2\sqrt{3}$  (2) 13

- (3)  $13 - 4\sqrt{3}$  (4) 1

11.  $64.5\% \text{ of } 800 + 36.4\% \text{ of } 1500 = (?)^2 + 38$

- (1) 32 (2) 38 (3) 42 (4) 48

(5) 34

12.  $348 \div 29 \times 15 + 156 = (?)^3 + 120$

- (1) 12 (2) 6 (3) 36 (4) 9

(5) None of these

13.  $(4 \times 4)^3 + (512 \div 8)^3 \times (32 \times 8)^4 = (2 \times 2)^{?+1}$

- (1) 8 (2) 12 (3) 6 (4) 14

(5) None of these

14.  $14 \times 627 \div \sqrt{1089} = (?)^3 + 141$

- (1)  $5\sqrt{5}$  (2)  $(125)^3$  (3) 25 (4) 5

(5) None of these

15.  $(80 \times 0.40)^3 \div (40 \times 1.6)^3 \times (128)^3 = 2^{?+7}$

- (1) 25 (2) 11 (3) 12 (4) 18

(5) None of these

## SIMPLIFICATION

23.  $45\% \text{ of } \sqrt{2025} \div 0.01 = (?)^3 + 25$

- (1) 3 (2)  $(81)^3$  (3) 225 (4) 9

(5) 12

24.  $3^{3.5} \times 21^2 \times 42^{2.5} \div 2^{2.5} \times 7^{3.5} = 21^?$

- (1) 8 (2) 10 (3) 12.5 (4) 6.5

(5) None of these

25.  $7072 \div (16\% \text{ of } 894) = 30 \times 1\frac{1}{12} \text{ of } (?) + 39$

- (1) 60 (2) 62 (3) 65 (4) 55

(5) 67

26.  $(562.5 \times 6)^6 \div (135 \div 9)^{10} + (37.5 \times 6)^7 = (3.75 \times 4)^{?+6}$

- (1) 0 (2) 2 (3) 3 (4) 4

(5) 5

27.  $(13 + 2\sqrt{5})^2 = ?\sqrt{5} + 189$

- (1) 26 (2) 25 (3) 52 (4) 130

(5) None of these

28.  $(7.9\% \text{ of } 134) - (3.4\% \text{ of } 79) = ?$

35. If a number is subtracted by two-third of 75 per cent of 500, the value so obtained is 320. What is the number?

- (1) 360 (2) 620 (3) 720 (4) 500

(5) None of these

36. Unit's digit of  $(7^{7^7} + 777^{777})$  is

- (1) 0 (2) 4 (3) 1 (4) 7

(5) None of these

37. The largest of 2 that divides  $12^{2008} + 10^{2008}$  is

- (1)  $2^{2008}$  (2)  $2^{2009}$  (3)  $2^{2010}$  (4)  $2^{2011}$

(5) None of these

38.  $0.34\overline{57} - 0.13\overline{33} = ?$

- (1) 0.48 (2)  $0.48\overline{1}$  (3)  $0.48$  (4)  $4.8$

(5) None of these

**Directions :** What will come in place of the question mark (?) in the following questions?



**Directions (45-46) :** What will come in place of the question mark (?) in the following questions?

45.  $\sqrt{7^2 \times 6 - (15^2)} + 10 = ?$

- (1) 23 (2)  $(519)^{1/2}$  (3) 519 (4) 529  
(5) None of these

46.  $3\frac{1}{4} - 2\frac{1}{2} - 1\frac{5}{6} = \frac{77^2}{10} + 1\frac{5}{12}$

- (1) 25 (2)  $\sqrt{5}$  (3) 625 (4) 15  
(5) 5

**Directions (47-48) :** What should come in place of the question mark (?) in the following questions?

47.  $(0.49)^4 \times (0.343)^4 + (0.2401)^4 = (70 + 100)^{7-3}$

- (1) 3 (2) 1 (3) 4 (4) 7  
(5) None of these

48.  $\sqrt{13^2 + 28} = 4 + (3)^2 - 107 = (?)^2$

- (1) 2

11. A General of Army wants to form a square from 26562 armies. After arrangement, he found some armies left. How many armies were left?

- (1) 81 (2) 36 (3) 97 (4) 65  
(5) None of these

12. If  $a - \frac{1}{a-3} = 5$ , then the value of  $(a-3)^3 - \frac{1}{(a-3)^3}$  is

- (1) 5 (2) 7 (3) 2 (4) 14  
(5) 13

13. The simplified value of  $\frac{\sqrt{3}-\sqrt{2}}{\sqrt{12}-\sqrt{18}} - \frac{1}{3} \times \sqrt{27}$

- $-\frac{1}{2} \times \sqrt[3]{27}$  is closest to

- (1)  $(\sqrt{2}-1)$

present ages of all of them is 125 years. What is the difference between the Raman's daughter's present age and Raman's mother's present age?

- (1) 45 years (2) 40 years  
(3) 50 years

(4) Cannot be determined

(5) None of these

18. On a school's Annual Day sweets were to be equally distributed amongst 112 children. But on that particular day, 32 children were absent. Thus the remaining children got 6 extra sweets. How many sweets was each child originally supposed to get?

- (1) 24 (2) 18 (3) 15  
(4) Cannot be determined  
(5) None of these

19. When all the students in a school are made to stand in rows of 54, 30 such rows are formed. If the students are made to stand in

**Directions (45-46) :** What will come in place of the question mark (?) in the following questions ?

45.  $\sqrt{7^2 \times 6 + (15^2)} + 10 = ?$

- (1) 23      (2)  $(519)^{1/2}$       (3) 519  
(5) None of these

46.  $3\frac{1}{4} + 2\frac{1}{2} - 1\frac{5}{6} = \frac{(?)^2}{10} + 1\frac{5}{12}$

- (1) 25      (2)  $\sqrt{5}$       (3) 625  
(5) 5

**Directions (47-48) :** What should come in place of the question mark (?) in the following questions?

47.  $(0.49)^4 \times (0.343)^4 \div (0.2401)^4 = (?)$

- (1) 3      (2) 1      (3) 4  
(5) None of these

48.  $\sqrt{13^2 + 28 + 4 - (3)^3} + 107 = (?)^2$

- (1) 2      (2) 16      (3) 256  
(5)  $(256)^2$

49. Select the missing number at the end of the series from the given responses.

$(123)^{25} \div (123)^7 = 123$

- (1) 0.5      (2) 1      (3) 2  
(5) None of these

50. Which one of the following will divide  $5^{71} + 5^{72} + 5^{73}$ ?

- (1) 150      (2) 160      (3) 155  
(5) None of these

## LEVEL-II

1. The value of  $\frac{(1.111)^3 + (2.212)^3}{(1.111)(2.212)(1.111 + 2.212)}$

between

- (1) -0.5 and 0      (2) 0 and 0.5  
(3) 0.5 and 1      (4) 2.5 and 3  
(5) None of these

2. If  $a^3b = abc = 180$ ,  $a$ ,  $b$ ,  $c$  are integers, then the value of  $c$  is

- (1) 110      (2) 1      (3) 4  
(5) None of these

3. In a division sum, the divisor is equal to the quotient and twice the remainder. If the dividend is 100, then the divisor and the remainder are respectively the divisor and the remainder.

(1)  $\frac{4b-a^2}{a} = 3$       (2)  $\frac{4b-2a}{a^2} = 3$

(3)  $(a+1)^2 = 4b$       (4)  $\frac{a(a+2)}{b} = 3$

- (5) None of these

24. There are some parrots in a forest. If the total number of legs of the parrots in the forest are 858, how many parrots are there in the forest?

- (1) 845 (2) 833  
(4) Cannot be determined  
(5) None of these

25. The bus fare for one person from Agra to Aligarh and return is Rs. 3.360. If the fare for the same places for one person by train is four times the bus fare for the same places. What is the fare for 2 persons travelling by train between Agra and Aligarh?

- (1) Rs. 3.360 (2) Rs. 3.440  
(3) Rs. 3.440 (4) Rs. 3.440  
(5) None of these

26. Each boy contributed rupees equal to the number of girls and each girl contributed rupees equal to the number of boys in a class of 60 students. If the total amount collected is Rs. 1600, how many boys are there in the class?

- (1) 30 (2) 25 (3) 20  
(4) Data Inadequate (5) None of these

27. A sum of three non-zero numbers is 14. The first number is  $p$  times the second number and  $p$  times the third number. The value of  $p$  is

- (1)  $\frac{11}{24}$  (2)  $\frac{24}{13}$  (3)  $\frac{13}{24}$   
(5) None of these

28. Raju runs 1250 metre on Monday. Other days he runs 15 km. How many kilometres will he run in a week (He does not run on Sunday)?

- (1) 12.5 km (2) 20 km  
(3) 8.5 km (4) 25 km  
(5) None of these

29. If  $N = 0.369369369369...$  and  $M = 0.531531531531...$ , then what is the value of

$$\left(\frac{1}{N}\right) + \left(\frac{1}{M}\right)?$$

- (1)  $\frac{11100}{2419}$  (2)  $\frac{111}{100}$  (3)  $\frac{1897}{3162}$

(5) None of these

22. If  $4 \oplus 3 \oplus 8 = 10$  and  $5 \oplus 18$ , stand for four fundamental operations of addition, multiplication, and division necessarily in this order then  $c \oplus 4 \oplus 2 \oplus d \oplus b \oplus 3$  is
- (1) 3 (2) 9 (3) 10  
(5) None of these
23. A number consists of 3 digits. 10. The middle digit is equal to the other two and the number is increased by 99 if its digits are reversed. The number is
- (1) 253 (2) 235 (3) 523  
(5) None of these
24. If  $4 = 10^{2m}$  and  $9 = 10^{2n}$ , then
- (1)  $10^{2m-2n}$  (2)  $10^{2m}$   
(3)  $10^{n-m+1}$  (4)  $10^{2n}$   
(5) None of these
25. If  $a \oplus b$  is defined as  $a^b + b^a$ , then
- I.  $2 \oplus x = 100$   
II.  $4 \oplus x = 145$   
III.  $3 \oplus x = 145$   
IV.  $6 \oplus x = 100$
- For which of the above, is  $x$  smallest?
- (1) I (2) II (3) III  
(5) None of these
26.  $m$  is the smallest positive integer for any integer  $n \leq m$ , the quantity  $11n - 5$  is positive. What is the value of  $m$ ?
- (1) 4 (2) 5 (3) 8  
(5) None of these
27. Number  $S$  is obtained by squaring the difference between the digits of a two digit number  $D$ . The difference between  $S$  and  $D$  is 27. The two digit number  $D$  is:
- (1) 24 (2) 54 (3) 34  
(5) None of these
28. A number  $x$  lies between 0 and 1. Which of the following is true?
- (1)  $x > \sqrt{x}$  (2)  $x > \frac{1}{x}$  (3)  $x^3 > x^2$   
(5) None of these
29. The no. of employees in Obelix Menzies is a Prime number and is less than 100. The ratio of the no. of employees who are males and above, to that of employees who are females and below, can possibly be—
- (1) 101 : 88 (2) 87 : 100  
(3) 110 : 111 (4) 85 : 98  
(5) 97 : 84

## EXPLANATIONS

## LEVEL-I

1. (5)  $\sqrt{1764} \times \sqrt{?}\%$

$= 149.8 - 112 = 37.8$

$\Rightarrow 42 \times \sqrt{?}\% = 37.8$

$\Rightarrow \sqrt{?}\% = \frac{37.8}{42}$

$\Rightarrow ?\% = \frac{37.8 \times 37.8}{42 \times 42}$

$\therefore ? = \frac{37.8 \times 37.8 \times 100}{42 \times 42} = 81$

2. (5)  $\sqrt{33124} \times \sqrt{2601} - (83)^2$

$= ?^2 + (37)^2$

$\Rightarrow 182 \times 51 - 6889 = ?^2 + 1369$

$\Rightarrow 2393 = ?^2 + 1369$

$\Rightarrow ?^2 = 2393 - 1369 = 1024$

$\Rightarrow ? = \sqrt{1024} = 32$

3. (1)  $(\sqrt{3} - 2)^2 = ? - \sqrt{12} - \sqrt{36}$

$\Rightarrow 3 + 4 - 4\sqrt{3} = ? - 2\sqrt{3} - 6$

$\Rightarrow 7 - 4\sqrt{3} = ? - 2\sqrt{3} - 6$

$\Rightarrow ? = 7 - 4\sqrt{3} + 2\sqrt{3} + 6$

$= 13 - 2\sqrt{3}$

4. (5)  $? = 0.001 - 0.001 \times \frac{1}{10} + 0.01$

$= 0.001 - 0.0001 + 0.01 = 0.0109$

5. (2)  $(0.4)^3 \times (0.4)^7$

$= (0.4)^2 \times (0.4)^4 \times 2$

$\Rightarrow (0.4)^{3+7} = (0.4)^2 \times (0.4)^8$

$\Rightarrow (0.4)^{10-8} = (0.4)^2$

$\Rightarrow ? = 2$

6. (4)  $(\sqrt{5} - 2)^2 = ? - \sqrt{80}$

$\Rightarrow 5 + 4 - 4\sqrt{5} = ? - 4\sqrt{5}$

$\Rightarrow 9 - 4\sqrt{5} = ? - 4\sqrt{5}$

$\therefore ? = 9$

7. (5)  $? = \frac{\sqrt{576}}{4^2} \times 7.4 + 7^3 - 231$

$= 11.1 + 343 - 231 = 123.1$

9. (

10. (

11. (1

12. (2)

13. (3)

[a<sup>m</sup>

14. (4)

15. (2)

$$25. (1) 7072 \div \left( \frac{884 \times 16}{100} \right)$$

$$= 30 \times \frac{13}{12} \times \frac{?}{39}$$

$$\Rightarrow 7072 \div 141.44 = \frac{5 \times ?}{6}$$

$$\Rightarrow 50 \times 6 = 5 \times ?$$

$$\Rightarrow ? = \frac{50 \times 6}{5} = 60$$

$$26. (1) (3375)^6 \div (15)^{10} \div (225)^7 = (15)^{? - 6}$$

$$\Rightarrow ((15)^3)^6 \div (15)^{10} \div (15^2)^7 = (15)^{? - 6}$$

$$\Rightarrow 15^{18-10-14} = (15)^{? - 6}$$

$$\Rightarrow -6 = ? - 6$$

$$\Rightarrow ? = 0$$

$$27. (3) (13 + 2\sqrt{5})^2 = ? \times \sqrt{5} + 189$$

$$\Rightarrow 169 + 20 + 2 \times 13 \times 2\sqrt{5}$$

$$= ? \times \sqrt{5} + 189$$

$$\Rightarrow 189 + 52 \times \sqrt{5}$$

$$= ? \times \sqrt{5} + 189$$

$$\Rightarrow ? = 52$$

$$28. (5) ? = \frac{134 \times 7.9}{100} - \frac{79 \times 3.4}{100}$$

$$= 10.586 - 2.686 = 7.9$$

$$29. (2) \frac{762 \times ?}{100} + \frac{568 \times 44}{100} = 524.24$$

$$\Rightarrow 762 \times ? + 24992 = 52424$$

$$\Rightarrow 762 \times ? = 52424 - 24992$$

$$= 27432$$

$$\Rightarrow ? = \frac{27432}{762} = 36$$

$$30. (4) \frac{400 \times 185}{100} + \frac{240 \times 35}{100} = \frac{1648 \times ?}{100}$$

$$\Rightarrow 74000 + 8400 = 1648 \times ?$$

$$\Rightarrow 82400 = 1648 \times ?$$

$$\therefore ? = \frac{82400}{1648} = 50$$

$$31. (4) \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \frac{1}{30} + \dots + \frac{1}{n(n+1)}$$

$$= \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \dots + \frac{1}{n(n+1)}$$

32.

33.

34.

35. (2)

36. (1)

 $7^2$  $7^3$  $7^4$  $7^5$  $= U$  $= U$  $\therefore U$ 

Unl

7 (R

 $\therefore R$ 

37. (2) 2

 $= 2^{20}$  $= 2$  $= 2^{200}$  $= 2^{200}$ 

38. (2) 0.346

$$= \frac{3467 -}{9900}$$



$$\begin{aligned}
 4. (3) \text{ Expression} &= 3011 \times 3012 \\
 &= 3011 (3011 + 1) \\
 &= (3011)^2 + 3011 \\
 \therefore \text{Required answer} &= 3011
 \end{aligned}$$

$$\begin{aligned}
 5. (2) \left(1 - \frac{1}{n+1}\right) &+ \left(1 - \frac{2}{n+1}\right) + \left(1 - \frac{3}{n+1}\right) \\
 &+ \dots + \left(1 - \frac{n}{n+1}\right) \\
 &= n - \left(\frac{1}{n+1} + \frac{2}{n+1} + \frac{3}{n+1} + \dots + \frac{n}{n+1}\right) \\
 &= n - \frac{1+2+3+\dots+n}{n+1} \\
 &= n - \frac{n(n+1)}{2(n+1)} = n - \frac{n}{2} = \frac{n}{2}
 \end{aligned}$$

$$6. (3) x = \sqrt{30 + \sqrt{30 + \sqrt{30 + \dots}}}$$

On squaring both sides,

$$x^2 = 30 + \sqrt{30 + \sqrt{30 + \sqrt{30 + \dots}}}$$

$$\Rightarrow x^2 = 30 + x$$

$$\Rightarrow x^2 - x - 30 = 0$$

$$\Rightarrow x^2 - 6x + 5x - 30 = 0$$

$$\Rightarrow x(x-6) + 5(x-6) = 0$$

$$\Rightarrow (x-6)(x+5) = 0$$

$$\Rightarrow x = 6 \text{ because } x \neq -5$$

$$7. (1) a = 64 \text{ and } b = 289$$

$$\therefore \sqrt{a} = \sqrt{64} = 8 \text{ and}$$

$$\sqrt{b} = \sqrt{289} = 17$$

$$\therefore \left(\sqrt{\sqrt{a} + \sqrt{b}} - \sqrt{\sqrt{b} - \sqrt{a}}\right)^2$$

$$= (\sqrt{8+17} - \sqrt{17-8})^2$$

$$= (\sqrt{25} - \sqrt{9})^2$$

$$= (5-3)^2 = (2)^2$$

$$8. (4) m = \sqrt{5 + \sqrt{5 + \sqrt{5 + \dots}}}$$

On squaring both sides,

$$m^2 = 5 + m \Rightarrow m^2 - m = 5 \dots (i)$$

Again,

$$n = \sqrt{5 - \sqrt{5 - \sqrt{5 - \dots}}}$$

On squaring both sides,

$$n^2 = 5 - n$$

$$\Rightarrow n^2 + n = 5$$

$$\therefore m^2 - m = n^2 + n \dots (ii)$$

$$9. (1)$$

$$\text{Sim}$$

$$=$$

$$=$$

$$=$$

$$\therefore E$$

$$+$$

$$+$$

$$=$$

$$= 3$$

$$10. (5)$$

$$= 4$$

$$=$$

$$=$$

$$=$$

$$=$$

$$=$$

$$=$$

$$=$$

$$=$$

$$=$$

$$=$$

$$11. (1)$$

$$=$$

$$=$$

$$=$$

$$\therefore \text{Num}$$

$$=$$

$$12. (4) a -$$

$$=$$

$$\Rightarrow (a -$$

$$\Rightarrow \frac{2x}{560} = 6 \Rightarrow x = \frac{560 \times 6}{2} = 1680$$

$$\therefore \text{Required answer} = \frac{1680}{112} = 15$$

19. (3) **Tricky Approach**

Number of students in the school  
 $= 54 \times 30 = 1620$

Number of rows of 45 students each  
 $36$

$$\begin{aligned} 20. (4) \text{ Total spent amount} &= \left( \frac{591}{3} + \frac{45}{60} \right) \times \\ &= (197 + 585) \text{ paise} \\ &= 782 \text{ Paise} \\ &= \text{Rs. } 7.82 \end{aligned}$$

21. (1) Number of extra sweets  $= 2 \times 480$   
 These sweets were to be distributed  
 120 children.

$$\therefore \text{Number of sweets to be given to each} \\ \text{originally} = \frac{960}{120} = 8$$

$$\begin{aligned} 22. (4) \text{ Total electric consumption} \\ &= (10 \times 16 \times 8 + 3 \times 10 \times 8) \text{ units} \\ &= (1280 + 240) \text{ units} = 1520 \text{ units} \end{aligned}$$

23. (3) Let the number of hens be  $x$  and  
 goats be  $y$ .

$$\therefore x + y = 90 \quad \dots (i)$$

$$2x + 4y = 248$$

$$\Rightarrow x + 2y = 124$$

$$\Rightarrow 90 - y + 2y = 124 \quad [\text{From equation (i)}]$$

$$\Rightarrow y = 124 - 90 = 34$$

24. (5) **Tricky Approach**

If the number of parrots in the forest  
 then number of tigers  $= 858 - x$

$$\therefore x \times 2 + (858 - x) \times 4 = 1746$$

$$\Rightarrow 2x = 3432 - 1746 = 1686$$

$$\Rightarrow x = \frac{1686}{2} = 843$$

25. (1) Bus fare = Rs. 420

$$\text{Train fare} = 2 \times 420 \times \frac{3}{4}$$

$$= \text{Rs. } 630$$

$\therefore$  Required total fare

$$= \text{Rs. } (2 \times 420 + 4 \times 630)$$

$$= \text{Rs. } (840 + 2520)$$

$$= \text{Rs. } 3360$$

26. (4) If the number of boys  $= x$  then number  
 girls  $= 60 - x$

$$\therefore (60 - x)x + (60 - x)x = 1600$$

8. (1)  $Y = \frac{1}{(|x-2|-4)}$ . This

be defined when the denominator is not zero. The denominator is zero when  $|x-2| = 4$ .

$\Rightarrow x = 6, -2$

$|x-2| = 4$  can be written as  $x-2 = 4$  or  $x-2 = -4$  and  $-(x-2) = 4 \Rightarrow -x+2 = 4$

Also  $x = 6$ , then  $|x-2| = 4$

$x = -2$ , then  $|x-2| = 4$

9. (2) Let the no. be  $xy$ , value

Reversed No  $\rightarrow yx$ , value

$(10y + x) - (10x + y) = 18$

$9y - 9x = 18$

$y = x + 2$

No.  $\rightarrow XY$

If

$x = 1 \quad y = 3 \quad \text{No.} \rightarrow 13$

$x = 2 \quad y = 4 \quad \text{No.} \rightarrow 24$

$x = 3 \quad y = 5 \quad \text{No.} \rightarrow 35$

$x = 4 \quad y = 6 \quad \text{No.} \rightarrow 46$

$x = 5 \quad y = 7 \quad \text{No.} \rightarrow 57$

$x = 6 \quad y = 8 \quad \text{No.} \rightarrow 68$

$x = 7 \quad y = 9 \quad \text{No.} \rightarrow 79$

We have to find numbers where

there are 6 other numbers

10. (3) We are given that the sum of the digits is 10. This means that the last digit must be zero.

The last digit of 4 odd numbers is 7, 9.

Sum of unit digit of 4 numbers is 10, so the unit digit of the sum is 0.

Now, if unit digits of 4 numbers are 1, 3, 5, 7, then  $1+3+5+7 = 16$ , unit digit is not zero.

Same as true for  $(3+5+7+9) = 24$ , unit digit is not zero.

Only possibility is  $(7+9+1+3) = 20$ , unit digit is zero.

Now check through options

If one of the numbers is 21, then the other three numbers would be 17, 19, 21, 23.

$17 + 19 + 21 + 23 = 80$  but  $\frac{80}{10} = 8$  is not a perfect square.

One of the numbers cannot be 21, so the unit digit can only be 7, 9.

If one of the no. is 41, then the other three numbers would be 37, 39, 41, 43.

$37 + 39 + 41 + 43 = 160$ ,  $\frac{160}{10} = 16$  is a perfect square.

perfect square.

19. (1) Let us start with smallest p

$$a = 2, a + 2 = 4 \rightarrow \text{not prime no.}$$

If  $a = 3$ , then we get 3, 5, 7  $\rightarrow$  a

$a = 3$  is one possible solution for

$$\text{If } a = 5, \rightarrow 5, 7, \textcircled{9}$$

$$a = 7 \rightarrow 7, \textcircled{9}, 11$$

$$a = 11 \rightarrow 11, 13, \textcircled{15}$$

$$a = 13 \rightarrow 13, \textcircled{15}, 17$$

We will have only one set of so  
ing the given relationship.

20. (4) Starting from the least num  
sider option (3) first. i.e. numb  
2.

We are given that no. of adults  
no. of girls > no. of families.

If no. of families = 2  $\Rightarrow$  no. of g  
then no. of boys is at least 4 a  
is at least 5, but in two famili  
max. 4 adults, hence option (3)

Now let no. of families = 3, the  
at least 4, no. of boys is at le  
adults is at least 6.

In 3 families we can have max  
hence minimum no. of families

21. (4) If  $x = 2$  (Prime no.) and  $y =$   
no.)

then  $xy$  is even hence (2) is ruled

Again  $y - x = 6 - 2 = 4 \rightarrow$  even  $\rightarrow$   
ruled out.

$$\frac{x+y}{x} = \frac{6+2}{2} = 4 \rightarrow \text{even}$$

$\rightarrow$  Hence option (3) is also ruled

22. (4) 4 c 3 d 8 a 10 b 5

$$\Rightarrow 4 \times 3 + 8 - 10 \div 5$$

$$\Rightarrow 12 + 8 - 2 = 18$$

Now,

$$3 \text{ c } 4 \text{ a } 2 \text{ d } 6 \text{ b } 3$$

$$\Rightarrow 3 \times 4 - 2 + 6 \div 3$$

$$\Rightarrow 12 - 2 + 2$$

$$\Rightarrow 14 - 2 = 12$$

23. (1) From Option

$$\text{Number} = 253$$

$$5 = 2 + 3 \text{ and}$$

$$352 - 253 = 99$$

**Method-2 :**

$$\text{Number} = 100x + 10y + z$$

$$x + y + z = 10; y = x + z$$

$$\Rightarrow x(x+64) - 63(x+64) = 0$$

$$\Rightarrow (x-63)(x+64) = 0$$

$$\Rightarrow x = 63, -64 \Rightarrow x = 63$$

Quantity I > Quantity II

3. (1) **Quantity I** : 9 mangoes and 4 apples

$$\therefore 4 \text{ apples} = 9 \text{ pine apples}$$

$$= 3 \times 48 = ₹ 144$$

$$[3 \text{ Pineapples} = ₹ 48]$$

$$\therefore 3 \text{ apples} = \frac{144 \times 3}{4} = ₹ 108$$

$$\therefore 9 \text{ mangoes} = ₹ 108$$

$$\therefore 1 \text{ mango} = \frac{108}{9} = ₹ 12$$

$$\text{Cost of 10 mangoes} = ₹ 120$$

**Quantity II** :

| $F_1$ | $F_2$ |
|-------|-------|
| +100  | +100  |
| -12   |       |
|       | -10   |
|       | +15   |
| +12   |       |
| +20   |       |
|       | +105  |

Quantity I > Quantity II

4. (3) **Quantity I** : Let the 3 consecutive numbers be  $(x-1)$ ,  $(x)$  and  $(x+1)$ . so

$$(x-1)(x)(x+1) = 3360$$

(1) If least no. = 17

$$\Rightarrow x-1 = 17$$

$$\Rightarrow x = 18$$

$$17 \times 18 \times 19 \text{ cannot end with } 0$$

(2) If  $(x-1) = 14 \Rightarrow x = 15$

$$14 \times 15 \times 16 = 210 \times 16 = 3360$$

**Quantity II** : Suppose the two consecutive integers be  $x$  and  $x+1$  respectively. According to question,

$$(x) \times (x+1) = 462$$

$$\Rightarrow x^2 + x - 462 = 0$$

$$\Rightarrow x^2 + 22x - 21x - 462 = 0$$

$$\Rightarrow x(x+22) - 21(x+22) = 0$$

$$\Rightarrow (x+22)(x-21) = 0$$

$$\therefore x = 21$$

Quantity II > Quantity I

5. (3) **Quantity I** : The no. is of the form  $45n$ . First divisor (425) is multiple of 17. Divisor (17).

$\therefore$  Required remainder

= Remainder obtained on dividing 45 by 17

**Quantity II** :  $n = 6q + 4$

$$2n = 12q + 8$$

Dividing 8 by 6 the remainder = 2

Quantity I > Quantity II

Suppose we are  
H.C.F. is to be deter  
steps.

**Step I :** Divide the  
number and find the

**Step II :** Now repeat  
remainder as divisor  
and continue this process  
remainder. Last divisor is

**Second Form :** Find  
divides 108 and 140 with  
Clearly, we are to find H.C.F.

**H.C.F. of more than two numbers :**  
H.C.F. of any two of the numbers  
find H.C.F. of the so found numbers  
and so on. The last H.C.F. is the  
H.C.F.

**To determine the H.C.F. of  
crete quantities.**

In order to find the H.C.F. of  
crete quantities, they must be in the  
same unit.

#### HCF OF DECIMALS

**Step I :** First of all, we find the  
of decimal places by putting the same  
given numbers. It is done by multiplying

have different decimal places.

**Step II :** Find the H.C.F. of the  
without taking decimals in the numbers  
integers.

**Step III :** Put in the result the same  
decimal places as there are in the  
bers after making them same.

#### LOWEST COMMON MULTIPLE

**Common Multiple :** It is a number  
which is exactly divisible by each of the  
under consideration.

**Example :** 60 is a common multiple of  
5, 6, 10, 12, 15, 20 and 30.

**Lowest Common Multiple :** The least  
est common multiple of two or more  
bers is the least or smallest number  
actly divisible by each of the numbers.  
more clearly we consider two numbers  
15 and 20.

Multiples of 15 are 15, 30, 45, 60, 75, 90, 105, 120, 135, 150, 165, 180, 195, 210, 225, 240, 255, 270, 285, 300, 315, 330, 345, 360, 375, 390, 405, 420, 435, 450, 465, 480, 495, 510, 525, 540, 555, 570, 585, 600, 615, 630, 645, 660, 675, 690, 705, 720, 735, 750, 765, 780, 795, 810, 825, 840, 855, 870, 885, 900, 915, 930, 945, 960, 975, 990, 1005, 1020, 1035, 1050, 1065, 1080, 1095, 1110, 1125, 1140, 1155, 1170, 1185, 1200, 1215, 1230, 1245, 1260, 1275, 1290, 1305, 1320, 1335, 1350, 1365, 1380, 1395, 1410, 1425, 1440, 1455, 1470, 1485, 1500, 1515, 1530, 1545, 1560, 1575, 1590, 1605, 1620, 1635, 1650, 1665, 1680, 1695, 1710, 1725, 1740, 1755, 1770, 1785, 1800, 1815, 1830, 1845, 1860, 1875, 1890, 1905, 1920, 1935, 1950, 1965, 1980, 1995, 2010, 2025, 2040, 2055, 2070, 2085, 2100, 2115, 2130, 2145, 2160, 2175, 2190, 2205, 2220, 2235, 2250, 2265, 2280, 2295, 2310, 2325, 2340, 2355, 2370, 2385, 2400, 2415, 2430, 2445, 2460, 2475, 2490, 2505, 2520, 2535, 2550, 2565, 2580, 2595, 2610, 2625, 2640, 2655, 2670, 2685, 2700, 2715, 2730, 2745, 2760, 2775, 2790, 2805, 2820, 2835, 2850, 2865, 2880, 2895, 2910, 2925, 2940, 2955, 2970, 2985, 3000, 3015, 3030, 3045, 3060, 3075, 3090, 3105, 3120, 3135, 3150, 3165, 3180, 3195, 3210, 3225, 3240, 3255, 3270, 3285, 3300, 3315, 3330, 3345, 3360, 3375, 3390, 3405, 3420, 3435, 3450, 3465, 3480, 3495, 3510, 3525, 3540, 3555, 3570, 3585, 3600, 3615, 3630, 3645, 3660, 3675, 3690, 3705, 3720, 3735, 3750, 3765, 3780, 3795, 3810, 3825, 3840, 3855, 3870, 3885, 3900, 3915, 3930, 3945, 3960, 3975, 3990, 4005, 4020, 4035, 4050, 4065, 4080, 4095, 4110, 4125, 4140, 4155, 4170, 4185, 4200, 4215, 4230, 4245, 4260, 4275, 4290, 4305, 4320, 4335, 4350, 4365, 4380, 4395, 4410, 4425, 4440, 4455, 4470, 4485, 4500, 4515, 4530, 4545, 4560, 4575, 4590, 4605, 4620, 4635, 4650, 4665, 4680, 4695, 4710, 4725, 4740, 4755, 4770, 4785, 4800, 4815, 4830, 4845, 4860, 4875, 4890, 4905, 4920, 4935, 4950, 4965, 4980, 4995, 5010, 5025, 5040, 5055, 5070, 5085, 5100, 5115, 5130, 5145, 5160, 5175, 5190, 5205, 5220, 5235, 5250, 5265, 5280, 5295, 5310, 5325, 5340, 5355, 5370, 5385, 5400, 5415, 5430, 5445, 5460, 5475, 5490, 5505, 5520, 5535, 5550, 5565, 5580, 5595, 5610, 5625, 5640, 5655, 5670, 5685, 5700, 5715, 5730, 5745, 5760, 5775, 5790, 5805, 5820, 5835, 5850, 5865, 5880, 5895, 5910, 5925, 5940, 5955, 5970, 5985, 6000, 6015, 6030, 6045, 6060, 6075, 6090, 6105, 6120, 6135, 6150, 6165, 6180, 6195, 6210, 6225, 6240, 6255, 6270, 6285, 6300, 6315, 6330, 6345, 6360, 6375, 6390, 6405, 6420, 6435, 6450, 6465, 6480, 6495, 6510, 6525, 6540, 6555, 6570, 6585, 6600, 6615, 6630, 6645, 6660, 6675, 6690, 6705, 6720, 6735, 6750, 6765, 6780, 6795, 6810, 6825, 6840, 6855, 6870, 6885, 6900, 6915, 6930, 6945, 6960, 6975, 6990, 7005, 7020, 7035, 7050, 7065, 7080, 7095, 7110, 7125, 7140, 7155, 7170, 7185, 7200, 7215, 7230, 7245, 7260, 7275, 7290, 7305, 7320, 7335, 7350, 7365, 7380, 7395, 7410, 7425, 7440, 7455, 7470, 7485, 7500, 7515, 7530, 7545, 7560, 7575, 7590, 7605, 7620, 7635, 7650, 7665, 7680, 7695, 7710, 7725, 7740, 7755, 7770, 7785, 7800, 7815, 7830, 7845, 7860, 7875, 7890, 7905, 7920, 7935, 7950, 7965, 7980, 7995, 8010, 8025, 8040, 8055, 8070, 8085, 8100, 8115, 8130, 8145, 8160, 8175, 8190, 8205, 8220, 8235, 8250, 8265, 8280, 8295, 8310, 8325, 8340, 8355, 8370, 8385, 8400, 8415, 8430, 8445, 8460, 8475, 8490, 8505, 8520, 8535, 8550, 8565, 8580, 8595, 8610, 8625, 8640, 8655, 8670, 8685, 8700, 8715, 8730, 8745, 8760, 8775, 8790, 8805, 8820, 8835, 8850, 8865, 8880, 8895, 8910, 8925, 8940, 8955, 8970, 8985, 9000, 9015, 9030, 9045, 9060, 9075, 9090, 9105, 9120, 9135, 9150, 9165, 9180, 9195, 9210, 9225, 9240, 9255, 9270, 9285, 9300, 9315, 9330, 9345, 9360, 9375, 9390, 9405, 9420, 9435, 9450, 9465, 9480, 9495, 9510, 9525, 9540, 9555, 9570, 9585, 9600, 9615, 9630, 9645, 9660, 9675, 9690, 9705, 9720, 9735, 9750, 9765, 9780, 9795, 9810, 9825, 9840, 9855, 9870, 9885, 9900, 9915, 9930, 9945, 9960, 9975, 9990, 10000.

The least common multiple of 15 and 20 is 60.

$\therefore$  L.C.M. of 15 and 20 is 60.



17. Find the greatest number which when divided by 12 leaves a remainder of 1.  
 (1) 99999 (2) 99958  
 (5) None of these
18. What is the LCM of  $(x+1)$  and  $(x-3)^2$   $(x-5)$ ?  
 (1)  $(x+5)(x+1)(x-3)$   
 (2)  $(x+5)^2(x+1)(x-3)$   
 (3)  $(x+5)(x-3)$   
 (4)  $(x+5)^2(x+1)^3(x-3)$   
 (5) None of these
19. A hall 50 m long and 45 m wide is to be paved with square tiles. Find the least number of tiles as its number in the hall so that the tiles exactly fit in the hall.  
 (1) 36 sq m and 80 tiles  
 (2) 16 sq m and 80 tiles  
 (3) 25 sq m and 90 tiles  
 (4) 36 sq m and 90 tiles  
 (5) None of these
20. How many numbers between 1 and 100 are divisible by 4, 5 and 6?  
 (1) 7 (2) 8  
 (5) None of these
21. A positive integer when divided by 17 leaves a remainder 45. When divided by 17, the remainder is:  
 (1) 11 (2) 8  
 (5) None of these
22. Find the largest 5-digit number which when added to 12345 makes a number divisible by 15, 21, 23 & 27.  
 (1) 84255 (2) 84250  
 (3) 99999 (4) 99990  
 (5) None of these
23. The product of two numbers is 100. If the HCF of the numbers is 3, then the LCM of the numbers is:  
 (1) 185 (2) 111 (3) 100  
 (5) None of these
24. The sum of two numbers is 100. If the HCF of the numbers is 4, how many pairs of numbers are possible?  
 (1) 1 (2) 2 (3) 3  
 (5) None of these
25. The number nearest to 10000 which is exactly divisible by each of 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100 is:  
 (1) 9240 (2) 10080 (3) 9999  
 (5) None of these

6. Two donkeys are s  
First donkey can r  
and the second ca  
donkeys run towa  
much time (in sec  
each other?  
(1) 60 (2) 80  
(5) None of these
7. M is the largest 4  
divided by 4, 5, 6  
2, 3, 4, and 5 resp  
remainder when M  
(1) 2 (2) 1  
(5) None of these
8. The HCF of two num  
is 221 times the H  
lies between 200 a  
the digits of the oth  
(1) 14 (2) 17  
(5) None of these
9. The LCM of two num  
their HCF. If their  
ence between the r  
= ?  
(1) 660 (2) 426  
(5) None of these
10. In finding the HCF e  
sion method, the last  
quotients are 1, 11 and  
is sum of the two num  
(1) 833 (2) 867  
(5) None of these
11. When 1062, 1134 and  
the greatest number x,  
case is y. What is the  
(1) 17 (2) 18  
(5) None of these
12. Let x be the greatest  
divides 955, 1027, 10  
each case is the same  
ing is NOT a factor of  
(1) 4 (2) 16  
(5) None of these
13. If the students of 9th  
rows of 6, 8, 12 or 16  
hind. Then the possib  
in the class is  
(1) 60 (2) 72  
(5) None of these
14. The GCD of 1.08, 0.36  
(1) 0.03 (2) 0.9  
(5) None of these

## QUANTITY BASED

**Directions (1-5) :** In each question, two quantities are given. Solve each question and choose the correct option :

- (1) Quantity I > Quantity II
- (2) Quantity I  $\geq$  Quantity II
- (3) Quantity I < Quantity II
- (4) Quantity II  $\geq$  Quantity I
- (5) Quantity I = Quantity II or no relation can be established

1. **Quantity I :** Find the number of 4-digits that can be divided by 3.

**Quantity II :** Find the number of 4-digits that can be divided by 4.

2. **Quantity I :** What is the remainder when 100 is divided by 1 and when divided by 5?

**Quantity II :** How many numbers between 10 and 1000 which will leave a remainder 1 and when divided by 3 will leave a remainder 3?

3. **Quantity I :** Which number is greater, 3026 and 5053, when divided by 13 respectively?

**Quantity II :** If the HCF of two numbers is 13, then the sum of the numbers is greater than 13?

4. **Quantity I :** What is the least number which 4320 be divided to form a perfect cube?

**Quantity II :** The least number which when added to 2497 so that the sum is divisible by 5, 6, 4 and 3.

5. **Quantity I :** When 'n' is divided by 5, the remainder is 2. What is the remainder when  $n^2$  is divided by 5?

**Quantity II :** The least number which when multiplied with 588 leads to a perfect square is

## DATA SUFFICIENCY

**Directions (1-5) :** Each question consists of a question and two statements given below it. You have to decide whether the data provided in the statements is sufficient to answer the question. Read the question and the statements and—

**Give answer (1)** If the data in the statements alone is sufficient to answer the question.

8. (5) The required no. should be the  
that can divide 420, 600 & 1050  
H.C.F. of (420, 600, 1050)  $\Rightarrow$   
 $420 = 7 \times 3 \times 2 \times 5 \times 2 = 5 \times 7 \times 3 \times 2$   
 $600 = 3 \times 2 \times 5 \times 2 \times 5 \times 2 = 3 \times 5 \times 2^3$   
 $1050 = 5 \times 2 \times 5 \times 3 \times 7 = 5^2 \times 2 \times 3 \times 7$   
H.C.F. =  $5 \times 3 \times 2 = 30$

9. (2) Since wastage is zero, hence  
cut squares of dimensions  
dimension of rectangle are  
divisible by dimension of square  
no. of squares are to be minimum.  
of the square should be greatest  
H.C.F. (36, 24) =  $2^2 \times 3 = 12$   
There would be 3 divisions along  
and 2 divisions along the breadth  
of squares would  $3 \times 2 = 6$ .

|    |    |    |    |
|----|----|----|----|
|    | 12 | 12 | 12 |
| 12 |    |    |    |
| 12 |    |    |    |

10. (3) Since each row should have eq  
balls, also, each row has balls of o  
only and the no. of such rows should  
mum, we have to find (H.C.F. of 1  
35) = 5

$$15 = 3 \times 5$$

$$20 = 5 \times 2^2$$

$$35 = 7 \times 5$$

Each row would have 5 balls, no. of

orange balls is  $\frac{15}{5} = 3$ , no. of white

$$\frac{20}{5} = 4 \text{ \& no. of green balls } = \frac{35}{5} = 7$$

No. of rows =  $3 + 4 + 7 = 14$  o

$$\frac{15+20+35}{5} = 14$$

11. (4) Product of numbers = H.C.F.  $\times$  L

$$12x = 2 \times 372$$

$$x = \frac{2 \times 372}{12} = 62$$

12. (2) H.C.F. of  $\left(\frac{10}{11}, \frac{3}{5}, \frac{4}{7}, \frac{6}{13}\right)$

$$= \frac{\text{H.C.F. of } (10, 3, 4, 6)}{\text{L.C.M. of } (11, 5, 7, 13)} = \frac{1}{5005}$$

28. (3) A makes one complete round of the circular track in  $\frac{5}{2}$  hours.

$$\text{lar track in } \frac{5}{2} = 2 \text{ hours.}$$

B in  $\frac{5}{3}$  hours and C in  $\frac{5}{2}$  hours.

That is after 2 hours A is at the starting point

B after  $\frac{5}{3}$  hours and C after  $\frac{5}{2}$  hours.

Hence the required time

$$= \text{LCM of } 2, \frac{5}{3} \text{ and } \frac{5}{2} \text{ hours}$$

$$= \frac{\text{LCM of } 2, 5, 5}{\text{HCF of } 3, 2} = \frac{10}{1} = 10 \text{ hours.}$$

29. (4) Let CP of each coconut be Re. 1.

$\therefore$  CP of 2500 coconuts

= Rs. 2500

SP of 2500 coconuts

= Rs. 2750

$$\therefore \text{Gain \%} = \frac{2750 - 2500}{2500} \times 100 = 10\%$$

30. (3) LCM

$$= \frac{\text{LCM of } 1, 5, 2, 4}{\text{HCF of } 3, 6, 9, 27} = \frac{20}{3}$$

31. (4)  $\text{LCM} \times \text{HCF} = \text{Product of two numbers}$

$$\therefore \text{Other number} = \frac{4800 \times 160}{480} = 1600$$

32. (3) Let the number be  $7x$  and  $8x$ .

$$\therefore \text{HCF} = x$$

$$\text{LCM} \times \text{HCF} = \text{Product of numbers}$$

$$\therefore 280 \times x = 56x^2$$

$$\Rightarrow x = 5$$

$\therefore$  Numbers are 35 and 40.

33. (3) Tricky approach

Let the number be  $13x$  and  $13y$  where  $x$  and  $y$  are prime to each other.

$$\therefore 13x \times 13y = 2028$$

$$\Rightarrow xy = \frac{2028}{13 \times 13} = 12 = 3 \times 4$$

$$\therefore \text{Numbers} = 13 \times 3 = 39 \text{ and}$$

$$13 \times 4 = 52$$

$\therefore$  Sum of numbers

$$= 39 + 52 = 91$$

34. (1) Let the numbers be  $x$  and  $y$ . Then

$$x + y = 45 \dots (i)$$

- $\therefore \text{LCM} = 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3 = 432 \text{ seconds}$   
 $= 7 \text{ minutes } 12 \text{ second}$   
 $\therefore \text{Required time}$   
 $= 10 : 07 : 12 \text{ hours}$
6. (2) Distance between both donkeys = 400 metre.  
 Relative speed =  $(3 + 2) \text{ m./sec.}$   
 $= 5 \text{ m./sec.}$   
 $\therefore \text{Required time}$   
 $= \frac{\text{Distance}}{\text{Relative speed}}$   
 $= \frac{400}{5} = 80 \text{ seconds}$
7. (2) Here, divisor - respective remainder = 2  
 Largest 4-digit number = 9999  
 LCM of 4, 5, 6 and 7 = 420  

$$\begin{array}{r} 2 \overline{) 4, 5, 6, 7} \\ \underline{2, 5, 3, 7} \end{array}$$
  
 $\therefore \text{LCM} = 2 \times 2 \times 5 \times 3 \times 7$   
 $= 420$   

$$\begin{array}{r} 420 \overline{) 9999} (23 \\ \underline{840} \\ 1599 \\ \underline{1260} \\ 339 \end{array}$$
  
 $\therefore \text{Divisible Number}$   
 $= 9999 - 339 = 9660$   
 Required number =  $9660 - 2$   
 $= 9658$   
 On dividing 9658 by 9, remainder = 1  
**Illustration :**  

$$\begin{array}{r} 9 \overline{) 9658} (1073 \\ \underline{9} \\ 65 \\ \underline{63} \\ 28 \\ \underline{27} \\ 1 \end{array}$$
8. (4) H.C.F. = 21  
 $\therefore \text{L.C.M.} = 21 \times 221$   
 $\therefore \text{First number} \times \text{second number} = 21 \times 21 \times 221$   
 $= 21 \times 21 \times 13 \times 17$   
 $= (21 \times 17) \times (21 \times 13)$   
 $= 357 \times 273$   
 First number = 270 ( $200 < 270 < 300$ )  
 $\therefore \text{Second number} = 357$   
 $\therefore \text{Sum of digits} = 3 + 5 + 7$   
 $= 15$

9

10

11.

Th

Re

$\Rightarrow$

$\therefore$

12. (2)

x is

95

$\Rightarrow$

1.0

1.0

1.0

HC

16

24



H.C.F. (2610, 1230)

$$2610 = 5 \times 2 \times 9 \times 29 = 5 \times 2 \times 3^2 \times 29$$

$$1230 = 5 \times 2 \times 3 \times 41$$

$$\text{H.C.F.} = 5 \times 2 \times 3 = 30$$

Now, we have to find no. of factors of 30.

$$\text{So, } 30 = 2 \times 3 \times 5$$

$$\text{No. of factors} = (1 + 1)(1 + 1)(1 + 1) = 2^3 = 8$$

$$[30 = 1 \times 30$$

$$2 \times 15$$

$$3 \times 10$$

$$5 \times 6]$$

$$21. (1) 3^1 = 3, 3^2 = 9.$$

$$3^3 = 27, 3^4 = 81$$

i.e. the unit's digit = odd number

 $\therefore$  Hence, both numbers are divisible by 2.

$$22. (3) \text{ Let the HCF of the numbers be } H \text{ and their LCM be } L. \text{ Then,}$$

$$L = 84 H$$

$$\text{and, } L + H = 680$$

$$\Rightarrow 84 H + H = 680$$

$$\Rightarrow H = \frac{680}{85} = 8$$

$$\therefore L = 84 \times 8 = 672$$

Product of numbers

$$= \text{HCF} \times \text{LCM}$$

$$\Rightarrow 56 \times \text{other number}$$

$$= 8 \times 672$$

$$\Rightarrow \text{Other number} = \frac{8 \times 672}{56} = 96$$

$$23. (2) \text{ Number of words typed by Dina and Dodo in first 2 minutes} = 30 + 25 = 55$$

Number of words typed in 20 minutes

$$= 55 \times 10 = 550$$

Now Dina type 30 words in 21st minute.

Number of words typed

$$= 550 + 30 = 580$$

Now it is the turn of Dodo who types remaining 20 words.

Time taken by Dodo

$$= \frac{60}{25} \times 20 = 48 \text{ seconds}$$

 $\therefore$  Total time taken

$$= 21 \text{ minutes } 48 \text{ seconds.}$$

$$\text{Remainder of } \frac{2600}{256} = \text{Remainder of } \frac{(2^8)^{75}}{256}$$

$$= \text{Remainder of } \frac{256^{75}}{256} = 0.$$

### QUANTITY BASED EXPLANATION

1. (1) **Quantity I** : We have to find a number that is a multiple of 24, 27 & 30, also it should be least  $\Rightarrow$  required number is L.C.M. of 24, 27, 30

$$\begin{array}{r|rrrr} 3 & 24 & 27 & 30 \\ 2 & 8 & 9 & 10 \\ \hline & 4 & 3 & 5 \end{array}$$

$$\text{L.C.M.} = 3 \times 2 \times 4 \times 3 \times 5 = 1080$$

$$6k = 8m + 2 \Rightarrow 3k = 4m + 1 \text{ or } k = \frac{4m+1}{3}$$

$$m = 2, k = 3, \text{ number i.e. } N = 19$$

General form of required number is

$$\text{L.C.M. of } (6, 8) + 19$$

$$24l + 19$$

Smallest number is 19 & greatest is less than 1000-

For  $l = 40$ , required no. is 979 [for  $l = 41$ , required no. is 1003 > 1000, hence inadmissible].

Total number would be from  $l = 0$  to  $l = 40$  i.e. 41 numbers in all.

**Quantity I** < **Quantity II**

3. (3) **Quantity I** :  $3026 - 11 = 3015$  and  $5053 - 13 = 5040$

Required number = HCF of 3015 and 5040

$$\begin{array}{r} 240 \\ 97 \\ \hline 60 \\ 37 \end{array}$$

$$\therefore \text{Required number} = (60 - 37) = 23$$

**Quantity I** < **Quantity II**

5. (1) **Quantity I** :  $n = 5 \times q + 2$

where  $q$  = quotient

On squaring both sides,

$$n^2 = (5q + 2)^2$$

$$= 25q^2 + 20q + 4$$

$$= 5(5q^2 + 4q) + 4$$

$$\therefore \text{Remainder} = 4$$

**Quantity II** : 2 | 588

$$\begin{array}{r|rr} 2 & 294 \\ 7 & 147 \\ \hline 7 & 21 \\ 3 & \end{array}$$

4. (4) We are given  $abc = 6$

From statement II,  $a, b$  and  $c$  are positive integers.

$$3 \times 2 \times 1 = 6$$

$$6 \times 1 \times 1 = 6$$

From statement I,  $(a+b+c)$  is not divisible by 8

In first case,  $a + b + c = 3 + 2 + 1 = 6 \rightarrow$  not div by 8.

In second case,  $a + b + c = 6 + 1 + 1 = 8 \rightarrow$  div by 8.

Combining I and II we get (3, 2, 1) is the only option. But we do not know which is  $a, b$  or  $c$ .

'a' could be 3, 2, 1. Same is for  $b$  and  $c$ .

5. (4) The data given in both statements are inadequate.

# 4

## Average

### IMPORTANT FACTS AND FORMULAE

**Average** : The average of a number of quantities of observations of the same kind is their sum divided by their number. The average is also called average value or mean value or more accurately an arithmetic mean.

Thus,

Average

$$= \frac{\text{Sum of quantities or observations}}{\text{Number of quantities or observations}}$$

It is so:

**Explanation** : As we know,

$$\text{Sum of the first } n \text{ natural numbers} = \frac{n(n+1)}{2}$$

Number of quantities =  $n$

$\therefore$  Average

$$= \frac{\frac{n(n+1)}{2}}{n} = \frac{n(n+1)}{2 \times n} = \frac{n+1}{2}$$

## AVERAGE

### C. Average related to odd numbers

1. Average of  $n$  first consecutive odd numbers =  $n$
2. The average of consecutive odd numbers to  $n$  is  $\frac{n+1}{2}$

Where  $n$  is the last odd number.

3. Average of squares of consecutive odd numbers to  $n$

$$= \frac{n(n+2)}{3}$$

(Where  $n$  = odd number)

4. Average of an arithmetic sequence with first and last number (term) known

$$\text{First number} + \text{Last number}$$

Above mentioned result can be summarized in the following way too :

If half of the journey is travelled at a speed of  $x$  km/hr and the rest half at a speed of  $y$  km/hr, then average speed during the whole journey is given by

$$\frac{2xy}{x+y} \text{ km/hr.}$$

### TYPE-IV : QUESTIONS RELATED TO HEIGHT, WEIGHT AND AGE

#### AVERAGE AGE

Tricks subject to certain conditions :

- A. When a person leaves a group and another person replaces him, then two cases arise :

Case I : When the sum of ages of all persons is constant.

## MODEL QUESTIONS

## LEVEL-1

1. Average weight of 25 boys in a class is 48 kgs. The average weight of the class of 40 students is 45 kgs. What is the average weight of the 15 girls in the class?  
(1) 44 kgs (2) 42 kgs (3) 40 kgs (4) 39 kgs (5) 42.5 kgs
2. The average marks in Science subject of a class of 20 students is 68. If the marks of two students were misread as 48 and 65 of the actual marks 72 and 61 respectively, then

8. The mean temperature of Monday and Tuesday was  $36^{\circ}\text{C}$ . If the temperature on Thursday was  $36^{\circ}\text{C}$ . If the temperature on Thursday was  $36^{\circ}\text{C}$ . If the temperature on Thursday was  $36^{\circ}\text{C}$ .

day was  $\frac{4}{5}$ th of that of Monday, then what was the temperature on Thursday?

(1)  $36.5^{\circ}\text{C}$  (2)  $32^{\circ}\text{C}$  (3)  $35.5^{\circ}\text{C}$  (4)  $34^{\circ}\text{C}$  (5) None of these

9. The average of the ages of Sumit, Krishna and Rishabh is 43 years and the average of the ages of Sumit, Rishabh and Rohit is 49 years. If Rohit is 54 years old, what is Krishna's age?  
(1) 45 years (2) 24 years

extra. The number of those who attended the picnic was —

(1) 8 (2) 16 (3) 12 (4) 24

(5) None of these

15. The average weight of a group of 75 girls was calculated as 47 kgs. It was later discovered that the weight of one of the girls was read as 45 kgs., whereas her actual weight was 25 kgs.

What is the actual average weight of the group of 75 girls? (Rounded off to two digits after decimal)

(1) 46.73 kgs (2) 46.64 kgs (3) 45.96 kgs (4) Cannot be determined (5) None of these

16. The average temperature of the first four days was  $38.6^{\circ}\text{C}$  and that of the last four days was  $35.8^{\circ}\text{C}$ . If the temperature on the 5th day was  $36.6^{\circ}\text{C}$ , what was the average temperature of the first four days?

(1)  $36.6^{\circ}\text{C}$  (2)  $35.8^{\circ}\text{C}$  (3)  $38.6^{\circ}\text{C}$  (4)  $35.8^{\circ}\text{C}$  (5) None of these

16. The average temperature of the first four days was  $38.6^{\circ}\text{C}$  and that of the last four days was  $35.8^{\circ}\text{C}$ . If the temperature on the 5th day was  $36.6^{\circ}\text{C}$ , what was the average temperature of the first four days?

20. The average of marks obtained by 120 students was 35. If the average of passed candidates was 39 and that of failed candidates is 15, the number of candidates who passed the examination is :

(1) 100 (2) 110 (3) 120 (4) 80 (5) None of these

21. The average weight of 45 students in a class was calculated as 36 kg. It was later found that the weight of two students in the class was wrongly calculated. The actual weight of one of the boys in the class was 32 kg., but it was calculated as 22 kg. What was the actual average weight of the class?

(1) 36 kg (2) 35 kg (3) 37 kg (4) 38 kg (5) None of these



# AVERAGE

26. The average salary of the entire staff in an office is Rs. 3200 per month. The average salary of officers is Rs. 6800 and that of non-officers is Rs. 2000. If the number of officers is 5, then find the number of non-officers in the office?  
 (1) 8 (2) 12 (3) 15 (4) 5  
 (5) None of these
27. Average rainfall on Monday, Tuesday, Wednesday and Thursday is 420.5 cm and average on Tuesday, Wednesday, Thursday and Friday is 440.5 cm. If the ratio of rainfall for Monday and Friday is 20 : 21, find the rainfall in cm on Monday and Friday.  
 (1) 1800, 1890 (2) 1600, 1690  
 (3) 1700, 1790 (4) 1500, 1590  
 (5) None of these
34. If the average of eight consecutive even numbers be 93, then the greatest number among them is  
 (1) 100 (2) 86 (3) 102 (4) 98  
 (5) None of these
35. The average of five consecutive positive integers is  $n$ . If the next two integers are also included, the average of all these integers will  
 (1) increase by 1.5 (2) increase by 1  
 (3) remain the same (4) increase by 2  
 (5) None of these

# AVERAGE

41. The average of three consecutive odd numbers is 52 more than  $\frac{1}{5}$ rd of the largest of these numbers. What is the smallest of these numbers?  
 (1) 79 (2) 75 (3) 81 (4) 77  
 (5) None of these
42. How many job applicants had applied if the ratio of selected to unselected was 19 : 17. If 1,200 less had applied and 800 less selected, then the ratio of selected to unselected would have been 1 : 1.  
 (1) 6000 (2) 7200 (3) 8400 (4) 4800  
 (5) None of these
48. Cinderella rode to the palace at speed of 105 kmph. On her way back to her house, she rode half the distance at speed of 85 kmph and the other half at speed of 90 kmph. The average speed of Cinderella for the entire journey is :  
 (1) 95.4 kmph (2) 94.5 kmph  
 (3) 89.5 kmph (4) 84.9 kmph  
 (5) None of these

## LEVEL-II

1. The average marks scored by three students in a test were 62. When two more students (A and B) were added, the new average of marks

AVERAGE

7. The present age of Rahul is 3 times his son's present age and  $\frac{4}{5}$ th of his father's present age. The average of the present ages of all of them is 62 years. What is the difference between the ages of grandfather and grandson?  
 (1) 56 years (2) 50 years  
 (3) 66 years (4) 60 years  
 (5) 76 years
8. Anit is 6 years older than his wife Soma. The present age of their son, Bunny is  $\frac{1}{3}$ rd of Soma's present age. If the sum of present ages of
13. The average age of five children of M and D is 10 years. If M's and D's ages are included, then the average age becomes 22 years. If M's age is 6 years more than D's age, what is the age of D?  
 (1) 49 years (2) 35 years  
 (3) 48 years (4) 38 years  
 (5) 34 years
14. The arithmetic mean of 100 observations is 24. 6 is added to each of the observations and then each of them is multiplied by 2.5. Find the new arithmetic mean.  
 (1) 30 (2) 75 (3) 35 (4) 60  
 (5) 55
15. After replacing an old member by a new member, it was found that the

20. A librarian purchased 60 story books for his library. But he found that he could get 4 extra books by spending Rs. 336 more and then the overall average price per book would be reduced by Re 1. The previous average price of each book was  
 (1) Rs. 84 (2) Rs. 83  
 (3) Rs. 68 (4) Rs. 100  
 (5) Rs. 90
21. In a one day match of 50 overs in an innings the team A had a run rate of 5.3 runs per over. Team B is playing and 5 overs are left and the required run rate to tie the match is 7.2 per over to match the score of Team A. What is team B's score?

26. The average age of 120 students in a group is 13.56 years. 35% of the number of students are girls and the rest are boys. If the ratio of the average age of boys and girls is 6 : 5, then what is the average age (in years) of the girls?  
 (1) 12 (2) 11.6 (3) 10 (4) 14.4  
 (5) 14
27. The average score in Mathematics of 90 students of section A and B of class IX was 63. The number of students in A were 10 more than those in B. The average score of students in A was 30% more than that of



# AVERAGE

2. A demographic survey of 100 families in which two parents were present revealed that the average age  $A$  of the oldest child is 20 yr less than  $\frac{1}{2}$  the sum of the ages of the two parents. If  $F$  represents the age of one parent and  $M$  the age of the other parent, then which of the following is equivalent to  $A$ ?
- (1)  $\frac{F + M - 20}{2}$  (2)  $\frac{F + M}{2} + 20$   
 (3)  $\frac{F + M}{2} - 20$  (4)  $F + M - 10$   
 (5) None of these
3. When six years are subtracted from the

# AVERAGE

13. The average age of four members A, B, C, & D of a Residents Welfare Association is 39 years. If B is 5 years younger than A, B was 33 years old when C joined at age of 25. A was 40 years old when D joined at the age of 30. Find the average age of A and C.
- (1) 39.5 years (2) 39 years  
 (3) 40.2 years (4) 42 years  
 (5) 40 years
14. The average age of a group of 8 friends is 25. Two friends, one aged 28 years and other 'x' years left the group while a third person aged 15 years joined the group. 2 years later, in the same month, the average age of all the 7 friends was calculated to be 23. Find the value of 'x'.
- (1) 30 yrs (2) 35 yrs (3) 40 yrs (4) 25 yrs

- Directions (19-21) :** There are 60 students in a class. These students are divided into 3 groups A, B and C of 15, 20 and 25 students each. The groups A and C are combined to form Group D.
19. What is the average weight of the students in Group D?
- (1) more than the average weight of A  
 (2) more than the average weight of C  
 (3) less than the average weight of C  
 (4) cannot be determined  
 (5) None of these
20. If one student from Group A is shifted to Group B, which of the following will be true?
- (1) The average weight of both the groups increases.  
 (2) The

## AVERAGE

- Quantity II :** The present age of Ronila is one fourth of that of her father. After 6 years the father's age will be twice the age of Kapil. If Kapil celebrated fifth birth day 8 years ago, what is Ronila's present age?
- Quantity I :** The average of five positive numbers is 213. The average of the first two numbers is 233.5 and the average of the last two numbers is 271. What is the third number?
- Quantity II :** The average of 6 numbers is 30. If the average of first four is 25 and that of last three is 35, the fourth number is
- Quantity I :** The average of
3. Is the average of the largest and the smallest of four given numbers greater than the average of the four numbers?
- The difference between the largest and the second largest number is greater than the difference between the second smallest and the smallest number.
  - The difference between the largest and the second largest number is less than the difference between the second largest and

## AVERAGE

23. What is the average speed (kmph) for the first 4 hours of the journey?
- 40
  - 50
  - 60
  - 80
  - 70
24. If the entire journey took 10 hours, then find the average speed (kmph) for the last 5 hours.
- 40 kmph
  - 50 kmph
  - 60 kmph
  - 80 kmph
  - 55 kmph
25. While calculating the average cost price of 8 items, a shopkeeper reversed the digits of one of the values, as a result of which the
30. According to the rules and regulations of DSCO club, the members of the club at any point of time must be at least 120 and at most 165. In the month of April, new members were made, which were  $\frac{1}{3}$ rd of the existing number of members. Also the total age of the new members was 1353 years. Find the new average age of all the current members if it is known that the average age which is an integral value decreased by 2 years, when new members joined the club.

# AVERAGE

## LEVEL-III

|         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|
| 1. (2)  | 2. (3)  | 3. (5)  | 4. (1)  | 5. (1)  | 6. (4)  |
| 7. (1)  | 8. (1)  | 9. (2)  | 10. (1) | 11. (2) | 12. (3) |
| 13. (1) | 14. (3) | 15. (2) | 16. (2) | 17. (1) | 18. (1) |
| 19. (4) | 20. (3) | 21. (3) | 22. (4) | 23. (1) | 24. (3) |
| 25. (3) | 26. (2) | 27. (1) | 28. (3) | 29. (4) | 30. (1) |

## QUANTITY BASED

|        |        |        |        |        |
|--------|--------|--------|--------|--------|
| 1. (3) | 2. (1) | 3. (3) | 4. (1) | 5. (3) |
|--------|--------|--------|--------|--------|

## DATA SUFFICIENCY

|        |        |        |        |        |
|--------|--------|--------|--------|--------|
| 1. (5) | 2. (4) | 3. (1) | 4. (4) | 5. (1) |
|--------|--------|--------|--------|--------|

## EXPLANATIONS

$$= 672 - 92 + 85 = 665$$

6. (1) Sum of 10 numbers = 402

Correct sum of 10 numbers

$$402 - 13 + 31 = 420$$

Hence, new average =  $\frac{402}{10} = 40.2$

7. (2) According to question, total weight of 45 students =  $45 \times 52 = 2340$

Total weight of 5 students who leave

$$= 5 \times 48 = 240$$

Total weight of 5 students who join

$$= 5 \times 54 = 270$$

Hence, new total weight of 45 students

$$= 2340 - 240 + 270 = 2370$$

$$\therefore \text{Average} = \frac{2370}{45} = 52\frac{2}{3} \text{ kg}$$

8. (2) Monday's Thursday =  $2 \times 36 = 72$

$$\Rightarrow 71 \times 4 = 284 + x$$

$$\Rightarrow 284 = 219 + x$$

$$\Rightarrow x = 284 - 219 = 65$$

12. (3) Total expenditure for the year

$$= (2200 \times 3 + 2250 \times 4 + 3120 \times 7)$$

$$= 6600 + 9000 + 21840$$

$$= ₹ 31200$$

Total saving = ₹ 1200

Total income = expenses + savings

$$= 31200 + 1200 = ₹ 32400$$

$$\therefore \text{Average income} = \frac{32400}{12} = ₹ 2700$$

13. (4) We are to determine the average weight of P, T, R, F and H.

Obviously, this can't be determined as we don't know the weight of H.

14. (1) Let there were  $x$  friends, then contribution of one friend

18. (3) According to question

$$A = \frac{P + M}{2}$$

$$19. (2) \text{ Let Serma's son's present age} = x \text{ years}$$

$$\therefore \text{Serma's present age} = 4x \text{ years}$$

$$\therefore \text{Serma's father's present age}$$

$$= 4x \times \frac{7}{4} = 7x \text{ years}$$

$$\therefore x + 4x + 7x = 52 \times 3$$

$$\Rightarrow 12x = 56 \Rightarrow x = \frac{56}{12} = 8$$

$$\therefore \text{Required difference}$$

$$= 7x - x = 6x$$

$$= 6 \times 8 = 48 \text{ years}$$

$$20. (1) \text{ If the number of successful candidates be}$$

$$\frac{5x+10}{5} = x+2$$

$$\text{New average} = \frac{5x+10+x+5+x+6}{7}$$

$$\frac{7x+21}{7} = x+3$$

$$\text{Difference} = x+3 - x-2 = 1$$

$$\therefore \text{Total age of initial 30 students} = 14 \text{ years } 4 \text{ months } \times 30$$

$$= 430 \text{ years}$$

$$\text{Total age of 35 students} = 13 \text{ years } 9 \text{ months } \times 35$$

$$= (435 + 26) \text{ years } 3 \text{ months}$$

$$= 481 \text{ years } 3 \text{ months}$$

$$\therefore \text{Total age of 4 new students} = 481 \text{ years } 3 \text{ months} - 430 \text{ years} - 9 \text{ years } 11 \text{ months}$$

$$= 481 \text{ years } 3 \text{ months} - 439 \text{ years } 11 \text{ months}$$

$$= 41 \text{ years } 4 \text{ months}$$

$$\therefore \text{Required average} = \frac{41 \text{ years } 4 \text{ months}}{4}$$

$$= 10 \text{ years } 4 \text{ months}$$

$$= \frac{154}{2} = 77$$

$$\Rightarrow a = \frac{154}{2} = 77$$

$$\begin{aligned} 24. (2) \text{ Number of girls} &= x \\ \text{Number of boys} &= 600 - x \\ \therefore 600 - x &\times 12 + 11x \\ &= 11 \times \frac{3}{4} \times 600 + \frac{47}{4} \times 600 \\ &= 7200 - 12x + 11x = 7050 \\ \Rightarrow x &= 7200 - 7050 = 150 \end{aligned}$$

$$\begin{aligned} 25. (1) m + m - 1 + m + 2 + m + 3 + m + 4 &= 5m \\ \Rightarrow m + 2 &= m \quad \dots (i) \\ \text{Required average} &= \frac{m + 2 + m + 3 + m + 4}{6} \\ &= \frac{6m + 27}{6} \\ &= \frac{2m + 9}{2} = \frac{2n - 21 + 9}{2} = \frac{2n - 12}{2} \end{aligned}$$

$$\begin{aligned} 26. (3) \text{ Number of non-officers} &= x \\ \text{According to the question,} \\ 3200 (x - 5) &= 6800 \times 5 + 2000x \\ \Rightarrow 32x + 160 &= 340 + 20x \\ \Rightarrow 32x - 20x &= 340 - 160 \\ \Rightarrow 12x &= 180 \end{aligned}$$

$$\begin{aligned} \therefore x + 3x &= 3 \times 114 \\ \Rightarrow 4x + 12x + 3x &= 342 \\ \Rightarrow 19x &= 342 \times 4 \\ \Rightarrow x &= \frac{342 \times 4}{19} = 72 \end{aligned}$$

$$\begin{aligned} \therefore \text{Largest number} &= 3x = 3 \times 72 = 216 \\ 30. (3) \text{ Sachin's new average} &= x \text{ runs} \\ \text{Total runs in 11 innings} &= 11 (x - 5) \\ \therefore 11 (x - 5) + 120 &= 12x \\ \therefore 12x - 11x &= 65 \\ \therefore x &= 65 \text{ runs} \end{aligned}$$

$$\begin{aligned} 31. (2) \text{ Required mean} &= \frac{1 \times 1 + 2 \times 2 + 3 \times 3 + 4 \times 4}{1 + 2 + 3 + 4 + 5 + 6 + 7} \\ &= \frac{1 + 2 + 3 + 4 + 5 + 6 + 7}{28} \\ &= \frac{140}{28} = 5 \end{aligned}$$

$$\begin{aligned} 32. (3) \text{ Total expenditure of man in a year} &= \text{Rs. } 180 \end{aligned}$$



$$\Rightarrow a = \frac{3}{4} \times 280 - \frac{3a}{4}$$

$$\Rightarrow a + \frac{3a}{4} = 210$$

$$\Rightarrow \frac{7a}{4} = 210$$

$$\Rightarrow a = \frac{210 \times 4}{7}$$

$$\Rightarrow a = 120$$

On putting the value of  $a$  in equation (1), we get  $b + c + d + e$

$$= \frac{4}{3} \times 120$$

$$\Rightarrow b + c + d + e = 160$$

Their average

$$= \frac{\text{Sum of numbers}}{\text{Total numbers}}$$

$$= \frac{160}{4} = 40$$

4.4. (2) Average age of kishore, his wife and child

315

**Method-2 :** Average marks in five subjects = 63, also sum total of marks in 6 subjects should be more by  $6 \times 2 = 12$ , hence marks in the 6th subject =  $63 + 12 = 75$  marks.

47. (1) Let Quantity of A, B & C be  $7x$ ,  $5x$  and  $4x$ . So,

Average Price of Mixture

$$= \frac{7x(35) + 5x(40) + 4x(42)}{7x + 5x + 4x}$$

$$= \frac{245x + 200x + 168x}{16x} = \frac{613}{16}$$

$$\Rightarrow 38.31 \text{ per kg}$$

48. (1) Average Speed for return journey

$$= \frac{2 \times 85 \times 90}{85 + 90} = \frac{15300}{175} = 87.42 \text{ kmph}$$

Speed for forward journey = 105 kmph & speed for return journey = 87.42 kmph, hence average speed for the entire journey

$$= \frac{2 \times 105 \times 87.42}{105 + 87.42}$$

4. (3) Total age of two new students  
=  $(32 \times 19 + 30 \times 18)$  years  
=  $(608 + 540)$  years  
= 648 years

$$\therefore \text{Required average} = \frac{648}{2}$$

$$= 324 \text{ years}$$

5. (2) Average weight of girls =  $x$  kg. According to the question,

$$58 \times 15 + 30x = \frac{712}{15} \times 45$$

$$\Rightarrow 870 + 30x = 2136$$

$$\Rightarrow 30x = 2136 - 870 = 1266$$

$$\Rightarrow x = \frac{1266}{30} = 42 \text{ kg}$$

6. (4) Sum of ages of A > B and C, 3 years ago  
=  $27 \times 3 = 81$  years

So after 3 years sum of their present ages

$$= 81 + (3 \times 3) = 90 \text{ years}$$

Similarly sum of present age of B and C

$$= (20 \times 2) + (5 \times 2) = 50 \text{ years}$$

$$\therefore \text{present age of A} = 90 - 50$$

Bunny's present age =  $\frac{x}{3}$  years  
According to the question,

$$x + 6 + \frac{x}{3} = 54$$

$$\Rightarrow x + \frac{x}{3} = 54 - 6 = 48$$

$$\Rightarrow \frac{4x}{3} = 48 \Rightarrow x = \frac{3}{4} \times 48$$

$$= 36 \text{ years}$$

$$\therefore \text{Bunny's present age} = \frac{36}{3}$$

$$= 12 \text{ years}$$

$$\therefore \text{Soma's age 12 years ago}$$

$$= 36 - 12 = 24 \text{ years}$$

9. (2) Binny's present age

$$= x \text{ years}$$

$$\therefore \text{Akki's present age}$$

$$= (x - 7) \text{ years}$$

According to the question,

$$x + 16 = 2(x - 7)$$

21. (4) Total runs scored by team A in 50 overs =  $50 \times 5.3 = 265$

Runs to be scored by team B in 5 overs = 7.2

$\therefore 5 = 35$

$\therefore$  Score of team B in 45 overs =  $265 - 35 = 229$

22. (2) 12th number =  $x$

$\therefore$  11th number =  $2x$

13th number =  $x + 3$

According to the question,

$2x + x + x + 3 = 13 \times 47 = 3 \times 39 = 7 \times 49$

$\therefore 4x + 3 = 611 - 117 = 343$

$\therefore 4x = 151 - 3 = 148$

$\therefore x = \frac{148}{4} = 37$

$\therefore$  Average of 11th and 13th number

$= \frac{2x + x + 3}{2} = \frac{3x + 3}{2}$

$= \frac{3 \times 37 + 3}{2} = \frac{114}{2} = 57$

23. (3) Number of boy students in the class

$= \frac{100}{100} = 25$

$= 5 \times 4 = 20$

According to the question

$5x - 2x = 48$

$\therefore 3x = 48$

$\therefore x = \frac{48}{3} = 16$

Total runs scored in three matches =  $5x + 4x$

$= 2x$

$= 11x = 11 \times 16 = 176$

$\therefore$  Required average =  $\frac{176}{3}$

$= 58 \frac{2}{3}$

12. (4) Sum of present ages of husband and wife

$= (23 \times 2 + 10)$  years

$= 56$  years

Sum of present ages of husband, wife and child

$= 60$  years

$\therefore$  Child's present age

$= 60 - 56 = 4$  years

13. (1) Sum of ages of children

$= 10 \times 5 = 50$

$\Rightarrow 10x + x + 206 = 14x$

$\Rightarrow 14x = 11x$

$\Rightarrow x = \frac{14x - 11x}{3} = 1.31 \text{ kg}$

17. (2) Total savings by 10 students

$= \text{Rs. } (10 \times 600) = \text{Rs. } 6000$

Three of the students have no savings.

Total savings by 6 students

$= \text{Rs. } 6000 - 1300$

$= \text{Rs. } 4700$

Let the savings by each of 5 students be

$\text{Rs. } 280.$

$\therefore$  Largest amount of savings

$= \text{Rs. } (4700 - 5 \times 280)$

$= \text{Rs. } (4700 - 1250)$

$= \text{Rs. } 3450$

18. (1) Let the number of members of Indian army be  $x$

$\therefore$  Number of European army =  $12000 - x$

According to the question,

$\frac{5}{12} \times x + 5 \times \frac{10}{12} \times (12000 - x) = 12000$



# AVERAGE

$$13 \times 52.8 + 13 \times 62.2 - x = 1350$$

$$686.4 + 808.6 - x = 1350$$

$$1495.0 - x = 1350$$

$$x = 1495 - 1350$$

$$x = 145$$

$$\Rightarrow 13\text{th number} = x$$

$$\text{Sum of 12 numbers}$$

$$= 1350 - 145 = 1205$$

$$\text{Their average} = \frac{1205}{12} = 50.2$$

29. (3) Number of students in section A = 40  
 Number of students in section B = 52  
 Total students = 40 + 52 = 92  
 Their average score = 75  
 $\Rightarrow$  Their total score  
 $= 92 \times 75$   
 $= 6900$   
 $\dots(1)$   
 Let the average score of students B = x  
 $\Rightarrow$  Average score of students B = x

# AVERAGE

8. (1) Let the no. of officers be x  
 then no. of workers = 400 - x  
 According to question,  
 $3000 + 400 = 10000 \times x + 2000$   
 $400 - x = 10000 \times x + 800000$   
 $100000 = 2000x$   
 $40000 = 8000x$   
 $\therefore x = 50$   
 $\therefore$  No. of officers = 50  
 $\therefore$  workers = 350

9. (2) Required average  
 $= \frac{(72.5 \times 8) - (80 \times 3) - (65 \times 2)}{3}$   
 $= \frac{580 - 240 - 130}{3} = 70$   
 $\therefore$  we know that average speed

(2)]

$$\Rightarrow x = 18 \text{ \& } a = 43$$

12. (3) In 1992, average age of 200 employees  
 $= 50$  years  
 In 1994 (just before leaving) average age of  
 200 employees would be  $50 + 2 = 52$  years  
 Average in 1994 (after 40 employees left)  
 $= \frac{52 \times 200 - 40 \times 58}{160} = \frac{10400 - 2320}{160} = 50.5$   
 Average in 1997 =  $50.5 + 3 = 53.5$   
 Average in 1997 (when new employees are re-  
 cruitd)  
 $= \frac{53.5 \times 160 + 45 \times 50}{210} = \frac{8560 + 2250}{210} = 51.476$   
 Average in the year 2000 =  $51.476 + 3 =$   
 $54.476$  yrs

13. (1) Total age of A, B, C & D is  $38 \times 4 = 152$

# AVERAGE

16. (2) 7 years ago,  
 Husband + wife  
 $\Rightarrow 2 \times 25 = 50$  years  
 Sum of present ages of husband and wife  
 $= 50 + 14 = 64$  years  
 Sum of present ages of husband, wife and child  
 $= 22 \times 3 = 66$  years  
 $\therefore$  Present age of child  $= 66 - 64 = 2$  years
17. (1) Let the no. of students in class X, Y and Z be  $p$ ,  $q$  and  $r$  respectively. The average score of the classes  $= \frac{83p + 76q + 85r}{p + q + r}$  ..... (i)

Also, we are given average score of class X

$$\begin{aligned} VC &= 5000 \text{ per boarder. Hence average expense per boarder when there are 100 boarders} \\ &= \frac{5000 \times 100}{100} = 5000 \end{aligned}$$

19. (4) We only know that Group D consists of 25 + 15 = 40 students, but the weight is unknown.

20. (3) Option (1) average weight of both the groups would increase if the weight of student who is being shifted is less than the average weight of group A and more than the average weight of group B. Option (2) is just the opposite of (1). Both the cases may or may not be true. However (3) would be correct because average weight of ... remains the same because ...

# AVERAGE

24. (3) For time slot of 8-10 i.e. total of 2 hours, speed is 60 kmph and for 6-8, speed is 40 kmph. and for 5-6 speed is 60 kmph, hence

$$\begin{aligned} \text{Average Speed} &= \frac{2 \times 80 + 2 \times 40 + 1 \times 60}{5} \\ &= \frac{160 + 80 + 60}{5} = 60 \text{ kmph} \end{aligned}$$

25. (3) Average increased by 13.5, hence sum increased by  $13.5 \times 8 = 108$ . Let the correct no. be  $abc$  where  $a$ ,  $b$  &  $c$  are the digits in hundreds, tens and unit place respectively. The reversed no. is  $cba$ . Now  $cba - abc = 108$  can be written as  $100c + 10b + a$  and  $cba$  can be written as  $100a + 10b + c$ , so

$$\begin{aligned} \frac{n(n+1)}{2} + x &= 24 \Rightarrow \frac{n^2 + n}{2} + x = 24n \\ \frac{n^2}{2} + x &= 24n - \frac{n}{2} \Rightarrow n^2 + 2x = 47n \\ 2x &= n(47 - n) \end{aligned}$$

$n$  cannot be zero or more than 47, because then  $x$  would be zero or negative, hence  $n < 47$ .

- when  $n = 46$ ,  $x = \frac{46}{2} = 23$  when  $n = 45$ ,  $x = 45$
30. (1) Let the no. of members initially  $= n$  and the average age  $= n$ . We are given that after

## AVERAGE

**Quantity II :** Total marks of 13 papers =  $13 \times 40 = 520$ . Total marks of first seven papers =  $42 \times 7 = 294$  and last seven papers =  $35 \times 7 = 245$ . When we add both the figures i.e.  $294 + 245$  which is equal to 539, the marks of seventh paper are included twice. So,  $539 - 520 = 19$  is the marks of the 7th paper.

**Quantity I < Quantity II**

2. (1) **Quantity I :** Age of teacher =  $25 \times 15 = 25 \times 13$   
=  $390 - 325 = 65$  years

**Quantity II :**  $\frac{24 \times 16 + T - B}{24} = 17$

$$\Rightarrow T - B = 17 \times 24 - 16 \times 24 = 24$$

Where  $T \rightarrow$  age of the teacher and  $B \rightarrow$  Actual age of the boy who is asked.

$$= 24 \text{ years}$$

**Quantity II :** Sixth result =  $6 \times 40 + 6 \times 50$   
 $= 11 \times 50$

$$= 204 + 312 = 550 = 56$$

**Quantity I < Quantity II**

## DATA SUFFICIENCY EXPLANATION

1. (5) According to statement I,

Akshay's weight is  $(15 \times 1.5)$  kg more than that of Sumil. According to statement II, Sumil's weight = 35 kg.

$$\text{Akshay's weight} = 35 + 15 (1.5) \text{ kg.}$$

Both the statements are required to answer the question.

Statement I only tells that Akshay is  $(15 \times 1.5)$  kg more than that of Sumil, but since we do not know Sumil's weight we cannot find the information is given.

# 5 Percentage

## IMPORTANT FACTS AND FORMULAE

The word 'Percentage' or 'Percent' means per hundred or for every one hundred or out of 100. It is a fraction whose denominator is 100. The numerator of the fraction is known as **rate percent**, it is usually denoted by %. Thus, x percent means x hundredths, written as x %.

$$\text{example : } 25\% = \frac{25}{100}$$

Apply it by 100 and put the sign of percentage (%) after it.

$$\text{Value in \%} = \left( \text{Fraction} \times 100 \right) \%$$

3. To find percent (%) of a given number :

Let us suppose that we have to find x % of a

## PERCENTAGE

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>VI.</b> If <math>A</math> is <math>r\%</math> more than <math>B</math>, then <math>B</math> is <math>\left(\frac{r}{100+r} \times 100\right)\%</math> less than <math>A</math>.</p> <p><b>VII.</b> If <math>A</math> is first increased by <math>x\%</math> and then decreased by <math>y\%</math>, the final value is given by <math>A \cdot \frac{100+x}{100} \times \frac{100-y}{100}</math> and net change (increase/decrease) is <math>\left( A \cdot \frac{(100+x)(100-y)}{(100)^2} - A \right) = A \cdot \frac{[100x - 100y - xy]}{(100)^2}</math> (Net change)</p> | <p><math>y</math> : % change in the consumption of the commodity.</p> <p>(ii) If the price of a commodity increases by <math>x\%</math>, then reduction in consumption so as not to increase the expenditure, is <math>\left[ \frac{x}{100+x} \times 100 \right]\%</math>.</p> <p>If the price decreases by <math>x\%</math>, then increase in consumption so as not to increase expenditure, is <math>\left[ \frac{x}{100-x} \times 100 \right]\%</math>.</p> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### IX. Marks in an Examination / Subject

(i) Let the maximum marks be  $M$ .

and the Pass marks be  $P\%$  =  $\frac{PM}{100}$

Marks obtained be  $x\%$  =  $\frac{xM}{100}$

## PERCENTAGE

**7. To convert fraction into % equivalent :** For converting a fraction into %, we simply multiply that fraction by 100, i.e.,  $\frac{a}{b} = \left( \frac{a}{b} \times 100 \right)\%$ .

**Increase/Decrease :** The percentage problems involving Increase or Decrease can be solved easily using common sense. 100 is a common factor in all such cases. The second factor needs to be decided. Whenever there is increase by  $x\%$ , the second factor has to be  $(100 + x)$ . These two factors are used as ratio to get the multiplying factor. It can be either  $\left( \frac{100+x}{100} \right)$  or  $\left( \frac{100}{100+x} \right)$ .

We use  $\left( \frac{100-x}{100} \right)$  when the unknown quan-

If the population decreases at the rate of  $r\%$  per annum, the population after  $n$  years is given by  $P \left( 1 - \frac{r}{100} \right)^n$ .

Population  $n$  years ago was  $\frac{P}{\left( 1 - \frac{r}{100} \right)^n}$ .

**II.** To increase a given number by a given  $r\%$ , multiply the given number by the factor  $\frac{100+r}{100}$  and to decrease replace plus sign by a minus sign in the factor i.e.,  $\frac{100-r}{100}$ .

**III.** If  $A$  and  $B$  are respectively  $a\%$  and  $b\%$  greater than  $C$ , then



## MODEL QUESTIONS

## LEVEL-1

- If  $30\%$  of  $(x - y) = 20\%$  of  $(x + y)$ , then what percent of  $x$  is  $y$ ?  
(1) 25 (2) 20 (3) 30 (4) 24  
(5) None of these
- $p$  is six times as large as  $q$ . The percent that  $q$  is less than  $p$  is  
(1)  $83\frac{1}{3}$  (2)  $16\frac{2}{3}$  (3) 90 (4) 60  
(5) None of these
- Product of one-third of a number and a second

- In a school there are 800 students out of whom 45 percent are girls. Monthly fee of each boy is ₹ 600 and monthly fee of each girl is 30 percent less than each boy. What is the total monthly fee of girls and boys together?  
(1) ₹ 4,25,400 (2) ₹ 4,14,600  
(3) ₹ 4,19,600 (4) ₹ 4,23,400  
(5) None of these
- Fresh cherries contain 99% water. Suppose you have 1 kg of fresh cherries. After a few hours in the sun, some water evaporates and the percentage of water in the cherries becomes 98%. The new weight (in kg) is

- P, Q and R have a certain amount of money with themselves. Q has 25% more than what P has, and R has  $\frac{1}{5}$ th of what Q has. If P, Q and R together have Rs. 150, then how much money does P alone have? (in Rs.)  
(1) 40 (2) 70 (3) 80 (4) 60  
(5) 50
- Rubina decided to donate 16% of her monthly salary to an NGO. On the day of donation she changed her mind and donated Rs. 6,567 which was 75% of what she had decided earlier. How much is Rubina's monthly salary?  
(1) Rs. 8,756 (2) Rs. 54,725  
(3) Rs. 6,56,700 (4) Rs. 42,880
- Sumitra has an average of 56% on her first 7 examinations. How much she should make on her eighth examination to obtain an average of 60% on 8 examinations?  
(1) 88% (2) 78% (3) 98%  
(4) Cannot be determined  
(5) None of these
- In a recent survey 40% houses contained two or more people. Of those houses containing only one person 25% were having only a male. What is the percentage of all houses, which contain exactly one female and no males?  
(1) 75 (2) 40 (3) 15  
(4) Cannot be determined



# PERCENTAGE

- there are 24% literate among men and 8% literate among women, the total number of literate persons in the town is  
 (1) 41,800 (2) 48,900  
 (3) 56,800 (4) 99,600  
 (5) 9995
27. A man had a certain amount with him. He spent 20% of that to buy an article and 5% of the remaining on transport. Then he gifted ₹ 120. If he is left with ₹ 1,400, the amount he spent on transport is  
 (1) ₹ 76 (2) ₹ 61 (3) ₹ 95 (4) ₹ 80  
 (5) None of these
28. In an examination, 52% of the candidates failed in English and 42% failed in Mathematics. If 17% failed in both the subjects,

34. The ratio of the number of boys and girls in a school is 3:2. If 20% of the boys and 25% of the girls are scholarship holders, the percentage of the school students who are not scholarship holders is  
 (1) 56 (2) 78 (3) 70 (4) 80  
 (5) None of these
35. A village lost 12% of its goats in a flood and 5% of remainder died from diseases. If the number left now is 8360, what was the original number before the flood?  
 (1) 10000 (2) 10000  
 (3) 1,00,000 (4) 8360  
 (5) None of these
36. A scored 72% in a cover with a maximum

# PERCENTAGE

41. Ravi is 12 years younger than Surya. Ravi's age is 40% of the sum of his and Surya's age. What will be Surya's age 9 years hence?  
 (1) 36 (2) 24 (3) 33 (4) 45  
 (5) None of these
42. Price of petrol increased from Rs. 40 per litre to Rs. 75 per litre. How much should the consumption of petrol be reduced in % so as to increase expenditure by only 10%?  
 (1) 12 (2) 20 (3) 15 (4) 18  
 (5) None of these
43. A man's annual income has increased by Rs. 1.2 lakhs but the tax on income that he has to pay has reduced from 12% to 10%. He now pays the same amount of tax as before. What is his increase in income?

49. Basil's working hours per day were increased by 15% and his wages per hour were increased by 20%. By how much percent did his daily earnings increase?  
 (1) 40 (2) 38 (3) 35 (4) 36  
 (5) None of these

## LEVEL-II

1. Pradeep invested 20% more than Mohit. Mohit invested 10% less than Raghu. If the total sum of their investment is ₹ 17,800, how much amount did Raghu invest?  
 (1) ₹ 6,000 (2) ₹ 8,000

# PERCENTAGE

6. In village A and B, out of their respective population (male + female),  $\frac{2}{5}$ th and  $\frac{3}{8}$ th are females. The number of males in village B is 25% more than those in village A. The number of females in village A and B together is 2125. What is the total population (male + female) of village A?

- (1) 3600 (2) 2500 (3) 3000 (4) 2000  
(5) 3500

7. Mrs. Sharma invests 15% of her monthly salary, i.e. Rs. 4554, in Mediclaim Policies. Later she invests 25% of her monthly salary in Child Education Policies; also she invests another 9% of her monthly salary in Mutual Funds. What is the total annual amount invested by Mrs. Pooja Pushpan?

- (1) Rs. 28952 (2) Rs. 43428  
(3) Rs. 347424 (4) Rs. 173712  
(5) None of these

23. Mr. Yadav spends 99% of his monthly salary on consumable items and 50% of the remaining on clothes and transport. He saves the

# PERCENTAGE

of the following is definitely true about the strength of the school in 2000 as compared to that in 1996?

- (1) Increase approximately by 2%  
(2) Decrease approximately by 2%  
(3) Increase approximately by 20%  
(4) Decrease approximately by 20%

(5) None of these

14. In an examination the percentage of students qualified to the number of students appeared from school 'A' is 70%. In school 'B' the number of students appeared is 20% more than that of school 'A'. The percentage of students

12. In an election only two candidates contested, 30% of the registered voters did not cast their votes and 160 votes were declared invalid. The winner got 484 votes more than his opponent. The number of valid votes received by the winner is 42% of the number of registered voters. How many registered voters cast their votes?

- (1) 2660 (2) 2260 (3) 2160 (4) 2800  
(5) 2520

13. Anurag's annual income is Rs. 2,96,000. He

# PERCENTAGE

- increased by 20% and that of Chotu decreased by 10%. Again two days later, no. of pencils with Cheeku decreased to  $66\frac{2}{3}$  and that of Chotu increased by  $33\frac{1}{3}\%$ . Then
- Cheeku has more pencils than Chotu
  - Chotu has more pencils than Cheeku
  - The no. of pencils with Cheeku & Chotu are same
  - Data insufficient
  - None of these
26. The consumption of an article increases by 20% and the price of the article increases by 10%. Then the percentage increase in the expenditure on the article is

# PERCENTAGE

1. The ratio of the prices of two houses A and B was 4 : 5 just year. This year, the price of A is increased by 25% and that of B by ₹ 50000. If their price are now in the ratio 9 : 10, the price of A last year was
- ₹ 360000
  - ₹ 450000
  - ₹ 480000
  - ₹ 500000
2. A worker earns a 5% raise. A year later, the worker receives a 2.5% cut in pay and now his salary is ₹ 22702.68. What was his salary to begin with?
- ₹ 22000
  - ₹ 22176
  - ₹ 25000
  - ₹ 22193
3. A wants to secure an annual income of ₹ 1500 by investing in 15% debentures of face value ₹ 100 each and available for ₹ 104 each. If the brokerage is 1%, then the sum of money he should invest is
- ₹ 22000
  - ₹ 22176
  - ₹ 25000
  - ₹ 22193
4. The ratio of the prices of two houses A and B was 4 : 5 just year. This year, the price of A is increased by 25% and that of B by ₹ 50000. If their price are now in the ratio 9 : 10, the price of A last year was
- ₹ 360000
  - ₹ 450000
  - ₹ 480000
  - ₹ 500000
5. A paper piece with 40 cm breadth and 24 cm length is taken. The broad side is folded so that the length is 20 cm. The area of the folded piece is
- $240\text{ cm}^2$
  - $480\text{ cm}^2$
  - $960\text{ cm}^2$
  - $1920\text{ cm}^2$
6. The weight of an empty bottle is 20% of the weight of bottle when filled with some liquid. Some of the liquid has been removed. Then, the bottle, along with the remaining liquid, weighed half of the original weight. What fractional part of the liquid has been removed?
- $\frac{2}{3}$
  - $\frac{3}{9}$
  - $\frac{2}{5}$
  - $\frac{1}{2}$
7. A paper piece with 40 cm breadth and 24 cm length is taken. The broad side is folded so that the length is 20 cm. The area of the folded piece is
- $240\text{ cm}^2$
  - $480\text{ cm}^2$
  - $960\text{ cm}^2$
  - $1920\text{ cm}^2$

# PERCENTAGE

- lines, such that each line consists of 70 characters. The percentage reduction in no. of sheets is closest to—  
 (1) 20 % (2) 5 % (3) 30 % (4) 35 %  
 (5) None of these
20. One bacteria splits into eight bacteria of the next generation. But due to environment, only 50% of one generation can produce the next generation. If the seventh generation number is 4096 million, what is the no. of the first generation?  
 (1) 1 million (2) 2 million  
 (3) 4 million (4) 8 million  
 (5) None of these
21. The length, breadth and height of a room are in the ratio 3 : 2 : 1. If the breadth & the height
24. Find the change in the number of marbles with A.  
 (1) increase by 10 (2) decrease by 10  
 (3) increase by 20 (4) decrease by 20  
 (5) None of these
25. % change in no. of marbles held by C is  
 (1) less than 100% (2) more than 100%  
 (3) equal to 100% (4) None of these  
 (5) None of these
26. Who has lost?  
 (1) A & B (2) B & C  
 (3) C alone (4) B alone  
 (5) None of these
27. ABC Ltd. had 'x' no. of female employees in year 1990, which increased by 20% in 1991.

# PERCENTAGE

- amount of 50% more than A. What % of the total amount paid by B was paid for guavas?  
 (1) more than 50% (2) less than 50%  
 (3) equal to 50% (4) Cannot be determined  
 (5) None of these
- QUANTITY BASED QUESTIONS**  
**Directions (1-5) :** In each of the question two quantities are given, solve each of them and choose the correct option as given below.  
 (1) Quantity I > Quantity II  
 (2) Quantity I > Quantity II  
 (3) Quantity I < Quantity II  
 (4) Quantity I > Quantity II

## DATA SUFFICIENCY QUESTIONS

**Directions (1-5) :** Each question consists of a question and two statements marked I and II.

each subject were 150. The overall percentage marks obtained by Akash in all the three subjects together were 54%. How many marks did he score in subject C?

**Quantity II :** A student was awarded certain marks in an examination. However, after re-evaluation, his marks were reduced by 40% of the marks that were originally awarded to him so that the new score now became 96. How many marks did the student lose after re-evaluation?



# PERCENTAGE

## PERCENTAGE

- I. A bought 20 % of what B bought.
- II. The vegetable seller had a total of 50Kg of potato.
4. Zakib spends 30% of his income on his children's education, 20% on recreation and 10% on healthcare. The corresponding percentages for Supriyo are 40%, 25% and 13%. Who spends more on children's education?
1. Zakib spends more on recreation than Supriyo.
- II. Supriyo spends more on healthcare than Zakib.
5. In a class of 30 students, Rashmi secured the third rank among the girls, while her brother Kumar studying in the same class, secured the sixth rank in the whole class. Between the two, who had a better overall rank?
- I. Kumar was among the top 25% of the boys
- ment list in the class in which 60% were

### LEVEL-III

|         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|
| 1. (3)  | 2. (2)  | 3. (3)  | 4. (1)  | 5. (4)  | 6. (3)  |
| 7. (3)  | 8. (1)  | 9. (2)  | 10. (4) | 11. (2) | 12. (3) |
| 13. (3) | 14. (5) | 15. (3) | 16. (3) | 17. (1) | 18. (3) |
| 19. (1) | 20. (1) | 21. (5) | 22. (3) | 23. (2) | 24. (2) |
| 25. (3) | 26. (1) | 27. (5) | 28. (1) | 29. (3) | 30. (1) |

### QUANTITY BASED

|        |        |        |        |        |
|--------|--------|--------|--------|--------|
| 1. (3) | 2. (3) | 3. (1) | 4. (3) | 5. (1) |
|--------|--------|--------|--------|--------|

### DATA SUFFICIENCY

|        |        |        |        |        |
|--------|--------|--------|--------|--------|
| 1. (4) | 2. (2) | 3. (4) | 4. (2) | 5. (2) |
|--------|--------|--------|--------|--------|

### EXPLANATIONS

$$\frac{405}{x} = 20\% \text{ of } x$$

$$\Rightarrow \frac{405}{x} = \frac{x}{5}$$

$$\Rightarrow x^2 = 405 \times 5$$

$$\Rightarrow x = \sqrt{405 \times 5}$$

$$\Rightarrow x = \sqrt{41 \times 5 \times 5} = 5 \times 5 = 45$$

$\therefore$  Required Number of sweets received by each child

$$\frac{405}{45} = 9$$

6. (d) Let Shamini's salary in June, 2009 be Rs.  $x$ .

$$\therefore 22385 = x \left( 1 + \frac{10}{100} \right)$$

$$\Rightarrow 22385 = x \times \frac{11 \times 11}{100}$$

$$= \text{Rs. } (440 \times 500 + 360 \times 420)$$

$$= \text{Rs. } (264000 + 151200)$$

$$= \text{Rs. } 415200$$

9. (d) Water Pulp

$$\frac{50x}{x} = 2y$$

$$\Rightarrow 50 = 2y$$

$$\therefore 50 \times 2y = 2y = 1 \text{ kg}$$

$$\Rightarrow 200y = 1 \text{ kg}$$

$$\therefore 100y = \frac{1}{200} \times 100$$

$$= \frac{1}{2} \text{ kg} = 500 \text{ g}$$

10. (2) Tricky approach

Monthly salary of Raj

$$= \frac{144 \times 60}{12 \times 100} = \text{Rs. } 0.072 \text{ lakh}$$

$\therefore$  Raj's monthly salary is Rs. 0.072 lakh



# PERCENTAGE

$$\Rightarrow \frac{72 \times x}{100} = 73656$$

$$\Rightarrow x = \frac{73656 \times 100}{72} = \text{Rs. } 102300$$

13. (2) Let the original income be Rs. 100

Expenses = Rs. 80

$\therefore$  Savings = Rs. 20

Increase in expenditure

$$= 37 \frac{1}{2} = \frac{75}{2} \%$$

$\therefore$  His present expenditure

$$= \text{Rs. } (80 + 30) = \text{Rs. } 110$$

Increase in his income =  $16 \frac{2}{3} \%$

$$\therefore \text{His present income} = \text{Rs. } \frac{350}{3}$$

$\therefore$  His present savings

$$= \text{Rs. } \left( \frac{350}{3} - 110 \right) = \text{Rs. } \frac{20}{3}$$

$$\Rightarrow x \times \frac{16}{100} \times \frac{75}{100} = 6567$$

$$\Rightarrow x = \frac{6567 \times 100 \times 100}{16 \times 75}$$

$$= \text{Rs. } 54725$$

16. (5) If the maximum marks be  $x$ , then

$$40\% \text{ of } x = 483 + 117$$

$$\Rightarrow \frac{x \times 40}{100} = 600$$

$$\Rightarrow x = \frac{600 \times 100}{40} = 1500$$

$\therefore$  Minimum passing marks for girls

$$= \frac{1500 \times 35}{100} = 525$$

17. (2) Let Vimal's expense be Rs. 100.

$\therefore$  Aman's expense = Rs. 130

and Raman's expense

$$= \frac{100 \times 100 \times \text{Rs. } 1000}{90} = \frac{100000}{9}$$

# PERCENTAGE

(4) The question cannot be answered because the total marks are not given.

Note: The answer choice (1) i.e., 88% marks obtained by following method would be wrong.

$$80 \times 8 = 64 \times 7 = 448$$

for example,

for maximum marks in each of the first 7 examinations be 100.

and in the 8th exam only 50.

Total maximum marks in first 7 exams

$$= 7 \times 100 = 700$$

Total marks obtained in first of exams

$$= 56 \times 100 = 5600$$

$$\therefore \frac{5600}{700} = 8$$

Total maximum marks obtained in 8 exams

$$= 700 + 50 = 750$$

for the average percentage in 8 exams to be 60%, the total marks obtained should be

$$= \frac{60}{100} \times 750 = 450$$

$$\therefore x \times 60 = 2055 \Rightarrow x = 3425$$

$$\therefore \frac{19900 \times 16}{100} = 25840$$

$$\therefore \text{Kaurhal's monthly salary} = \frac{(25840 \times 100)}{12}$$

$$= \text{Rs. } 215333$$

$$24. (4) \text{ Girls} \Rightarrow \frac{2000 \times 3\%}{100} = 720$$

$$\text{Boys} \Rightarrow 2000 - 720 = 1280$$

$$\text{Each girl's fee} = 480 \times \frac{75}{100} = \text{Rs. } 360$$

$$\therefore \text{Total monthly fee} = \text{Rs. } (1280 \times 480 + 720 \times 360)$$

$$= \text{Rs. } (614400 + 259200)$$

$$= \text{Rs. } 873600$$

25. (1) If the number of persons included in survey be  $x$ , then

$$x \times 60 = 2055 \Rightarrow x = 3425$$



$$= \frac{a}{100} = 12\%$$

43. (2) Initial annual income of person = Rs.  $x$  lakhs

According to the question,

$$x + 1.21 \times \frac{10}{100} = \frac{12x}{100}$$

$$\Rightarrow 10x + 12 = 12x$$

$$\Rightarrow 2x = 12 \Rightarrow x = \text{Rs. 6 lakhs}$$

$\therefore$  Required income

$$= \text{Rs. } (6 + 1.2) \text{ lakhs}$$

$$= \text{Rs. 7.2 lakhs}$$

$$44. (2) b = \frac{a \times 2}{100} = \frac{2a}{100}$$

$$\therefore b \% \text{ of } 10 = 10 \times \frac{2a}{100} \%$$

$$= 20\% \text{ of } \frac{a}{100}$$

45. (2) Let the total amount with the man be Rs. 1000.

Donations to charity = Rs. 300

Remaining amount = Rs. 700

Amount given to wife

$$= \frac{700 \times 30}{100} = \text{Rs. 210}$$

Amount given to son

$$= \frac{700 \times 25}{100} = \text{Rs. 175}$$

Remaining amount

$$= \text{Rs. } (700 - 210 - 175)$$

$$= \text{Rs. 315}$$

$$\therefore \text{Amount received by each daughter} = \frac{315}{3}$$

$$= \text{Rs. 105}$$

When a daughter receive Rs. 105, total amount = Rs. 1000

When a daughter get Rs. 4200000 total amount

$$= \text{Rs. } \left( \frac{1000}{105} \times 4200000 \right)$$

$$= \text{Rs. 400 lakhs}$$

46. (2) Resultant increase of 16% and 14%

$$= \left( 16 + 14 + \frac{16 \times 14}{100} \right) \%$$

$$= (30 + 2.24)\% = 32.24\%$$

Percentage change of 32.24% and - 30%

$$= 99 \times \frac{110}{100} = 108.9$$

Strength in 2000

$$= 108.9 \times \frac{90}{100} = 98.01$$

$$100 - 98.01 = 2\% \text{ decrease}$$

(3) Let the number of students appeared in school A = 100

∴ Number of students qualified in school A = 70

∴ According to question,

$$= 120$$

Number of students appeared in School B

Number of students qualified in School B

$$= 70 + 50\% \text{ of } 70 = 70 + 35 = 105$$

∴ Required percentage ratio

$$= \frac{105 \times 100}{120} = 87.5\%$$

$$= \frac{120}{6} = 20$$

$$= \frac{2100 \times 100}{35000} = 6$$

$$= \frac{23}{100} \times 35000 \times 12 = 96600$$

(1) No. of candidates appeared in state A

No. of candidates appeared in state B = x

$$\frac{7x}{100} - \frac{6}{100}x = 80$$

$$x = 8000$$

(2) Original value of car and house

$$Rs. (500000 + 700000)$$

$$Rs. 12,00,000$$

After 3 years,

$$New \text{ cost of car} = 500000 \times 85\% \times 87\% \times 89\%$$

$$Rs. 329077.5$$

$$New \text{ cost of house} = 700000 \times 110\% \times 112\% \times 114\%$$

$$Rs. 983136$$

$$Total \text{ new cost}$$

$$Rs. (329077.5 + 983136) = Rs. 13,12,213.5$$

$$Approximate \text{ increase}$$

$$13,12,214 - 12,00,000 = Rs. 1,12,214$$

(3) Let Ms. Pooja Pushpan's monthly salary

$$Rs. x$$

# PERCENTAGE

According to the question,  
13% of the  $x = \text{Rs. } 8554$

$$\Rightarrow x = \text{Rs. } \left( \frac{8554 \times 100}{13} \right)$$

$$= \text{Rs. } 65800$$

$$\text{Total monthly investment in percentage} = 13 + 23 + 8 = 44$$

$$\therefore \text{Total monthly investment} = 44\% \text{ of Rs. } 65800$$

$$= \text{Rs. } \left( \frac{44 \times 65800}{100} \right) = \text{Rs. } 28952$$

$$\therefore \text{Total annual investment} = \text{Rs. } (12 \times 28952)$$

$$= \text{Rs. } 347424$$

23. (5) Let Mr. Yadav's annual Salary be  $x$ .

$$\text{Amount spent on consumables} = 0.80x$$

$$\text{Amount spent on clothes and transport} = 0.50(x - 0.80x)$$

$$= 0.50 \times 0.20x$$

$$u = \left( 1 + \frac{1}{3} \right) x \rightarrow u \text{ is } \frac{1}{3} \text{ more than } x$$

Putting in eq. (1)

$$u + x = 56$$

$$x + \frac{1}{3}x + x = 56, x = 56 \times \frac{3}{7} = \text{Rs. } 24$$

$$\text{New Price of Potato} = 150\% \text{ of } 24 = 24 + 12 = \text{Rs. } 36$$

26. (3) Let cost of cauliflower =  $\text{Rs. } 100/\text{kg}$

$$\text{Transportation cost} = \text{Rs. } 15 \text{ (i.e., } 15\% \text{ of } 100)$$

$$\text{Total cost} = 100 + 15 = 115$$

$$\text{New cost of cauliflower} = 70\% \text{ of } 100 = 70$$

$$\text{New cost of transportation} = 14.5\% \text{ of } 15 = 21.75$$

$$\text{Total new cost} = 70 + 21.75 = 91.75$$

$$\% \text{ change (decrease) in total cost} = \frac{115 - 91.75}{115} \times 100 = 20.2\%$$

# PERCENTAGE

Let  $x$  be amount over Rs. above  $\text{Rs. } 15000$   
8% of  $x + 2\% \text{ of } x = \text{Rs. } 500$

$$\frac{100}{10}x = \text{Rs. } 5000 \text{ and}$$

$$x = 500 \times \frac{10}{100} = \text{Rs. } 500$$

Let  $x$  be the total no. of votes on the voter's

part, then

$$\text{Total valid votes casted} = \frac{85}{100}x - 80$$

$\therefore$  15% of voters did not vote while 80 votes were declared invalid

Sum of votes of Shatrughan & Amitabh is equal to total valid votes casted.

Amitabh got 45% as per question of total votes

$$\frac{45}{100}x$$

$$\text{Shatrughan got } 150 \text{ less than Amitabh}$$

$$\text{Shatrughan Votes} = \frac{45}{100}x - 150$$

$$\text{Amitabh Votes} = \frac{45}{100}x - 150$$

$$\therefore x = 10000$$

$\therefore$  Available value of debentures

$$= \left( \frac{104}{100} \times 10000 \right)$$

$$= \text{Rs. } 10400$$

New brokerage = 1% of  $\text{Rs. } 10400$

$$= \frac{1}{100} \times 10400 = \text{Rs. } 104$$

$\therefore$  Sum of money invested

$$= \text{Rs. } (10400 + 104) = \text{Rs. } 10504$$

4. (1) Let the average monthly income of an officer be Rs.  $x$ .

$$\therefore \text{Monthly income of a clerk} = \frac{150x}{140} = \text{Rs. } \frac{5x}{7}$$

$$\text{Monthly income of a manager} = \frac{100x}{50}$$



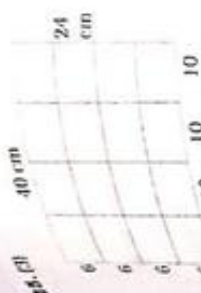
## PERCENTAGE

When  $c = 3$ ,  $a = 74$  and  $50$ .  
 $\rightarrow$  let  $c = 1$  then no. of marks from  $c = 1 \times 3$   
 and  
 $a = 76\%$  no. of marks from  $a = 76 \times 1 = 76$   
 $\therefore$  % of marks from Group A  
 $= \frac{76}{76 + 46 + 3} \times 100 = \frac{76}{125} \times 100 = 60.8\%$   
 $\rightarrow$  when  $c = 2$  and  $a = 75$   
 no. of marks from  $c = 2 \times 3 = 6$   
 no. of marks from  $a = 75 \times 1 = 75$   
 $\therefore$  % of marks from Group A  
 $= \frac{75}{75 + 46 + 6} \times 100 = \frac{75}{127} \times 100 = 59\%$

Hence,  $c = 1$  is the only possibility.  
 $\therefore$  Otherwise no. of questions in Group C will increase, total marks from Group C will increase which would mean decrease in total

fraction part of the liquid removed

$$\frac{50}{100 - 20} = \frac{5}{8}$$



When breadth (40 cm.) is folded twice, it divides the breadth by 4

## PERCENTAGE

Marks obtained by Rohit

$$= 65 - 45 = 20$$

Marks obtained by Rameen

$$= 20 + 25 = 45$$

$\therefore$  Marks obtained by Ravi

$$= 45 + 25 = 70$$

$\therefore$  If the maximum marks be  $x$ , then

$$x - 50 = 70$$

$$\Rightarrow x = 120$$

$\therefore$  Percentage of marks obtained by Ravi

$$= \frac{70}{120} \times 100 = 58.3\%$$

$$= \frac{70}{120} \times 100 = 58.3\%$$

12. (3) Let the number of students in college be

$$100.$$

8. (1) Let the prices of two houses be ₹  $4x$  and ₹  $5x$

$$x \times 1.1 \times 1.08 \times 0.9 = 26730$$

$$\Rightarrow x = \frac{26730}{1.1 \times 1.08 \times 0.9} = 25000$$

7. (3) Dividend for preferred share of company

$$= 50000 \times 10 \times \frac{20}{100}$$

$$= 100000$$

Rest dividend amount of company

$$= 180000 - 100000 = 80000$$

$$= ₹ 50000$$

$\therefore$  Dividend percent for common share

$$= \frac{50000 \times 100}{200000} = 25\%$$

# PERCENTAGE

# PERCENTAGE

Hence 16 sheets are required i.e.

$$\% \text{ change} = \frac{20-16}{20} \times 100 = 20\%$$

20. (1) Let no. of bacteria initially be 10.  
No. of bacteria in 2nd, 3rd, 4th, ... generation would be

$$= \left( \frac{10}{2} \times 8 \right) \left( 8 \times \frac{40}{2} \right) \left( 8 \times \frac{160}{2} \right)$$

$$= 40, 160, 640, 8 \dots$$

Series becomes 10, 40, 160, 640, ...

This is a Geometric Progression with common ratio as 4.

Hence 7th term of GP =  $ar^6$  where  $|a_n = ar^{n-1}|$   
 $a \rightarrow 1st \text{ term}$

$r \rightarrow$  common ratio

$$= 10(4)^6 = 10 \times 4096$$

When 7th generation bacteria is  $10 \times 4096$ , initial bacteria is 10.

When 7th generation bacteria is 4096 million,

Initial ba

$$\left[ \frac{2,60,000 - x}{100} \times 125 \right] \times \frac{25}{100} (2,60,000 - x)$$

$$\left[ x - \frac{25}{100} (2,60,000 - x) \right]$$

$$= 2y$$

$$1.5 (2,60,000 - x) - x = 2y$$

$$3,90,000 = 2y + 2.5x \dots (2)$$

Solving (1) & (2) simultaneously, we get

$$A = 1,40,000$$

Note : Votes got can also be calculated as

$$\left[ \frac{2,60,000 - 125}{100} (2,60,000 - x) \right], \text{ as } 2,60,000$$

votes were cast each time.

$$23. (2) \text{ Pages in } X = 120$$

$$\text{Pages in } Y = \frac{120 \times 110}{100} = 132$$

$$120 \times 90$$

$$120 \times 90$$

$$120 \times 90$$

$$120 \times 90$$

$$120 \times 90$$

$$120 \times 90$$

$$120 \times 90$$

$$120 \times 90$$

$$120 \times 90$$

$$120 \times 90$$

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$$120 \times 90$$

$$x : y = \frac{1}{2}$$

$$\frac{3}{2}C : A : B : C = 2C : \frac{3}{2}C : C = 4 : 3 : 2$$

$$\text{in the end, } A + B = \frac{5}{4}C \text{ and } B + C = 2A$$

$$\text{on solving we get } A = \frac{3}{4}C \text{ \& } B = \frac{2}{4}C$$

$$A : B : C = \frac{3}{4}C : \frac{2}{4}C : C = 3 : 2 : 4$$

$$\text{Initial Ratio} = 4 : 3 : 2$$

$$\text{New Ratio} = 3 : 2 : 4$$

$$B \text{ has lost } 3x - 2x = x \text{ marbles, given to us as}$$

$$10.$$

$$\text{So } x = 10$$

$$\text{Initial no. } A : B : C = 40 : 30 : 20$$

$$100 - \frac{20}{100} (100) - \frac{20}{100} (100 - \frac{20}{100} (100))$$

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$$100 - \frac{20}{100} (100) - \frac{20}{100} (100 - \frac{20}{100} (100))$$

$$\% \text{ total employees} = (22x) \frac{100}{55} = 40x$$

Each organisation has equal no. of employees hence

$$\text{Each organisation has } \frac{40}{4}x = 10x \text{ employees}$$

% of employees going from 1st, 2nd, 3rd & 4th organisation respectively

$$= \frac{4x}{10x} \times 100 = \frac{5x}{10x} \times 100 = \frac{6x}{10x} \times 100 = \frac{7x}{10x} \times 100$$

$$= 40\%, 50\%, 60\%, 70\%$$

29. (3) for sake of simplicity let us assume that there were 100 goats to start with. Then,

In case of option (1)  $p = q$ , let  $p = q = 20\%$

In that case of no. goats in the end would be

$$100 - \frac{20}{100} (100) - \frac{20}{100} (100 - \frac{20}{100} (100))$$

$$100 - \frac{20}{100} (100) - \frac{20}{100} (100 - \frac{20}{100} (100))$$

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$$100 - \frac{20}{100} (100) - \frac{20}{100} (100 - \frac{20}{100} (100))$$

This means  $\frac{2}{3}$ rd of number is 60  
ber

$$= 60 \times \frac{3}{2} = 90$$

**Quantity II :** Let the number be

$$\therefore \frac{x \times 20}{100} = 120$$

$$\Rightarrow x = 120 \times 5 = 600$$

$$\therefore 120\% \text{ of } 600$$

$$= \frac{600 \times 120}{100} = 720$$

Quantity I < Quantity II

2. (3) **Quantity I :** Let there be a students in the class.

Those who failed in Physics

$$= 50\% \text{ of } 100 = 50$$

Those who failed in Chemistry

$$= 40\% \text{ of } 100 = 40$$

Those who failed in both = 10% of

Those who failed in Physics include students who have failed in both Chemistry.

Similarly, same is true for those who failed in Chemistry, hence total no. of students failed in one or both subjects

= students who failed in Physics  
student who failed in chemistry  
student who failed in both.

$$= (50 - 10) + (40 - 10) + 10 = 80$$

Students who passed in both = 100 - 80  
= 20 or 20%

OR

**Method-2 :**

$$(50\% - 10\%) = 40\% \text{ failed in Physics only}$$

$$(40\% - 10\%) = 30\% \text{ failed in Chemistry only}$$

$$10\% \text{ failed in both}$$

$$\text{Total students who failed} = 40\% + 30\% + 10\% = 80\%$$

$$\text{Total students who passed} = 100 - 80 = 20\%$$

**Quantity II :**  $n(A) = 35$ ,  $n(B) = 45$ ,

$$n(A \cap B) = 20$$

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$= 35 + 45 - 20 = 60$$

$\therefore$  Percentage of failed students in English or both = 60%

$\therefore$  Successful students = 40%

Quantity I < Quantity II

# 6

## Profit and Loss (Including Discounts)

### IMPORTANT FACTS AND FORMULAE

1. **Cost Price (C.P.)** : It is the price at which an article is purchased. Profit and loss both are marked at cost price.
2. **Selling Price (S.P.)** : It is the price at which the article is sold.
- 3.(i) **Profit or Gain** : If S.P. is greater than the C.P. there is profit or gain. Profit = S.P. - C.P.  
(ii) **Loss** : If S.P. is less than the C.P. there is loss.

**Type-2 :**

1. If  $N_1$  and  $N_2$  both are the rates of gain or both are the rates of loss, then, C.P.

$$= \left( \frac{100}{N_1 - N_2} \right) \times \text{Amount of difference between selling prices.}$$

## PROFIT AND LOSS (INCLUDING DISCOUNTS)

V. If A sells a thing to B at a loss of  $R_1\%$ , B sells it to C at loss of  $R_2\%$  and C sells it to D at a loss of  $R_3\%$  then

Cost price of D = Cost price of

$$A \left( 1 - \frac{R_1}{100} \right) \left( 1 - \frac{R_2}{100} \right) \left( 1 - \frac{R_3}{100} \right)$$

VI. An article is sold at a profit of  $x\%$ . Had it sold for Rs. a more,  $y\%$  would have been gained. Then,

$$\text{More gain} \times 100$$

$$\text{C.P.} = \frac{\text{Difference in percentage profit}}{\text{More gain} \times 100}$$

VII. If C.P. of  $x$  articles is equal to the selling price of  $y$  articles, then profit percentage =  $\frac{x-y}{y} \times 100\%$

VIII. A dealer purchases a certain number of articles at  $x$  a rupee and the same number at  $y$  a rupee. He mixes them together and sells them at  $z$  a rupee. Then his gain or loss%

$$= \left( \frac{2xy}{x+y} - 1 \right) \times 100 \text{ according to +ve or -ve sign.}$$

IX. If a shopkeeper marks his goods at  $x\%$  above his C.P. and allows buyers a discount of  $y\%$  for cash payment, then there is  $\left( x - y - \frac{xy}{100} \right)\%$  profit or loss according to +ve or -ve sign.



# PROFIT AND LOSS (INCLUDING DISCOUNTS)

## PROFIT AND LOSS (INCLUDING DISCOUNTS)

11. Manoj sold an article for Rs. 15,000. Had he offered a discount of 10% on the selling price he would have earned a profit of 8%. What is the cost price?
- (1) Rs. 12,500 (2) Rs. 13,500  
(3) Rs. 12,250 (4) Rs. 13,250  
(5) None of these
12. A shopkeeper sold a T.V. set for Rs. 17,940/- with a discount of 8% and gained 19.6% If no discount is allowed, what will be his gain percent?
- (1) 25% (2) 26.4% (3) 24.8%  
(4) Cannot be determined
18. Rajni purchased a mobile phone and a refrigerator for Rs. 12,000 and Rs. 10,000 respectively. She sold the refrigerator at a loss of 12 per cent and mobile phone at a profit of 8 per cent. What is her overall loss/profit?
- (1) Loss of Rs. 280 (2) Profit of Rs. 2,100  
(3) Loss of Rs. 240 (4) Profit of Rs. 2,060  
(5) None of these
19. The value of machine depreciates at the rate of 12 per cent per annum. It was purchased three years ago. Its present value is Rs. 29644.032, what was the purchase price of the machine?
- (1) Rs. 48700 (2) Rs. 43500

20. A shopkeeper marks his goods at such a price that after allowing a discount of 12.5% on the marked price, he still earns a profit of 10%. The marked price of an article which costs him Rs. 4,900 is
- (1) Rs. 5,300 (2) Rs. 6,160  
(3) Rs. 5,490 (4) Rs. 6,260  
(5) None of these
26. A discount of 15% is given on the marked price of an article. The shopkeeper charges sales tax of 6% on the discounted price. If the selling price be Rs. 1081.20, what is the marked price of the article?
- (1) Rs. 1185.20 (2) Rs. 1250.20

33. An article was purchased for ₹ 78,350. Its price was marked up by 30%. It was sold at a discount of 20% on the marked up price. What was the profit percent on the cost price?
- (1) 4% (2) 7% (3) 5% (4) 3%  
(5) 6%
34. A shopkeeper buys 144 items for
- of the goods and gets a net gain of 20% of the cost. By how much above the cost should the goods be marked for the sale?
- (1) 40% (2) 50% (3) 60% (4) 70%  
(5) None of these



# PROFIT AND LOSS (INCLUDING DISCOUNTS)

39. A reduction of 21% in the price of an item enables a person to buy 3 kg more for ₹ 100. The reduced price of item per kg is  
 (1) ₹ 5.50 (2) ₹ 7.50 (3) ₹ 10.50 (4) ₹ 7.00  
 (5) None of these
40. A sold a horse to B for ₹ 4800 by losing 20%. B sells it to C at a price which would have given A a profit of 15%. B's gain is  
 (1) ₹ 1800 (2) ₹ 1900 (3) ₹ 2000 (4) ₹ 2100  
 (5) None of these
41. Two toys are sold at ₹ 504 each. One toy brings the dealer a gain of 12% and the other a loss of 4%. The gain or loss percent by selling both the toys is

47. A shopkeeper purchased six identical shirts at the rate of Rs. 220 each. He spent Rs. 1800 on transport and packaging and fixed a marked price of Rs. 450 for each shirt. However, he decided to give discount of 20% on the marked price of each shirt. If he could sell all the shirts, what is the approximate percent profit earned by him?  
 (1) 4.4 (2) 39 (3) 35 (4) 31  
 (5) 48
48. 'A' bought a certain quantity of oranges at total cost of Rs. 1200. He sold 1/3rd of those oranges at 20% loss. If 'A' earns an overall

# PROFIT AND LOSS (INCLUDING DISCOUNTS)

3. A dealer when sells an article at five times the cost price of the article it exceeds twice the cost price equal to three times the profit made in selling the article. What is the profit percentage made on the article?  
 (1) 60% (2) 55% (3) 45% (4) 50%  
 (5) 65%
4. Man sold two articles-A (at a profit of 40%) and B (at a loss of 20%). He incurred a total profit of Rs. 18 in the whole deal. If article A costs Rs. 140 less than article B, what is the price of article B?
10. The total cost price of two products is Rs. 4400. First product is sold at 15% profit while the second product is sold at 18% loss. If neither profit nor loss occurs then what is the cost price of the first product?

maintains the same, then find decrement in the profit is what per cent of the selling price of the article?  
 (1) 25% (2) 30% (3) 40% (4) 27.5%  
 (5) None of these

# PROFIT AND LOSS (INCLUDING DISCOUNTS)

15. A certain product C is made of two ingredients A and B in the ratio of 2 : 5. The price of A is three times that of B. The overall cost of C is Rs. 5.20 per kg including labour charge of 80 paise per kg. Find the cost of B per kg.
- Rs. 8.40
  - Rs. 4.20
  - Rs. 4.80
  - Rs. 2.80
  - None of these
16. Of the two varieties of rice available, variety A is bought at ₹ 32 per kg. and variety B at ₹ 80 per kg. Two varieties of rice are mixed together in the respective ratio of 8 : 5 and the mixture is sold at ₹ 72 per kg. What per-
- the percent profit earned in 1996 increased by 6%. What was his investment in 1995 ?
- Rs. 1,02,000
  - Rs. 1,50,500
  - Rs. 1,05,000
  - Data inadequate
  - None of these
21. A sells an item at 20% profit to B. B sells it to C at 10% profit. C sells it to D at Rs. 116 profit. Difference between the cost price of D and Cost price of A was Rs. 500. How much did B pay to A for the item ?
- Rs. 1,240
  - Rs. 1,250
  - Rs. 1,400
  - Rs. 1,300
  - Rs. 1,200

# PROFIT AND LOSS (INCLUDING DISCOUNTS)

17. A wholesaler had 200 dozens of mangoes. He sold some of these mangoes at 20% profit and the rest at 10% profit, so that he made 13% profit on selling all the mangoes. How many mangoes (in dozens) did he sell at 20% profit?
- 140
  - 60
  - 80
  - 120
  - None of these
18. Radha marks her goods 25% above the cost price. She sells 35% of goods at the marked price, 40% at 15% discount and the remaining goods at discount. What is her overall profit or loss?
4. Vishal goes to a shop to buy a radio costing ₹ 2558. The rate of sales tax is 7%. He tells the shopkeeper to reduce the price of the radio to such an extent that he has to pay ₹ 2563, inclusive of sales tax. Find the reduction needed in the price of the radio.
- ₹ 179.76
  - ₹ 170
  - ₹ 168
  - ₹ 169
  - None of these
5. A dairyman pays ₹ 6.4 per litre of milk. He adds water and sells the mixture at ₹ 7.2 per litre. What is his profit or loss per litre of milk?

# PROFIT AND LOSS (INCLUDING DISCOUNTS)

10. A company produces a bike at the cost of Rs. 16,000. It fixes Rs. 24,000 as selling price in show room. The company allows the show room to include tax and profit but the selling price must not exceed Rs. 25,000 per bike. If the profit and tax ratio is 6 : 5, what is the profit of show room out of profit of the company per bike?

- (1) 55% (2) 60% (3) 75% (4) 80% (5) 85%

11. A firm of readymade garments makes both men's and women's shirts. Its average profit is 6% of the sales. Its profit in men's shirts average 8% of the sales and women's shirts comprise 60% of the output. The average profit per sales rupee in women's shirts is

- (1) 0.0466 (2) 0.0666 (3) 0.0866 (4) 0.1066

Direction (16-17): Mr. David manufactures and sells a single product at a fixed price in a free market. The selling price of each unit is ₹ 30. On the other hand, the cost in rupees of producing  $x$  units is  $240 + 6x + x^2$ , where  $x$  and  $y$  are some constants. Mr. David noted that doubling the daily production from 20 to 40 units, increases the daily production cost by  $\frac{2}{3}\%$ . However, an increase in daily production from 40 to 60 units results in an increase of only 50% in the daily production cost. Assume that demand is unlimited and that Mr. David can sell as much as he can produce. His objective is to maximise profit.

16. How many units should Mr. David produce daily?

- (1) 100 (2) 70 (3) 150 (4) 130

17. By  $\frac{2}{3}\%$ . However, an increase in daily production from 40 to 60 units results in an increase of only 50% in the daily production cost. Assume that demand is unlimited and that Mr. David can sell as much as he can produce. His objective is to maximise profit.

18. How many units should Mr. David produce daily?

- (1) 100 (2) 70 (3) 150 (4) 130

# PROFIT AND LOSS (INCLUDING DISCOUNTS)

19. 'True Value' a shop for second hand cars brought 3 cars for ₹ 1,00,000 each. They spent a total of ₹ 1,25,000 on repairs of all the three cars and then sold them to 3 different customers. If one of the cars was sold for ₹ 1,45,000 and it is known that a total profit of 40% was made on all the 3 cars, then find the price at which the other 2 cars were sold if both the remaining cars were sold for same price.

- (1) ₹ 2,25,000 (2) ₹ 1,45,000 (3) ₹ 3,05,000 (4) ₹ 1,55,000 (5) None of these

20. A readymade garment store marks up his merchandise in such a way that a profit of 10% is earned even after giving a discount of 6% to the customer.

21. A shopkeeper cheats both his customers as well as his suppliers. While he purchases 20% more from the supplier, he sells 15% less to his customers. Find his profit percentage if he claims to be selling at no profit, no loss?

- (1)  $\frac{700}{17}\%$  (2)  $\frac{600}{13}\%$  (3)  $\frac{650}{19}\%$  (4)  $\frac{620}{15}\%$

22. Given that cost price of two articles is the same. While one is sold at a profit of 20%, the other is sold for ₹ 110 more than the first. If the overall profit is 25%, then find the cost price of each article.

- (1) ₹ 1000 (2) ₹ 1100 (3) ₹ 1200 (4) ₹ 1500 (5) None of these

23. A shopkeeper cheats both his customers as well as his suppliers. While he purchases 20% more from the supplier, he sells 15% less to his customers. Find his profit percentage if he claims to be selling at no profit, no loss?

- (1)  $\frac{700}{17}\%$  (2)  $\frac{600}{13}\%$  (3)  $\frac{650}{19}\%$  (4)  $\frac{620}{15}\%$

## PROFIT AND LOSS (INCLUDING DISCOUNTS)

### QUANTITY BASED QUESTIONS

**Directions (1-5) :** In each of the question two quantities are given, solve each of them and choose the correct option as given below.

- (1) Quantity I > Quantity II
- (2) Quantity I ≥ Quantity II
- (3) Quantity I < Quantity II
- (4) Quantity II ≥ Quantity I
- (5) Quantity I = Quantity II or relation can't be established

1. **Quantity I :** A trader sells his goods at 15% discount and increases his profit by 20%.

### DATA SUFFICIENCY QUESTIONS

**Directions (1-5) :** Each question consists of a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statements is sufficient to answer the question. Read the question and both the statements and—

**Give answer (1)** If the data in statement I alone is sufficient to answer the question while the data in statement II alone is not sufficient to answer the question.

**Give answer (2)** If the data in statement II

## PROFIT AND LOSS (INCLUDING DISCOUNTS)

### SHORT ANSWERS

#### LEVEL-I

|         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|
| 1. (5)  | 2. (2)  | 3. (1)  | 4. (1)  | 5. (3)  | 6. (3)  |
| 7. (1)  | 8. (3)  | 9. (3)  | 10. (2) | 11. (1) | 12. (5) |
| 13. (2) | 14. (5) | 15. (4) | 16. (1) | 17. (3) | 18. (3) |
| 19. (2) | 20. (5) | 21. (2) | 22. (2) | 23. (3) | 24. (4) |
| 25. (2) | 26. (4) | 27. (2) | 28. (3) | 29. (4) | 30. (5) |
| 31. (5) | 32. (3) | 33. (1) | 34. (3) | 35. (4) | 36. (4) |

$$= \frac{50000 - 40000}{50000} \times 100$$

$$= \frac{4}{50} \times 100 = 8\%$$

2. (2) Cost price of 145m long cloth

$$= \text{Rs. } (12225 - 145 \times 10)$$

$$= \text{Rs. } 10875$$

∴ CP of 1 metre long cloth

$$= \frac{10875}{145} = \text{Rs. } 75$$

3. (1) Rs. 60



# PROFIT AND LOSS (INCLUDING DISCOUNTS)

9. (3) Total marked Price of article =  $25 \times 45 =$  Rs. 1125/-  
 Selling Price (Giving 10% discount)  
 $= \frac{90}{100} \times 1125 = 1012.5$   
 Let cost price = Rs.  $x$   
 Profit% = 50  
 $\therefore \frac{150}{100} x = 1012.5 \Rightarrow x = 675$   
 Now the selling price is Rs. 1125  
 then profit  
 $= 1125 - 675 = 450$   
 $\therefore$  profit =  $\frac{450}{675} \times 100 = 66\frac{2}{3}\%$   
 10. (2) After a discount of 20%  
 cost of the book = 80% of Rs. 150  
 $= \frac{150 \times 80}{100} = \text{Rs. } 120$   
 Let the additional discount be  $x\%$   
 Second discount =  $120 - 108 = \text{Rs. } 12$   
 $\therefore x\%$  of Rs. 120 = 12  
 $\Rightarrow \frac{120 \times x}{100} = 12$   
 $\Rightarrow x = \frac{12 \times 100}{120} = 10\%$   
 11. (1) Let the cost price be Rs.  $x$   
 $\therefore 90\%$  of 15000 = 108% of  $x$

12. (3) Let the C.P. of the watch be Rs.  $x$   
 First S.P. =  $\frac{120x}{100} = \text{Rs. } \frac{6x}{5}$   
 Case II  
 C.P. = Rs. 45 = 1000  
 S.P. = Rs.  $\left( \frac{6x}{5} - 1000 \right)$   
 Profit =  $\left[ \frac{6x}{5} - x \right] = \text{Rs. } \frac{x}{5}$   
 $\therefore \frac{x}{5} \times \frac{100}{100} = 100 - 25$   
 $\Rightarrow 25x - 2500 = 20x$   
 $\Rightarrow 5x = 2500$   
 $\Rightarrow x = \frac{2500}{5} = \text{Rs. } 500$   
 14. (5) Let the C.P. of each bed be Rs.  $x$   
 S.P. of first bed = Rs.  $\frac{130x}{100}$   
 $= \text{Rs. } \frac{13x}{10}$   
 S.P. of second bed  
 $= \text{Rs. } \left( \frac{13x}{10} - 5504 \right)$   
 According to the question,

# PROFIT AND LOSS (INCLUDING DISCOUNTS)

22. (2) Profit percent  
 $= \frac{54}{876} \times 100 = 6\%$   
 23. (3) Let the C.P. of article be Rs. 100.  
 $\therefore$  Its marked price = Rs. 120  
 S.P. = Rs. 110  
 $\therefore$  Discount Rs.  $120 - 110$   
 $= \text{Rs. } 10$   
 Let the discount be  $x\%$   
 $\therefore \frac{x \times 120}{100} = 10$   
 $\therefore x = \frac{1000}{120}$   
 $= \frac{25}{3} = 8\frac{1}{3}\%$   
 24. (4) Single equivalent discount for successive  
 discounts of 20% and 4%  
 $= \left( 20 + 4 - \frac{20 \times 4}{100} \right)\%$   
 $= 38.56\%$   
 $\therefore$  Difference  
 $= 500 \times \left( 40 - 38.56\% \right)$   
 $= \frac{500 \times 1.44}{100}$

17. (3) Let the C.P. of a cellphone be ₹  $x$   
 According to the question,  
 132% of  $x = 6600$   
 $\Rightarrow \frac{x \times 132}{100} = 6600$   
 $\Rightarrow x = \frac{6600 \times 100}{132} = ₹ 5000$   
 18. (3) Total C.P. = Rs.  $(12000 + 10000) = \text{Rs. } 22000$   
 Total S.P.  
 $= \text{Rs. } \left( \frac{12000 \times 108}{100} + \frac{10000 \times 88}{100} \right)$   
 $= \text{Rs. } 11280 + 8800 = \text{Rs. } 20080$



# PROFIT AND LOSS (INCLUDING DISCOUNTS)

27. (2)

Profit on first part 8% Profit on second part 11%



Ratio of first and second parts  
= 4 : 6 = 2 : 3

∴ Quantity of 2nd part

$$= \left( \frac{2}{5} \times 1000 \right) \text{ kg}$$

= 600 kg

28. (3) If the marked price of T.V. be Rs. x, then

$$\frac{4x}{5} - \frac{3x}{4} = 500$$

$$\Rightarrow \frac{16x - 15x}{20} = 500$$

$$\Rightarrow \frac{x}{20} = 500$$

$$\Rightarrow x = 10000$$

∴ Required cost

Single equivalent discount

$$= \left( 20 + \frac{25}{4} - \frac{20 \times 25}{400} \right) \%$$

$$= 25\%$$

$$\therefore x \times \frac{75}{100} = 120$$

$$\Rightarrow x = \frac{120 \times 100}{75} = \text{Rs. } 160$$

Required % = 60%

33. (1) Gain percent (% change)

$$= \left( 30 - 20 - \frac{30 \times 20}{100} \right) \% = 4\%$$

34. (3) 20 items are broken out of 144 items.

∴ C.P. of 124 items

$$= \text{Rs. } \left( \frac{144 \times 90}{100} \right) = \text{Rs. } 129.60$$

$$\text{Total S.P.} = \text{Rs. } (1.20 \times 124)$$

$$= \text{Rs. } 148.8$$

$$\therefore \text{Gain} = \text{Rs. } (148.80 - 129.60)$$

$$= \text{Rs. } 19.20$$

∴ Gain per cent

# PROFIT AND LOSS (INCLUDING DISCOUNTS)

41. (1) C.P. of first toy = Rs. x

C.P. of second toy = Rs. y

$$\therefore \frac{x \times 112}{100} = 504$$

$$\Rightarrow x = \frac{504 \times 100}{112} = \text{Rs. } 450$$

$$\text{Again, } y \times \frac{96}{100} = 504$$

$$\Rightarrow y = \frac{504 \times 100}{96} = \text{Rs. } 525$$

$$\text{Total C.P.} = \text{Rs. } (450 + 525)$$

$$= \text{Rs. } 975$$

$$\text{Total S.P.} = 2 \times 504$$

$$= \text{Rs. } 1008$$

$$\text{Gain} = 1008 - 975 = \text{Rs. } 33$$

$$\therefore \text{Profit percent} = \frac{33 \times 100}{975}$$

$$= \frac{44}{13} = 3\frac{5}{13}\%$$

42. (1) C.P. of article = Rs. x

$$\text{S.P.} = \frac{120x}{100} = \text{Rs. } \frac{6x}{5}$$

# PROFIT AND LOSS (INCLUDING DISCOUNTS)

$$\Rightarrow \frac{30x - 27x}{25} = 30$$

$$\Rightarrow 3x = 30 \times 25$$

$$\Rightarrow x = \frac{30 \times 25}{3} = \text{Rs. } 250$$

44. (3) C.P. of mobile = Rs. A (let)

$$\therefore \frac{x \times 112}{100} = P$$

$$\text{and, } \frac{96x}{100} = Q$$

$$\therefore Q : P = \frac{96x}{100} : \frac{112x}{100}$$

$$= 96 : 112 = 6 : 7$$

45. (1) C.P. of 25 windows

$$= \frac{120000 \times 75}{100} = \text{Rs. } 90,000$$

After additional discount,

C.P. for builder

$$= \text{Rs. } (90000 - 7500)$$

$$= \text{Rs. } 82500$$

$\therefore$  Cost of each window

$$= \frac{82500}{25}$$

According to the question,

S.P. of 12 oranges

$$= \frac{1200 \times 110}{100} = \text{Rs. } 1320$$

4 oranges are sold on 20% loss.

$$\therefore \text{Their S.P.} = \frac{400 \times 80}{100} = \text{Rs. } 320$$

$\therefore$  Required S.P. of remaining 8 oranges

$$1320 - 320$$

$$= \text{Rs. } 1000$$

$\therefore$  Required profit percent

$$= \left( \frac{1000 - 800}{800} \right) \times 100 = 25\%$$

49. (1) Let the C.P. of article be Rs. x

$$\therefore \text{Marked price} = \text{Rs. } \frac{112x}{100}$$

According to the question,

$$\frac{112x}{100} \times \frac{95}{100} = 532$$

$$\Rightarrow x = \frac{532 \times 10000}{112 \times 95} = \text{Rs. } 500$$

$$= \frac{(2x + 100) \times 100}{5}$$

$$= 40x + 4000 - 40x = 10x + 500$$

$$\Rightarrow 40x + 4000 - 500 = 3500$$

$$\Rightarrow 10x = 3500 - 4000 = -500$$

$$\Rightarrow x = \frac{-500}{10} = \text{Rs. } -50$$

9. (1) According to the question,

$$3 (\text{SP} - \text{CP}) = 5\text{CP} - 2\text{SP}$$

$$\Rightarrow 5\text{CP} + 3\text{CP} = 5\text{SP}$$

$$\Rightarrow \left( \frac{8\text{CP}}{5} \right) = \text{SP}$$

$$\text{Profit} = \frac{8\text{CP}}{5} - \text{CP} = \frac{3\text{CP}}{5}$$

Required profit percentage

$$= 100 \times \left( \frac{3}{5} \right) = 60\%$$

4. (5) Let cost price of article B be Rs. x.

$\therefore$  Cost price of article A

$$= \text{Rs. } (x - 140)$$

According to the question,

$$20x$$

$$\therefore \text{Difference} = \text{Rs. } 3$$

$$\therefore \text{Difference} = \text{Rs. } 5$$

$$= \frac{100}{3} \times 5 = \text{Rs. } 200$$

7. (1) C.P. of Item A = Rs. x

$\therefore$  C.P. of Item B = Rs.  $(x + 180)$

$$\text{S.P. of Item A} = \frac{5}{4}$$

$$\text{S.P. of Item B} = \frac{5}{4}$$

$$= \frac{120x}{100}$$

$$\Rightarrow \frac{100}{(x + 180) \times 50} = \frac{5}{4}$$

$$= \frac{2x}{x + 180} = \frac{5}{4}$$

$$\Rightarrow 8x = 5x + 900$$

$$\Rightarrow 8x - 5x = 900$$

$$\Rightarrow 3x = 900$$

$$\Rightarrow x = \frac{900}{3} = \text{Rs. } 300$$

# PROFIT AND LOSS (INCLUDING DISCOUNTS)

10. (1) C.P. of first product = Rs.  $x$

$\therefore$  C.P. of second product

$$= \text{Rs. } (4400 - x)$$

According to the question,

$$x \times 115 \times \frac{82}{100} + (4400 - x) \times \frac{82}{100} = 4400$$

$$\Rightarrow \frac{115x}{100} + \frac{82x}{100} - \frac{82x}{100} = 4400$$

$$\Rightarrow \frac{33x}{100} = 4400 - 3608 = 792$$

$$\Rightarrow x = \frac{792 \times 100}{33} = \text{Rs. } 2400$$

11. (5) Since MP  $\times (100 - 14)$

$$= \text{CP} \times (100 + 20)$$

$$\Rightarrow \text{MP} \times 86 = \text{CP} \times 120$$

$$\Rightarrow \frac{\text{MP}}{\text{CP}} = \frac{120}{86} = \frac{60}{43}$$

$$\therefore \frac{\text{MP} - \text{CP}}{\text{CP}} = \frac{60 - 43}{43} = \frac{17}{43}$$

$$\Rightarrow \frac{850}{43} = \frac{17}{43}$$

$$\Rightarrow 850 \times 43$$

14. (4) C.P. of article B = Rs.  $x$

$\therefore$  C.P. of article A

$$= \text{Rs. } (x + 100)$$

According to the question,

$$(x + 100) \times 100 + 100$$

$$= (2x + 100) \times \frac{105}{100}$$

$$\Rightarrow 140x + 14000 + 60x$$

$$\Rightarrow 210x + 10500$$

$$\Rightarrow 210x - 200x$$

$$\Rightarrow 10x = 3500$$

$$\Rightarrow \text{Rs. } x = 350$$

OR

$$(x + 100) \times 40 - 40x$$

$$= (2x + 100) \times 5$$

$$\Rightarrow 4000 = (2x + 100) \times 5$$

$$\Rightarrow 2x + 100 = 800$$

$$\Rightarrow 2x = 700$$

$$\Rightarrow x = \text{Rs. } 350$$

15. (4) C.P. of ingredient B

# PROFIT AND LOSS (INCLUDING DISCOUNTS)

According to the question,

$$\frac{7x \times 70}{5 \times 100} + 35$$

$$= \frac{105}{100}$$

$$\Rightarrow 980x + 3500 \times 5 = 210x \times 5 \Rightarrow 1050x$$

$$980x = 17500$$

$$\Rightarrow 70x = 17500$$

$$\Rightarrow x = \frac{17500}{70} = \text{Rs. } 250$$

Alter :

Let, C.P. of each article = ₹100

$\therefore$  M.P. of each article = ₹140

Now, S.P. of article 'A'

$$= 140 \times \frac{70}{100} = ₹ 98$$

$$\text{Total profit} = 200 \times \frac{5}{100} = ₹ 10$$

$$\therefore \text{Total S.P.} = 200 + 10 = ₹ 210$$

$$\therefore \text{S.P. of article 'B'} = 210 - 98$$

$$= ₹ 112$$

Accor

$$\therefore 1.25(x - 5000) = 1.20x$$

$$0.05x = 1.25 \times 5000$$

$$x = \frac{6250}{0.05}$$

$$x = \text{Rs. } 105000$$

21. (3) C.P. for A = Rs.  $x$   
C.P. for D

$$= x \times \frac{120}{100} \times \frac{110}{100} + 115$$

$$= 1.32x + 115$$

$\therefore$  According to the question,  
 $1.32x + 115 - x = 500$

$$\Rightarrow 0.32x = 500 - 115 = 384$$

$$\Rightarrow x = \frac{384}{0.32} = \text{Rs. } 1200$$

$$\therefore \text{C.P. for B} = \frac{1200 \times 120}{100}$$

$$= \text{Rs. } 1440$$

22. (1) C.P. of article A = Rs.  $x$

# PROFIT AND LOSS (INCLUDING DISCOUNTS)

$$= x \times \frac{9}{5} = 72000$$

$$= x = \frac{72000 \times 5}{9} = \text{Rs. } 40000$$

25. (3) At the rate of 50 paise per metre, C.P. of 250 metre of wire

$$= \text{Rs. } \left( \frac{250 \times 50}{100} \right) = \text{Rs. } 125$$

$$\text{C.P. of 500 metre of wire} = \text{Rs. } 250$$

$\therefore$  For 10% overall profit,

$$\text{Total S.P.} = \text{Rs. } \left( \frac{250 \times 110}{100} \right) = \text{Rs. } 275$$

$$\text{S.P. of 250 metre of wire}$$

$$= \text{Rs. } \left( \frac{125 \times 105}{100} \right) = \text{Rs. } 131.25$$

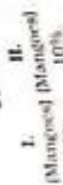
$$\therefore \text{S.P. of remaining 250 metre wire} = \text{Rs. } (275 - 131.25)$$

$$= \text{Rs. } 143.75$$

$\therefore$  Required profit percent

$$\Rightarrow x = \frac{600}{10} = 60 \text{ dozens}$$

OR



Required number of Mangoes

$$= \frac{3}{(3+7)} \times 200 = 60 \text{ dozens.}$$

26. (1) C.P. of 100 articles = Rs. 100

Their marked price = Rs. 125

$\therefore$  S.P. of 35 articles

$$= \text{Rs. } (1.25 \times 35)$$

$$= \text{Rs. } 43.75$$

S.P. of 40 articles

# PROFIT AND LOSS (INCLUDING DISCOUNTS)

$$\Rightarrow x + \frac{7x}{100} = 2568$$

$$\Rightarrow x = \frac{256800}{107}$$

$$\Rightarrow x = 2400$$

$\therefore$  Reduction needed in the price of radio =  $(2568 - 2400) = \text{₹ } 168$

5. (2) Let the quantity of milk purchase be  $x$  and quantity of water added be  $y$ .

Then, ratio of water to milk be  $y : x$ .

$$\text{C.P.} = 6.4x$$

$$\text{S.P.} = 8(x + y)$$

Profit percent = 37.5%

$$\therefore 8(x + y) = 6.4x \times 1.375$$

$$8x + 8y = 8.8x$$

$$\Rightarrow 8y = 0.8x$$

$$\Rightarrow \frac{x}{y} = \frac{80}{8} = \frac{10}{1}$$

$$\Rightarrow y : x = 1 : 10$$

6. (2) According to a prof.

## LEVEL-III

1. (1) According to question, S.P. of 12 marbles = ₹1, loss = 20%

$\therefore$  C.P. of 12 marbles

$$= ₹ \frac{1}{0.8} = ₹ 1.25$$

Now, S.P. of 12 marbles at a gain of 20%

$$\text{C.P.} \times 1.2 = 1.25 \times 1.2 = ₹ 1.5$$

It means in order to gain 20%, he should sell 12 marbles for ₹1.5

i.e. For ₹1, he should sell  $\frac{12}{15} = 8$  marbles

2. (2) Let the original cost price be ₹100.

Then, profit = ₹320 and S.P. = ₹420



# PROFIT AND LOSS (INCLUDING DISCOUNTS)

9. (4) Required number of items

$$= \frac{(2000 - 1000)}{(60 - 40)} = \frac{1000}{20} = 50$$

10. (3) Profit of the company per bike = Rs. (24,000

$$- 16,000)$$

$$= \text{Rs. } 8,000$$

Profit of the show room

$$= (35,000 - 24,000) \times \frac{6}{11}$$

$$= \frac{11000 \times 6}{11} = \text{Rs. } 6,000$$

$$\text{Required percentage} = \frac{6000}{8000} \times 100 = 75\%$$

11. (1) According to question, women's shirts

comprise 60% of the output.

∴ Men's shirts comprise 40%

∴ Average profit from men's

14. (2) Cost of producing 1500 watches

$$= 150 \times 1500 + 30,000 = 255,000$$

Sales revenue from 1500 watches

$$= (1200 \times 250) + (300 \times 100) = 3,30,000$$

$$\text{Profit earned} = 3,30,000 - 255,000 = \text{₹ } 75,000$$

15. (2) Cost of producing 1500 watches = 2,55,000

let the no. of watches sold during the season

= x, so no. of watches sold otherwise (at ₹ 100/

unit)

$$= 1500 - x$$

Break-even means no profit no loss. Hence Cp

= SP

$$250x + 100 (1500 - x) = 2,55,000$$

$$150x = 1,05,000, x = 700 \text{ watches}$$

16. (1) We know  $06 \frac{2}{3}\% = \frac{2}{3}$

Production cost when 20 units are produced

$$= 240 + 20b + (20)^2c = 240 + 20b + 400c$$

# PROFIT AND LOSS (INCLUDING DISCOUNTS)

Using options, we find profit is maximum

when x

$$= 100$$

23. (5) Ratio is 1 : 4

| Pencil | Eraser |
|--------|--------|
| X      | a      |
| Y      | b      |

% P/unit

C.P./Unit

When 20 pencils are sold →

$$20x - 20y = 5a \text{ i.e. } 20x - 10b = 5a$$

When 20 erasers are sold

$$20b - 20a = 8x \text{ i.e. } 20b - 10a = 8x$$

$$\text{Also } \% \text{ Profit} = \frac{5a}{20y} \times 100 \text{ and } \% \text{ loss} =$$

$$\frac{8x}{20y} \times 100$$

$$\frac{5a}{20y} = 2 \left[ \frac{5x}{20b} \right] \Rightarrow \frac{5a}{y} = \frac{8x}{b} \dots (i)$$

$$\text{Also if } b = 3, 2 \Rightarrow \frac{5a}{y} = \frac{8x}{3} \text{ or } 2a = \frac{8}{3}x \text{ or } 2a = \frac{8}{3}x$$



# PROFIT AND LOSS (INCLUDING DISCOUNTS)

26. (2) Let CP of each article = ₹  $x$ , then sales Revenue  
 $1.2x + [1.2x + 110] = 1.25(2x)$   
 $2.4x + 110 = 2.5x \Rightarrow 0.1x = 110, x = ₹ 1100$
27. (1) Let C.P. per unit = ₹ 1, but actually he is paying Re. 1 for 1.2 units (getting 20% more).  
 So, his actual C.P./unit =  $\frac{1}{1.2} = \frac{10}{12}$  per unit  
 C.P. for 0.85 units =  $\frac{10}{12} \times 0.85 = \frac{17}{24}$   
 [because he is giving 15% less]  
 Actual S.P. charged from the customer—  
 S.P./unit = ₹ 1 [Since he is selling at no profit & no loss] actually he is charging ₹ 1 for 0.85 units].

17

Pr

# PROFIT AND LOSS (INCLUDING DISCOUNTS)

30. (1) Quantity I : Net gain percent

$$= \left[ \frac{20 \times 15}{20 - 15 - 100} \right] \%$$

Quantity II : Let C.P. of one article = ₹ 1  
 ⇒ C.P. of 15 articles = ₹ 15 and C.P. of 12 articles = ₹ 12.  
 S.P. of 12 articles = C.P. of 15 articles = ₹ 15

$$\text{Gain \%} = \left[ \frac{\text{S.P. of 12 articles} - \text{C.P. of 12 articles}}{\text{C.P. of 12 articles}} \right] \times 100$$

$$= \left[ \frac{15 - 12}{12} \right] 100 = \frac{1}{4} \times 100 = 25\%$$

Quantity I < Quantity II

Quantity I : If the initial C.P. of car be ₹  $x$

## DATA SUFFICIENCY EXPLANATION

1. (4) From Statement I  $S.P. = \frac{2}{3} C.P.$

From Statement II  $\frac{C.P. - S.P.}{C.P.} \times 100 = 33.33$

$$\Rightarrow \frac{C.P. - S.P.}{C.P.} = \frac{33.33}{100}$$

$$\Rightarrow \frac{C.P. - S.P.}{C.P.} = \frac{1}{3}$$

$$\Rightarrow C.P. - S.P. = \frac{1}{3} C.P.$$

# 7

## Ratio & Proportion

### IMPORTANT FACTS AND FORMULAE

1. **Ratio** : The ratio of two quantities of the same kind and in the same units is a comparison by division of the measure of two quantities.  
In other words, ratio of two quantities of the same kind is the relation between their measures and determines how many times the one quantity is greater than or less than the other quantity.
- (ii) **Ratio of less inequality** : A ratio  $a : b$  is said to be a ratio of less inequality if  $a < b$ .  
**Ex.**  $4 : 7$ ,  $5 : 9$ ,  $2 : 3$ .
- (iii) **Ratio of equality** : A ratio  $a : b$  is said to be a ratio of equality if  $a = b$ .  
**Ex.**  $2 : 2$ ,  $4 : 4$ ,  $5 : 5$ .

### RATIO AND PROPORTION

**Reciprocal or Inverse Ratio** : If the antecedent and consequent of a ratio interchange their places, the new ratio is called the inverse ratio of the first ratio.

In other words, if  $a \neq 0$ ,  $b \neq 0$  then the reciprocal ratio of  $a : b$  is  $\frac{1}{a} : \frac{1}{b}$ . Clearly,  $\frac{1}{a} : \frac{1}{b}$  is same as  $b : a$ .

Thus the reciprocal ratio of  $a : b$  is  $b : a$ .

**Ex.**  $\frac{2}{7}$  is the inverse ratio of  $\frac{7}{2}$ .

**5. Proportion** : (i) If two ratios are equal, they make a proportion or the equality of two ratios constitutes a proportion. Thus  $\frac{2}{3} = \frac{4}{6}$  or  $2 : 3 :: 4 : 6$  and is read as '2 is to 3 as 4 is to 6'.

Thus third proportion of  $a, b$  is  $\frac{b^2}{a}$ .

(v) If  $a : x :: x : b$ ,  $x$  is called the mean or second proportion of  $a, b$ .

We have  $x^2 = a \times b$  or  $x = \sqrt{a \times b}$ .

$\therefore$  Mean proportion of  $a$  and  $b = \sqrt{ab}$ .

In this case,  $a, x, b$  are also called in continued proportion.

**6. Chain Rule.**

**Direct proportion** : Two quantities are said to be directly proportional if on the increase (or decrease) of the one, the other increases (or decreases) in the same ratio.

**Example** : If the number of workers increases, the work done also increases in the same ratio.

**Indirect proportion** : Two quantities are said to be indirectly proportional if on the increase (or decrease) of the one, the other decreases (or increases) in the same ratio.

**Example** : If the speed of a car increases, the time taken to reach a destination decreases in the same ratio.

# MODEL QUESTIONS

## LEVEL-1

1. If  $x : y = 4 : 5$ , then  $(3x + 4y) : (5x + 3y) =$   
 (1)  $3 : 5$  (2)  $5 : 3$  (3)  $17 : 35$  (4)  $35 : 17$   
 (5) None of these
2. If  $A : B = 1 : 3$ ,  $B : C = 5 : 7$ ,  
 $C : D = 9 : 7$ , then  $A : B : C : D = ?$   
 (1)  $15 : 45 : 63 : 49$  (2)  $15 : 49 : 45 : 63$   
 (3)  $45 : 15 : 63 : 49$  (4)  $49 : 15 : 45 : 63$   
 (5) None of these
3. When a number is added to a second number,  
 the sum is  $33\frac{1}{3}$  percent of the second

9. There is a ratio of 5 : 4 between two num-  
 bers. If 40 percent of the first is 12, then 50%  
 of the second number is  
 (1) 12 (2) 24 (3) 18 (4) 20  
 (5) None of these
10. The ratio of the sum of two numbers and their  
 difference is 5 : 1. The ratio of the greater  
 number to the smaller number is  
 (1) 2 : 3 (2) 3 : 2 (3) 5 : 1 (4) 1 : 5  
 (5) None of these
11. An employer reduces the number of his  
 employees in the ratio 9 : 8 and  
 their wages in the ratio 14 : 15. %

19. If a sum of Rs. 1,170 was distributed among  
 A, B and C in the ratio 2 : 3 : 4, by mistake, in  
 place of  $\frac{1}{2} : \frac{1}{3} : \frac{1}{4}$ , who was benefited most  
 and by how much?  
 (1) B, Rs. 220 (2) C, Rs. 250  
 (3) B, Rs. 270 (4) A, Rs. 280  
 (5) None of these
20. Find the fraction which bears the same ratio  
 1 : 3 as 3 : 5
21. Priya's marks in History and Geography are in  
 the ratio 5 : 7. If she got 14 marks more in Ge-  
 ography than in History, what are her History  
 marks?  
 (1) 49 (2) 42 (3) 55 (4) 35  
 (5) None of these
22. The ratio of present ages of Rahul and his  
 sister is 3 : 4. 10 years ago, the ratio of their  
 ages was 13 : 19. What is Rahul's present  
 age (in years)?  
 (1) 25 (2) 48 (3) 42 (4) 54  
 (5) None of these
23. According to the will, the wealth of D.

# RATIO AND PROPORTION

34. Find two numbers such that their mean proportional is 18 and the third proportional to them is 144  
 (1) 6 and 42 (2) 9 and 36  
 (3) 3 and 18 (4) 6 and 12  
 (5) None of these
35. A purse has Rs. 34.5 in the form of 1-rupee, 50-paise and 10-paise coins in the ratio of 6 : 9 : 10. Find the number of 10-paise coins.  
 (1) 10 (2) 30 (3) 20 (4) 40  
 (5) None of these
36. What number should be added to each of the numbers 103, 135, 110 and 144 so that the resulting numbers are in proportion?  
 (1) 12 (2) 15 (3) 9 (4) 6  
 (5) None of these
37. Two numbers

# RATIO AND PROPORTION

38. Given that 24 carat gold is pure gold, 18 carat gold is  $\frac{3}{4}$  gold and 20 carat gold is  $\frac{5}{6}$  gold, the ratio of the pure gold in 18 carat gold to the pure gold in 20 carat gold is  
 (1) 5 : 8 (2) 9 : 10 (3) 15 : 24 (4) 8 : 5  
 (5) None of these
39. The ratio between the angles of a quadrilateral is 3 : 4 : 6 : 7. Half the second largest angle of the quadrilateral is equal to the smaller angle of a parallelogram. What is the value of adjacent angle of the parallelogram?  
 (1) 135° (2) 125° (3) 94° (4) 96°  
 (5) None of these
40. The largest and the second largest angles of a triangle are in the ratio of 4 : 3 respectively. The smallest angle is half the largest angle. What is the difference between the smallest and the largest angles of the triangle?  
 (1) 36° (2) 60° (3) 40° (4) 20°  
 (5) None of these
41. The respective ratio between the present ages of Ram, Rohan and Raj is 3 : 4 : 5. If the average of their present ages is 28 years, then what would be the sum of the ages of Ram and Rohan together after 5 years?  
 (1) 45 years (2) 55 years  
 (3) 50 years (4) 58 years

## LEVEL-II

42. When x is added to each of 9, 15, 21 and 31, the numbers so obtained are in proportion. What is the mean proportional between the numbers (3x - 2) and (5x - 4)?  
 (1) 4.2 (2) 3.5 (3) 2.0 (4) 3.0  
 (5) None of these
43. If  $a : b :: c : d$  then  $\frac{a}{b} : \frac{c}{d} :: ?$   
 (1) 12 : 9 : 8  
 (2) 9 : 8 : 24  
 (3) 8 : 9 : 24  
 (4) 9 : 12 : 8  
 (5) None of these
44. Raju eat  $\frac{3}{8}$  part of a pizza and Adam ate  $\frac{3}{16}$



RATIO AND PROPORTION

11. At present, Pia is 6 years older to Ray. The respective ratio between the present ages of Pia and Mint is 3 : 4. At present Ray is 14 years younger to Mint. What is Ray's present age ?  
 (1) 16 years (2) 20 years  
 (3) 14 years (4) 18 years  
 (5) 24 years
12. Three years ago, Raj's age was equal to the sum of the present ages of his only son and only daughter. Three years ago, the respective ratio between the ages of his daughter and his son at that time was 12:13. If Raj's present age is 48 years, what is the present age of his daughter?  
 (1) 12 years (2) 15 years  
 (3) 18 years (4) 21 years  
 (5) 24 years
13. Cost of a diamond varies directly as the square of its weight. A diamond broke into four pieces with their weights in the ratio 1 : 2 : 3 : 4. If the loss in the total value of the diamond was Rs. 70,000, the price of the original diamond was  
 (1) Rs. 1,00,000 (2) Rs. 1,40,000  
 (3) Rs. 1,20,000 (4) Rs. 1,60,000
14. A person has some hens and some cows. If the number of heads is 48 and the number of legs is 140, then the number of cows is  
 (1) 10 (2) 12  
 (3) 14 (4) 16  
 (5) 18
15. A person has some hens and some cows. If the number of heads is 48 and the number of legs is 140, then the number of cows is  
 (1) 10 (2) 12  
 (3) 14 (4) 16  
 (5) 18
16. A person has some hens and some cows. If the number of heads is 48 and the number of legs is 140, then the number of cows is  
 (1) 10 (2) 12  
 (3) 14 (4) 16  
 (5) 18
17. Cost of a diamond varies directly as the square of its weight. A diamond broke into four pieces with their weights in the ratio 1 : 2 : 3 : 4. If the loss in the total value of the diamond was Rs. 70,000, the price of the original diamond was  
 (1) Rs. 1,00,000 (2) Rs. 1,40,000  
 (3) Rs. 1,20,000 (4) Rs. 1,60,000

18. At present, the respective ratio between the ages of A and B is 3 : 4 and that between A and C is 1 : 2. Six years hence, the sum of the ages of A, B and C will be 96 years. What is the present age of A?  
 (1) 12 years (2) 21 years  
 (3) 16 years (4) 15 years  
 (5) 9 years
19. Salaries of A, B and C are in the ratio of 2 : 3 : 5 respectively. If their salaries were increased by 15%, 10% and 20% respectively, what will be the new respective ratio of their salaries?  
 (1) 3 : 3 : 10 (2) 23 : 33 : 60  
 (3) 11 : 11 : 20 (4) 11 : 11 : 20
20. The average age of 4 members of the family was 30 years, what will be son's age in 2021? (in years)  
 (1) 39 (2) 41 (3) 43 (4) 45  
 (5) None of these
21. The cost of making an article is divided between materials, labour and overheads in the ratio of 3:4:1. If the material costs ₹ 234, then the labour cost is:  
 (1) ₹ 176 (2) ₹ 312 (3) ₹ 78 (4) ₹ 390  
 (5) None of these
22. A camel pursues an elephant and takes 5 leaps for every 7 leaps of the elephant, but 5 leaps of elephant are equal to 3 leaps of camel. What is the ratio of speeds of camel and elephant?  
 (1) 3 : 4 (2) 4 : 3 (3) 5 : 4 (4) 4 : 5  
 (5) None of these



4. Seats for Maths, Physics and Chemistry are in the ratio of 5 : 7 : 8. A proposal to increase the seats for Maths by 25% and 75% respectively. The respective ratio of increased seats will be  
(1) 2 : 3 : 4 (2) 6 : 7 : 8  
(3) 5 : 7 : 8 (4) Cannot be determined  
(5) None of these
5. Mr. X invested a certain sum in Equity funds in the ratio of 3 : 2. At the end of one year, he received a dividend of 30% on his investment. In the second year, he reinvested the dividend in the ratio of 6 : 5. If the amount invested in Equity Funds was ₹ 94,500, the amount invested in Equity Funds in the second year was  
(1) ₹ 75,000  
(2) ₹ 60,000  
(3) ₹ 45,000  
(4) ₹ 30,000  
(5) None of these
6. The ratio between the interior angles of a parallelogram is 2 : 3. The smaller angle of the parallelogram is equal to the smallest angle of a quadrilateral. The sum of the other three angles of the quadrilateral is  
(1) 252° (2) 226°  
(3) 216° (4) 180°  
(5) None of these
7. The ratio between the interior angles of a parallelogram is 7 : 8. The ratio between the angles of a triangle is 6 : 7 : 12. What is the sum of the angles of the parallelogram and the triangle?  
(1) 168° (2) 228°  
(3) 216° (4) 180°  
(5) None of these
8. The respective ratio between two numbers 'X' and 'Y' is 4 : 3. If 'X' is increased by 25% and 'Y' is increased by 50%, the ratio of the resultant 'X' and 'Y' becomes 10 : 9. What is the original value of 'Y'?  
(1) 24 (2) 12 (3) 18  
(4) 16 (5) 14
9. Poorvi's age 8 years ago is equal to the present age of her son. 5 years hence, the ratio between Poorvi's age and her son's age will be 11 : 5 respectively. Poorvi's husband's age is 10 years more than her. His husband's present age is  
(1) 30 years (2) 35 years  
(3) 40 years (4) 45 years  
(5) 50 years

## RATIO AND PROPORTION

### QUANTITY BASED QUESTIONS

**Directions (1-5) :** In each of the questions, quantities are given, solve each of them and choose the correct option as given below.

- (1) Quantity I > Quantity II
- (2) Quantity I ≥ Quantity II
- (3) Quantity I < Quantity II
- (4) Quantity II ≥ Quantity I
- (5) Quantity I = Quantity II or relation cannot be established

26. The concentration of alcohol in Mixture A, B and C is 50%, 60% and 80% respectively. If 2 litres, 3 litres and 4 litres of mixture A, B, C are taken respectively and mixed, then what is the ratio of alcohol to the other constituent of the mixture?

- (1) 1 : 2
- (2) 2 : 1
- (3) 2 : 3
- (4) 3 : 2
- (5) None of these

27. There are two containers A and B. Container A is filled with 30 litres of kerosene while container B is filled with 30 litres of water.

## RATIO AND PROPORTION

5. What is Ratish's present age?  
I. Ratish is 5 years older than Suresh.  
II. The ratio between the ages of Suresh and Ratish is 4 : 5 respectively.

### DATA SUFFICIENCY QUESTIONS

**Directions (1-5) :** Each question consists of a question and two statements numbered I and II below it. You have to decide whether the data given in the statements is sufficient to answer the question.

### SHORT ANSWERS

### LEVEL-I

- |        |        |        |        |        |
|--------|--------|--------|--------|--------|
| 1. (3) | 2. (1) | 3. (3) | 4. (1) | 5. (2) |
|--------|--------|--------|--------|--------|

EXPLANATIONS

LEVEL-I

1. (3)  $\frac{x}{y} = \frac{4}{5}$

$$\frac{3x+y}{5x+3y} = \frac{\frac{3 \times 4}{5} + 1}{5 \times \frac{4}{5} + 3}$$

$$\frac{3 \times 4 + 5}{5 \times 4 + 3} = \frac{12+5}{20+3} = \frac{17}{23}$$

2. (1) A : B = 1 : 3 = 15 : 45  
B : C = 5 : 7 = 15 : 21  
= 45 : 63

C : D = 9 : 7 = 27 : 21  
= 63 : 49

$\therefore$  A : B : C : D = 15 : 45 : 63 : 49

3. (3) Let the first number be x and the second number be y.

$x + y = 333 \frac{1}{3}\%$  of y

$\Rightarrow \frac{11x + 17y}{x} = \frac{37}{3}$

$\Rightarrow 37x = 33x + 52y \Rightarrow 4x = 52y$

$\Rightarrow x = \frac{52y}{4} = 13y$

6. (5) According to the question,

$\therefore \frac{55x}{100} : \frac{85y}{100} = 1 : 2$

$\Rightarrow 55x : 85y = 1 : 2$

$\Rightarrow \frac{5x}{8y} = \frac{1}{2} \Rightarrow \frac{x}{y} = \frac{8}{2} = \frac{4}{1}$

$\therefore x = 4a, y = 5a$

$\therefore 5a - 4a = 10 \Rightarrow a = 10$

$\therefore x = 40$  and  $y = 50$

$\therefore 3x + y = 3 \times 40 + 50 = 170$

7. (1) Let, Numbers = 2x and 3x

$\therefore \frac{2x+4}{3x+4} = \frac{5}{7}$

$\Rightarrow 15x + 20 = 14x + 28$

$\Rightarrow x = 28 - 20 = 8$

= Difference between numbers.

= Total number of students be 4

11. (2) Required ratio  
 $= 9 : 14$  B : 15  
 $= 21 : 20$

12. (3) Third proportional of 12 and 18 = x

$\therefore 12 : 18 = 18 : x$

$\Rightarrow x = \frac{18 \times 18}{12} = 27$

13. (3)  $A \times \frac{1}{2} = B \times \frac{1}{3} = C \times \frac{1}{4}$

$\Rightarrow \frac{A}{2} = \frac{B}{3} = \frac{C}{4}$

$\therefore A : B : C = 2 : 3 : 4$

$\therefore A = \frac{2}{2} \times 900 = \text{Rs. } 900$

$B = \frac{3}{3} \times 900 = \text{Rs. } 900$

$C = \frac{4}{4} \times 900 = \text{Rs. } 900$

$\therefore A : B : C = 2 : 3 : 4$

$\therefore A : B : C = 2 : 3 : 4$

$\therefore A : B : C = 2 : 3 : 4$

$\therefore A : B : C = 2 : 3 : 4$

$\therefore A : B : C = 2 : 3 : 4$

$\therefore A : B : C = 2 : 3 : 4$

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$\therefore A : B : C = 2 : 3 : 4$

$\therefore A : B : C = 2 : 3 : 4$

$\therefore A : B : C = 2 : 3 : 4$

18. (2) Distribution among A, B and C :

Actual ratio =  $\frac{1}{2} : \frac{1}{3} : \frac{1}{4}$

$= \left( \frac{1}{2} \times 12 \right) : \left( \frac{1}{3} \times 12 \right) : \left( \frac{1}{4} \times 12 \right)$

$= 6 : 4 : 3 = 54 : 36 : 27$  (multiplied by 9)

Wrong Ratio = 2 : 3 : 4 = 26 : 39 : 52

(multiplied by 13)

Clearly, C gained

Gain

$= \text{Rs. } \left[ \frac{4}{9} \times 1170 - \frac{3}{13} \times 1170 \right]$

$= \text{Rs. } (520 - 270) = \text{Rs. } 250$

19. (3) A : B = 2 : 3

B : C = 3 : 7

$\therefore$  A : B : C = 2 : 3 : 7

$\therefore$  A = 2k, B = 3k, C = 7k

$\therefore$  A + B = 5k, B + C = 10k

C + A = 9k

$\therefore$  Required ratio = 5k : 10k : 9k

$= 5 : 10 : 9$

20. (2) Let the required fraction be x

According to the question,

23. (i) Successful students

$$= \frac{2}{11} \times 132 = 24$$

Unsuccessful students

$$= \frac{9}{11} \times 132 = 108$$

When 4 more students succeed

$$\text{Required ratio} = \frac{24}{108 + 4} = \frac{24}{112} = \frac{3}{14}$$

24. (i) Before battle

$$\text{Officers} = 3x$$

$$\text{Soldiers} = 31x$$

According to the question

$$\text{After battle}$$

$$\frac{3x - 6}{31x - 22} = \frac{1}{13}$$

$$39x - 78 = 31x - 22$$

$$8x = 56 \Rightarrow x = 7$$

$$\therefore \text{Required number of officers} = 3 \times 7 = 21$$

$$25. (i) 6A = 4B = 9C$$

$$\frac{6A}{36} = \frac{4B}{36} = \frac{9C}{36}$$

$$\therefore \frac{A}{6} = \frac{B}{9} = \frac{C}{4}$$

$$\therefore A : B : C = 6 : 9 : 4$$

$$6A = 4B = 9C$$

$$\frac{A}{6} = \frac{B}{9} = \frac{C}{4} = \frac{1}{60}$$

$$\therefore A = \frac{1}{60} \times 6 = \frac{1}{10}$$

$$\therefore B = \frac{1}{60} \times 9 = \frac{3}{20}$$

$$\therefore C = \frac{1}{60} \times 4 = \frac{1}{15}$$

$$\therefore A : B : C = \frac{1}{10} : \frac{3}{20} : \frac{1}{15} = 2 : 3 : 4$$

$$\therefore \text{L.C.M. of } 6, 4 \text{ and } 9 = 36$$

$$\therefore \text{L.C.M. of } 6, 4 \text{ and } 9 = 36$$

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$$\therefore \text{L.C.M. of } 6, 4 \text{ and } 9 = 36$$

Fourth proportional = 22

$$= \frac{275 \times 150}{100} = 412.5$$

$$= 412.5$$

$$27. (i) \text{History} = 5, 7$$

$$17 - 50 \text{ units} = 14$$

$$14 \times 5 = 70$$

$$14 \times 7 = 98$$

$$14 \times 14 = 196$$

$$14 \times 17 = 238$$

$$14 \times 22 = 308$$

$$14 \times 27 = 378$$

$$14 \times 35 = 490$$

$$14 \times 49 = 686$$

$$14 \times 77 = 1078$$

$$14 \times 121 = 1694$$

$$14 \times 147 = 2058$$

$$14 \times 196 = 2744$$

$$14 \times 238 = 3332$$

$$14 \times 308 = 4312$$

$$14 \times 378 = 5292$$

$$14 \times 490 = 6860$$

$$14 \times 686 = 9604$$

$$14 \times 1078 = 15092$$

$$14 \times 1694 = 23716$$

$$14 \times 2744 = 38416$$

$$14 \times 3332 = 46648$$

$$14 \times 4312 = 60368$$

$$14 \times 5292 = 74088$$

$$14 \times 6860 = 96040$$

$$14 \times 9604 = 134456$$

$$14 \times 15092 = 211288$$

$$14 \times 23716 = 332024$$

$$14 \times 38416 = 537824$$

$$14 \times 60368 = 845152$$

$$14 \times 96040 = 1344560$$

$$14 \times 134456 = 1882384$$

$$14 \times 211288 = 2958032$$

$$14 \times 332024 = 4648336$$

$$14 \times 537824 = 7529536$$

$$14 \times 845152 = 11832128$$

$$14 \times 1344560 = 18823840$$

$$14 \times 2958032 = 41412448$$

$$14 \times 4648336 = 65076704$$

$$14 \times 7529536 = 105413504$$

$$14 \times 11832128 = 165650792$$

$$14 \times 18823840 = 263533760$$

$$14 \times 29580320 = 414124480$$

$$14 \times 41412448 = 579774272$$

$$14 \times 579774272 = 8116939808$$

$$14 \times 8116939808 = 113637157312$$

$$14 \times 113637157312 = 1590920202368$$

$$14 \times 1590920202368 = 22272882833152$$

$$14 \times 22272882833152 = 311820359664128$$

$$14 \times 311820359664128 = 4365485035297792$$

(ii) Number of candies = 7x

$$\text{Number of chewing gums} = 13x$$

$$\text{According to the question}$$

$$7x - 8 = 13x - 11$$

$$13x - 16 = 13x - 11$$

$$13x - 13x = 16 - 11$$

$$0 = 5$$

$$\text{Number of remaining candies}$$

$$7x - 8 = 35 - 8 = 27$$

$$7 \times 5 - 8 = 35 - 8 = 27$$

$$7 \times 5 - 8 = 35 - 8 = 27$$

$$7 \times 5 - 8 = 35 - 8 = 27$$

$$7 \times 5 - 8 = 35 - 8 = 27$$

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(iii) According to the question

$$a + b + c = 21 \times 3 = 63$$

$$\text{Again}$$

$$a = \frac{3}{8} (b + c)$$

$$\Rightarrow \frac{8}{3} a = b + c$$

$$\text{From equations (i) and (ii)}$$

$$\frac{8}{3} a + 3a = 63$$

$$\Rightarrow \frac{8a + 9a}{3} = 63$$

$$\Rightarrow 17a = 189 \times 3$$

$$\Rightarrow 17a = 567 \times 3$$

$$\Rightarrow a = \frac{567 \times 3}{17} = 100.82$$

$$\therefore \text{Mean proportional} = \sqrt{ab}$$

$$\Rightarrow \sqrt{ab} = 18$$

$$\Rightarrow ab = 324$$

$$\therefore \text{Third proportional} = \frac{b^2}{a}$$

$$\Rightarrow \frac{b^2}{a} = 144$$

$$\therefore \text{Third proportional} = 144$$

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# RATIO AND PROPORTION

40. Let father's age =  $x$  years  
 Son's age =  $52 - x$  years

After four years,  
 Father's age =  $(x + 4)$  years  
 Son's age =  $(52 - x + 4)$  years

According to the question,  
 $\frac{x + 4}{52 - x + 4} = \frac{3}{4}$

$4(x + 4) = 3(52 - x + 4)$   
 $4x + 16 = 156 - 3x + 12$   
 $4x + 3x = 156 - 16 - 12$   
 $7x = 128$   
 $x = \frac{128}{7}$

41. Let father's age =  $x$  years  
 Son's age =  $52 - x$  years

After four years,  
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 $4x + 16 = 156 - 3x + 12$   
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43. Let father's age =  $x$  years  
 Son's age =  $52 - x$  years

After four years,  
 Father's age =  $(x + 4)$  years  
 Son's age =  $(52 - x + 4)$  years

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 $4x + 16 = 156 - 3x + 12$   
 $4x + 3x = 156 - 16 - 12$   
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 $x = \frac{128}{7}$

44. Let father's age =  $x$  years  
 Son's age =  $52 - x$  years

After four years,  
 Father's age =  $(x + 4)$  years  
 Son's age =  $(52 - x + 4)$  years

According to the question,  
 $\frac{x + 4}{52 - x + 4} = \frac{3}{4}$

$4(x + 4) = 3(52 - x + 4)$   
 $4x + 16 = 156 - 3x + 12$   
 $4x + 3x = 156 - 16 - 12$   
 $7x = 128$   
 $x = \frac{128}{7}$

45. Let father's age =  $x$  years  
 Son's age =  $52 - x$  years

After four years,  
 Father's age =  $(x + 4)$  years  
 Son's age =  $(52 - x + 4)$  years

According to the question,  
 $\frac{x + 4}{52 - x + 4} = \frac{3}{4}$

# RATIO AND PROPORTION

Required numbers = 9 and 36

OR

Directly from options numbers will be 6 and 36 as Mean proportion = 18

$\sqrt{6 \times 36} = 18$

35. (2) Ratio of the number of 1 rupee, 50 paise and 10 paise coins = 6 : 9 : 10

Respective ratio of their values  
 $= 6 \times \frac{1}{100} : 9 \times \frac{1}{200} : 10 \times \frac{1}{1000}$   
 $= \frac{6}{100} : \frac{9}{200} : \frac{10}{1000}$   
 $= \frac{6}{100} : \frac{9}{200} : \frac{1}{100}$   
 $= 6 : 4.5 : 1$

Sum of the terms of ratio  
 $= 6 + 4.5 + 1 = 11.5$

Value of 10 paise coins  
 $= \text{Rs. } \left( \frac{2}{23} \times 34.5 \right) = \text{Rs. } 3$

36. (3) Let the number to be added be  $x$ .  
 According to the question,  
 $\frac{135 - x}{103 - x} = \frac{144 - x}{110 - x}$

$(135 - x)(110 - x) = (144 - x)(103 - x)$   
 $14850 - 135x - 110x + x^2 = 14832 - 144x - 103x + x^2$   
 $14850 - 245x + x^2 = 14832 - 247x + x^2$   
 $14850 - 245x = 14832 - 247x$   
 $14850 - 14832 = 245x - 247x$   
 $18 = -2x$   
 $x = -9$

46. Let father's age =  $x$  years  
 Son's age =  $52 - x$  years

After four years,  
 Father's age =  $(x + 4)$  years  
 Son's age =  $(52 - x + 4)$  years

According to the question,  
 $\frac{x + 4}{52 - x + 4} = \frac{3}{4}$

$4(x + 4) = 3(52 - x + 4)$   
 $4x + 16 = 156 - 3x + 12$   
 $4x + 3x = 156 - 16 - 12$   
 $7x = 128$   
 $x = \frac{128}{7}$

47. Let father's age =  $x$  years  
 Son's age =  $52 - x$  years

After four years,  
 Father's age =  $(x + 4)$  years  
 Son's age =  $(52 - x + 4)$  years

According to the question,  
 $\frac{x + 4}{52 - x + 4} = \frac{3}{4}$

$4(x + 4) = 3(52 - x + 4)$   
 $4x + 16 = 156 - 3x + 12$   
 $4x + 3x = 156 - 16 - 12$   
 $7x = 128$   
 $x = \frac{128}{7}$

48. Let father's age =  $x$  years  
 Son's age =  $52 - x$  years

After four years,  
 Father's age =  $(x + 4)$  years  
 Son's age =  $(52 - x + 4)$  years

According to the question,  
 $\frac{x + 4}{52 - x + 4} = \frac{3}{4}$

$4(x + 4) = 3(52 - x + 4)$   
 $4x + 16 = 156 - 3x + 12$   
 $4x + 3x = 156 - 16 - 12$   
 $7x = 128$   
 $x = \frac{128}{7}$

49. Let father's age =  $x$  years  
 Son's age =  $52 - x$  years

After four years,  
 Father's age =  $(x + 4)$  years  
 Son's age =  $(52 - x + 4)$  years

# RATIO AND PROPORTION

40. Let father's age =  $x$  years  
 Son's age =  $52 - x$  years

After four years,  
 Father's age =  $(x + 4)$  years  
 Son's age =  $(52 - x + 4)$  years

According to the question,  
 $\frac{x + 4}{52 - x + 4} = \frac{3}{4}$

$4(x + 4) = 3(52 - x + 4)$   
 $4x + 16 = 156 - 3x + 12$   
 $4x + 3x = 156 - 16 - 12$   
 $7x = 128$   
 $x = \frac{128}{7}$

41. Let father's age =  $x$  years  
 Son's age =  $52 - x$  years

After four years,  
 Father's age =  $(x + 4)$  years  
 Son's age =  $(52 - x + 4)$  years

According to the question,  
 $\frac{x + 4}{52 - x + 4} = \frac{3}{4}$

$4(x + 4) = 3(52 - x + 4)$   
 $4x + 16 = 156 - 3x + 12$   
 $4x + 3x = 156 - 16 - 12$   
 $7x = 128$   
 $x = \frac{128}{7}$

42. Let father's age =  $x$  years  
 Son's age =  $52 - x$  years

After four years,  
 Father's age =  $(x + 4)$  years  
 Son's age =  $(52 - x + 4)$  years

According to the question,  
 $\frac{x + 4}{52 - x + 4} = \frac{3}{4}$

$4(x + 4) = 3(52 - x + 4)$   
 $4x + 16 = 156 - 3x + 12$   
 $4x + 3x = 156 - 16 - 12$   
 $7x = 128$   
 $x = \frac{128}{7}$

43. Let father's age =  $x$  years  
 Son's age =  $52 - x$  years

After four years,  
 Father's age =  $(x + 4)$  years  
 Son's age =  $(52 - x + 4)$  years

According to the question,  
 $\frac{x + 4}{52 - x + 4} = \frac{3}{4}$

$4(x + 4) = 3(52 - x + 4)$   
 $4x + 16 = 156 - 3x + 12$   
 $4x + 3x = 156 - 16 - 12$   
 $7x = 128$   
 $x = \frac{128}{7}$

44. Let father's age =  $x$  years  
 Son's age =  $52 - x$  years

After four years,  
 Father's age =  $(x + 4)$  years  
 Son's age =  $(52 - x + 4)$  years

According to the question,  
 $\frac{x + 4}{52 - x + 4} = \frac{3}{4}$

$4(x + 4) = 3(52 - x + 4)$   
 $4x + 16 = 156 - 3x + 12$   
 $4x + 3x = 156 - 16 - 12$   
 $7x = 128$   
 $x = \frac{128}{7}$

45. Let father's age =  $x$  years  
 Son's age =  $52 - x$  years

After four years,  
 Father's age =  $(x + 4)$  years  
 Son's age =  $(52 - x + 4)$  years

According to the question,  
 $\frac{x + 4}{52 - x + 4} = \frac{3}{4}$



# RATIO AND PROPORTION

45. (4) Let A's present age =  $3x$  years  
B's present age =  $5x$  years  
Ratio of their ages after 5 years  
 $\frac{3x+5}{5x+5} = \frac{12}{20}$   
 $60x + 100 = 65x + 65$   
 $5x = 35$   
 $x = 7$   
 $\therefore$  Percentage of B = 35 years

46. (2) Let the numbers be  $3x$  and  $4x$ . Then,  
 $16x^2 = 8 \times (9x^2 - 224)$   
 $= 16x^2 = 72x^2 - 224$   
 $= 56x^2 = 224$   
 $= x^2 = 4$   
 $= x = 2$   
Hence, numbers are 6, 8.

47. (1) According to question in a proportion,  
multiplication of 1st and 4th term =  
Multiplication of 2nd and 3rd terms  
 $\therefore 40 = 2.5x$   
 $\therefore x = 16$

## LEVEL-II

1. (1) Let the present ages of Ram, Rahim and Raj be  $3x$ ,  $4x$  and  $5x$  years respectively.  
 $\therefore 3x + 4x + 5x = 3 \times 29$   
 $\Rightarrow 12x = 87 \therefore x = 7$   
Sum of the ages of Ram and Rahim after 5 years  
 $= 3x + 4x + 10 = 7x + 10$   
 $= 7 \times 7 + 10 = 59$  years

2. (2) Number of students :  
A =  $5x$   
B =  $6x$   
C =  $8x$   
 $\therefore$  Required ratio =  $\frac{5x \times 130}{100} : \frac{6x \times 125}{100} : \frac{8x \times 125}{100}$   
 $= 650 : 750 : 1000$   
 $= 13 : 15 : 20$

3. (4) Let the earnings of A and B be Rs.  $4x$  and  $7x$  respectively.  
After 50% increase, A's earnings = 150% of  $4x$

# RATIO AND PROPORTION

Part of work done by B  
 $= 1 - \frac{5}{6} = \frac{1}{6}$

$\therefore$  Time taken by B in doing  $\frac{1}{6}$  work = 3 days  
 $\therefore$  Time taken by B in doing whole work =  $3 \times 6 = 18$  days

10. (1) Dex's sister's Monthly salary

= Rs. (12000  $\div$  4)

Dex's father's Monthly salary

= Rs.  $\left( \frac{51200 \times 5}{2} \right)$

= Rs. 125000

$\therefore$  Dex's Monthly salary

=  $125000 \times \frac{1}{5} =$  Rs. 25000

Dex's Monthly Savings

= Rs. 25000

Dex's Monthly Savings

6. (2)  $3x + 4x + 6x + 7x = 360^\circ$   
 $\therefore 20x = 360^\circ \therefore x = 18^\circ$   
 $\therefore$  smallest angle of the parallelogram

$\therefore 6x - 3x = 54^\circ$   
 $\therefore \frac{6x}{2} = 27^\circ$

$\therefore$  adjacent angle of parallelogram

$\therefore 180^\circ - 54^\circ = 126^\circ$

$\therefore$  The smallest angle of triangle is half of the

largest angle.

$\therefore$  Ratio of three angles = 4 : 3 : 2

$\therefore$  Ratio of three angles = 180

$\therefore 4x + 3x + 2x = 180$

$\therefore 9x = 180 \therefore x = 20$

$\therefore$  Required difference

$\therefore 4x - 2x = 2x$

$\therefore 2 \times 20 = 40^\circ$

8. (3) Om's monthly income = Rs.  $7x$

Ravi's monthly income = Rs.  $9x$

Om's savings =  $\frac{20 \times 7x}{100}$

= Rs.  $\frac{7x}{5}$

# RATIO AND PROPORTION

13. (4) Let the present ages of father and son be  $5x$  and  $2x$  years respectively.  
After 4 years,

Son's age =  $y$  years  
and mother's age =  $2y$  years

$$\therefore y = 2x + 4 \Rightarrow x = \frac{y-4}{2}$$

$\therefore$  Father's present age

$$= 5 \left( \frac{y-4}{2} \right) \text{ years}$$

Mother's present age =  $(2y - 4)$  years

Clearly, data are inadequate.

14. (5) Let the original number of students in Arts and Science faculties be  $5x$  and  $8x$  respectively.

According to the question,

$$\frac{5x - 150}{8x - 80} = \frac{3}{4}$$

$$\Rightarrow 24x + 240 = 20x + 600$$

$$\Rightarrow 4x = 360$$

$$\Rightarrow x = \frac{360}{4} = 90$$

$$\begin{aligned} \therefore k \cdot 70x^2 &= 70000 \\ \Rightarrow kx^2 &= \text{Rs. } 1000 \\ \therefore \text{Cost of the original diamond} &= k \cdot 1000x^2 \\ &= 1000 \times 100 = \text{Rs. } 100000 \end{aligned}$$

18. (3) Actual price of diamond

$$= k \times 6^2$$

= 36k where  $k$  is constant of proportionality.

New price of diamond

$$= kx_1^2 + kx_2^2$$

$$\Rightarrow 36k \times \frac{5}{8} = k(x_1^2 + x_2^2)$$

$$\Rightarrow \frac{45}{2} = x_1^2 + x_2^2$$

$$\Rightarrow 22.5 = x_1^2 + x_2^2$$

$$\Rightarrow (4.5)^2 + (1.5)^2 = x_1^2 + x_2^2$$

19. (4) Time taken by B =  $x$  minutes.

Time taken by A =  $(x + 10)$  minutes.

$$\frac{2}{x}$$

Pranab's annual income

$$\text{Rs. } 700000 = \text{Pranab's monthly salary} \times \frac{12}{12} = \text{Rs. } 60000$$

$$\therefore \text{Pranab's monthly salary} = \frac{700000}{12} = \text{Rs. } 60000$$

$$\therefore (1)x = y + 52$$

$$\Rightarrow x = y - 26$$

$$\therefore x + y + z = 221$$

$$\Rightarrow y + 52 + y + y - 26 = 221$$

$$\Rightarrow 3y = 221 - 26 = 195$$

$$\Rightarrow y = \frac{195}{3} = 65$$

$$\therefore x = 65 + 52 = 117$$

$$\Rightarrow z = 65 - 26 = 39$$

$$\therefore x : y : z = 117 : 65 : 39$$

$$\Rightarrow 9 : 5 : 3$$

$$\therefore A : B : C = 3 : 4 : 6$$

$$\therefore A : B : C = 3 : 4 : 6$$

According to the question,

After 6 years,

Sum of ages of A, B and C = 96

$$\Rightarrow 3x + 4x + 6x + 18 = 96$$

25. (1) Required answer

$$= \frac{29900}{632} = \frac{40}{11}$$

26. (3) Joe's present age =  $x$  years (let)

$\therefore$  His father's present age

$$= \frac{7x}{2} \text{ years}$$

Joe's brother's age

$$= (x + 3) \text{ years}$$

According to the question,

$$\frac{7x}{2} = \frac{14}{5} \Rightarrow \frac{7x}{2 \times 5} = \frac{14}{5}$$

$$\Rightarrow 35x = 28 \times 2$$

$$\Rightarrow 7x = 84$$

$$\Rightarrow x = \frac{84}{7} = 12 \text{ years}$$

27. (1) Ami's present age

$$= 2x \text{ years}$$

Dio's present age =  $x$  years

Camel's present age

# RATIO AND PROPORTION

$$\therefore \text{Labour cost} = \frac{4}{5} \times 624 = ₹ 312$$

**Alter :**

$$\text{labour cost} = \frac{4}{5} \times 234 = ₹ 312$$

$$30. (2) \text{ Required ratio} = \frac{5}{3} : \frac{7}{5}$$

$$= \frac{5}{3} \times 15 : \frac{7}{5} \times 15 = 25 : 21$$

## LEVEL-III

1. (2) Number of boys in class

$$= \left( \frac{5}{5+7} \right) \times 168 = 70$$

$$\text{Number of girls} = 168 - 70 = 98$$

$$\text{Number of students speaking Hindi only} =$$

$$\frac{168}{2} = 84$$

$$= \text{Number of students speaking English only}$$

$$= \frac{16}{21} \times 84 = 64$$

$$\text{Number of boys speak}$$

$$= 4 \times 5 = 20$$

$$\therefore \text{Required sum}$$

$$= 240^\circ + 20^\circ = 260^\circ$$

$$3. (1) 20x + 73x + 83x$$

$$= 950 : 25 = 15 : 30$$

$$= \frac{880}{176} = 5$$

$$\therefore 176x = 880 \Rightarrow x = 5$$

$$\therefore \text{Number of coins got by Anita}$$

$$= 73x + 15 = 73 \times 5 + 15 = 380$$

$$4. (1) \text{ Let the initial seats for Maths, Physics and}$$

$$\text{Biology be } 5x, 7x \text{ and } 8x \text{ respectively.}$$

$$\text{Now, new seats for Maths}$$

$$= \frac{5x \times 140}{100}$$

$$\text{For Physics} = \frac{7x \times 150}{100}$$

$$\text{and for Biology} = \frac{8x \times 175}{100}$$

$$\therefore \text{Required ratio} = \frac{5x \times 140}{100} : \frac{7x \times 150}{100} : \frac{8x \times 175}{100}$$

$$= 5 \times 140 : 7 \times 150 : 8 \times 175$$

$$= 2 : 3 : 4$$

# RATIO AND PROPORTION

$$\therefore \text{Required sum} = 144 + 72 = 216^\circ$$

$$7. (1) \text{ Let the adjacent angles be } 7x^\circ \text{ and } 8x^\circ.$$

$$\therefore 7x + 8x = 180$$

$$\therefore 15x = 180$$

$$\therefore x = 12$$

$$\therefore \text{Smaller angle} = 7 \times 12 = 84^\circ$$

$$\therefore \text{Again, } 5y + 6y + 7y + 12y = 360^\circ$$

$$\therefore 30y = 360^\circ$$

$$\therefore y = 12^\circ$$

$$\therefore \text{Second largest angle of the quadrilateral}$$

$$= 7 \times 12 = 84^\circ$$

$$\therefore \text{required sum} = 84 + 84 = 168^\circ$$

$$8. (3) \text{ Let first number } X \text{ be } 4a,$$

$$\therefore \text{second number } Y = 7a$$

$$\text{According to the question,}$$

$$\left( \frac{125}{100} + 3 \right) : (14a + 3)$$

$$= 2 : 5$$

$$\therefore \frac{5a + 3}{14a + 3} = \frac{2}{5}$$

$$\therefore 28a + 6 = 25a + 15$$

$$\therefore 28a - 25a = 15 - 6$$

$$\therefore 3a = 9$$

$$\therefore a = 3$$

$$\therefore X = 4a = 12$$

$$\therefore Y = 7a = 21$$

$$\therefore \text{Required difference}$$

$$= \frac{164}{3} - 44$$

$$= \frac{164 - 132}{3}$$

$$= \frac{32}{3}$$

$$\therefore C = A + 3$$

$$\text{and } A + 4 = B - 1$$

$$\therefore B = A + 5$$

$$\text{From equations (i) and (ii)}$$

$$A + A + 5 = 87$$

$$\therefore 2A = 87 - 5 = 82$$

$$\therefore A = 41 \text{ years}$$

$$\therefore B = 41 + 5 = 46 \text{ years}$$

$$C = 41 + 3 = 44 \text{ years}$$

$$\therefore \frac{A}{B} = \frac{41}{46}$$

$$\therefore D = \frac{4}{3} \times 41$$

$$= \frac{164}{3} \text{ years}$$

$$\therefore \text{Required difference}$$

$$= \frac{164}{3} - 44$$

$$= \frac{164 - 132}{3}$$

$$= \frac{32}{3}$$

# RATIO AND PROPORTION

Amount given to the mother

$$= \text{Rs. } \frac{x}{5}$$

Remaining amount

$$= \text{Rs. } \left[ x - \frac{x}{5} \right] = \text{Rs. } \frac{4x}{5}$$

Amount spent in insurance and PPF = Rs.  $\frac{4x}{5} \times 2$

$$= \text{Rs. } \frac{2x}{5}$$

$\therefore$  Amount deposited in PPF

$$= \frac{x}{5} \times \frac{2x}{5} = \text{Rs. } \frac{3x}{25}$$

According to the question,

$$\frac{x}{5} \times \frac{3x}{25} = 12600$$

$$= \frac{4x \times 3x}{25} = 12600$$

Savings

$$= 100 - (30 + 14 + 24 + 16)$$

$$= \text{Rs. } (100 - 84) = \text{Rs. } 16$$

When savings is Rs. 16, monthly salary,

Rs. 100

When savings is Rs. 4400, monthly salary

$$= \text{Rs. } \left( \frac{100 \times 4400}{16} \right)$$

$$= \text{Rs. } 27500$$

$$15. (3) A : B = 4 : 5 = 4x : 5x$$

$$C : D = 7 : 5 = 7y : 5y$$

$$\therefore A + B + C + D = 27 \times 4 = 108$$

$$\Rightarrow 4x + 5x + 7y + 5y = 108$$

$$\Rightarrow 9x + 12y = 108$$

$$\Rightarrow 3x + 4y = 36 \dots (i)$$

$$\text{Again, } 5y + 3 = 4x - 14$$

$$\Rightarrow 4x - 5y = 17 \dots (ii)$$

By equation (i)  $\times 5 + (ii) \times 4$ ,

# RATIO AND PROPORTION

Amount collected from 1st class fares

$$= 50 \times 1000 = \text{₹ } 320$$

$\frac{1}{50}$  of the four wall is  $2h$  ( $h \rightarrow$  height)

$\frac{1}{50}$  of the four wall is  $2b$  ( $b \rightarrow$  breadth)

$\frac{1}{50}$  of the four wall is  $2l$  ( $l \rightarrow$  length)

$\frac{1}{50}$  of the four wall is  $2h$  ( $h \rightarrow$  height)

$\frac{1}{50}$  of the four wall is  $2b$  ( $b \rightarrow$  breadth)

$\frac{1}{50}$  of the four wall is  $2l$  ( $l \rightarrow$  length)

$\frac{1}{50}$  of the four wall is  $2h$  ( $h \rightarrow$  height)

$\frac{1}{50}$  of the four wall is  $2b$  ( $b \rightarrow$  breadth)

$\frac{1}{50}$  of the four wall is  $2l$  ( $l \rightarrow$  length)

$\frac{1}{50}$  of the four wall is  $2h$  ( $h \rightarrow$  height)

$\frac{1}{50}$  of the four wall is  $2b$  ( $b \rightarrow$  breadth)

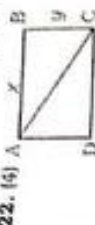
$\frac{1}{50}$  of the four wall is  $2l$  ( $l \rightarrow$  length)

$\frac{1}{50}$  of the four wall is  $2h$  ( $h \rightarrow$  height)

$\frac{1}{50}$  of the four wall is  $2b$  ( $b \rightarrow$  breadth)

$\frac{1}{50}$  of the four wall is  $2l$  ( $l \rightarrow$  length)

$\frac{1}{50}$  of the four wall is  $2h$  ( $h \rightarrow$  height)



Let ABCD be the rectangular field and let AB

Similarly Little paid  $\frac{1}{4}$ th of sum paid by others, let amount paid by other =  $x$ , then  $x + \frac{1}{4}x = 60000$

$x = 60000 \times \frac{4}{5} = 48000$  and amount paid by Little

$$= \frac{48000}{4} = 12000$$

Total amount paid by Jaspal =  $60000 - 20000 - 15000 - 12000 = 13000$

22. (4) A



# RATIO AND PROPORTION

$$\Rightarrow A = \frac{1}{2}B + \frac{1}{2}D$$

Now,  $\frac{1}{2}D < \frac{1}{2}B$  because  $D < B$ . So,  $A < B$ .

B's income is the highest.

24. (2) Average marks of girls = A

$$\therefore A = \frac{3 \times (A + 1) + 10}{3 + 1}$$

$$\Rightarrow 4A = 3A + 3 + 10$$

$$\Rightarrow A = 4A - 3A - 3 = A - 3$$

25. (2) Let Nish invests ₹ x.

$$\therefore \text{Vikas's investment} = ₹ 2x$$

$$\text{Manav's investment} = ₹ 3x$$

Ratio of the equivalent capitals for 1 year, 6 months and 4 months of Nish, Vikas and Manav respectively

$$= x \times 12 : 2x \times 6 : 4 \times 3x = 1 : 1 : 1$$

Clearly, the profit will be shared equally.

$$\therefore \text{Profit} = ₹ 12000$$

$$\therefore \text{Nish's share} = ₹ 4000$$

$$\therefore \text{Vikas's share} = ₹ 4000$$

$$\therefore \text{Manav's share} = ₹ 4000$$

Variables A

Kerosene

White

Kerosene

White

Kerosene

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Kerosene

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Kerosene

Ratio of Kerosene in A to B = 23 : 7

28. (1) Let the 3 pieces be x, 2x and 3x kg respectively.

Total weight of the stone = x + 2x + 3x = 6x

Value when stone is broken in ratio of 1 : 2 : 3

$$= x^2 + 2(2x)^2 + 3(3x)^2 = x^2 + 4x^2 + 9x^2 = 14x^2$$

Value when stone is broken in the ratio of 4 : 1 : 1

$$= (4x)^2 + (x)^2 + (x)^2 = 18x^2$$

Also, we are given  $18x^2 - 14x^2 = 6400$

$$4x^2 = 6400$$

$$x^2 = 1600 \text{ and } x = 40$$

$$\text{Original value} = 6x^2 = 6 \times 40^2 = 36 \times 40 \times 40 = ₹ 57,600$$

$$\therefore \text{Ratio of profit distribution} = 1 : 2 : 3$$

$$\therefore \text{Nish's share} = ₹ 19,200$$

$$\therefore \text{Vikas's share} = ₹ 19,200$$

$$\therefore \text{Manav's share} = ₹ 19,200$$

$$\therefore \text{Total share} = ₹ 57,600$$

$$\therefore \text{Ratio of profit distribution} = 1 : 2 : 3$$

$$\therefore \text{Nish's share} = ₹ 19,200$$

$$\therefore \text{Vikas's share} = ₹ 19,200$$

$$\therefore \text{Manav's share} = ₹ 19,200$$

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$$\therefore \text{Manav's share} = ₹ 19,200$$

$$\therefore \text{Total share} = ₹ 57,600$$

$$\therefore \text{Ratio of profit distribution} = 1 : 2 : 3$$

$$\therefore \text{Nish's share} = ₹ 19,200$$

$$\therefore \text{Vikas's share} = ₹ 19,200$$

$$\therefore \text{Manav's share} = ₹ 19,200$$

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$$\therefore \text{Total share} = ₹ 57,600$$

$$\frac{2y}{3} = \frac{A}{3}$$

$$\Rightarrow 2y = A$$

$$\Rightarrow y = \frac{A}{2}$$

$$\Rightarrow \frac{A}{2} = \frac{A}{3}$$

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$$\Rightarrow \frac{A}{2} = \frac{A}{3}$$

Quantity II : Rahul's present age = 2x years

Deepak's present age = 3x years

After five years,

Rahul's age = 31 years

$$\therefore 2x + 5 = 31$$

$$\Rightarrow 2x = 31 - 5 = 26 \Rightarrow x = 13$$

Deepak's age after 3 years

$$= 3x + 3 = 3 \times 13 + 3 = 42 \text{ years}$$

Quantity I = Quantity II

3. (3) Quantity I : incomes of A and B = ₹ 6x and 5x

Expenses of A and B = ₹ 4y and 3y

$$\therefore 6x - 4y = 400 \quad \dots \text{ii}$$

$$5x - 3y = 400 \quad \dots \text{iii}$$

By equation iii  $\times 3 - \text{ii} \times 4$

$$\Rightarrow 15x - 12y - 20x + 12y$$

$$= 1200 - 1600$$

$$\Rightarrow 2x = 400 \Rightarrow x = 200$$

$\therefore$  Total income

$$= 6x + 5x = 11x$$

## QUANTITY BASED EXPLANATION

1. (1) Quantity I : Let the present age of A and B be a & b respectively.

then, five years from now

$$\frac{a+5}{b+5} = \frac{11}{9} \text{ or}$$

$$\frac{a+5}{b+5} = \frac{11}{9} \Rightarrow 9a + 45 = 11b + 55$$

$$9a - 11b = 10 \quad \dots (1)$$

five years ago, A's age = (a - 5) and B's age = (b - 5)

$$\frac{a-5}{b-5} = \frac{3}{2} \Rightarrow 2a - 10 = 3b - 15$$

$$2a - 3b = -5 \quad \dots (2)$$

Solving the two equations simultaneously, we

$$3b - 2a = 5$$

$$2a - 3b = -5$$

$$3b - 2a = 5$$

$$2a - 3b = -5$$

$$3b - 2a = 5$$

$$2a - 3b = -5$$

$$3b - 2a = 5$$

$$2a - 3b = -5$$

$$3b - 2a = 5$$

$$2a - 3b = -5$$

$$3b - 2a = 5$$

$$2a - 3b = -5$$

$$3b - 2a = 5$$

$$2a - 3b = -5$$



$$\Rightarrow 20x + 40 = 18x + 48$$

$$\Rightarrow 2x = 48 - 40 = 8$$

$$\Rightarrow x = 4$$

B's present age = 12 years  
 $\therefore$  A's required age =  $4x + 5$   
 $= 16 + 5 = 21$  years

Quantity I > Quantity II

5. (3) **Quantity I** : Let the amount be Rs.  $x$ .

$$\Rightarrow \left( \frac{5}{9} - \frac{2}{14} \right) x = 40$$

$$\Rightarrow \left( \frac{1}{3} - \frac{1}{7} \right) x = 40$$

$$\Rightarrow \left( \frac{7-3}{21} \right) x = 40$$

$$\Rightarrow x = \frac{40 \times 21}{4} = \text{Rs. } 210$$

**Quantity II** :  $A \times 0.5 = B \times 0.6 = C \times 0.75$

$$= \frac{A \times 5}{10} = \frac{B \times 6}{10} = \frac{C \times 75}{100}$$

$$\Rightarrow \frac{21}{5} (y - 5y) = 2 \Rightarrow \frac{21y}{5} - 21y = 2$$

$$\Rightarrow -4y = -10$$

$$\Rightarrow y = 2.5$$

$$\text{and } x = 1.5$$

From here we find  $y$  and then  $x$ .

2. (5) Both the statements are required.

Let Katrina's age and Priyanka's age, five years before be  $6x$  and  $5x$  respectively. Their present age are  $6x + 5$  and  $5x + 5$ . Five years later their age would be  $6x + 10$ ,  $5x + 10$ .

$$\frac{6x + 10}{5x + 10} = \frac{8}{7} \Rightarrow 42x + 70 = 40x + 80$$

$$2x = 10, x = 5$$

$$\text{Katrina's present age} = 6x + 5 = 35$$

$$\text{Priyanka's present age} = 5x + 5 = 30.$$

3. (4) From statement I,  $\frac{p}{q} = \frac{1}{4}, q > 0 \Rightarrow p > 0$

From statement II,  $\frac{3p}{2q} = \frac{6}{16}, p > 0 \Rightarrow q > 0$

# 8 Alligation or Mixture

## IMPORTANT FACTS AND FORMULAE

1. The literal meaning of the word 'Alligation' is 'mixing'.

2. As a matter of fact, alligation may be under-

stood as a rule that enables us

(i) to find out the mean or average value of mix-  
 ture when the prices of two or more ingredients  
 are given, and the proportion in which they are  
 mixed is given. This is known as 'Alligation Medial'.

(ii) to find out the proportion in which two ingre-  
 dients of the same nature of given prices should  
 be mixed so as to produce a mixture worth a de-  
 sired price. This is known as 'Alligation Alternate'.

Quantity of cheaper

Quantity of dearer

Price of dearer - Mean price

Mean price - Price of cheaper

We represent it as

C, P of a unit

Quantity of cheaper

C, P of a unit

Quantity of dearer

C, P of a unit

Quantity of dearer

# ALLIGATION OR MIXTURE

- is 3 : 1, in the second container 5 : 2. If they are mixed up, the ratio of milk to water in the mixture will be
- (1) 28 : 41 (2) 41 : 28  
(3) 15 : 41 (4) 41 : 15  
(5) 17 : 41

6. Two types of tea costing ₹ 180 per kg and ₹ 280 per kg should be mixed in the ratio so that the mixture obtained, sold at ₹ 320 per kg to earn a profit of 20% is

- (1) 3:13 (2) 1:13 (3) 4:13 (4) 2:13  
(5) 5:13

7. In two blends of mixed tea, the ratios of Darjeeling and Assam tea are 4 : 7 and 2 : 5. The

12. Three glasses of equal volume contain acid mixed with water. The ratios of acid and water are 2 : 3, 3 : 4 and 4 : 5 respectively. Contents of these glasses are poured in a large vessel. The ratio of acid and water in the large vessel is

- (1) 411 : 540 (2) 401 : 544  
(3) 417 : 564 (4) 407 : 560  
(5) 405 : 541

13. Two bottles contain acid and water in the ratio 2 : 3 and 1 : 2 respectively. These are mixed in the ratio 1 : 3. What is the ratio of acid and water in the new mixture ?

- (1) 7:13 (2) 11:57 (3) 23:37 (4) 1:3

10. Solution A contains 10% acid and solution B contains 30% acid. In what ratio should solution A be mixed with solution B to obtain a mixture with 25% acid?

- (1) 1 : 2 (2) 3 : 1 (3) 1 : 3 (4) 2 : 1  
(5) 3 : 2

11. In what ratio should coffee powder costing Rs. 2500 per kg be mixed with coffee powder costing Rs. 1500 per kg so that the cost of the mixture is Rs. 2250 per kg?

- (1) 1 : 4 (2) 4 : 1 (3) 3 : 1 (4) 1 : 3  
(5) 2 : 3

12. A mixture is composed of 11 parts of pure milk and 2 parts of water. If 35 litres of water are added to the

# ALLIGATION OR MIXTURE

27. An alloy is made by mixing metal A costing Rs. 2000/kg. and metal B costing Rs. 400/kg in the ratio A : B = 3 : 1. What is the cost (in Rs.) of 8 kilograms of this alloy ?

- (1) 1500 (2) 9800 (3) 6400 (4) 12800  
(5) 12500

28. A jar contains a blend of a fruit juice and water in the ratio 5 : x. When 1 litre of water is added to 4 litres of the blend the ratio of fruit juice to water becomes 1 : 1. What is the value of x?

- (1) 3 (2) 1 (3) 2 (4) 4

# ALLIGATION OR MIXTURE

35. A and B are solutions of acid and water. The ratios of water and acid in A and B are 4 : 5 and 1 : 2, respectively. If x litres of A is mixed with y litres of B, then the ratio of water and acid in the mixture becomes 8 : 13. What is  $\frac{x}{y}$ ?
- (1) 5 : 6 (2) 2 : 5 (3) 3 : 4 (4) 2 : 3  
(5) 4 : 3
36. Alloy A contains metals x and y only in the ratio 5 : 2 and alloy B contains these metals in the ratio 3 : 4. Alloy C is prepared by mixing A and B in the ratio 4 : 5. The percentage of x in alloy C is :
- (1) 45 (2) 55  $\frac{5}{9}$  (3) 44  $\frac{4}{9}$  (4) 56
41. A vessel contains 64 litres of mixture of milk and water in the ratio 7 : 3 respectively. 8 litres of mixture is replaced by 12 litres of milk. What is the ratio of milk and water in the resulting mixture?
- (1) 64 : 21 (2) 35 : 22  
(3) 64 : 23 (4) 65 : 21  
(5) None of these
42. Vessel A contains a mixture of milk and water in the respective quantities of 20 litres and 4 litres. Vessel B contains a mixture of milk and water in the respective ratio of 3 : 1. Mixtures from vessel A and B are both mixed together in vessel C. If the resultant percentage of water in vessel C was 20%, what was the initial quantity of mixture in vessel B? (in litres)
- (1) 16 (2) 20 (3) 12 (4) 24

# ALLIGATION OR MIXTURE

47. A jar was containing 60 litres of mixture of milk and water. The respective ratio of milk and water in the mixture was 7 : 5. From this jar 12 litres of mixture was taken out and 8 litres of pure milk was added. What is the respective ratio of milk and water in the mixture after the final operation?
- (1) 10 : 15 (2) 7 : 5 (3) 9 : 7 (4) 9 : 5  
(5) 9 : 4
48. To m litres of a n% solution of acid, x litres of water is mixed to yield  $(m - 10)\%$  solution of acid. If  $m > 25$ , then x equals
- (1)  $\frac{10m}{m-10}$  (2)  $\frac{5m}{m-10}$  (3)  $\frac{2m}{m-10}$  (4)  $\frac{m}{m-10}$   
(5) None of these
49. Manish bought 25 kg of rice at
5. A mixture of milk and water in a jar contains x litre milk and 8 litre water. x litre milk and 8 litre water are mixed to form a new mixture. If 44% of the new mixture is 20 litre, find the value of x? (in litres)
- (1) 7 (2) 8 (3) 6 (4) 5  
(5) 9
5. A mixture of milk and water contains 15% water. What quantity of pure milk should be added to 70 litres of mixture to reduce the quantity of water to 12%? (in litres)
- (1) 1.25 (2) 12.5 (3) 1.5 (4) 2.25  
(5) 2.75
6. A mixture contains wine and water in the ratio

# ALLIGATION OR MIXTURE

vessel. The ratio of milk and water now becomes 13 : 2 respectively. Again 15 litres of mixture is taken out. What is the quantity of milk in the resulting mixture? (in litres)

- (1) 85 litres
- (2) 80 litres
- (3) 81 litres
- (4) 91 litres
- (5) None of these

12. In jar A, 180 litre milk was mixed with 36 litre water. Some of this mixture was taken out from jar A and put it in jar B. If after adding 6 litres of water in the mixture, the respective ratio between milk and water in jar B was 5 : 2 respectively, what was the amount of mixture that was taken out from jar A? (in litres)

17. 30 litre of pure water was mixed with 150 litre of pure milk, then 30 litre of this mixture was then sold and more quantity of pure milk and pure water was added in the respective ratio of 5 : 6. If the final quantity of milk was 500% of the initial quantity of water, what was the quantity of pure water that was added? (in litre)

- (1) 12
- (2) 30
- (3) 18
- (4) 24
- (5) 36

18. In a mixture of 150 litres, the ratio of milk to water is 2 : 1. What amount of water should be further added to the mixture so that the ratio of the milk :

# ALLIGATION OR MIXTURE

23. A chemist has 10 L of a solution that is 10% sulfuric acid by volume. He wants to dilute the solution to 4% strength by adding water. How many litres of water must be added?

- (1) 15 L
- (2) 20 L
- (3) 18 L
- (4) 25 L
- (5) None of these

24. Liquid A and liquid B were mixed in an equal proportion in vessel X to form a 200 litre mixture. 50 litre of liquid C was added to the mixture in vessel X. 100 litres of this mixture was taken out and put in vessel Y. If 20 litres of liquid D was added to the mixture in vessel Y, what is the percentage of liquid C in the mixture?

25. A mixture of milk and water in the mixture was 4 : 1 respectively. Again 23 litres of the mixture was taken out and replaced with 27 litres of water. What is the respective ratio of milk and water in the resultant mixture?

- (1) 358 : 372
- (2) 116 : 69
- (3) 69 : 43
- (4) 101 : 37
- (5) 368 : 227

30. A solution of sugar syrup has 15% sugar. Another solution has 5% sugar. How many litres of the second solution must be added to 20 L of the first solution to make a solution of 20% sugar?



ALLIGATION OR MIXTURE

5. One type of liquid contains 25% of milk, the other contains 30% of milk. A container is filled with 6 parts of the first liquid and 4 parts of the second liquid. The percentage of milk in the mixture is —  
 (1) 27% (2) 31% (3) 29% (4) 33%  
 (5) None of these
6. One test tube contains some acid and another test tube contains an equal quantity of water. To prepare a solution, 20 g of the acid is poured into the second test tube. Then, two-third of the so-formed solution is poured from the second tube into the first. If the fluid

17. 15 litres of a mixture contains 20% alcohol and rest water. If 3 litres of water is added in it, then percentage of alcohol in the new mixture will be :  
 (1) 25% (2) 20% (3) 10.67% (4) 33.33%  
 (5) 9 : 2
18. The proportion of 1st and 2nd liquids in a mixture is 2 : 3 while in another mixture their ratio is 5 : 4. In what ratio should the two mixtures be mixed, so that the new mixture contains equal amount of the two liquids?  
 (1) 3 : 7 (2) 5 : 9 (3) 3 : 4 (4) 2 : 5  
 (5) 3 : 5
19. In a mixture of syrup and water, the ratio of syrup and water is 5 : 17.

2. **Quantity I :** A vessel contains mixture with 70% milk. A part of it is taken out and replaced by mixture with 50% milk. The resultant mixture has 55% milk only. Find the amount of mixture replaced.  
**Quantity II :** 15 litres of a mixture contains 20% alcohol and rest water. If 3 litres of water is added in it, then percentage of alcohol in the new mixture will be :  
 3. **Quantity I :** In a poultry farm, there are some hens and some cows. The total no. of heads are 30 while the total no. of legs of both cows



# ALLIGATION OR MIXTURE

**Give answer (3)** If the data either in statement I alone or in statement II alone is sufficient to answer the question.

**Give answer (4)** If the data even in both the statements I and II together is not sufficient to answer the question.

**Give answer (5)** If the data in both the statements I and II together are necessary to answer the question.

1. What was the initial quantity of mixture of juice and water?

I. Juice and water were in the ratio of 8 : 3 respectively in the mixture initially.

II. When 11 litres of mixture is taken out and 6 litres of juice is added, the ratio between juice and water becomes 10 : 3 respectively.

2. What was the initial quantity of juice in the mixture of juice and water?

I. Initial quantity of the mixture was 25 litres.

II. Initially the quantity of juice in the mixture was 70%, after adding 10 litres of water.

## LEVEL-II

|         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|
| 1. (2)  | 2. (5)  | 3. (2)  | 4. (1)  | 5. (2)  | 6. (5)  |
| 7. (3)  | 8. (1)  | 9. (5)  | 10. (2) | 11. (4) | 12. (4) |
| 13. (5) | 14. (2) | 15. (3) | 16. (1) | 17. (2) | 18. (4) |
| 19. (4) | 20. (5) | 21. (3) | 22. (3) | 23. (1) | 24. (1) |
| 25. (5) | 26. (3) | 27. (3) | 28. (2) | 29. (5) | 30. (4) |

## LEVEL-III

|         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|
| 1. (4)  | 2. (5)  | 3. (3)  | 4. (3)  | 5. (1)  | 6. (4)  |
| 7. (1)  | 8. (1)  | 9. (2)  | 10. (1) | 11. (4) | 12. (3) |
| 13. (2) | 14. (1) | 15. (4) | 16. (1) | 17. (3) | 18. (2) |
| 19. (2) | 20. (3) |         |         |         |         |

## QUANTITY BASED

$$\text{Required S.P.} = \text{Rs.} \left( \frac{695 \times 100}{100} \right)$$

$$= \text{Rs.} 695$$

$$\text{Rate per kg} = 695 \times 100$$

$$= \text{Rs.} 13.7$$

$$\therefore \text{In 20 litres of mixture,}$$

$$\text{Alcohol} = \frac{20 \times 20}{100} = 4 \text{ litres}$$

$$\text{Water} = 20 - 4 = 16 \text{ litres}$$

$$\text{On adding 4 litres of water,}$$

$$\text{Quantity of water} = 16 + 4$$

$$= 20 \text{ litres}$$

$$\text{Quantity of mixture} = 24 \text{ litres}$$

$$\therefore \text{Required percent}$$

$$= \frac{4}{24} \times 100 = 16\frac{2}{3}\%$$

$$= \text{Rs.} \frac{500}{3} \text{ per kg.}$$

$$\text{By rule of alligation,}$$

$$\begin{array}{l} \text{Varely I} \\ \text{Rs. 100} \end{array} \quad \begin{array}{l} \text{Varely II} \\ \text{Rs. 250} \end{array}$$

$$\frac{250 - 500}{3} = \frac{500 - 250}{3}$$

$$= \frac{250}{3}$$

$$= \frac{250}{3}$$

$$= \frac{250}{3}$$

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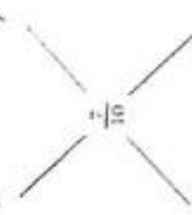
$$= \frac{250}{3}$$

$$= \frac{250}{3}$$

8. (3) By the rule of alligation

liquid 1 in mixture 1. liquid 1 in mixture II.

$$\frac{3}{8} \quad \frac{6}{7}$$



$$\frac{6}{7} - \frac{7}{10} = \frac{11}{70}$$

$$\frac{60 - 49}{70} = \frac{11}{70}$$

$$\therefore \text{Required ratio} = \frac{11}{70} : \frac{13}{40}$$

$$= 11 \times 4 : 13 \times 7$$

$$= 44 : 91$$

9. (3) Remaining acid = Initial quantity

$$\left( 1 - \frac{\text{Quantity taken out}}{\text{Original quantity}} \right)$$

$$= 20 \left( 1 - \frac{4}{20} \right)^2$$

$$= 20 \left( 1 - \frac{1}{5} \right)^2$$

$$= \frac{5 \times 5 \times 10}{8 \times 5 \times 10} = 9$$

$$= \frac{35}{40} = \frac{7}{8}$$

$$= 40 \times 9 \times 5$$

$$= 1800$$

$\therefore$  Required quantity of third variety of tea

$$= 8x + y = 8 \times 5 + 5 = 45 \text{ kg.}$$

11. (2) In 60 kg of alloy A.

$$\text{Lead} = \frac{3}{5} \times 60 = 36 \text{ kg.}$$

$$\text{Tin} = \frac{2}{5} \times 60 = 24 \text{ kg.}$$

$$\text{In 100 kg of alloy B,}$$

$$\text{Tin} = \frac{1}{5} \times 100 = 20 \text{ kg.}$$

$$\text{In 160 kg of new alloy,}$$

$$\text{Tin} = 24 + 20 = 44 \text{ kg.}$$

12. (2) Let the capacity of each glass be 1 litre.

On mixing all three mixtures together,

$$\text{Acid} = \frac{2}{5} + \frac{3}{7} + \frac{4}{9}$$

$$= \frac{126 + 135 + 140}{315}$$

$$= \frac{401}{315}$$

$$= 401 \text{ litre}$$

$$= \left( \frac{2}{5} + 1 \right) : \left( \frac{3}{5} + 2 \right)$$

$$= \frac{2+5}{5} : \frac{3+10}{5}$$

$$= 7 : 13$$

14. (4)

$$\text{Type-I} \quad \text{Type-II}$$

$$\text{Copper} = \frac{8}{11} \quad \text{Copper} = \frac{15}{22}$$

$$\frac{5}{7} - \frac{15}{22} = \frac{5}{154}$$

$$\frac{110-105}{154} = \frac{5}{154}$$

$$\therefore \text{Required ratio} = \frac{5}{154} : \frac{1}{77}$$

$$= 5 : 2$$

15. (2) In 49 kg. of mixture,

$$\text{Tea of Assam} = \left( \frac{5}{7} \times 49 \right) \text{ kg.}$$

$$= 35 \text{ kg.}$$

$$\text{Tea of Darjeeling} = (49 - 35) \text{ kg.}$$

$$= 14 \text{ kg.}$$

Let x kg. of Darjeeling tea be added,

$$35 : 2$$

Container-II

$$\text{Milk} = \frac{4}{4} = 1 \text{ litre.}$$

$$\text{Water} = 1 \text{ litre}$$

Container-III

$$\text{Milk} = \frac{5}{7} = \frac{25}{7} \text{ litres}$$

$$\text{Water} = \frac{10}{7} \text{ litres}$$

$\therefore$  Required ratio in container-IV

$$= \left( \frac{12}{5} + 3 - \frac{25}{7} \right) : \left( \frac{3}{5} + 1 - \frac{10}{7} \right)$$

$$= \left( \frac{84 + 105 - 125}{35} \right) : \left( \frac{21 + 35 - 50}{35} \right)$$

$$= \frac{314}{35} : \frac{106}{35}$$

$$= 157 : 53$$

18. (3) In coloured picture,

$$\text{Blue part} = \frac{4}{7}$$

$$\text{Yellow part} = \frac{3}{7}$$

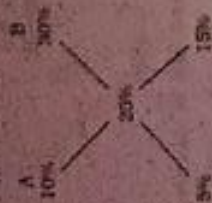
In upper half,

$$\text{Blue part} = \frac{2}{5 \times 2} = \frac{1}{5}$$

# ALLIGATION OR MIXTURE

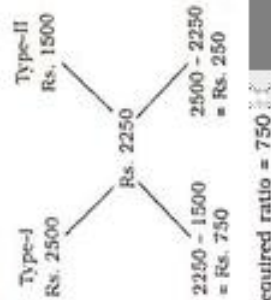
∴ Required ratio = 5 : 15  
= 1 : 3

OR

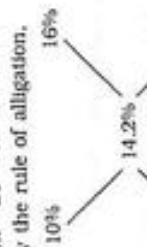


Required ratio = 5 : 15 = 1 : 3  
[Note : Such type of questions easily solved by alligation method]

20. (3) By the rule of alligation.

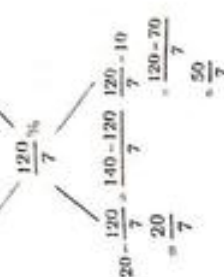


Milk : Water = 11x : 2x  
On adding 35 litres of water,  
⇒ 11x = 2 (2x + 35)  
11x = 4x + 70  
7x = 70  
⇒ x = 10  
∴ Quantity of pure milk  
= 11 × 10 = 110  
22. (1) In 300 ml of mixture :  
Acid : 300  
1 + x, water = 300x  
1 + x  
∴ 300 : 300x + 50 = 2 : 5  
⇒ 300 / (300x + 50) = 2 / 5  
⇒ 6 / (7x + 1) = 2 / 5  
⇒ 14x + 2 = 30  
⇒ 14x = 28 ⇒ x = 2  
23. (2) By the rule of alligation.



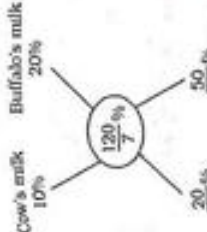
# ALLIGATION OR MIXTURE

24. (1) By the rule of alligation.  
Cow's milk 10%  
Buffalo's milk 20%



∴ Required ratio = 20 : 50 = 2 : 5

OR



Cow's milk : Buffalo's milk

26. (3) Remaining quantity of ethanol = Initial quantity

$\left(1 - \frac{\text{quantity taken out}}{\text{Initial quantity}}\right)$

$= 80 \left(1 - \frac{20}{80}\right)$

$= 80 \times \left(1 - \frac{1}{4}\right)$

$= 80 \times \frac{3}{4} = 45 \text{ litres}$

∴ Quantity of water

$= (80 - 45) \text{ litres} = 35 \text{ litres}$

27. (4) A : B = 3 : 1 = 6 : 2

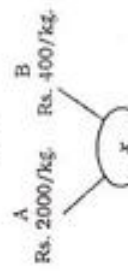
∴ Cost of 8 kg. of alloy

$= \text{Rs. } 16 \times 2000 = 2 \times 400$

$= \text{Rs. } (12000 + 800)$

$= \text{Rs. } 12800$

OR



$$\frac{20}{5+x} = \frac{4x}{5+x} + 1$$

$$\Rightarrow \frac{20}{5+x} = \frac{4x+5+x}{5+x}$$

$$\Rightarrow 20 = 5x + 5$$

$$\Rightarrow 5x = 15$$

$$\Rightarrow x = 3$$

29. (4) Initially,

$$\text{Copper} = 3x \text{ gm.}$$

$$\text{Tin} = 2x \text{ gm.}$$

According to the question,

On adding 250 gm. of copper,

$$3x + 250 = 2 \times 2x$$

$$\Rightarrow 4x - 3x = 250$$

$$\Rightarrow x = 250$$

$\therefore$  Quantity of tin

$$= 2x = (2 \times 250) \text{ gm} = 500 \text{ gm.}$$

OR

$$\text{Copper : Tin} = 3 : 2$$

$$\text{Copper} = 3x$$

$$\text{Tin} = 2x$$

$$\text{Now } (3x + 250) = 2 \times 2x$$

$$3x + 250 = 4x$$

$$\Rightarrow x = 250 \text{ gm.}$$

$$\therefore \text{Amount of tin} = 250 \times 2$$

$$= 500 \text{ gm.}$$

$$\Rightarrow 30x = 50y$$

$$\Rightarrow \frac{x}{y} = \frac{50}{30} = \frac{5}{3}$$

31. (3) C.P. of mixture =  $\frac{50 \times 100}{112}$

$$= \text{Rs. } 50 \text{ per kg.}$$

By the rule of alligation,



32. (4) In 5 kg of alloy -A,

$$\text{Quantity of zinc} = 5 \times \frac{3}{7} = \frac{15}{7} \text{ kg.}$$

In 6 kg of alloy -B,

$$\text{Quantity of zinc} = \frac{6 \times 2}{7} = \frac{12}{7} \text{ kg.}$$

Total quantity of zinc

$$= \frac{15}{7} + \frac{12}{7} = \frac{27}{7} \text{ kg}$$

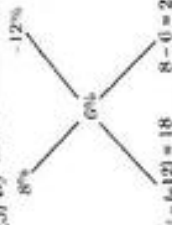
$\therefore$  Required percent

$$= \frac{27}{7} \times \frac{100}{27}$$

By the rule of alligation.



34. (3) By the rule of alligation,



Required ratio = 18 : 2 = 9 : 1  
 $\therefore$  Objects sold at 8% profit

$$= \frac{9}{10} \times 80 = 72$$

35. (3) Water: Acid Total Quantity

$$A \quad 4 : 5 \quad 9$$

$$B \quad 1 : 2 \quad 3$$

$$13 (4x + 3y) = 8 (5x + 6y)$$

$$52x + 39y = 40x + 48y$$

$$12x = 9y$$

$$\frac{x}{y} = \frac{9}{12}$$

$$\Rightarrow x : y = 3 : 4$$

36. (2)  $x : y$  Quantity

$$A \quad 5 : 2 \quad 7$$

$$B \quad 3 : 4 \quad 7$$

A and B are mixed in such a way that their ratio is 4 : 5

(i) Quantity of A =  $7 \times 4 = 28$  units

Quantity of x in A =  $5 \times 4 = 20$  units

Quantity of y in A =  $2 \times 4 = 8$  units

(ii) Quantity of B =  $7 \times 5 = 35$  units

Quantity of x in B =  $3 \times 5 = 15$  units

Quantity of y in B =  $4 \times 5 = 20$  units

Percentage of x in alloy C

$$= \left( \frac{20+15}{28+35} \right) \times 100\%$$

$$= \frac{35}{63} \times 100\%$$

$$= \frac{500}{9}$$



38. (4) If there be 1 litre of acid in 4 litres of mixture, then in case I,

$$\text{Percentage of acid} = \frac{1}{4} \times 100 = 25\%$$

Case II,

$$\text{Percentage of acid} = \frac{2}{6} \times 100 = 33\frac{1}{3}\%$$

∴ Percentage of acid in original mixture = 25%  
 39. (3) Mixture : 2 kg of rice at Rs. 15/kg + 3 kg of rice at Rs. 13/kg

$$\text{Total weight} = 2 + 3 = 5 \text{ kg}$$

Total cost price

$$= (2 \times 15) + (3 \times 13)$$

$$= 30 + 39 = \text{Rs. } 69$$

Cost price per kg of the mixture

$$= \frac{69}{5} = \text{Rs. } 13.80$$

Selling price to get  $33\frac{1}{3}\%$  profit

$$= \frac{100 - 33\frac{1}{3}}{100} \times \text{Rs. } 13.80$$

In resulting mixture,  
 Milk = 44.8 - 5.6 + 1.2

$$= 51.2 \text{ litres}$$

$$\text{Water} = 19.2 - 2.4$$

$$= 16.8 \text{ litres}$$

$$\therefore \text{Required ratio} = 51.2 : 16.8$$

$$= 64 : 21$$

42. (1) In vessel B,

$$\text{Quantity of milk} = 3x \text{ litres}$$

$$\text{Quantity of water} = x \text{ litres}$$

In vessel C,

$$\text{Quantity of water}$$

$$= (4 + x) \text{ litres}$$

$$\text{Total quantity} = (24 + 4x) \text{ litres}$$

According to the question,

$$\frac{4 + x}{24 + 4x} = \frac{20}{100} = \frac{1}{5}$$

$$\Rightarrow 20 + 5x = 24 + 4x$$

$$\Rightarrow 5x - 4x = 24 - 20$$

$$\Rightarrow x = 4$$

$$\therefore \text{Quantity of mixture in vessel B} = 4 \times 4 = 16 \text{ litres}$$

43. (4) In original mixture

$$\Rightarrow 20x = 360$$

$$\Rightarrow \frac{360}{20} = 18 \text{ litres}$$

45. (1) Quantity of water in jar B = x litres, (let).

∴ Quantity of milk

$$= (20 - x) \text{ litres}$$

In jar A,

$$\text{Quantity of milk} = \frac{5}{9} \times 36$$

$$= 20 \text{ litres}$$

$$\text{Quantity of water} = \frac{4}{9} \times 36$$

$$= 16 \text{ litres}$$

According to the question,

$$\frac{20 + 20 - x}{16 + x} = \frac{5}{3}$$

$$\Rightarrow \frac{40 - x}{16 + x} = \frac{5}{3}$$

$$\Rightarrow 80 + 5x = 120 - 3x$$

$$\Rightarrow 8x = 120 - 80 = 40$$

$$\Rightarrow x = 5 \text{ litres}$$

$$\Rightarrow 40 - x = 35$$

$$\Rightarrow 8x = 120 - 80 = 40$$

$$\Rightarrow x = 5 \text{ litres}$$

On adding 8 litres of pure milk,  
 Required ratio = (28 + 8) : 20  
 = 36 : 20 = 9 : 5

48. (1) [Tricky approach]

$$\text{Acid} = \frac{m}{100} \times m = \frac{m^2}{100}$$

$$\text{New percentage} = \frac{m^2 \times 100}{(m + x) \times 100}$$

$$\Rightarrow \frac{m^2}{m + x} = m - 10$$

$$\Rightarrow m^2 - 10m + mx = m^2$$

$$\Rightarrow x(m - 10) = 10m$$

$$\Rightarrow x = \frac{10m}{m - 10}$$

$$\Rightarrow x = \frac{10m}{m - 10}$$

$$\Rightarrow x = \frac{10m}{m - 10}$$

$$\Rightarrow x = \frac{10m}{m - 10}$$

49. (4) C.P. of 40 kg of mixture

$$= ₹ [25 \times 32] + [15 \times 36]$$

$$= ₹ (800 + 540) = ₹ 1340$$

S.P. of 40 kg of mixture

$$= ₹ (40 \times 40.2) = ₹ 1608$$

$$\text{Profit} = ₹ (1608 - 1340)$$

$$= ₹ 268$$

# ALLIGATION OR MIXTURE

$$\begin{aligned} \Rightarrow \frac{x^2}{x+y} &= x-10 \\ \Rightarrow x^2 &= x^2 - 10x + y(x-10) \\ \Rightarrow y &= \frac{10x}{x-10} \end{aligned}$$

2. (5) In remaining (78 - 26) = 52 litres of mixture.

$$\begin{aligned} \text{Quantity of milk} &= \frac{6}{13} \times 52 \\ &= 24 \text{ litres} \end{aligned}$$

$$\begin{aligned} \text{Quantity of water} &= \frac{7}{13} \times 52 \\ &= 28 \text{ litres} \end{aligned}$$

Let x litres of milk be added.

$$\begin{aligned} \therefore \frac{26}{52+x} &= \frac{40}{100} = \frac{2}{5} \\ \Rightarrow 104 + 2x &= 140 \\ \Rightarrow 2x &= 140 - 104 = 36 \\ \Rightarrow x &= \frac{36}{2} = 18 \text{ litres} \end{aligned}$$

3. (2) Remaining mixture after giving 26 litres of

$$\begin{aligned} \text{B. (2) Let } x \text{ litres of pure milk be added} \\ \text{Quantity of water} &\Rightarrow \text{from } 15\% \text{ to } 12\% \\ \therefore \frac{50 \times 15}{100} &= (50+x) \times 12 \\ \Rightarrow 750 &= 600 + 12x \\ \Rightarrow 12x &= 750 - 600 = 150 \\ \Rightarrow x &= \frac{150}{12} = 12.5 \text{ litres} \end{aligned}$$

6. (5) Quantity of wine = 5x litres  
Quantity of water = x litres.  
On adding 5 litres of water,

$$\begin{aligned} \frac{5x}{x+5} &= \frac{5}{2} \\ \Rightarrow 2x &= x+5 \Rightarrow x=5 \\ \therefore \text{Quantity of wine} &= 5 \times 5 \\ &= 25 \text{ litres} \end{aligned}$$

7. (3) Suppose a container contains x units of liquid from which y units are taken out and replaced by water. After n operations, the quantity of pure liquid

$$= x \left( 1 - \frac{y}{x} \right)^n \text{ units}$$

# ALLIGATION OR MIXTURE

9. (4) In one pure mixture,

$$\begin{aligned} \text{Alcohol} &\Rightarrow 0.7 \text{ litre} \\ \text{Water} &\Rightarrow 0.3 \text{ litre} \\ \text{Let } x \text{ litre of alcohol be added.} \\ \therefore \frac{0.3}{1+x} &= \frac{15}{100} = \frac{3}{20} \end{aligned}$$

$$\begin{aligned} \Rightarrow 6 &= 3 + 3x \\ \Rightarrow 3x &= 6 - 3 = 3 \\ \Rightarrow x &= 1 \text{ litre} \end{aligned}$$

10. (2) In 20 litres of mixture

$$\begin{aligned} \text{Milk} &= \frac{3}{5} \times 20 = 12 \text{ litres} \\ \text{Water} &= 8 \text{ litres} \\ \text{In 10 litres of mixture,} \\ \text{Milk} &= 6 \text{ litres} \\ \text{Water} &= 4 \text{ litres} \end{aligned}$$

On adding 10 litres of milk,  
Milk  $\Rightarrow 12 + 6 + 10 = 28$  litres  
Water  $\Rightarrow 8 - 4 = 4$  litres  
Again, in 10 litres of mixture,

$$\text{Milk} = \frac{4}{10} \times 10 = 4 \text{ litres}$$

12. (4) In jar A,

Milk : Water = 180 : 36 = 5 : 1  
Let x litres of mixture be taken out.

$$\text{Quantity of milk} = \frac{5x}{6} \text{ litres}$$

$$\text{Quantity of water} = \frac{x}{6} \text{ litres}$$

According to the question.

On adding 5 litres of water in jar B,

$$\frac{5x}{6} - \frac{5}{2} = \frac{5}{2} \Rightarrow x = 5$$

$$\Rightarrow \frac{5x}{x+36} = \frac{5}{2}$$

$$\Rightarrow 10x = 5x + 180 \Rightarrow 5x = 180$$

$$x = \frac{180}{5} = 36 \text{ litres}$$

13. (5) In container A,

Milk : Water = 80 : 20 = 4 : 1

ALLIGATION OR MIXTURE

On adding 6 litres of water,  
Total quantity of water = 41 litres  
Quantity of mixture =  $126 + 6 = 132$  litres  
 $\therefore$  Percentage of water

$$= \frac{41}{132} \times 100 \approx 31$$

15. (3) Initial ratio = Water : Milk

$$= 1 : 4$$

In 20% of mixture

Water = 0.5 litres

Milk = 2 litres

$\therefore$  In remaining 10 litres of mixture,

Water = 2 litres

Milk = 8 litres

$$\therefore \frac{\text{Water}}{\text{Milk}} = \frac{2+x}{8} = \frac{4}{1}$$

$$\Rightarrow x + 2 = 32 \Rightarrow x = 30$$

$$\text{Again, } \frac{\text{Water}}{\text{Milk}} = \frac{2+30}{8+y} = \frac{1}{4}$$

$$\Rightarrow \frac{32}{8+y} = \frac{1}{4}$$

$$\Rightarrow 5x = 150 - 125 = 25$$

$$\Rightarrow x = \frac{25}{5} = 5$$

$\therefore$  Quantity of water added

$$= 6 \times 5 = 30 \text{ litres}$$

18. (4) In 150 litres of mixture,

$$\text{Milk} = \frac{2}{3} \times 150 = 100 \text{ litres}$$

Water = 50 litres

Again, Milk : Water = 1 : 2

$$= 100 : 200$$

$\therefore$  Quantity of water to be added =  $(200 - 50)$  litres

$$= 150 \text{ litres}$$

19. (4) In 26 litres of mixture,

$$\text{Milk} = \frac{10}{13} \times 26 = 20 \text{ litres}$$

Water = 6 litres

In original mixture,

$$\text{Milk} = 10x \text{ litres}$$

$$\text{Water} = 3x \text{ litres}$$

21. (3) Case I,

$$\text{Liquid A} = 2y \text{ litres}$$

$$\text{Liquid B} = 3y \text{ litres}$$

Case II,

$$3y + 6 - 2y - 6 = 18$$

$$\Rightarrow y = 18$$

$$\therefore x = 5y = 5 \times 18 = 90 \text{ litres}$$

$\therefore$  Initial quantity of milk

$$= 3x \text{ litres}$$

Initial quantity of water =  $x$  litres

According to the question,

$$\frac{24}{25} \times 4x \text{ litres of mixture}$$

$$\text{In } \frac{24}{25} \times 4x \text{ litres of mixture}$$

$$\text{Quantity of milk} = \frac{96x}{25} \times \frac{3}{4}$$

$$= \frac{72x}{25} \text{ litres}$$

$$\text{Quantity of water} = \frac{24x}{25} \text{ litres}$$

On adding 24 litres of water,

$$\frac{72x}{25} = \left( \frac{24x}{25} + 24 \right) \times 2$$

Let  $x$  L of water be added.

$$\text{Then, } (10+x) \times \frac{4}{100} = 1$$

$$\Rightarrow x = 15 \text{ L}$$

24. (1) Ratio of liquids A, B and C in vessel X

$$= 100 : 100 : 50 = 2 : 2 : 1$$

In vessel Y,

$$\text{Liquid-A} = 40 \text{ litres} \left( = \frac{2}{5} \times 100 \right)$$

$$\text{Liquid-B} = 40 \text{ litres} \left( = \frac{2}{5} \times 100 \right)$$

$$\text{Liquid-C} = 20 \text{ litres} \left( = \frac{1}{5} \times 100 \right)$$

$$\text{Liquid-D} = 20 \text{ litres}$$

$$\text{Total quantity} = 120 \text{ litres}$$

$\therefore$  Percentage of liquid-C

$$= \frac{20}{120} \times 100$$

$$= \frac{50}{3} \%$$

# ALLIGATION OR MIXTURE

26. (3) Initially,

Quantity of milk = 3x litres

Quantity of water = x litres

In four litres of mixture,

Milk = 3 litres

Water = 1 litre

According to the question,

$$\frac{3x - 3 + 3}{x - 1} = \frac{4}{1}$$

$$\Rightarrow 4x - 4 = 3x \Rightarrow x = 4 \text{ litres}$$

27. (3) According to question, quantity of gold in alloy

$$\frac{80}{100} \times 50 = 40 \text{ g}$$

Let x g of gold is mixed with this alloy.

$$\frac{40 + x}{50 + x} \times 100 = 95$$

$$\Rightarrow 4000 + 100x = 4750 + 95x$$

$$\Rightarrow x = \frac{750}{5} = 150 \text{ g}$$

$\therefore$  Required ratio  
= 73.6 : (18.4 + 27)

$$= 73.6 : 45.4 = 368 : 227$$

30. (4) Let x L of second solution must be added.

$$\text{Then, } \frac{15 \times 20 + 5 \times x}{20 + x} = 10$$

$$\Rightarrow 300 + 5x = 200 + 10x$$

$$\Rightarrow x = \frac{100}{5} = 20 \text{ L}$$

## LEVEL-III

1. (4) Solution B = 100 litres and Solution A = 25 litres

Cost price of 1 litre of solution B = Rs. 10 and

Cost price of 1 litre of solution A = Rs. 5

Total cost price

$$= (100 \times 10) + (25 \times 5)$$

$$= \text{Rs. } 1125$$

Total selling price

$$= (100 + 25) \times 10 = \text{Rs. } 1250$$

$$\text{So, profit} = 1250 - 1125 = 125$$

$$\therefore \left(1 - \frac{x}{54}\right)^2 = \frac{24}{54} = \frac{4}{9}$$

$$\therefore \left(1 - \frac{x}{54}\right) = \frac{2}{3}$$

$$\therefore \frac{x}{54} = \frac{1}{3} = 18 \text{ L}$$

5. (1) According to question, milk in mixture

$$\frac{25}{6 \times 100} + 4 \times \frac{30}{100} = \frac{270}{100}$$

$$\text{for 10 parts} = \frac{270 \times 10}{100} = 27\%$$

6. (4) Initially, let x g of water was taken.

1st process

$$\text{First test tube } (x - 20)$$

$$\text{Second test tube } (x + 20)$$

2nd process

$$\text{1st } \left[ (x - 20) + \frac{2}{3}(x + 20) \right]$$

# ALLIGATION OR MIXTURE

$$30 \text{ gm of Zinc} + 30 \text{ gm of Copper} = 60$$

$$1 \text{ gm of Zinc} + 1 \text{ gm of Copper} = \frac{60}{30} = ₹ 18 \text{ per gram}$$

1 gm of Zinc is given to be ₹ 12 per gram, hence

$$1 \text{ gram of Copper} = 18 - 12 = ₹ 6 \text{ per gram}$$

$$6 \times \frac{12}{6} = 12$$

$$2 \times \frac{4}{2} = 4$$

$$\text{Ratio} = 2 : 4 \text{ or } 1 : 2$$

$$\text{Weight of Copper} = \frac{1}{3} \times 30 = 10 \text{ gm}$$

9. (2) Ratio of milk to water originally = 17 : 68

$$= 1 : 4$$

$$\frac{1}{5} \times \frac{75}{100} = \frac{3}{4}$$

$$\frac{1}{4} \times \frac{11}{20} = \frac{11}{80}$$



# ALLIGATION OR MIXTURE

= 18 km and 20 kmph

In 108 minutes =  $20 \times \frac{108}{60}$  = 36 km

Ratio of distance = 18 : 36 = 1 : 2

12.(3) We are given,

$$\left(1 - \frac{10}{k}\right) \times \frac{25}{36} = \frac{25}{36} \text{ (where, } k \text{ is quantity of wine)}$$

$$= 1 - \frac{10}{k} = \frac{5}{6} \Rightarrow k = 60 \text{ litres}$$

$$13.(2) \begin{array}{ccc} 65 & & 75 \\ & \searrow & \nearrow \\ & 72 & \\ & \nearrow & \searrow \\ 3 & & 7 \end{array}$$

Ratio is 3 : 7. The bartender stole  $\frac{3}{10}$  th of the

original alcohol &  $\frac{3}{10} = 30\%$ .

14.(1) Since average weight of members of all 4 groups taken together is 50 kg and average weight of members of Group B & D is also 50kg. This means average weight of members of Group A & C must be 50 kg. Their individual averages are 35 kg & 52 kg respectively.

On adding 3 litres of water

Percentage of alcohol =  $\frac{3}{18} \times 100 = 16.67\%$

18. (2)

$$\begin{array}{ccc} 2 & & 5 \\ & \searrow & \nearrow \\ & 1 & \\ & \nearrow & \searrow \\ 5 & & 2 \end{array}$$

Required ratio

$$= \frac{1}{18} : \frac{1}{10} : 18 = 5 : 9$$

19. (2) Original quantity of mixture = 5x litres

Syrup = 3x litres; Water = 2x litres

On adding 5 litres of syrup,

Percentage of water

$$= \frac{2x}{5x+5} \times 100$$

$$\therefore \frac{2x}{5x+5} \times 100 = 35$$

$$= \frac{2x}{5}$$

# ALLIGATION OR MIXTURE

$$\frac{500}{500} = 1$$

$$\text{proportion of water in } 10\text{L} = \frac{10}{500} \times 100 = \frac{10}{51}$$

$$\frac{500}{500} = 1$$

proportion of alcohol in 2 [1]

$$\frac{10 - 10 \times \frac{1}{51}}{10 + 10 \times \frac{1}{51}} = \frac{500}{500} = 1$$

Hence, A = 1

## QUANTITY BASED EXPLANATION

1.(3) Quantity 1 : Fraction of water in mixture =

$$\frac{1}{4}$$

Fraction of water in water = 1

Fraction of water required =  $\frac{1}{3}$

$$\frac{1}{4} \quad \frac{1}{3}$$

mixture containing 70% milk is mixed with  
3 th of mixture containing 50% milk. Hence

4 th of mixture was replaced and  $\frac{3}{4} = 75\%$

Quantity II : In 15 litres of mixture,

$$\text{Alcohol} = \frac{15 \times 20}{100} = 3 \text{ litres}$$

$$\text{Water} = 15 - 3 = 12 \text{ litres}$$

On adding 3 litres of water,

$$\text{Percentage of alcohol} = \frac{3}{18} \times 100 = 16.67\%$$

Quantity I > Quantity II

3. (3) Quantity 1 : We know that a hen has two legs while a cow has 4 legs. Also average no. of legs when both are considered

$$= \frac{84}{30} = \frac{14}{5}$$

$$2 \quad \frac{14}{5}$$

$$\begin{aligned} \Rightarrow \frac{2x}{5x+5} \times 20 &= 7 \\ \Rightarrow 40x &= 35x+35 \\ \Rightarrow 5x &= 35 \\ \Rightarrow x &= 7 \end{aligned}$$

∴ Original quantity of mixture =  $5 \times 7 = 35$  litres

Quantity 1 > Quantity 2

5. (1) **Quantity 1** : Let the price of the third variety

be ₹  $x$  per kg.

$$\therefore (126 \times 1 + 135 \times 1 + 2 \times x) = 153 \times 4$$

$$\text{or } 261 + 2x = 612$$

$$\text{or } 2x = 612 - 261 = 351$$

$$\therefore x = \frac{351}{2} = 175.50$$

∴ The required price = ₹ 175.50 per kg.

**Quantity 2** : C.P. of 16 kg of the mixture = ₹  $(12 \times 16 + 4 \times 2)$

$$\Rightarrow \frac{7x}{10x+10} = \frac{5}{10}$$

$$\Rightarrow 7x = 5x + 50$$

$$\Rightarrow 2x = 50 \Rightarrow x = 25 \text{ litres}$$

$$\Rightarrow \text{Juice} = \frac{7 \times 25}{10} = 17.5 \text{ litres}$$

3. (5) In  $x$  litres of mixture,

$$\text{Milk} = \frac{4x}{5} \text{ litres}$$

$$\text{Water} = \frac{x}{5} \text{ litres}$$

From both statements,

Remaining milk

$$= 4x - \frac{4x}{5} = \frac{16x}{5} \text{ litres}$$

# Partnership

## 9

### IMPORTANT FACTS AND FORMULAE

**Partnership** is a form of association of two or more persons who contribute resources like non-physical together in order to carry out a business. Those of invest money are known as **partners**. The total investment is known as capital.

**Kinds of Partnership** : There are two kinds of partnership :

(i) **Simple Partnership** : It is one in which the

total profit before its distribution among partners. A working partner receives a fixed salary from the profits for managing the business. In order to solve the problems regarding partnership, remember this special trick :

$$\frac{A's \text{ Capital} \times A's \text{ time in partnership}}{B's \text{ Capital} \times B's \text{ time in partnership}} = \frac{A's \text{ profit}}{B's \text{ profit}}$$

(ii) **Partnership**

# PARTNERSHIP

5. A, B and C invest to start a restaurant. The total investment was Rs. 3 lakhs. B invested Rs. 50,000 more than A and C invested Rs. 25,000 less than B. If the profit at the end of the year was Rs. 14,400 then what is C's share of the profit (in Rs.) ?  
 (1) 3600 (2) 4800 (3) 6000 (4) 7200  
 (5) 4700
6. A starts a cement trading business by investing Rs. 5 lakhs. After 2 months, B joins the business by investing Rs. 10 lakhs and then 4 months after B joined C too joins them by investing Rs. 20 lakhs. 1 year after A started the business they make Rs. 3,50,000 in profit. What is B's share of the profit (in Rs.) ?  
 (1) 75000 (2) 1,25,000  
 (3) 1,50,000 (4) 1,00,000
7. A and B started a business together by investing Rs. 30,000 and Rs. 42,000 respectively. After 6 months, C joined them by investing a fourth of their capital. They earned an annual profit of Rs. 39,200 out of which C's share is Rs. 8,000. Find out C's investment ?  
 (1) Rs. 30 thousand (2) Rs. 17 thousand  
 (3) Rs. 31 thousand (4) Rs. 3 thousand  
 (5) None of these
8. A started a business. After 4 months from the start of the business, B and C joined. The respective ratio between the investments of A, B and C was 4 : 6 : 5. If A's share in an-

17. A and B started a business together by investing Rs. 30,000 and Rs. 42,000 respectively. After 6 months, C joined them by investing a fourth of their capital. They earned an annual profit of Rs. 39,200 out of which C's share is Rs. 8,000. Find out C's investment ?  
 (1) Rs. 30 thousand (2) Rs. 17 thousand  
 (3) Rs. 31 thousand (4) Rs. 3 thousand  
 (5) None of these
18. A started a business. After 4 months from the start of the business, B and C joined. The respective ratio between the investments of A, B and C was 4 : 6 : 5. If A's share in an-

22. Gita invests ₹ 48,000 to start a business. Four months later Shyam joins her by investing ₹ 62,000 and another two months later Deepika joins them both by investing ₹ 80,000. At the end of one year the business earns a profit of ₹ 20,000. What is Deepika's share in the profit ?  
 (1) ₹ 7,000 (2) ₹ 6,000  
 (3) ₹ 7,200 (4) ₹ 6,300  
 (5) None of these
23. A and B started a partnership business investing some amount in the ratio of 3 : 5. C joined them after six months with an amount equal to that of B; profit at the end of one year should be distributed in what proportion among A, B and C?  
 (1) 5 : 16 : 7



# PARTNERSHIP

three years, they earned a profit of Rs. 71,20,000. What is Raksha's share in the profit?

- (1) Rs. 38,20,000
- (2) Rs. 26,40,000
- (3) Rs. 27,00,000
- (4) Rs. 38,40,000
- (5) None of these

28. A and B both start a small business with an investment of Rs. 3500 and Rs. 4000 respectively. At the end of few months from the start of the business, A withdrew from the business completely and B remained for the years. If the annual profit was divided between A and B in the respective ratio of 7 : 12, then after how many months from the start of the business?

C joined and the respective ratio between the investments of B and C was 5 : 3. If the annual profit earned by them was Rs. 48,000, what was A's share in the profit?

- (1) Rs. 9,200
- (2) Rs. 12,800
- (3) Rs. 10,800
- (4) Rs. 8,400
- (5) Rs. 11,600

34. A started a business by investing Rs. 25000. At the end of 4th month from the start of the business B joined with Rs. 15000 and at the end of 6th month from the start of the business, C joined with Rs. 20000. If A's share in profit at the end of one year was Rs. 7,750, what was the total profit received?

- (1) Rs. 12,050
- (2) Rs. 13,810

# PARTNERSHIP

3. A started a business. After 4 months from the start of the business, B and C joined. The respective ratio between the amounts invested by A, B and C was 6 : 11 : 12. If the C's share in annual profit was Rs. 720 more than A's share, what was the total annual profit earned?

- (1) Rs. 7080
- (2) Rs. 7590
- (3) Rs. 7080
- (4) Rs. 8940

(5) None of these

4. A and B entered into a partnership by investing some amounts. The investment of A is twice of the investment of B. Another person C joined them after 4 months. At the end of a year, the profit shares of A and C are

The respective ratio between the investments of A, B and C was 3 : 7 : 4. If B share in annual profit was Rs. 2000 more than that of A. What was the total amount of profit earned?

- (1) Rs. 10000
- (2) Rs. 20000
- (3) Rs. 14000
- (4) Rs. 15000
- (5) Rs. 16000

9. Tom started a business with Rs. 52000 and after 4 months Harry joined him with Rs. 39000. At the end of the year out of the total profits Harry received total Rs. 20000 including  $\frac{1}{4}$ th of the profits as commission for managing.



## PARTNERSHIP

13. A starts a business with Rs. 40,000 capital. After 2 months, B joined him with Rs. 60,000 capital. C joined with them after some more time with Rs. 1,20,000 capital. At the end of the year, out of a total profit of Rs. 3,75,000, C gets Rs. 1,50,000 as his share. After how many months, did B joining the business did the partner C join?

- (1) 4 months  
(2) 6 months  
(3) 3 months  
(4) 5 months  
(5) 2 months

14. A, B and C have a certain amount of money with themselves. C has  $\frac{3}{4}$  of what A has and B has Rs. 50 less than

than who invests Rs. 30,000. At the end of the year they divide the profit in the ratio 5 : 4. After how many months did Mohan join Ravi?

- (1) 4 (2) 5 (3) 6 (4) 2  
(5) 7

19. A, B and C invest in a business in the ratio 4 : 5 : 7. C is a sleeping partner, so his share of profit will be half of what it would have been if he were a working partner. If they make Rs. 36,000 as annual profit of which 25% is to be invested in the business, how much does B get (in Rs.)?

- (1) 7560 (2) 10800 (3) 8640 (4) 9200  
(5) 10,000

## PARTNERSHIP

23. Minad, Vikas and Manav enter into a partnership. Minad invests some amount at the beginning. Vikas invests double the amount after 6 months and Manav invests thrice the amount invested by Minad after 8 months. They earn a profit of ₹ 45,000 at the end of the year. What is Manav's share in the profit?

- (1) ₹ 25,000 (2) ₹ 15,000  
(3) ₹ 12,000 (4) ₹ 9,000

(5) None of these

24. Radhika and Saugveda enter into partnership with ₹ 10,000 and ₹ 8000 respectively. After

1. of the time, Radhika contributes

3. ₹ 770 have been divided among A, B and C such that A receives  $\frac{2}{9}$ th of what B and C together receive. Then, A's share is

- (1) ₹ 140 (2) ₹ 154 (3) ₹ 165 (4) ₹ 170  
(5) None of these

4. Three persons A, B and C together start a business by investing capitals in the ratio 8 : 5 : 7 respectively. After 3 months A withdraws 25% of his investment. One month later B makes additional investment of 20% of his initial amount. Further, 2 months later,

C increases his investment by  $\frac{1}{10}$ th of his

8. A started a business. After 4 months from the start of the business, B and C joined. The respective ratio between the amounts invested by A, B and C was 6 : 11 : 12. If the C's share in annual profit was Rs. 720 more than A's share, what was the total annual profit earned?
- (1) Rs. 7680 (2) Rs. 7580  
(3) Rs. 7860 (4) Rs. 8680
- (5) None of these
9. Priya, Riya & Supriya invested in the ratio of 2 : 3 : 5 in a business. After 3 months Priya invested additional  $33\frac{1}{3}\%$  of her original investment. If the total profit at the end of the year was Rs. 1,50,000, what is the share of Riya in the profit?
10. A and B started a business with investments of Rs. 1,00,000 and Rs. 1,50,000 respectively. After 6 months, C joined them with an investment of Rs. 2,00,000. If the total profit at the end of the year was Rs. 4,50,000, what is the share of C in the profit?
11. A and B started a business with investments of Rs. 1,00,000 and Rs. 1,50,000 respectively. After 6 months, C joined them with an investment of Rs. 2,00,000. If the total profit at the end of the year was Rs. 4,50,000, what is the share of C in the profit?

12. A and B started a business with investments of Rs. 1,00,000 and Rs. 1,50,000 respectively. After 6 months, C joined them with an investment of Rs. 2,00,000. If the total profit at the end of the year was Rs. 4,50,000, what is the share of C in the profit?
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16. A and B started a business with investments of Rs. 1,00,000 and Rs. 1,50,000 respectively. After 6 months, C joined them with an investment of Rs. 2,00,000. If the total profit at the end of the year was Rs. 4,50,000, what is the share of C in the profit?
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20. A and B started a business with investments of Rs. 1,00,000 and Rs. 1,50,000 respectively. After 6 months, C joined them with an investment of Rs. 2,00,000. If the total profit at the end of the year was Rs. 4,50,000, what is the share of C in the profit?

## SHORT ANSWERS

## LEVEL-I

|         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|
| 1. (1)  | 2. (4)  | 3. (3)  | 4. (1)  | 5. (2)  | 6. (2)  |
| 7. (3)  | 8. (1)  | 9. (3)  | 10. (1) | 11. (1) | 12. (2) |
| 13. (1) | 14. (2) | 15. (5) | 16. (3) | 17. (1) | 18. (3) |
| 19. (1) | 20. (5) | 21. (3) | 22. (4) | 23. (1) | 24. (2) |
| 25. (2) | 26. (4) | 27. (1) | 28. (1) | 29. (2) | 30. (4) |
| 31. (1) | 32. (4) | 33. (3) | 34. (1) | 35. (3) |         |

B's capital is twice C's capital

$$\frac{B}{C} = 2$$

$$\frac{B}{C} = 2$$

$$B : C = 2 : 1$$

$$\therefore A : B : C = 4 : 2 : 3 \times 2 = 3 : 1$$

$$= 8 : 6 : 3$$

3. (3) Ratio of the equivalent capitals of A and B for 1 month

$$= (4000 \times 8 + 6000 \times 4) : (5000 \times 9 + 3000 \times 3)$$

$$= (32000 + 24000) : (45000 + 9000)$$

$$= 56000 : 54000$$

$$= 28 : 27$$

Sum of the terms of ratio =  $28 + 27 = 55$

A is an active partner.

Allowance got by A in 1 year = Rs. 1200

Remaining profit

$$= \text{Rs. } (6700 - 1200) = \text{Rs. } 5500$$

$$\therefore \text{B's share} = \text{Rs. } \left( \frac{27}{55} \times 5500 \right)$$

$$= \text{Rs. } 2700$$

4. (1) Ratio of equivalent capitals of A and B for 1 month

$$= (4 \times 12) : (5 \times 10)$$

$$= 24 : 25$$

Sum of the terms of ratio =  $24 + 25 = 49$

$$\therefore \text{B's share} = \text{Rs. } \left( \frac{25}{49} \times 49000 \right)$$

$$= 3 : 5 : 6$$

Sum of the terms of ratio =  $3 + 5 + 6 = 14$

$$\therefore \text{B's share} = \text{Rs. } \left( \frac{5}{14} \times 250000 \right) = \text{Rs. } 125000$$

7. (3) Ratio of shares =  $3 : 7$

Amount distributed as profit

$$= \text{Rs. } \left( \frac{60000 \times 60}{100} \right) = \text{Rs. } 36000$$

$$\therefore \text{A's share} = \text{Rs. } \left( \frac{3}{10} \times 36000 \right)$$

$$= \text{Rs. } 10800$$

8. (1) Ratio of the investments of A, B and C =  $2 : 3 : 5$

Ratio of the periods of their investments =  $4 : 2 : 3$

$\therefore$  Ratio of shares

$$= 2 \times 4 : 3 \times 2 : 5 \times 3$$

$$= 8 : 6 : 15$$

According to the question,

$$\therefore (15 - 8) = \text{Rs. } 147000$$

$$\therefore 6 = \text{Rs. } \left( \frac{147000 \times 6}{7} \right) = \text{Rs. } 126000$$

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$$\therefore \text{X's total investment} = 3x \times 5 + 13x \times \frac{3x}{2} \times 7$$

$$= \frac{1}{2} \left( 60x + \frac{273x^2}{2} \right)$$

$$= \frac{63x}{2}$$

$$= \frac{15x^2}{2}$$

$$= \frac{93x}{2}$$

$$= \frac{93x}{2}$$

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$$= \frac{93x}{2}$$

$$\therefore \text{Y's total investment} = 5x \times 5 + (5x - 5x \times \frac{3}{5}) \times 7$$

$$= 25x + 14x = \text{Rs. } 39x$$

$$\therefore \text{X's share} = \frac{93x}{2} \times \frac{3}{5} = \text{Rs. } 6.84$$

$$= \frac{93x}{2} \times \frac{3}{5} = \text{Rs. } 6.84$$

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$$= \frac{93x}{2} \times \frac{3}{5} = \text{Rs. } 6.84$$

$$\therefore \text{C's share} = \frac{x}{78} \times \frac{x}{15000}$$

$$= \frac{15000x}{78 \times x} = 7600x$$

$$= 15000x - 7600x = 7400x$$

$$= 15000x - 7600x = 7400x$$

$$= 15000x - 7600x = 7400x$$

$$= 15000x - 7600x = 7400x$$

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$$= 15000x - 7600x = 7400x$$

# PARTNERSHIP

17. (1) A and B's equivalent capital for 1 month

$$= \text{Rs. } (12 \times 36 + 12 \times 42) \text{ thousands}$$

$$= \text{Rs. } (432 + 504) \text{ thousand}$$

$$= \text{Rs. } 936 \text{ thousand}$$

Let C's investment be Rs.  $x$  thousand

$$\text{A and B's share}$$

$$= 39200 - 8000 = \text{Rs. } 31200$$

$$\therefore \frac{936}{9x} = \frac{31200}{8000}$$

$$\therefore \frac{104}{x} = \frac{312}{80} = x = \frac{80 \times 104}{312}$$

$$= \text{Rs. } \frac{80}{3} \text{ thousand}$$

18. (3) Ratio of the equivalent capitals of A, B and

C for 1 month

$$= (12 \times 4) : (6 \times 8) : (5 \times 8)$$

$$= 48 : 48 : 40 = 6 : 6 : 5$$

$$\text{Sum of ratios} = 6 + 6 + 5 = 17$$

If total annual profit be Rs.  $x$ , then

A's

$$= 50 = 1500 \times 9$$

$$\therefore 9 = \frac{1500 \times 9}{5}$$

$$= \text{Rs. } 2700$$

$$\therefore X + Y = \text{Rs. } (1800 + 2700)$$

$$= \text{Rs. } 4500$$

21. (3) Initial investments of A, B and C respectively = Rs. 125, Rs. 750 and Rs. 250 (Rs.)

$$\text{Ratio of their equivalent capitals for 1 month}$$

$$= (125 \times 8 + 500 \times 4) : (750 \times 12) : (250 \times 8 + 125 \times 4)$$

$$= (1000 + 2000) : (9000) : (2000 + 500) = 3000 : 9000 : 2500$$

$$= 30 : 90 : 25 = 6 : 18 : 5$$

$$\text{Sum of ratios} = 6 + 18 + 5 = 29$$

$$\therefore \text{B's share} = \frac{18}{29} \times 5800$$

$$= \text{Rs. } 3600$$

# PARTNERSHIP

Clearly, the profit will be shared equally.

$$\therefore \text{Munav's share} = \text{Rs. } \frac{1}{3} \times 45000 = \text{Rs. } 15000$$

25. (2) Ratio of the equivalent capitals of Mrudul and Shubaka for one month =  $295000 \times 24$  :

$$33500 \times 20 = 354 : 335$$

$$\text{Sum of the ratios} = 354 + 335$$

$$= 689$$

$$\therefore \text{Mrudul's share}$$

$$= \text{Rs. } \left( \frac{354}{689} \times 120575 \right)$$

$$= \text{Rs. } 61950$$

26. (4) Share of profit = Geeta : Shweta : Sita =

$$4200 : 8400 : 5400 = 7 : 14 : 9$$

$$\text{Sita's share}$$

$$= \frac{9}{7+14+9} \times 24000$$

$$= \frac{9}{30} \times 24000 = \text{Rs. } 7200$$

remaining amount after investing 32%

$$= \text{Rs. } \frac{1000}{9}$$

27. Ratio of Menka's, Sonu's and Raghu's investments

$$= 100 : 110 : \frac{1000}{9} = 90 : 99 : 100$$

$$\text{Sum of ratios} = 90 + 99 + 100 = 289$$

$$\therefore \text{Raghu's investment}$$

$$= \text{Rs. } \left( \frac{100}{289} \times 5799 \right) = \text{Rs. } 2000$$

30. (4) B's investment = Rs.  $x$

$$\therefore 12000 \times 12 = 5x$$

$$\Rightarrow x = \frac{12000 \times 12}{5} = \text{Rs. } 28800$$

31. (1) Ratio of equivalent capitals for 1 month =

$$(18000 \times 12) : (24000 \times 8) : (30000 \times 6) = 18 : 12 : (24 \times 6) = 18 : 16 : 15$$

$$\text{Sum of ratios} = 18 + 16 + 15 = 49$$



# PARTNERSHIP

34. (1) Ratio of the equivalent capitals of A, B and C for 1 month

$$= 12 \times 25000 : 8 \times 15000 : 6 \times 20000$$

$$= 12 \times 25 : 8 \times 15 : 6 \times 20$$

$$= 5 : 2 : 2$$

Sum of the terms of ratio

$$= 5 + 2 + 2 = 9$$

If the total profit at the end of year be Rs. x, then

A's share = Rs. 7750

$$= \frac{5x}{9} = 7750$$

$$\Rightarrow x = \frac{7750 \times 9}{5} = \text{Rs. } 13950$$

35. (3) Ratio of the shares of A, B and C = Ratio of the equivalent capitals of A, B and C for 1 month

$$= 4000 \times 12 : 4500 \times 6 : 6000 \times 3$$

$$= 40 \times 12 : 45 \times 6 : 60 \times 3$$

$$= 16 : 9 : 6$$

Sum of the terms of ratio

$$= 16 + 9 + 6 = 31$$

If total annual profit be Rs. x, then

$$\Rightarrow x = \frac{7750 \times 9}{5} = \text{Rs. } 13950$$

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$$= \frac{20000(1.21 - 1)}{20000 \times 0.21}$$

$$= \text{Rs. } 4200$$

2. (3) Ratio of equivalent capitals of A, B and C for 1 month

$$= 43500 \times 12 : 25000 \times 12 : 60000 \times 8$$

$$= 35 \times 12 : 25 \times 12 : 60 \times 8$$

$$= 7 : 5 : 8$$

Sum of the terms of the ratio = 7 + 5 + 8 = 20

According to the question,

If the total annual profit be Rs. x, then

difference of shares of C and B = Rs. 1977

$$\Rightarrow \left( \frac{8}{20} - \frac{5}{20} \right) x = 1977$$

$$\Rightarrow \frac{3x}{20} = 1977$$

$$\Rightarrow 3x = 1977 \times 20$$

$$\Rightarrow x = \frac{1977 \times 20}{3} = \text{Rs. } 13180$$

3. (1) Ratio of the equivalent capitals of A, B and C for 1 month

$$= 6 \times 12 : 11 \times 8 : 10 \times 6$$

# PARTNERSHIP

9. (2) Ratio of the equivalent capitals of A, B and C for 1 month

$$= (10000 \times 6 + 20000 \times 6) : (12000 \times 6) : (18000 \times 6)$$

$$= 100 : 120 : 180$$

$$= 5 : 6 : 9$$

Sum of the terms of ratio

$$= 5 + 6 + 9 = 20$$

If the total annual profit be Rs. x, then

According to the question,

difference between the shares of A and B = Rs. 6000

$$\Rightarrow \left( \frac{5}{20} - \frac{6}{20} \right) x = 6000$$

$$\Rightarrow \frac{-x}{4} = 6000$$

$$\Rightarrow x = -24000$$

$$\Rightarrow x = 24000$$

$$\Rightarrow x = 24000$$

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# PARTNERSHIP

$$\begin{aligned}
 &= 1350 : x : x \\
 &\text{Sum of the terms of ratio} \\
 &= 2x + 1350 \\
 &\therefore \text{A's share} \\
 &= \frac{1350}{2x + 1350} \times 14000 \\
 &= 5040 = \frac{1350 \times 14000}{2x + 1350} \\
 &= 2x + 1350 = \frac{1350 \times 14000}{5040} \\
 &= 3750 \\
 &= 2x = 3750 - 1350 = 2400 \\
 &\therefore \text{C's investment} = \text{Rs. } 2400
 \end{aligned}$$

13. (1) Let C remain in the business for  $x$  months

Ratio of shares

$$= (40000 \times 12) : (60000 \times 10) : (120000 \times x) = 4 : 5 : x$$

C's share  $\rightarrow$

$$\left( \frac{375000x}{9+x} \right) = 1,50,000$$

$$\left( \frac{375x}{9+x} \right) = 150$$

$$\left( \frac{2}{3} + \frac{1}{6} \right) = 10260$$

$$\Rightarrow \frac{1}{2} = 10260$$

$$\Rightarrow \frac{1}{6} = \frac{10260 \times 2}{6} = \text{Rs. } 3420$$

16. (3) A's investment = Rs.  $x$

$\therefore$  B's investment = Rs.  $2x$

C's investment = Rs.  $2.5x$

$\therefore$  Ratio of their shares

$$= 4x : 8x : 12x = 2.5x : 4 : 12 : 30 = 2 : 6 : 15$$

Sum of the terms of ratio =  $2 + 6 + 15 = 23$

$$\therefore \text{B's share} = \text{Rs. } \left( \frac{6}{23} \times 5819 \right) = \text{Rs. } 1518$$

Ratio of the new investment of A be Rs.  $x$

Ratio of the equivalent capitals of A and B for 1 month

$$= (2000 \times 2) + (2000 \times x) : 2 : 4000 \times 12$$

$$= 4000 + 4000x : 24000$$

$$= 4000 + x : 24000$$

# PARTNERSHIP

10. (3) Ratio of the shares of A, B and C = Ratio of the equivalent capitals of A, B and C for 1 month

$$= x \times 12 : x \times 8 : 3.5 \times 8$$

Where A's investment = Rs.  $x$  lakhs

$$= 3x : 2x : 7$$

Sum of the terms of ratio

$$= 3x + 2x + 7 = 5x + 7$$

$$\therefore \text{A's share} = \frac{3x}{5x+7} \times 62400$$

$$\Rightarrow \frac{3x}{5x+7} \times 62400 = 24000$$

$$\Rightarrow \frac{3x \times 26}{5x+7} = 10$$

$$\Rightarrow 78x = 50x + 70$$

$$\Rightarrow 28x = 70$$

$$\Rightarrow x = \frac{70}{28} = \text{Rs. } 2.5 \text{ lakhs}$$

$$\therefore \text{C's share} = \frac{7}{5x+7} \times 62400$$

$$\therefore \text{Wife's share} = \frac{2}{5} \times 55 = \text{Rs. } 2.2 \text{ lakhs}$$

Share of three grandchildren

$$= 3 \times 0.125 = \text{Rs. } 0.375 \text{ lakh}$$

$\therefore$  Required answer

$$= \text{Rs. } (2.2 + 0.375) \text{ lakh}$$

$$= \text{Rs. } 2.575 \text{ lakhs}$$

$$= \text{Rs. } 257500$$

23. (2) Let Nindad invests ₹  $x$ .

$\therefore$  Vikas's investment = ₹  $2x$

Manav's investment = ₹  $3x$

Ratio of the equivalent capitals for 1 year, 6 months and 4 months of Nindad, Vikas and Manav respectively

$$= x \times 12 : 2x \times 6 : 4 \times 3x = 1 : 1 : 1$$

Clearly, the profit will be shared equally.

$$\therefore \text{Manav's share} = \text{Rs. } \frac{1}{3} \times 45000 = ₹ 15000$$

24. (2) Ratio of profit distribution =

Radhika : Sandhya : Vikas

Arjun : Anam : Suman : Geeta  
= 8 : 15 : 24 : 30  
Part of rent paid by Suman

$$= \frac{24}{8+15+24+30} \times 77$$

2. (4) According to question,

$$\frac{A}{B} = \frac{3 \times 2}{4 \times 3} = \frac{9}{12} \quad \frac{B}{C} = \frac{3 \times 4}{4 \times 4} = \frac{12}{16}$$

$$\therefore A : B : C = 9 : 12 : 16$$

$$\text{Sum of Ratio} = 9 + 12 + 16 = 37$$

$$\therefore A's \text{ share} = \frac{9}{37} \times 370 = ₹ 90$$

3. (1) According to question,

$$A+B+C = ₹ 770$$

$$A = \frac{2}{9}(B+C)$$

From Eqs. (i) and (ii), we get

$$A + \frac{2}{9}A = ₹ 770$$

$$\Rightarrow 11A = ₹ 770 \times 2$$

$$\Rightarrow A = ₹ 140$$

Thus, A's share is ₹ 140

7. (2) N's investment = ₹ 17000, 31

$$\left( \frac{7000 \times \frac{3}{4} \times 3 \right) + \left( \frac{5000 \times 2000 \times 3}{N} \right) = 6 = ₹ 7320$$

$$Y's \text{ Investment} = 6000 \times 12 = ₹ 7200$$

$$X's \text{ share from profit}$$

$$= \frac{7200}{(7200 + 7200)} \times 720 = ₹ 360$$

8. (1) Ratio of the equivalent capitals of A, B and

C for 1 month

$$= 6x \times 12 : 11x \times 8 : 12x \times 8$$

$$= 9 : 11 : 12$$

$$\text{Sum of the terms of ratio} = 9 + 11 + 12 = 32$$

If total annual profit be Rs. y, then

$$\frac{12y}{32} - \frac{9y}{32} = 720 \Rightarrow \frac{3y}{32} = 720$$

$$\Rightarrow y = \frac{720 \times 32}{3} = ₹ 7680$$

9. (1) Let the initial investment of Priya, Riya and Supriya be 2x, 3x and 5x respectively.

Ratio of their investment at end of year would be  $\rightarrow$  Investment of Priya = 2x for 3 months

## QUANTITY BASED EXPLANATION

1. (1) **Quantity I :** Ratio in which profit should be divided

$$= 15000 (12) : 10000 (7)$$

$$150 : 70 = 15 : 7$$

$$\text{Minal's share} = \frac{7}{(15+7)} \times 5000 = ₹ 1400$$

**Quantity II :** Let amount invested by Arjun

$$= x, \text{ then Karan} = x - 5000 \text{ and Tarun}$$

$$= x + 5000$$

$$x + (x - 5000) + (x + 5000) = 45000$$

$$x = 15000$$

Ratio of investment of Karan, Tarun and Arjun

$$= 10000 : 20000 : 15000 = 2 : 4 : 3$$

$$\text{Arjun's share of profit} = \frac{3}{9} \times 10900 = ₹ 3666$$

**Quantity I < Quantity II**

2. (1) **Quantity I :** Let Ninal invests ₹ x.

$$\therefore \text{Vikas's investment} = ₹ 2x$$

$$\text{Manav's investment} = ₹ 3x$$

Ratio of the

$$\text{It's share} = \frac{2}{7} \times \frac{8x}{5} = ₹ \frac{8x}{35}$$

$\therefore$  Difference

$$= \frac{27x}{35} - \frac{8x}{35} = \frac{19x}{35}$$

$$\therefore \frac{19x}{35} = 36000$$

$$\Rightarrow x = \frac{36000 \times 35}{19}$$

$$= ₹ 70000$$

**Quantity II :** Son : wife = 3 : 1 = 9 : 3

Wife : daughter = 3 : 1

$$\therefore \text{Son : wife : daughter} = 9 : 3 : 1$$

$$\text{Sum of ratios} = 9 + 3 + 1 = 13$$

If total wealth be Rs. x, then

Son's share = daughter's share

$$= ₹ 10,000$$

$$\Rightarrow \frac{9x}{13} - \frac{x}{13} = 10,000$$

$$\therefore \frac{7x}{26} = 87500$$

$$\Rightarrow x = \frac{87500 \times 26}{7} = \text{Rs. } 325000$$

5. (1) Quantity 1 < Quantity II  
 Expenditure = Rs. 8x  
 Savings = 20%

$$\therefore \text{A's income} = \frac{8x \times 100}{20}$$

$$= \text{Rs. } 40x$$

$$\text{Similarly, B's savings} = \text{Rs. } 9x$$

$$\therefore \text{B's income} = \frac{9x \times 100}{15} = \text{Rs. } 60x$$

$$\text{C's savings} = \text{Rs. } 20x$$

$$\therefore \text{C's income} = \frac{20x \times 100}{25} = \text{Rs. } 80x$$

$$= \frac{23,000 \times 31,500}{53,000}$$

$$= 6200 \times 23$$

$$\text{Share of A} = \text{Rs. } 13,800$$

$$\text{Quantity I} > \text{Quantity II}$$

#### DATA SUFFICIENCY EXPLANATION

1. (1) Statement I alone is sufficient. We only want to know the ratio of investments and not actual investment.

$$\text{Ratio of Profit} = 7 : 8$$

Let investment made by A and B and 'a' and 'b' respectively. While A invested for 12 months, B invested for 12 - 4 = 8 months. Hence, Ratio of Profit 7 : 8 = 12a : 8b

$$\frac{7}{8} = \frac{12a}{8b} \Rightarrow \frac{a}{b} = \frac{7}{12}$$

# 10 Time and Work

## IMPORTANT FACTS AND FORMULAE

Certain general rules are given below. The candidates will find them quite useful while solving problems related to Time and Work.

1. While solving problems related to Time and Work we have to compare the work of several working agents (men, women and children). To solve this purpose, we need to find out the amount of work done by each agent in a common unit time. This is called as 'work done'.

complete the work = 1 : 3 i.e. A will take  $\frac{1}{3}$  of the time taken by B to complete that work.

9. The time required to complete a certain piece of work is increased (or decreased) in the same ratio in which the number of men engaged to do that piece of work is decreased (or increased).



## PIPES AND CISTERN

The method of solving the questions on pipes and cistern is almost same as that of solving the questions of Time and Work, but with a bit difference as negative work comes into play due to leakage of pipes and cistern.

**Important Points :**

**Inlet :** A pipe connected with a cistern or a tank or a reservoir that fills it is known as inlet.

**Outlet :** A pipe connected with a tank or a cistern or a reservoir to empty it, is known as an outlet.

**Important Formulae,**

1. If a pipe can fill a tank in  $x$  hours then part of tank filled in 1 hour =  $\frac{1}{x}$
2. If a pipe can empty a full tank in  $y$  hours, then part of tank emptied in 1 hour =  $\frac{1}{y}$
3. If a pipe can fill a tank in  $x$  hours and another pipe can empty the full tank in  $y$  hours (where  $y > x$ ), then on opening both the pipes at the same time, the net part filled in 1 hour =  $\left(\frac{1}{x} - \frac{1}{y}\right)$

## MODEL QUESTIONS

10. B and C together can complete a work in 8 days. A and B together can complete the same work in 12 days and A and C together can complete the same work in 16 days. In how many days can A, B and C together complete the same work ?  
 (1)  $3\frac{9}{13}$  (2)  $7\frac{5}{13}$  (3)  $7\frac{5}{12}$  (4)  $3\frac{5}{12}$   
 (5) None of these
11. 24 men can complete a piece of work in 18 days while 12 women can complete the same piece of work in 28 days. 27 men start working and are replaced by 14 women after 8 days. In how many days will 14 women finish the remaining work ?  
 (1) 12 days (2) 14 days

pleted in 18 days. In how many days the whole project was completed?

- (1)  $24\frac{1}{2}$  (2) 25 (3) 22 (4)  $21\frac{1}{2}$   
 (5) 20

16. A project manager hired 16 men to complete a project in 38 days. However, after 30 days,

he realized that only  $\frac{5}{9}$  of the work is complete. How many more men does he need to hire to complete the project on time ?  
 (1) 48 (2) 24 (3) 32 (4) 16  
 (5) 36

17. 8 men and 4 women together can complete a piece of work in 6 days. Work done by a man

# TIME AND WORK

- of the work together. Amount shall be paid to B as  
 (1) ₹ 210 (2) ₹ 100 (3) ₹ 200 (4) ₹ 475  
 (5) ₹ 225
22. A, B and C can do a job in 6 days, 12 days and 15 days respectively. After  $\frac{1}{5}$  of the work is completed, C leaves the job. Rest of the work is done by A and B together. Time taken to finish the work is  
 (1)  $5\frac{5}{6}$  days (2)  $5\frac{1}{4}$  days  
 (3)  $3\frac{1}{2}$  days (4)  $3\frac{3}{4}$  days  
 (5)  $3\frac{1}{3}$  days
28. A can do as much work in 4 days as B can do in 5 days, and B can do as much work in 6 days as C in 7 days. In what time will C do a piece of work which A can do in a week?  
 (1)  $10\frac{5}{24}$  days (2)  $4\frac{4}{5}$  days  
 (3)  $6\frac{8}{15}$  days (4)  $12\frac{6}{19}$  days  
 (5)  $4\frac{9}{13}$  days
29. A man does double the work done by a boy in the same time. The number of days that 3 men and 4 boys will take to finish a work which can be done by 10 men in 8 days is  
 (1) 8 (2) 7 (3)  $7\frac{3}{4}$  (4)  $8\frac{1}{2}$

32. The time taken by 4 men to complete a job is double the time taken by 5 children to complete the same job. Each man is twice as fast as a woman. How long will 12 men, 10 children and 8 women take to complete a job, given that a child would finish the job in 20 days?  
 (1) 4 days (2) 3 days (3) 2 days (4) 1 day  
 (5) 5 days
34. The labourers A, B, C were given a contract of Rs. 750 for doing a certain piece of work. All the three together can finish the work in 8 days. A and C together can do it in 12 days, while A and B together can do it in  $13\frac{1}{3}$  days. The money will be divided in the ratio.  
 (1) 4 : 5 : 6 (2) 4 : 7 : 5  
 (3) 5 : 7 : 4 (4) 5 : 6 : 8

40. C is 5 times as productive as B. A takes 60 days to complete a task. If A, B and C work together they can complete the task in 12 days. In how many days can B complete the task if he worked alone?  
 (1) 18 (2) 27 (3) 90 (4) 72  
 (5) 60
41. A can complete 50% of a job in 9 days and B can complete 25% of the job in 9 days if they worked alone. If they worked together how much of the job in  $\frac{3}{4}$  can they complete in 9 days?  
 (1) 80 (2) 90 (3) 75 (4) 100  
 (5) 25
42. A can do  $\frac{4}{5}$  of a work in 12 days, whereas B can do  $\frac{5}{6}$  of the same work in 15 days. Both work together for 10 days. C completes

48. Three men, four women and six children can complete a work in 7 days. A woman does double the work a man does and a child does half the work a man does. How many women alone can complete this work in 7 days?
- (1) 8 (2) 7 (3) 12 (4) Cannot be determined (5) None of these
49. A man, a woman and a boy together complete a piece of work in 3 days. If a man alone can do it in 6 days and a boy alone in 18 days, how long will a woman take to complete the work?
- (1) 9 days (2) 21 days (3) 24 days (4) 27 days (5) None of these
50. Anil and Sujit together can complete an assignment of data entry in 5 days. Sujit's speed is 80% of Anil's speed and the total key depressions

51. A man can complete a piece of work in 4 days, 12 women can complete the same piece of work in 4 days, whereas 6 children can complete the same piece of work in 8 days. 2 men, 8 children and 3 women work together for 2 days. If only women were to finish the remaining work in 2 days, how many total women would be required?
- (1) 12 (2) 18 (3) 24 (4) 20 (5) None of these
52. A project requires 12 women to complete it in 16 days. 12 women started working and after a few days from the start of the project, 4 women left. If the remaining project was completed in 18 days, in how many days the whole project was completed?
- (1)  $24\frac{1}{2}$  (2) 26 (3) 22 (4)  $21\frac{1}{2}$  (5) 20

53. Efficiency of A is two times more than efficiency of B. A and B started working alternatively. A started at 11 A.M. and B alone can complete the work in 17 days. In how many days will A and B together complete the work?
- (1) 21 days (2) 20 days (3) 24 days (4) 26 days (5) 28 days
54. 24 girls work for 4 days on a project and complete one-fourth of it. 30 boys work for the next  $x$  days and complete one-third of it. If after  $x + 4$  days, 24 girls and 30 boys work together and complete the remaining work in  $\frac{6}{5}$  days, how many days will 30 boys take to complete the whole work?
- (1) 21 (2) 20 (3) 18 (4) 22 (5) 23
55. Let day, then in which day work 2 will be completed?
- (1) 3rd day (2) 4th day (3) 5th day (4) 2nd day (5) 6th day
56. In how many hours will A and B together complete work 2?
- (1) 20 (2) 25 (3) 15 (4) 30 (5) 35
57. Work 1 and 2 together are assigned to labours B and C individually. By what percentage C is more efficient than B?
- (1) 30% (2) 45% (3) 35% (4) 44% (5) 32%
58. E takes work 2 and leaves after 2 days. If B continues on the remaining work 2, then in how many hours the remaining work 2 will be completed?



18. A does one-fifth of a work in a week. B finishes the same in a fortnight. B starts the work and works only for 3 days. Thereafter A completes the job. He will finish it in
- (1) 10 days                      (2) 7 days  
(3) 12 days                      (4) 28 days  
(5) 20 days
19. A can do a piece of work in 20 days and B in 30 days. They work together for 7 days and then both leave the work. Then C alone finishes the remaining work in 10 days. In how many days will C finish the full work?
- (1) 25 days                      (2) 30 days  
(3) 24 days                      (4) 20 days  
(5) 22 days
20. A and B can do a piece of work in 45 and 40

- 25.** A can do 20% of a work in 4 days, it can do  $\frac{33\frac{1}{3}}{3}$  % of the same work in 10 days. They worked together for 9 days, C completed the remaining work in 6 days B and C together will complete 75% of the same work in ;
- (1) 9 days                  (2) 15 days  
(3) 10 days                (4) 12 days  
(5) 11 days
- 26.** Two men and 7 women can complete a work in 28 days, whereas 6 men and 16 women can do the same work in 11 days. In how many days will 5 men and 4 women, working together, complete the same work ?
- (1) 20      (2) 18      (3) 14      (4) 22  
(5) 25

## TIME AND WORK

### LEVEL-III

1. The efficiency of Ajit is 25% more than that of Gaur and Gaur takes 25 days to complete a set of works. Ajit started a work alone and then Gaur joined him 5 days before actual completion of the work. How many days Ajit did work for alone in this process?
- (1) 9 days                      (2) 10 days  
(3) 11 days                    (4) 12 days  
(5) 13 days
2. Two coal loading machines each working 12 h per day for 8 days handle 9000 tonnes of coal with an efficiency of 100%. What is the efficiency of one machine?

6. 5 men can complete 5 road projects in 200 days. With the help of one machine they can complete one project in 15 days. In how many days, 20 men with 2 machines can complete 10 road projects?
- (1)  $5\frac{5}{11}$  (2)  $54\frac{5}{11}$  (3)  $9\frac{8}{11}$  (4)  $47\frac{5}{11}$
- (5)  $11\frac{4}{11}$
7. 15 men can complete a work in 210 days. They started the work but at the end of 10 days 15 additional men, with double efficiency, were introduced.



# TIME AND WORK

man from D, 1 man from R and 1 man from F will complete the work?

- (1) 18 days
- (2) 25 days
- (3) 28 days
- (4) 15 days
- (5) 24 days

12. A can do a piece of work in 10 days, while B alone can do it in 15 days. They work together for 5 days and the rest of the work is done by C in 2 days. If they get ₹ 450 for the whole work, how should they divide the money?

- (1) ₹ 225, ₹ 150, ₹ 75
- (2) ₹ 250, ₹ 100, ₹ 100
- (3) ₹ 200, ₹ 150, ₹ 100
- (4) ₹ 175, ₹ 175, ₹ 100
- (5) None of these

18. Two men x and y started working for a certain company at similar jobs on January 1, 1950. x asked for an initial salary of ₹ 300 with an annual increment of ₹ 30. y asked for an initial salary of ₹ 200 with a rise of ₹ 15 every six months. Assume that the arrangements remained unaltered till Dec, 31, 1959. Salary is paid on the last day of the month. What is the total amount paid to them as salary during the period?

in groups of two each. One job takes two third the time needed by the second pair to complete the work. Which is the first pair?

- (1) A, B
- (2) A, C
- (3) B, C
- (4) A, D
- (5) B, D

# TIME AND WORK

22. Ahmed, Anwar and Ali are three ironmen. Ahmed, Anwar and Ali can iron 117 shirts in 13 hours. Working together, Ahmed can iron as many less shirts than Anwar as Anwar can iron less than Ali. Also total shirts ironed by iron less than Ali. Ahmed in 2 hours of total shirts ironed by Ali in two hours. Find the no. of shirts ironed by Anwar in 3 hours.

- (1) 15
- (2) 9
- (3) 3
- (4) 12
- (5) 10

23. A and B together can complete a task in 20 days. B and C together can complete the same task in 30 days. A and C together can complete the same task in 40 days.

same time, they should fill the tank in what fraction of the time it would have taken the large pump alone?

- (1)  $\frac{4}{7}$
- (2)  $\frac{1}{3}$
- (3)  $\frac{2}{3}$
- (4)  $\frac{3}{4}$
- (5)  $\frac{1}{4}$

28. There is a leak in the bottom of the tank. This leak can empty a full tank in 8 hours. When the tank is full, a tap is opened into the tank which makes water at rate of 6 L per hour and the tank is filled in 12 hours.

(5) Quantity I = Quantity II or relation can't be established

**1. Quantity I :** A can do a work in 10 days, while B can do it in 15 days. Find the time required to finish the work if both work together.

**Quantity II :** A and B do a work in 10 days for which B and C take 20 days and A and C take 15 days. How much time would B alone take to do the same work?

**2. Quantity I :** 4 men or 10 women or 14 girls can do a job in 55 days. In how many days would 3 men and 5 women and 10 girls do the same job?

**Quantity II :** 4 men and 2 women can complete one-third of the work in 4 days, while 5 men and 4 women can complete the whole work in 8 days. In how many days

the question. Read the question and both the statements and—

**Give answer (1)** if the data in statement I alone is sufficient to answer the question while the data in statement II alone is not sufficient to answer the question.

**Give answer (2)** if the data in statement II alone is sufficient to answer the question, while the data in statement I alone is not sufficient to answer the question.

**Give answer (3)** if the data either in statement I alone or in statement II alone is sufficient to answer the question.

**Give answer (4)** if the data even in both the statements I and II together is not sufficient to answer the question.

- I. A is twice as efficient as B.  
II. A and B together can finish a piece of work in  $10\frac{2}{3}$  days. B and C together can finish the same piece of work in  $17\frac{7}{9}$  days. A and C together can finish the same piece of work in  $11\frac{3}{7}$  days.

## SHORT ANSWERS

## LEVEL-I

|        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|
| 1. (5) | 2. (1) | 3. (3) | 4. (4) | 5. (3) | 6. (1) |
|--------|--------|--------|--------|--------|--------|

## EXPLANATIONS

## LEVEL-I

|        |      |     |
|--------|------|-----|
| 1. (5) | Days | Men |
|        | 16   | 12  |
|        | 8    | x   |

Where  $x$  = number of men

$$\therefore 8 : 16 :: 12 : x$$

$$\Rightarrow 8 \times x = 16 \times 12$$

$$\Rightarrow x = \frac{16 \times 12}{8} = 24$$

|        |     |           |      |
|--------|-----|-----------|------|
| 2. (1) | Men | Hours/Day | Days |
|--------|-----|-----------|------|

$$\begin{aligned} \Rightarrow 2W &= 1M \\ \therefore 2M + 5W &= 6W \\ \therefore M_1 D_1 &= M_2 D_2 \\ \Rightarrow 9 \times 12 &= 3 \times D_2 \\ \Rightarrow D_2 &= \frac{9 \times 12}{3} = 36 \text{ days} \end{aligned}$$

7. (2) In 36 days 12 men can do 1 complete work.

In 36 days 12 women can do  $\frac{3}{4}$ th of the work  
Since time and the no. of persons is the same  
is both cases.

1 woman's daily work =  $\frac{3}{4}$ th of 1 man's daily  
work

8 women's daily work

$$= \frac{3}{4} \times 8 = 6 \text{ men's daily work}$$

(10 men + 8 women's daily work)

$$= (10 \text{ men} + 6 \text{ men})$$

= 16 men's daily work.

12 men can do the work in 36 days

$$\therefore 16 \text{ men can do the work in } 36 \times \frac{12}{16} = 27 \text{ days.}$$

8. (4)  $\therefore$  A completes  $\frac{6}{7}$  work in 12 days.

$\therefore$  Time taken by A in doing whole work =

$$\begin{aligned} (A+B)'s \text{ 1 day's work} &= \frac{1}{16} + \frac{1}{40} \\ \text{On adding all these three equations,} \\ 2(A+B+C)'s \text{ 1 day's work} &= \end{aligned}$$

$$= \frac{1}{8} + \frac{1}{12} + \frac{1}{16} = \frac{6+4+3}{48} = \frac{13}{48}$$

$$\Rightarrow (A+B+C)'s \text{ 1 day's work} = \frac{13}{96}$$

$\therefore$  A, B and C together can complete the work  
in  $\frac{96}{13} = 7\frac{5}{13}$  days

11. (1) 24 men complete 1 work in 18 days.

What part of work will 27 men do in 8 days?

$$\frac{M_1 D_1}{W_1} = \frac{M_2 D_2}{W_2}$$

$$\Rightarrow \frac{24 \times 18}{1} = \frac{27 \times 8}{W_2}$$

$$\Rightarrow W_2 = \frac{27 \times 8}{24 \times 18} = \frac{1}{2}$$

$$\text{Remaining work} = 1 - \frac{1}{2} = \frac{1}{2}$$

$$\Rightarrow 1 = \frac{9 \times 12 \times 2 \times 30}{2 \times 1 \times 1 \times 6} = \frac{36}{1} = 36$$

Required number of days = 36

13. (2) According to the question,

$$9 \times 360 \text{ children} = 18 \times 72 \text{ men} = 12 \times 162 \text{ women}$$

$\Rightarrow$  5 children = 2 men = 3 women

$\therefore$  4 men + 12 women + 10 children

= 10 children + 20 children + 10 children =

40 children

$$\therefore M_1 D_1 = M_2 D_2$$

$$\Rightarrow 9 \times 360 = 40 \times D_2$$

$$\Rightarrow D_2 = \frac{9 \times 360}{40} = 81 \text{ days.}$$

$$14. (5) \frac{M_1 D_1}{W_1} = \frac{M_2 D_2}{W_2}$$

$$\frac{6 \times 6}{W_1} = \frac{9 \times 10}{1}$$

$$\Rightarrow W_1 = \frac{6 \times 6}{9 \times 10} = \frac{2}{5}$$

$$16. (3) \text{Remaining work} = 1 - \frac{5}{9} = \frac{4}{9}$$

$$\frac{M_1 D_1}{W_1} = \frac{M_2 D_2}{W_2}$$

$$\Rightarrow \frac{16 \times 30}{5} = \frac{M_2 \times 8}{9}$$

$$\Rightarrow M_2 \times 8 \times 5 = 16 \times 30 \times 4$$

$$\Rightarrow M_2 = \frac{16 \times 30 \times 4}{8 \times 5} = 48$$

$\therefore$  Additional men

$$= 48 - 16 = 32$$

17. (1) According to the question,  
1 man = 2 women  
 $\therefore$  8 men = 4 women

$$= (16 + 4) \text{ women} = 20 \text{ women}$$

$$4 \text{ men} + 8 \text{ women} = 16 \text{ women}$$

20 women's 2 days' work

$$= \frac{2}{6} = \frac{1}{3} \text{ part}$$

$$\text{Remaining work} = 1 - \frac{1}{3} = \frac{2}{3}$$

$$(C + A)'s 1 \text{ day's work} = \frac{1}{40}$$

On adding,

$$2(A + B + C)'s 1 \text{ day's work}$$

$$= \frac{1}{20} + \frac{1}{30} + \frac{1}{40}$$

$$= \frac{6 + 4 + 3}{120} = \frac{13}{120}$$

$$\therefore (A + B + C)'s 1 \text{ day's work} = \frac{13}{240}$$

$\therefore A's 1 \text{ day's work}$

$$= \frac{13}{240} - \frac{1}{30} = \frac{13 - 8}{240}$$

$$= \frac{5}{240} = \frac{1}{48}$$

$C's 1 \text{ day's work}$

$$= \frac{13}{240} - \frac{1}{20} = \frac{13 - 12}{240}$$

$$= \frac{1}{240}$$

$$\therefore \text{Required ratio} = 48 : 240$$

$$= \frac{1}{6} + \frac{1}{12} + \frac{2 + 1}{12} = \frac{3}{12} = \frac{1}{4}$$

$\therefore$  Time taken in doing  $\frac{7}{8}$  part of work

$$= \frac{7}{8} \times 4 = \frac{7}{2} = 3\frac{1}{2} \text{ days}$$

23. (1) Number of pages copied by  $x$  in hour

$$\frac{80}{20} = 4$$

Number of pages copied by  $x$  and  $y$  in 1 hour

$$= \frac{135}{27} = 5$$

$\therefore$  Number of pages copied by  $y$  in 1 hour

$$= 5 - 4 = 1$$

$\therefore$  Required time = 20 hours.

24. (4) Earning in the first one hour =  $\frac{2000}{50}$

$$\text{Rs. } 40$$

Earnings for additional 5 hours

$$= 40 \times \frac{3}{2} \times 5 = \text{Rs. } 300$$

25. (1) 200 workers do  $\frac{1}{4}$  work in 50 days.

How many workers will do

$$= \frac{4 - 1}{8} = \frac{3}{8} \text{ parts}$$

$\therefore$  Time taken by George

$$= \frac{8 \times \frac{5}{8}}{3} = \frac{40}{3} \text{ hours}$$

Work done by all three in 1 hour =  $\frac{1}{6} + \frac{1}{8} +$

$$\frac{3}{40}$$

$$= \frac{20 + 15 + 9}{120} = \frac{44}{120}$$

$$= \frac{11}{30}$$

$$\therefore \text{Required time} = \frac{30}{11} = 2\frac{8}{11} \text{ hours}$$

31. (3)  $\therefore P$  does  $\frac{1}{4}$ th work in 10 days.

$$\therefore P \text{ will do 1 work in } 10 \times 4$$

$$= 40 \text{ days}$$

$$\therefore Q \text{ does } 40\% \text{ part of work in 40 days}$$

$$\therefore Q \text{ will do 100\% work in } \frac{40 \times 100}{40} = 100 \text{ days}$$

27. (3) 12 men + 1 woman's 14 days' work

= 4 women + 2 men's 8 days' work

$$\Rightarrow 28 \text{ men} + 14 \text{ women} = 32 \text{ women} + 16 \text{ men}$$

$$\Rightarrow (28 - 16) = 12 \text{ men} = (32 - 14) = 18 \text{ women}$$

$$\Rightarrow 2 \text{ men} = 3 \text{ women}$$

$$\therefore 1 \text{ woman} = \frac{2}{3} \text{ man}$$

$$\therefore \text{Wages per day of 1 man} = \text{Rs. } 180$$

$$\therefore \text{Wages per day of 1 woman}$$

$$= \frac{2}{3} \times 180 = \text{Rs. } 120$$

28. (1) A's 4 days' work = B's 5 days' work

$$\Rightarrow A : B = 4 : 5$$

Again,  $B : C = 6 : 7$

$$\therefore A : B : C = 4 \times 6 : 5 \times 6 : 5 \times 7$$

$$= 24 : 30 : 35$$

$$\therefore \text{Time taken by } A = 7 \text{ days}$$



# TIME AND WORK

$$\therefore 12 \text{ men} = \frac{5}{8} \times 12$$

$$= \frac{15}{2} \text{ children}$$

Again, 1 man = 2 women

$$\therefore 2 \text{ women} = \frac{5}{8} \text{ child}$$

$$\therefore 8 \text{ women} = \frac{5}{2} \text{ children}$$

$$\therefore 12 \text{ men} + 10 \text{ children} + 8 \text{ women} =$$

$$\left( \frac{15}{2} + 10 + \frac{5}{2} \right)$$

$$= 20 \text{ children}$$

$$\therefore 1 \text{ child does 1 work in 20 days}$$

$$\therefore \text{Time taken by 20 children} = \frac{20}{20} = 1 \text{ day.}$$

$$34. (i) (A + B + C)'s 1 \text{ day's work}$$

$$= \frac{1}{8}$$

$$(A + C)'s 1 \text{ day's work}$$

$$= \frac{1}{12}$$

$$\therefore B's 1 \text{ day's work} = \frac{1}{8} - \frac{1}{12}$$

$$35. (i) (A + B)'s 1 \text{ day's work}$$

$$= \frac{1}{12}$$

$$\therefore (A + B)'s 5 \text{ days' work} = \frac{5}{12}$$

$$\text{Remaining work} = 1 - \frac{5}{12} = \frac{7}{12}$$

$$\therefore A \text{ does } \frac{7}{12} \text{ work in 14 days}$$

$$\therefore A \text{ will do 1 work in}$$

$$= \frac{14 \times 12}{7} = 24 \text{ days}$$

$$36. (4) \text{Time taken by } C = x \text{ days}$$

$$\therefore \text{Time taken by } A = 3x \text{ days}$$

$$\therefore (A + B + C)'s 1 \text{ day's work}$$

$$= \frac{1}{7.5}$$

$$= \frac{1}{x} + \frac{1}{3x} + \frac{1}{15}$$

$$= \frac{10}{75} = \frac{2}{15}$$

$$= \frac{1}{x} + \frac{1}{3x} = \frac{2}{15} - \frac{1}{15} = \frac{1}{15}$$

# TIME AND WORK

$$\text{Time taken by A and C} = \frac{15}{\frac{1}{4} + \frac{1}{5}}$$

$$= 15 \text{ days.}$$

$$\therefore (A + B + C)'s 1 \text{ hour's work} = \frac{1}{12}$$

$$37. (i) C's 1 hour's work$$

$$= \frac{1}{12} - \frac{1}{27} = \frac{1}{54}$$

$$= \frac{9-4-2}{108} = \frac{3}{108} = \frac{1}{36}$$

$$\therefore \text{Ratio of A's, B's, and C's 1 hour's work}$$

$$= \frac{1}{27} : \frac{1}{54} : \frac{1}{36}$$

$$= \left( \frac{1}{27} \times 108 \right) : \left( \frac{1}{54} \times 108 \right) : \left( \frac{1}{36} \times 108 \right)$$

$$= 4 : 2 : 3$$

$$\text{Sum of the terms of ratio}$$

$$= 4 + 2 + 3 = 9$$

$$\therefore C's share$$

$$= \frac{1}{x} + \frac{1}{x+10} = \frac{1}{12}$$

$$= \frac{x+10+x}{x(x+10)} = \frac{1}{12}$$

$$\Rightarrow x(x+10) = 24x + 120$$

$$\Rightarrow x^2 + 10x - 24x - 120 = 0$$

$$\Rightarrow x^2 - 14x - 120 = 0$$

$$\Rightarrow x^2 - 20x + 6x - 120 = 0$$

$$\Rightarrow x(x-20) + 6(x-20) = 0$$

$$\Rightarrow (x-20)(x+6) = 0$$

$$\Rightarrow x = 20, x = -6$$

$$\therefore \text{Time taken by B in doing } \frac{1}{2} \text{ work} = 15 \text{ hours}$$

$$39. (3) \text{Time taken by A in painting the house} = x \text{ days.}$$

$$\therefore \text{Time taken by B} = 3x \text{ days}$$

$$\therefore (A + B + C)'s 1 \text{ day's work} = \frac{1}{15}$$

$$= \frac{1}{x} + \frac{1}{3x} + \frac{1}{15} = \frac{1}{15}$$

Also,  $A : B = 3x : x$

$$A + B + C = 3$$

$$\Rightarrow A + B = 2$$

$$\Rightarrow 4x + 2 = x + \frac{1}{2}$$

$$\text{Time taken by B} = 48 \times 2 = 96 \text{ days}$$

40. (3) Time taken by C =  $x$  days

$$\therefore \text{Time taken by B} = 5x \text{ days}$$

$$(A + B + C)'s \text{ 1 day's work}$$

$$= \frac{1}{12}$$

$$= \frac{1}{60} + \frac{1}{5x} + \frac{1}{x} = \frac{1}{12}$$

$$= \frac{1}{5x} + \frac{1}{x} = \frac{1}{12} - \frac{1}{60}$$

$$\Rightarrow \frac{1+5}{5x} = \frac{5-1}{60} = \frac{4}{60}$$

$$= \frac{6}{5x} = \frac{1}{15}$$

$$\Rightarrow 5x = 15 \times 6 = 90 \text{ days} = \text{Time taken by days B}$$

OR

$$C : B = 5x : x$$

T.W

41. (3) A does 50% work in 9 days

$$\text{Time taken by A in doing 1 work} = 9 \times 2 = 18 \text{ days}$$

Similarly,

$$\text{Time taken by B in doing 1 work} = 9 \times 4 = 36 \text{ days}$$

$$\therefore (A + B)'s \text{ 1 day's work}$$

$$= \frac{1}{18} + \frac{1}{36}$$

$$= \frac{2+1}{36} = \frac{3}{36} = \frac{1}{12}$$

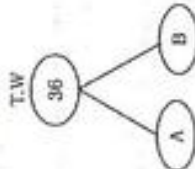
$$\therefore (A + B)'s \text{ 9 days' work}$$

$$= \frac{9}{12} = \frac{3}{4}$$

$$\therefore \text{Required percent}$$

$$= \frac{3}{4} \times 100 = 75\%$$

OR



$$\text{Days : } 9 \times 2 = 18 \quad 9 \times 4 = 36$$

$$= 10 \left( \frac{5+4}{150} \right) = 15$$

$$\text{Remaining work} = 1 - \frac{11}{15} = \frac{4}{15}$$

This part of work is done by C in 4 days

$$\therefore \text{Time taken by C in doing 1 work} = 15 \text{ days}$$

$$\therefore (A + B + C)'s \text{ 1 day's work}$$

$$= \frac{1}{30} + \frac{1}{25} + \frac{1}{15}$$

$$= \frac{5+6+10}{150} = \frac{21}{150} = \frac{7}{50} = \frac{14}{100}$$

$$\therefore \text{Their 2 days' work} = \frac{28}{100} \text{ i.e. } 28\%$$

$$43. (3) \text{ Time taken by A in doing 1 work} = 45 \text{ days}$$

$$\text{Time taken by B in doing 1 work}$$

$$= 18 \times \frac{4}{3} = 24 \text{ days}$$

$$(B + C)'s \text{ 8 days' work}$$

$$= 8 \left( \frac{1}{24} + \frac{1}{36} \right)$$

$$= 8 \left( \frac{3+2}{72} \right) = \frac{5}{9}$$

$$\text{Remaining work} = 1 - \frac{5}{9} = \frac{4}{9}$$

$$\Rightarrow \frac{36 \times 40}{3} = M_2 \times \frac{2}{4}$$

$$\Rightarrow 36 \times 40 = M_2 \times 2$$

$$\Rightarrow M_2 = \frac{36 \times 40}{2} = 60$$

$\therefore$  Additional men

$$= 60 - 36 = 24$$

46. (2) 2 men = 1 woman

$$\therefore 1 \text{ man} = \frac{1}{2} \text{ woman}$$

$$\therefore 3 \text{ men} = \frac{3}{2} \text{ women}$$

Again, 2 children = 1 man

$$= \frac{1}{2} \text{ woman}$$

$$\therefore 1 \text{ child} = \frac{1}{4} \text{ woman}$$

$$\therefore 6 \text{ children} = \frac{6}{4} = \frac{3}{2} \text{ women}$$

Now, three men, four women and six children

$$= \frac{3}{2} + 4 + \frac{3}{2} = 7 \text{ women}$$

Hence, 7 women complete the work in 7 days.

$$47. (1) 1 \text{ man} + 1 \text{ woman} + 1 \text{ child's 1 day's work}$$

$$= \frac{1}{1} + \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$



∴ Time taken by A and B in doing  $\frac{2}{3}$ rd work

$$= \frac{2}{3} \times 6 = 4 \text{ days}$$

∴ A and B worked together for 4 days.

11. (2) For work 1, (A + B)'s 1 day's work =  $\frac{1}{3} +$

$$\frac{1}{2} = \frac{5}{6}$$

∴ Time taken =  $\frac{6}{5}$  days

= 1 day and 2 hours

For work 2, (A + B)'s 1 day's work =  $\frac{1}{5} + \frac{1}{5} =$

$$\frac{2}{5}$$

∴ Time taken =  $\frac{5}{2}$  days

= 2 days and 5 hours

Total = 3 days 7 hours. Since Work 2 is done in 2nd order, the last day (4th day) will be dedicated to work 2

15. (1) All labours complete work 1 in = 1

$$\left( \frac{1}{3} + \frac{1}{5} + \frac{1}{8} + \frac{1}{4} \right)$$

$$= 1 + \frac{40 + 60 + 24 + 15 + 30}{120}$$

$$= 1 + \frac{169}{120} = \frac{169}{120} = 0.7$$

Hours taken =  $10 \times 0.7$

= 7 hours

All labours complete work 2 in

$$= 1 + \left( \frac{1}{5} + \frac{1}{8} + \frac{1}{4} + \frac{1}{10} \right)$$

$$= 1 + \left( \frac{8 + 5 + 10 + 4}{40} \right)$$

$$= 1 + \frac{35}{40} = \frac{40}{35} = \frac{8}{7} = 1.1$$

Hours taken =  $1.1 \times 10 = 11$  hours

Required ratio = 7:11

16. (2) Ratio of the efficiencies of A and B = 3:4

∴ Ratio of time taken = 4:3

∴ (A + B)'s 1 day's work =  $\frac{1}{8}$

Let time taken by C alone in doing whole work be  $x$  days.

(B + C)'s 6 days' work =  $\frac{3}{4}$

$$\Rightarrow \frac{6}{14} + \frac{6}{x} = \frac{3}{4} \Rightarrow \frac{6}{x} = \frac{3}{4} - \frac{6}{14}$$

$$\Rightarrow \frac{6}{x} = \frac{21 - 12}{28} \Rightarrow \frac{6}{x} = \frac{9}{28}$$

$$\Rightarrow x = \frac{28 \times 6}{9} = \frac{56}{3} \text{ days}$$

17. (1) Work done in first two days

$$= \frac{2}{30} + \frac{1}{20} + \frac{1}{15} + \frac{1}{20} + \frac{1}{10}$$

$$= \frac{4 + 3 + 6 + 13}{60} = \frac{13}{60}$$

Work done in first 8 days =  $\frac{52}{60}$

$$\text{Remaining work} = 1 - \frac{52}{60} = \frac{8}{60} = \frac{2}{15}$$

No

$$\text{Remaining work} = 1 - \frac{1}{5} = \frac{4}{5}$$

∴ Time taken by A in finishing the remaining work

$$= 35 \times \frac{4}{5} = 28 \text{ days}$$

19. (2) Work done by A and B in 7 days

$$= \frac{7}{20} + \frac{7}{30} = \frac{21 + 14}{60} = \frac{35}{60} = \frac{7}{12}$$

$$\text{Remaining work} = 1 - \frac{7}{12} = \frac{5}{12}$$

∴ Time taken by C =  $\frac{12}{5} \times 10 = 24$  days

20. (2) Let A left the work after  $x$  days.

According to the question.

Work done by A in  $x$  days + work done by B in  $(23 + x)$  days = 1

$$\Rightarrow \frac{x}{45} + \frac{23 + x}{40} = 1$$

$$\Rightarrow \frac{8x + 207 + 9x}{40} = 1$$



$$\therefore \text{Their 10 day's work} = \frac{90}{100} = \frac{9}{10}$$

$$\therefore \text{Remaining work} = 1 - \frac{9}{10} = \frac{1}{10}$$

$$\therefore \text{Suresh does } \frac{1}{10} \text{ work in 3 days.}$$

$$\therefore \text{Time taken by Suresh in doing 1 work} = 3 \times 10 = 30 \text{ days}$$

$$23. (1) \text{Time taken by A} = \frac{65}{35} = 18 \text{ days}$$

$$\text{Time taken by B} = \frac{75}{25} = 30 \text{ days}$$

$$(A + B)'s 1 \text{ day's work} = \frac{1}{18} + \frac{1}{30}$$

$$= \frac{5-3}{90} = \frac{2}{90} = \frac{1}{45}$$

$$\therefore \text{Required time} = \frac{45}{2} \text{ days}$$

$\therefore$  Total wages

$$= \text{Rs. } \frac{45}{2} \times (3.50 + 2.50)$$

$$= \text{Rs. } \left( \frac{45}{2} \times 6 \right) = \text{Rs. } 67.5$$

$$24. (2) \text{Time taken by A} = x \text{ days}$$

$$\therefore \text{Time taken in doing one-third work} = \frac{9}{3} = 3 \text{ days}$$

[OR]

$$A = 3C$$

$$\frac{22.5}{22.5} \text{ TW}$$

$$\frac{A+C}{A+C}$$

$$\text{Days : } 22.5$$

$$\text{Efficiency : } \frac{22.5}{22.5} = 1$$

$$A + C = 1$$

$$\Rightarrow 3C + C = 1$$

$$\Rightarrow C = \frac{1}{4}$$

$$\Rightarrow A = \frac{3}{4}$$

$$\text{Their 15 days work} = 1 \times 15 = 15$$

$$\text{Work left} = 22.5 - 15 = 7.5$$

$$\text{Time taken by B to do this work} = 15 \text{ days.}$$

$$\text{Their total efficiency} = 3 + 2 = 5$$

$$\text{Part of work completed in 9 days} = 9 \times 5 = 45$$

$$\text{Work left} = 60 - 45 = 15$$

$$\text{This work is finished by C in 6 days.}$$

$$\text{Efficiency of C} = \frac{15}{6}$$

$$(B + C)'s \text{ total efficiency} = \frac{12+15}{6} = \frac{27}{6}$$

$$\text{Time taken to complete 75\% of Job} = \frac{60 \times \frac{3}{4}}{\frac{27}{6}}$$

$$= \frac{60 \times 3 \times 6}{27 \times 4} = 10 \text{ days}$$

$$26. (4) (2 \text{ men} + 7 \text{ women})'s 1 \text{ day work} = \frac{1}{28}$$

$$\Rightarrow 2m + 7w = \frac{1}{28} \quad \dots(1)$$

$$(6 \text{ men} + 16 \text{ women})'s 1 \text{ day work} = \frac{1}{11}$$

$$\Rightarrow 6m + 16w = \frac{1}{11} \quad \dots(2)$$

Multiplying

$$= \frac{2 \times 5}{154} = \frac{7}{154} = \frac{1}{22}$$

$\therefore 15w + 3m$  will complete the job in 22 days.  
Let the number of books distributed by a man, a woman and a child per day be  $5x$ ,  $4x$  and  $2x$  respectively

Books distributed on the first day

$$= 7 \times 5x + 5 \times 4x + 8 \times 2x$$

$$= 35x + 20x + 16x = 71x$$

Books distributed on the second day =  $7 \times 5x$

$$+ 3 \times 4x + 5 \times 2x$$

$$= 35x + 12x + 10x = 57x$$

Books distributed on the third day =  $4 \times 5x +$

$$5 \times 4x + 3 \times 2x$$

$$= 46x$$

$$\therefore 71x + 57x + 46x = 2000$$

$$\Rightarrow 174x = 2000$$

$$\therefore 57x = \frac{2000}{174} \times 57x = 650$$

28. (3) Part of the tank filled in 1 hour by pipe A =

$$\frac{1}{2}$$

Part of the tank filled by both pipes in 1 hour

$$= \frac{1}{2} + \frac{1}{3} = \frac{5}{6}$$

# TIME AND WORK

$$\Rightarrow x = \frac{9 \times 20}{4 \times 3} = 15 \text{ hours}$$

30. (1) Part of the cistern filled in 2 hours by pipe

$$A = \frac{2}{3}$$

Part of the cistern filled in 1 hour by pipe B

$$= \frac{1}{4}$$

$$\therefore \text{Total part filled} = \frac{2}{3} + \frac{1}{4} = \frac{8+3}{12} = \frac{11}{12}$$

When all three pipes are opened, the part filled in one hour

$$= \frac{1}{3} + \frac{1}{4} + \frac{4+3-12}{12} = \frac{-5}{12}$$

L.e.  $\frac{5}{12}$  part will be emptied per hour.

$\therefore$  Time taken to empty  $\frac{11}{12}$  part

$$= \frac{11}{12} \times \frac{12}{5} = \frac{11}{5} \text{ hours}$$

$$= 2 \text{ hours } 12 \text{ minutes}$$

$$\therefore \text{Required time} = 5 + 2 = 12$$

$$R = 6 \text{ days}$$

So, the third daughter can complete the work

$$\text{in } = \frac{72}{6} = 12 \text{ days}$$

4. (3) Let 'C' complete the work in x days.

Work done by (A+B) in 1 day =  $\frac{1}{10}$  work.

Work done by (B+C) in 1 day =  $\frac{1}{18}$  work

A's 5 day's work + B's 10 day's work + C's 15 day's work = 1

$$\therefore (A+B)'s 5 \text{ day's work} + (B+C)'s 5 \text{ day's work} + C's 10 \text{ day's work} = 1$$

$$\frac{5}{10} + \frac{5}{18} + \frac{10}{x} = 1$$

$$\Rightarrow x = 45 \text{ days}$$

5. (3) According to question,

| A | B | C |
|---|---|---|
|---|---|---|

$$\text{Efficiency} \quad 3 : 2 : 6$$

$$\text{No. of days} \quad 2 : 3 : 1$$

$$\text{Number of days taken by A} = 12.$$

# TIME AND WORK

Therefore, total time taken to complete the work

$$= 5 + \frac{1}{9} = 5\frac{1}{9} \text{ days}$$

$$\therefore 25 \text{ men complete one project in } = \frac{200}{5} = 40 \text{ days}$$

$$\text{One man completes one project in } = \frac{200}{5} = 200 \text{ days}$$

5 men complete, with the help of one machine

= 15 days

Let, one machine complete one project in x days.

$$\therefore \left(\frac{1}{40}\right) + \left(\frac{1}{x}\right) = \frac{1}{15} \Rightarrow \frac{1}{x} = \frac{1}{120}$$

$$\therefore \frac{1}{15} - \frac{1}{40} = \frac{8-3}{120} = \frac{5}{120}$$

$$\frac{1}{x} = \frac{5}{120} \Rightarrow x = 24 \text{ days}$$

$$20 \text{ men ca}$$

8. (2) Let one man takes x days to complete the work and one woman takes y days to complete the work independently.

$$\text{Then, } \frac{4 \times 4}{x} + \frac{16 \times 4}{y} = \frac{1}{3}$$

$$\text{and } \frac{6 \times 2}{x} + \frac{12 \times 2}{y} = \frac{2}{9}$$

Solving above equations, we get

$$x = 108, y = 216$$

Let x women be added to complete the work in 3 days.

$$\text{Then, } \frac{9 \times 3}{108} + \frac{3(12-x)}{216}$$

$$= 1 - \left(\frac{1}{3} - \frac{2}{9}\right) = \frac{4}{9}$$

$$\Rightarrow 35 - 36 - 3x = \frac{216 \times 4}{9} = 96$$

$$\Rightarrow 3x = 96 - 72 = 24 \Rightarrow x = 8$$

9. (3) Let one worker of category I can finish the work in  $\frac{2m}{3h}$ .

$$7F + 3R = \frac{1}{8} \Rightarrow 1 \text{ day's work } 7D + 7F + 7R =$$

$$\frac{7}{24}$$

$\Rightarrow 1 \text{ day's work}$

$$1D + 1F + 1R = \frac{1}{24}$$

$\Rightarrow 1 \text{ day's work}$

$\Rightarrow$  They all will complete the work in 24 days.

12. (1) According to question, work done by A and

$$B \text{ in 5 days} = \left( \frac{1}{10} + \frac{1}{15} \right) \times 5 = \frac{5}{6}$$

$$\therefore \text{Rest Work} = 1 - \frac{5}{6} = \frac{1}{6}$$

$\therefore C$  alone can do the work in  $6 \times 2 = 12$  days

$$\text{Ratio of their work} = \frac{5}{10} : \frac{5}{15} : \frac{2}{12}$$

$$= 3 : 2 : 1$$

$$\text{Share of money} = \frac{3}{6} \times 450 =$$

$$\frac{2}{6} \times 450 : \frac{1}{6} \times 450$$

$$₹ 225, ₹ 150, ₹ 75$$

13. (4) According to question, after one day

$$\text{contents left} = \left( 1 - \frac{1}{3} \right) = \frac{2}{3}$$

$$\text{Second day contents evaporated} = \frac{2}{3} \times \frac{3}{4} = \frac{6}{12}$$

$\therefore$  Contents left after second day

$$\frac{2}{3} - \frac{6}{12} = \frac{8-6}{12} = \frac{2}{12} = \frac{1}{6}$$

14. (2) Work of inlet pipe in an hour =  $\frac{1}{8} - \frac{1}{12} =$

$$\frac{1}{24}$$

Without leak, the inlet pipe will fill in the tank in 24 hours.

$\therefore$  Capacity of the tank

$$= 4 \times 24 \times 60 = 5760 \text{ litres}$$

**Method-2 :** The outlet pipe has to work for 4 more hours. Work done by the outlet pipe in

$$4 \text{ hours} = 4 \times \frac{1}{8} = \frac{1}{2} \text{ i.e. } \frac{1}{2} \text{ of the tank is emp-}$$

ty by the outlet pipe in 4 hours. This  $\frac{1}{2}$  of

the tank is filled by the inlet pipe in 12 hours.

15.

16.

17.

**Option (3)**  $x = 10, y = 8$ ,  
then  $\frac{3}{10} + \frac{2}{8} = \frac{11}{20}$ , check eq. (2)  $\frac{10}{2} + \frac{8}{2} = 1$ ,  
hence option (3) is the correct choice.

**21. (2)** Let the first typist take  $x$  minutes and the second typist  $y$  minutes in doing the work.

$$\begin{aligned}\therefore \frac{1}{x} + \frac{1}{y} &= \frac{1}{6} \\ \Rightarrow \frac{y+x}{xy} &= \frac{1}{6} \\ \Rightarrow 6x+6y &= xy \quad \dots(i) \\ \text{Again, } \frac{4}{x} + \frac{6}{y} &= \frac{4}{5} \\ \Rightarrow \frac{4y+6x}{xy} &= \frac{4}{5} \\ \Rightarrow 20y+30x &= 4xy \quad \dots(ii) \\ \text{By equation (i) } \times 4 - (ii), \\ 24x+24y-20y-30x &= -4xy-4xy=0 \\ \Rightarrow 6x=4y &\Rightarrow 3x=2y \quad \dots(iii)\end{aligned}$$

**22. (2)** Let Ali iron  $x$  shirts in one hour then Ansari iron  $(x-10)$  shirts, and Ahmed iron  $(x-y-10)$  or  $(x-20)$  shirts.

$$\begin{aligned}\text{Also, } 4(x-20) &= 2(x-10) \\ \Rightarrow 4x-80 &= 2x-20 \\ \Rightarrow \frac{16}{5}x &= 8y \Rightarrow x = \frac{5}{2}y\end{aligned}$$

No. of shirts ironed in 1 hour by Ali =  $\frac{5}{2}y$ .

$$\text{Ansari} = \frac{3}{2}y \text{ and Ahmed} = \frac{1}{2}y$$

As per question,  
 $\frac{5}{2}y + \frac{3}{2}y + \frac{1}{2}y = \frac{117}{13}$

$$\begin{aligned}\Rightarrow \frac{9}{2}y &= 9 \Rightarrow y = 2, \text{ so } x = 5\end{aligned}$$

4x and let there be  $4y$  males and  $5y$  females in AIC Ltd.]

$$\begin{aligned}\text{Total wages of males} &= 5 \times 4y = 20xy \\ \text{Total wages of females} &= 4 \times 5y = 20xy \\ \text{Ratio of sum of wages of male to females} &= 20 : 20 = 1 : 1\end{aligned}$$

$$\begin{aligned}\therefore \text{Amount received by all women} \\ &= 1,55,000 \times \frac{2}{5} = \text{Rs } 62,000\end{aligned}$$

**25. (3) Method-1** : Total no. of man days required to finish the work is  $4x$ . Also, total time taken

$$\text{actually is } 4 + \frac{3(4)}{4} = 7 \text{ days}$$

$x$  employees worked for 1st day, number of man days =  $x$

$$\rightarrow (x-5) \text{ employees worked for 2nd day, number of man days}$$

$$= (x-5)$$

$$\rightarrow (x-10) \text{ employees worked for 3rd day, number of man days}$$

$$= (x-10)$$

We have to compare the time taken by 3 small pumps and 1 large pump i.e. 3 large pumps with time taken by 1 large pump alone.

$$\begin{aligned}\Rightarrow 3 \text{ large pumps will take } \frac{1}{3} \text{ rd time taken by } \\ \text{1 large pump}\end{aligned}$$

**Method-2** : Ratio of efficiency of large : small pump =  $1 : \frac{2}{3}$  or  $3 : 2$  hence ratio of time =  $2 : 3$

Let one large pump take 2 hours, then one small pump would take 3 hours.

Work done in 1 hour by 1 large & 3 small

$$\text{pumps} = \frac{1}{2} + \left(\frac{1}{3} \times 3\right) = \frac{3}{2}$$

$$\text{Hence, total time taken} = \frac{2}{3} \text{ hour.}$$



Wages per male =  $250 + 15 \times 40 = ₹ 850$   
 Wages per female =  $300 + 10 \times 50 = ₹ 800$

**Checking options****Option (1)**

When male operators (M) = 15,  
 calls handled =  $15 \times 40 = 600$   
 Remaining calls =  $1000 - 600 = 400$

So, no. of females =  $\frac{400}{50} = 8$

Total Cost =  $8(800) + 15(850) = 6400 + 12750$   
 $= ₹ 19150$

**Option (2)**

When M = 14,

Calls handled =  $14 \times 40 = 560$   
 Remaining calls =  $1000 - 560 = 440$

440 calls would

$= 8500 + 1600 = 10,100$

Hence, total cost is minimum in option (4)

30. (4) Let conical tank contain 'x' litres of liquid then cylindrical tank would hold  $(x + 1600)$  litres. So,

$$1x - 200/2 = x + 500 - 200$$

$$2x - 400 = x + 300 \Rightarrow x = 700$$

Hence, cylindrical tank would hold  $700 + 500 = 1200$  L.

**QUANTITY BASED EXPLANATION**

1. (3) **Quantity I**: Efficiency of A =  $\frac{1}{10} \times 100 = 10\%$

Efficiency of B =  $\frac{1}{15} \times 100 = 6.67\%$

Efficiency of (A + B) =  $(10 + 6.67)\% = 16.67\%$

$\therefore 16.67\%$  is done in 1 day.

$\therefore 100\%$  of work would be done in

A men can do the job in 55 days. Let  $\frac{55}{7}$  men take 'n' days. Then,

$$4 \times 55 = \frac{55}{7} \times n$$

$$\Rightarrow n = 4 \times 55 \times \frac{7}{55} = 28 \text{ days}$$

**Quantity II**: 4 men + 2 women take 4 days to

complete  $\frac{1}{3}$  rd of work, hence they will take 4

$\times 3 = 12$  days to complete the entire work.

$$4M + 2W \rightarrow 12 \text{ days,}$$

12 (4M + 2W)  $\rightarrow$  1 day to complete the whole work.

$$\text{i.e. } 48M + 24W \rightarrow 1 \text{ day} \dots\dots (i)$$

Similarly 5M + 4W take 8 days to do whole work

4. (1) **Quantity I**: Let Pipe C take x hours to empty the tank, then when all 3 are opened simultaneously, work done in one-hour =

$$\frac{1}{10} + \frac{1}{12} + \frac{1}{60} = \frac{1}{x}$$

$$\frac{11}{60} + \frac{1}{15} = \frac{7}{60} + \frac{1}{x} \Rightarrow x = \frac{60}{7} \text{ hours}$$

$$= \frac{8}{7} \text{ hours}$$

**Quantity II**: 1 hour's work of Pipe A and B

$$= \frac{1}{10} + \frac{1}{12} = \frac{9}{60}$$

Hence both the pipes would take  $\frac{40}{9}$  hours to fill the tank.

Now, Pipe B has worked for a total of 6 hours, and, let pipe A work for a total of 'x' hours

Substituting the values,

$$\therefore B's 1 \text{ hour's work} = \frac{5}{48} = \frac{1}{16}$$

$$= \frac{5-3}{48} = \frac{1}{24}$$

$\therefore B$  alone will finish the work in 24 hours

**Quantity II :**  $(A + B + C)$ 's 1 day's work

$$= \frac{1}{20} + \frac{1}{30} + \frac{1}{60} = \frac{3+2+1}{60} = \frac{1}{10}$$

$$A's 2 \text{ days' work} = \frac{2}{20} = \frac{1}{10}$$

Work done in first three days

$$= \frac{1}{10} + \frac{1}{10} + \frac{2}{20} = \frac{1}{5}$$

Hence, the work will be finished in 15 days.

Quantity I > Quantity II

#### DATA SUFFICIENCY EXPLANATION

1. (4) Both the statements are  $\text{?}$

From statement II, if A takes 'a' days, B will take  $\frac{a}{2}$  days to finish the work.

$$\frac{1}{a} + \frac{2}{a} = \frac{1}{18} \Rightarrow \frac{3}{a} = \frac{1}{18} \Rightarrow a = 54 \text{ days}$$

B will take  $\frac{54}{2} = 27$  days to finish the work.

Both the statements are required.

4. (3) From statement I,

$$(A + B + C)'s 1 \text{ day's work} = \frac{13}{48}$$

$$(A + B)'s 1 \text{ day work} = \frac{5}{24}$$

$$\Rightarrow C's 1 \text{ day work} = \frac{13}{48} - \frac{5}{24} = \frac{3}{48}$$

Also given that  $(B + C)$ 's 1 day work =  $\frac{7}{48}$

$$\frac{7}{48} = \frac{3}{48} + \frac{1}{12}$$

# 11 Time & Distance

## IMPORTANT FACTS AND FORMULAE

1. **Speed :** Speed is the distance covered by an object per unit time. Thus, if a person travels a distance of 25 kilometres in 5 hours, then his speed is 5 km/hour.

$$\text{Distance} = \text{Speed} \times \text{Time}$$

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

**Basic Rules :**

More clearly,  $1 \text{ kmph} = \frac{1 \text{ km}}{1 \text{ hour}}$

$$= \frac{1000 \text{ m}}{60 \times 60 \text{ sec.}} = \frac{5}{18} \text{ m/sec.}$$

Thus, a speed of 18 kilometres per hour = speed of 5 metres per second

6. The speed of a body relative to the other body is called its relative speed.

7. If two bodies are moving in the same direction, then their relative velocity will be

train to cover a distance equal to its own length.

13. Time taken by a train in passing a platform or a bridge or a train (stationary objects of some length) is equal to the time taken by the train to cover a distance equal to its own length plus the length of the platform or bridge or another train.
14. If two trains of length  $x$  km and  $y$  km are moving in the same direction at  $u$  kmph and  $v$  kmph where  $u > v$ , then their relative velocity  $= (u - v)$  kmph and the time taken by faster train to cross the slower train

$$= \frac{x + y}{u - v} \text{ hrs.}$$

$t$  hours in reaching B and A respectively, then, A's speed  $= B$ 's speed  $- (d/b - d/a)$

#### FACTS ON BOATS AND STREAMS

17. In water, the direction along the stream is known as downstream. And the direction against the stream is called upstream.
18. If speed of a boat in still water be  $u$  km/hr. and the speed of the stream be  $v$  km/hr then :  
Speed of the boat along the stream (downstream)  
 $= (u + v)$  km/hr.  
 $=$  speed of boat in still water  $+$  speed of stream.  
and Speed of the boat against the stream (upstream)

distance at 10 km/hour and the remaining distance at 12 km/hour. The distance between P and Q is  
(1) 72 km (2) 96 km (3) 160 km (4) 70 km (5) 85 km

8. A bus travels 150 km in 3 hours and then travels next 2 hours at 60 km/hr. Then the average speed of the bus will be  
(1) 55 km/hr. (2) 54 km/hr.  
(3) 50 km/hr. (4) 60 km/hr.  
(5) 45 km/hr
9. A man can cover a certain distance in 3 hours 36 minutes if he walks at the rate of 5 km/hr. If he covers the same distance on cycle at the rate of 24 km/hr, then the time taken by him in minutes is

taken to cross a pole at the same speed. What is the speed of the train in metre/second ?  
(1) 16 (2) 10 (3) 8  
(4) Cannot be determined  
(5) None of these

15. The speed of a car is 1.5 times the speed of a bus. If the car travels at the speed of 60 km/hr., what will be the difference in the time taken by the bus and the time taken by the car to cover 720 km ?  
(1) 5 hours (2) 6 hours  
(3) 4 hours (4) 8 hours  
(5) None of these

16. A 300 metre long train moving with an average speed of 126 km/hr. crosses a platform in 24 seconds. A man crosses the same 300



- from the moment they meet, train B, 222m long and travelling at 54kmph in opposite direction to which train A is travelling?
- (1) 16 seconds  
(2) 18 seconds  
(3) 12 seconds  
(4) 14 seconds  
(5) 10 seconds

21. Train A which is 320m long can cross a pole in 16 seconds. If it takes 5 times each time for exactly 18 minutes, how many hours will it take to cover a distance of 576 kms? (in hours)

- (1) 8  
(2)  $10\frac{1}{2}$   
(3)  $8\frac{1}{2}$   
(4) 9

- (5)  $9\frac{1}{2}$

26. A train B speeding with 120 kmph crosses another train C running in the same direction, in 2 minutes. If the lengths of the trains B and C be 100m and 200m respectively, what is the speed (in kmph) of the train C?
- (1) 111  
(2) 123  
(3) 127  
(4) 129  
(5) None of these

27. An employee may claim Rs. 7.00 for each km when he travels by taxi and Rs. 6.00 for each km if he drives his own car. If in one week he claimed Rs. 675 for travelling 90 km, how many kms did he travel by taxi?

- (1) 135  
(2) 155  
(3) 162  
(4) 170  
(5) None of these

28. A plane left 30 minutes later than the scheduled time and in order to reach the destination 1500 km away in time, it had to increase

- and if they travel in same direction from A towards B then A meet B after 6 hours. What is the speed of car starting from A?
- (1) 70 kmph  
(2) 120 kmph  
(3) 60 kmph  
(4) Data inadequate  
(5) None of these

32. A bus covered a certain distance from village A to village B at the speed of 60 km/hr. However on its return journey it got stuck in traffic and covered the same distance at the speed of 40 km/hr and took 2 hours more to reach its destination. What is the distance covered between village A and B?

(1) 240 km.  
(2) 200 km.  
(3) 200 km.  
(4) Cannot be determined  
(5) None of these

38. A person has to travel from point B in certain time. Travelling at a speed of 5 kmph he reaches 45 minutes late and while travelling at a speed of 8 kmph he reaches 15 minutes early. What is the distance from point A to point B?
- (1) 15 kms  
(2) 9 kms  
(3) 12 kms  
(4) 18 kms  
(5) 14 kms

39. Sumit left for city X from city Y at 5.50a.m. He travelled at the speed of 60 kms/hr. for 2 hours 15 minutes. After that the speed was reduced to 40 kms/hour. If the distance between two cities is 350 kms., at what time did Sumit reach the city X?



# TIME AND DISTANCE

44. A 280 metre long train moving with an average speed of 108 km/hr crosses a platform in 12 seconds. A man crosses the same platform in 10 seconds. What is the speed of the man in metre/second?

(1) 5 m/s (2) 8 m/s (3) 12 m/s  
(4) Cannot be determined  
(5) None of these

45. Due to inclement weather, an air plane reduced its speed by 300 km/hr. and reached the destination of 1200 km late by 2hrs. Then the schedule duration of the flight was

(1) 1 hour (2) 1.5 hours

## LEVEL-II

1. Ram and Shyam are travelling from point A to B, which are 60km apart. Travelling at a certain speed Ram takes one hour more than Shyam to reach point B. If Ram doubles his speed he will take 30 minutes less than Shyam to reach point B. At what speed was Ram driving from point A to B?

(1) 15 kmph (2) 35 kmph  
(3) 30 kmph (4) 25 kmph  
(5) 20 kmph

2. An express train travelled at an average speed of 100 km/hr. The distance between two cities is 200 km. The train reached the city at 12:30 pm. At what time did it start?

# TIME AND DISTANCE

12. Anna left for city A from city B at 5:20 am. She travelled at the speed of 80 kmph for 2 hours 15 minutes. After that speed was reduced to 60 kmph. If the distance between two cities be 370 km, at what time did Anna reach city A?

(1) 9:20 am (2) 9:25 am  
(3) 9:35 am (4) 10:05 am  
(5) None of these

13. Train A of length 120 metre can cross a platform of length 240 metre in 18 seconds. The ratio of speed of train A and Train B is 4:5. Then find the time taken by Train B to cross the same platform.

(1) 140 km (2) 120 km  
(3) 160 km (4) 130 km  
(5) 140 km

7. The distance between two places A and B is 140 km. The first scooter departs from place A to B at a speed of 50 kmph at 10 am. The second scooter departs from place B to A at a speed of 30 kmph at 12 pm. At what time will both the scooters meet each other?

(1) 12:30 pm (2) 12:40 pm  
(3) 12:50 pm (4) 1:00 pm  
(5) 1:10 pm

# TIME AND DISTANCE

17. The most economical speed for this journey is :  
 (1) 20 kmph (2) 30 kmph  
 (3) 35 kmph (4) 40 kmph  
 (5) None of these
18. The total cost for the journey at this most economical speed is :  
 (1) Rs. 6000 (2) Rs. 8000  
 (3) Rs. 10000 (4) Rs. 11000  
 (5) None of these
19. A boat takes 3 hours less time in rowing from point X to point Y in downstream than that in rowing from point Y to point Z in upstream. The distance between X and Y is 20 km which is half of the distance between Y and Z. Speed of boat in still water is 10 kmph. The speed

23. Places A and B are 300 km apart. Train X leaves from A for B and train Y leaves from B for A at the same time on the same day on parallel tracks. Both trains meet after 5.1 hours. The speed of train Y is 10 km/h more than that of train X. What is the speed in km/h of train Y?  
 (1) 41 (2) 54 (3) 31 (4) 56  
 (5) 50
24. A and B start moving towards each other from places X and Y, respectively, at the same time on the same day. The speed of A is 20% more than that of B. After meeting on the way, A and B take :

# TIME AND DISTANCE

28. Two cars start from places A and B, 100 km apart, towards each other. Both cars start simultaneously. A later, starting on one car, starts at the same time towards the other car, and as soon as it reaches the second car, it flies back to the first car and it continues in this manner flying backwards and forwards from one car to the other until the cars meet. Both cars travel at a speed of 50 kmph and the bird flies at 100 kmph. Total distance covered by the bird will be :  
 (1) 50 km (2) 100 km  
 (3) 200 km (4) 300 km  
 (5) None of these
29. A, B, C walk 1 km in 5 minutes, 8 minutes and 10 minutes respectively. The first one starts with a circumference of the circle is 300 m after what time will they meet at the starting point?  
 (1) 20 s (2) 100 s (3) 300 s (4) 200 s  
 (5) None of these
4. A train can travel 20% faster than a car. Both start from the point A at the same time and reach point B, 75 km away from A at the same time. On the way however the train lost about 12.5 min while stopping at the stations. The speed of the car is  
 (1) 50 km/h (2) 55 km/h  
 (3) 60 km/h (4) 65 km/h  
 (5) None of these
5. Two cars start together in the same direction from the same place. The first car goes with a

# TIME AND DISTANCE

9. A small aeroplane can travel at 320 km/h in still air. The wind is blowing at a constant speed of 40 km/h. The total time for a journey against the wind is 135 min. What will be the time in minutes for the return journey with the wind? (ignore take off and landing time for the aeroplane)
- 94.5 min.
  - 108.125 min.
  - 120 min.
  - None of these

10. Arun, Barun and Kiranmala starts from the same place and travel in the same direction at speed of 30, 40 and 60 kmph respectively. Barun starts two hours after Arun. If Barun and Kiranmala overtake Arun at the same

some point at the end of the 6th minute, by the first time, after the start of the race. All the runners have the same starting point and each runner maintains a uniform speed throughout the race. If the fastest runner runs at twice the speed of the slowest runner, then what is the time taken by the fastest runner to finish the race?

- 20 min
- 15 min
- 10 min
- 5 min
- 12 min

**Directions (15-16) :** The petrol consumption rate of a new model car 'Patio' depends on its speed and may be described by the adjoining graph :

# TIME AND DISTANCE

17. The average speed for the entire journey
- 32 kmph
  - 37.5 kmph
  - 45 kmph
  - None of these
  - 30 kmph

18. The total distance travelled by Sachitra in first 10 hours of her journey is

- 270 km
- 320 km
- 350 km
- 400 km
- 200 km

19. If Sachitra takes 'x' hours to complete her entire journey, then x =

- 10 hrs
- 12 hrs
- 15 hrs
- 25 hrs
- 20 hrs

20. G

- Red & Alshwarya would meet least frequently
- Alshwarya & Bigasha would meet most frequently
- None of these
- Cannot be determined

**Directions (23-24) :** Sharmila starts from point A, on a circular path, 600m long, at an average speed of 15m/s. Ferose starts from point B, which is diametrically opposite to point A and goes towards Sharmila at an average speed of 15m/s. They meet for the first time at a point C after a certain time. Point C is between point A & B on the circular path.

23. Find the time from the start when both

runner B finishes 8 metres ahead of runner C. Each runner travels the entire distance at a constant speed.

What was the length of the race?

- (1) 36 metres  
(2) 48 metres  
(3) 60 metres  
(4) 72 metres  
(5) 70 metres

28. Navjivan Express from Ahmedabad to Chennai leaves Ahmedabad at 6:30 am and travels at 50 km/hr towards Baroda. Situated 100 km away. At 7:00 am, Howrah Express leaves Baroda towards Ahmedabad and travels at 40 km/hr. At Ahmedabad and Baroda, the traffic controller at

## QUANTITY BASED QUESTIONS

Directions (1-5) : In each of the questions the quantities are given, solve each of them and choose the correct option as given below.

- (1) Quantity I > Quantity II  
(2) Quantity I < Quantity II  
(3) Quantity I = Quantity II  
(4) Quantity I > Quantity II  
(5) Quantity I < Quantity II

be established

1. Quantity I : A train 100 metres long needs a man going in opposite direction at 5 km/hr and passes him in  $7\frac{1}{5}$  seconds. What is the

respectively, when they meet it is found that one train has travelled 40 km more than the other. The distance between the two stations (in km) is :

6. Quantity I : Two trains, each of length 125 metre, are running in parallel tracks in opposite directions. One train is running at a speed of 5 km/hour and they cross each other in 6 seconds. The speed of the other train is :

Quantity II : A train can travel 50% faster than a car. Both start from point A at the same time and reach point B, 75 kms away from A at the same time. On the way, however, the train lost about 12.5 minutes while

stopped at the stations. The speed of the car is :

1. The length of both the trains together is 450 metres

2. Train A is slower than train B.

3. A train crosses a signal post in X seconds.

What is the length of the train?

1. The train crosses a platform of 100 metre in 7 seconds

2. The train is running at the speed of 80 km/hr.

4. A and B are running around a circular track. Both start at the same point and at the same time. Find the time required for their first meeting.

1. Speed of A is 4 m/s while speed of B is 3 m/s for the first second



5. (3) Rate downstream of boat =  $\frac{48}{6} = 8 \text{ km/h}$

Rate upstream =  $\frac{24}{6} = 4 \text{ km/h}$

$\therefore$  Speed of current =  $\frac{1}{2}$  (Rate down stream - rate upstream) =  $\frac{1}{2} (8 - 4) = 2 \text{ km/h}$

6. (2) Speed of boat in still water =  $\frac{1}{2}$  (Rate downstream + rate upstream) =  $\frac{1}{2} (15 + 9) \text{ km/h}$

=  $\frac{1}{2} (24) = 12 \text{ km/h}$

7. (3) Let the total distance be  $4x \text{ km}$ .

Time =  $\frac{\text{Distance}}{\text{Speed}}$

According to the question,

$\frac{x}{10} + \frac{3x}{12} = 7$

$\Rightarrow \frac{x}{10} + \frac{3x}{12} = 7$

$2 \times 30 \times 20$

1. (4) Required average speed =  $\frac{2 \times 30 \times 20}{30 + 20}$

=  $\frac{120}{5} = 24 \text{ km/h}$

=  $\left( \frac{10}{24} \times 60 \right) \text{ minutes}$

= 45 minutes

10. (3) We know that

Speed =  $\frac{\text{Distance}}{\text{Time}}$

=  $\left( \frac{200 \times 120}{19} \right) \text{ m/sec.}$

=  $\frac{380}{19} \text{ m/sec.} = 20 \text{ m/sec.}$

=  $\frac{20 \times 18}{5} \text{ km/hr} = 72 \text{ km/hr.}$

11. (5) Initial speed =  $80 \text{ km/hr}$   
Total distance =  $80 \times 10 = 800 \text{ km}$ .

New speed =  $\frac{800}{4} = 200 \text{ km/hr.}$

Increase in speed

=  $200 - 80 = 120 \text{ km/hr.}$

14. (3) Let the speed of the train be  $x \text{ m/sec.}$

$\frac{220 \times 3}{x} - \frac{220}{x} = 80$

$\Rightarrow \frac{220 \times 3}{x} - \frac{220}{x} = 80$

$\Rightarrow x = \frac{220 \times 3}{80} = 8.25 \text{ m/sec.}$

= speed of train

15. (2) Speed of train

=  $\frac{60}{15} = 40 \text{ kmph}$

$\therefore$  Required difference

=  $\left( \frac{720}{40} - \frac{720}{60} \right) \text{ hours}$

=  $(18 - 12) \text{ hours} = 6 \text{ hours}$

16. (1) Let the length of the platform be  $x \text{ metre.}$   
Speed of train =  $126 \text{ kmph}$

=  $\left( \frac{126 \times 5}{18} \right) \text{ m/sec.}$

## LEVEL-III

1. (2) 2. (2) 3. (3) 4. (3) 5. (1) 6. (3)

7. (2) 8. (2) 9. (2) 10. (3) 11. (4) 12. (2)

13. (2) 14. (3) 15. (2) 16. (2) 17. (1) 18. (1)

19. (3) 20. (3) 21. (5) 22. (4) 23. (3) 24. (2)

25. (3) 26. (2) 27. (2) 28. (2) 29. (1) 30. (3)

## QUANTITY BASED

1. (1) 2. (3) 3. (1) 4. (1) 5. (3)

## DATA SUFFICIENCY

1. (1) 2. (4) 3. (4) 4. (5) 5. (1)

## EXPLANATIONS

## LEVEL-I

1. (4) Required average speed =  $\frac{2 \times 30 \times 20}{30 + 20}$

$$18. (3) \text{ Speed of car } = \frac{\text{Distance}}{\text{Time}}$$

$$= \frac{540}{9} = 60 \text{ kmph}$$

$$\text{Speed of train} = 2 \times 60 = 120 \text{ kmph}$$

$$\therefore \text{Speed of bike}$$

$$= \frac{2}{3} \times 120 = 80 \text{ kmph}$$

$$\therefore \text{Distance covered by bike in 5 hours}$$

$$= 5 \times 80 = 400 \text{ km}$$

19. (1) If the length of train A be  $x$  metre, then length of train B =  $2x$  metre.

When a train crosses a pole, it covers a distance equal to its own length.

$$\therefore \text{Required ratio} = \frac{x}{25} : \frac{2x}{75}$$

$$= \frac{1}{25} \times 75 : \frac{2}{75} \times 75$$

$$= 3 : 2$$

20. (3) Let the length of train A be  $x$  metre.

$$\text{Speed of train A} = 63 \text{ kmph}$$

$$= \left( \frac{35 \times 30}{2} \right) \text{ metre/second}$$

$$= \frac{65}{2} \text{ metre/second}$$

$$\therefore \text{Required time} = \frac{\text{Length of trains A and B}}{\text{Relative Speed}}$$

$$= \left( \frac{168 + 222}{\frac{65}{2}} \right) \text{ seconds}$$

$$= \left( \frac{390 \times 2}{65} \right) \text{ seconds}$$

$$= 12 \text{ seconds}$$

21. (5) Speed of train

$$= \frac{\text{Length of train}}{\text{Time taken in crossing}}$$

$$= \frac{320}{16} = 20 \text{ m/sec.}$$

$$= \left( 20 \times \frac{18}{5} \right) \text{ kmph}$$

$$\therefore \text{Speed of the car} = \frac{15 \times 90}{16} = 75 \text{ kmph}$$

$$\therefore \text{Required distance} = \text{Speed} \times \text{time}$$

$$= 75 \times 4 = 450 \text{ km.}$$

24. (5) According to the question,

$$\text{Distance covered by A in 8 hrs.}$$

$$= 6 \times 8 = 48 \text{ km}$$

$$\therefore \text{Distance covered by B in 8 hrs.}$$

$$= (114 - 48) \text{ km} = 66 \text{ km}$$

$$\therefore \text{Required speed}$$

$$= \frac{66}{8} \text{ kmph} = 8 \frac{1}{4} \text{ kmph}$$

25. (2) Nitin will take rest seven times i.e. 70 minutes in covering 150 km.

$$\text{Time taken to cover 150 km at 15 kmph}$$

$$= \frac{150}{15} = 10 \text{ hours}$$

$$\therefore \text{Total time}$$

$$= 10 \text{ hours} + 70 \text{ minutes}$$

$$= 11 \text{ hours } 10 \text{ minutes}$$

26. (1) Let the speed of train C be  $x$  kmph.

Speed

$$\Rightarrow 7x + 540 - 6x = 575$$

$$x = 575 - 540 = 135$$

$$\therefore \text{Required distance} = 135 \text{ km.}$$

28. (4) Usual speed of plane =  $x$  kmph

$$\therefore \text{New speed} = 1x + 250 \text{ kmph}$$

$$\frac{1500}{x} - \frac{1500}{x + 250} = \frac{20}{60}$$

$$\Rightarrow \frac{1500}{x} - \frac{1500}{x + 250} = \frac{1}{3}$$

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29. (2) Required time = LCM of 18, 24 and 32 seconds.

$$= 288 \text{ seconds}$$

30. (4) If the required distance be  $x$  km, then

$$\frac{x}{5} - \frac{x}{6} = \frac{30 - 5}{60}$$

$$= \frac{3x - 24}{120} = 2$$

$$= x = 2 \times 120 = 240 \text{ km}$$

34. (3) Let the speed of boat in still water be  $x$  kmph and that of current be  $y$  kmph

$$\therefore x + y = \frac{4.8 \times 60}{8} = 36$$

$$= x + y = 36 \quad \text{--- (i)}$$

$$\text{and } x - y = \frac{4.8 \times 60}{9} = 32$$

$$= x - y = 32 \quad \text{--- (ii)}$$

By equation (i) - (ii),

$$x + y - x + y = 36 - 32 = 4$$

$$= 2y = 4 \Rightarrow y = \frac{4}{2} = 2 \text{ kmph}$$

35. (2) PQ =  $x$  km

$$\therefore QR = (200 - x) \text{ km}$$

$$\text{Total distance} = 200 \text{ km}$$

$$\text{Total time} = \left( \frac{x}{90} + \frac{200 - x}{60} \right) \text{ hours}$$

$$= \left( \frac{2x + 600 - 3x}{180} \right) \text{ hours}$$

$$= 90 \text{ minutes} = \frac{3}{2} \text{ hours}$$

$$\text{Total time taken in covering a distance of } 576 \text{ km}$$

$$= \left( \frac{576 \times 3}{72} + \frac{3}{2} \right) \text{ hours}$$

$$= 8 + \frac{3}{2} = 9 \frac{1}{2} \text{ hours}$$

37. (2) Let the distance between point A and point B be  $x$  km.

$$\text{Difference of time} = 2 \text{ hours}$$

$$\therefore \frac{x}{12} - \frac{x}{15} = 2$$

$$= \frac{5x - 4x}{60} = 2$$

$$\Rightarrow \frac{x}{60} = 2 \Rightarrow x = 60 \times 2$$

$$= 120 \text{ km.}$$

$$\therefore \text{Time taken at } 15 \text{ kmph to cover } 120 \text{ km} =$$

$$\frac{120}{15} = 8 \text{ hours}$$

$$\therefore \text{Required speed to cover } 120 \text{ km at } 9 \text{ am, i.e. in } 7 \text{ hours}$$

$$= 180 + 2.5 \text{ km} = 180 \text{ km}$$

$$\text{Remaining distance} = 360 - 180 = 180 \text{ km}$$

$$\text{Speed} = 60 \text{ km/hr}$$

$$\text{Time taken to reach in city } = \frac{180}{60} = 3 \text{ hr}$$

$$\text{Total time} = 2.5 + 3 = 5.5 \text{ hr}$$

$$= 5.5 \text{ hr} = 330 \text{ minutes}$$

$$= 6 \text{ hr } 30 \text{ minutes}$$

$$\text{Sam started journey at } 5.50 \text{ a.m. } \therefore \text{ finished the journey at } 11.55 \text{ a.m.}$$

40. (3) Let both cars meet each other after  $t$  hours from 11 am.

$$\text{Distance} = \text{Speed} \times \text{Time}$$

$$\therefore 36 \times t + 44 \times (t - 2) = 592$$

$$\Rightarrow 36t + 44t - 88 = 592$$

$$\Rightarrow 80t = 592 + 88 = 680$$

$$\Rightarrow t = \frac{680}{80} = 8.5 \text{ hours i.e.,}$$

$$7:30 \text{ pm}$$

$$41. (1) \text{ Speed of train} = \frac{222}{6}$$

$$= 37 \text{ m/sec.}$$

$$\text{If the speed of man be } x \text{ m/sec, then } \frac{220}{x}$$

$$\text{If the length of platform be } x \text{ metre, then}$$

$$\frac{x + 260}{12} = 30$$

$$\Rightarrow x + 260 = 360 \Rightarrow x = 100$$

$$\Rightarrow x = 360 - 260 = 100 \text{ metre}$$

$$\therefore \text{Man's speed} = \frac{\text{Distance}}{\text{Time}} = \frac{100}{10} = 10 \text{ m/sec}$$

45. (3) Let the original speed of aeroplane be  $x$  kmph.

According to the question,

$$\frac{1200}{x - 30} - \frac{1200}{x} = 2$$

$$\Rightarrow 1200 \left( \frac{x - x + 30}{x(x - 30)} \right) = 2$$

$$\Rightarrow x(x - 30) = \frac{1200 \times 30}{2}$$

$$\Rightarrow x(x - 30) = 1800 \times 30$$

$$\Rightarrow x(x - 30) = 1800 \times 30$$

$$\Rightarrow x = 1800 \text{ kmph.}$$

$$\therefore \text{Scheduled duration of flight} = \frac{1200}{600} = 2 \text{ hours}$$

46. (3) Speed of racing car = 108 kmph.

$$\therefore \text{Length of racing track} = \text{Speed} \times \text{Time}$$

$$\frac{\text{Distance}}{\text{Speed}} = \text{Time}$$

$$\Rightarrow \frac{450}{x} - \frac{450}{y} = \frac{45}{60}$$

$$\Rightarrow \frac{450}{x} - \frac{450}{y} = \frac{3}{4}$$

Again, new speed of train B

$$= \frac{3y}{4} \text{ kmph.}$$

$$\frac{450}{\frac{3y}{4}} - \frac{450}{x} = \frac{30}{60}$$

$$\Rightarrow \frac{450 \times 4}{3y} - \frac{450}{x} = \frac{1}{2}$$

$$\Rightarrow \frac{600}{y} - \frac{450}{x} = \frac{1}{2} \quad \dots \text{(ii)}$$

On adding equations (i) and (ii),

$$\frac{600}{y} - \frac{450}{y} = \frac{3}{4} + \frac{1}{2}$$

$$\frac{150}{y} = \frac{5}{4}$$

50. (3) Difference between speeds

$$= (60) - (45) \text{ kmph.}$$

$$= 15 \text{ kmph.}$$

$$\therefore \text{Stoppage per hour}$$

$$= \left( \frac{15}{\text{Faster speed} \times 60} \right) \text{ minutes}$$

$$= \left( \frac{15 \times 60}{60} \right) = 15 \text{ minutes}$$

### LEVEL-II

1. (5) AB = 60 km

$$\text{Ram's speed} = x \text{ kmph}$$

$$\text{Shyam's speed} = y \text{ kmph}$$

Case I,

$$\frac{60}{x} - \frac{60}{y} = 1$$

Case II,

$$\frac{60}{y} - \frac{60}{2x} = \frac{30}{60} = \frac{1}{2} \quad \dots \text{(ii)}$$

On adding equations (i) and (ii),

$$\frac{60}{y} - \frac{60}{2y} = 1 + \frac{1}{2}$$

Distance covered in 6 minutes

$$\frac{50}{60} \times 6 = 5 \text{ km}$$

$$\therefore \text{Distance covered in } (31 \times 12) \text{ minutes} = 12 \times 5$$

$$25 = 300 \text{ km}$$

$$\text{Total distance} = 300 + 7.5 = 307.5 \text{ km}$$

$$\therefore \text{Speed of tractor} = \frac{\text{Distance}}{\text{Time}}$$

$$= \frac{575}{23} = 25 \text{ kmph}$$

$$\therefore \text{Speed of bus} = 50 \text{ kmph}$$

$$\therefore \text{Speed of car} = \frac{9}{5} \times 50 = 90 \text{ kmph}$$

$$\therefore \text{Distance covered by car in 4 hours}$$

$$= 4 \times 90 = 360 \text{ km}$$

4. (3) Let the length of platform be  $x$  metre.

$$\therefore \frac{432 + x}{42} = 38$$

$$\Rightarrow 432 + x = 1596$$

$$\Rightarrow x = 1596 - 432$$

$$= 1164 \text{ metre}$$

Let  $n$  boxes be added,

$$\therefore 432 + 1164 + 42n = 42 \times 52$$

$$\Rightarrow 1596 + 42n = 2184$$

$$\therefore \text{Distance} = \text{Speed} \times \text{Time}$$

$$\therefore \text{Distance covered by scooter A in } t \text{ hours} + \text{Distance covered by scooter B in } (t-2) \text{ hour} = 140$$

$$\Rightarrow 50t + 30(t-2) = 140$$

$$\Rightarrow 80t - 60 = 140 \Rightarrow 80t = 200 \Rightarrow t = \frac{200}{80} = \frac{5}{2}$$

hours

i.e. after 2 hours 30 minutes i.e. at 12 : 30 p.m.

8. (3) Speed of first train = 72 kmph.

$$= \left( \frac{72 \times 5}{18} \right) \text{ metre/second}$$

$$= 20 \text{ metre/second}$$

$$\text{Its length} = x \text{ metre}$$

$$\therefore \text{Length of platform} = 2x \text{ metre}$$

According to the question,

$$\frac{x + 2x}{20} = 45$$

$$\Rightarrow 3x = 45 \times 20 = 900$$

$$\Rightarrow x = \frac{900}{3} = 300 \text{ metre}$$

Second case.

$$\text{Speed of second train} = 54 \text{ kmph.}$$



## 10. (4) Time or Speed

$$\therefore \frac{x+20}{x} = \frac{4}{3}$$

$$\Rightarrow 4x = 3x + 60$$

$$\Rightarrow x = 60 \text{ kmph}$$

Case II.

$$\text{Speed} = x + 30 = 90 \text{ kmph}$$

$$\text{Distance} = 480 \text{ km.}$$

$\therefore$  Required time

$$= \frac{480}{90} = \frac{16}{3} \text{ hours}$$

$$= 5 \text{ hours } 20 \text{ minutes}$$

11. (4) When R started, P is ahead by  $= 40 \times 5 = 200$  metre

As P and R run in the same direction, relative speed of R

$$= 60 - 40 = 20 \text{ metre/minute}$$

$$\text{So, the time taken by R to catch P} = \frac{200}{20} = 10 \text{ minutes}$$

$$\therefore \text{Required distance covered by R} = 10 \times 60 = 600 \text{ metre}$$

12. (5) Distance covered in 2 hours 15 minutes

$$\text{i.e. } 2\frac{1}{4} \text{ hours}$$

$$= 60 \times \frac{9}{4} = 180 \text{ km}$$

Speed of train B

$$= \frac{\text{Length of train B}}{\text{Time taken in crossing}}$$

$$= 25 = \frac{x}{12}$$

$$\Rightarrow x = 25 \times 12 = 300 \text{ metre}$$

$\therefore$  Velocity of the stream

$$= 2 \text{ kmph}$$

Speed of the boat in still water is 15 kmph

Speed downstream

$$= (15 + 2) \text{ kmph} = 17 \text{ kmph}$$

Speed upstream

$$= (15 - 2) \text{ kmph} = 13 \text{ kmph}$$

Let the distance between A and B be  $x$  km.

$$\therefore \left( \frac{x}{17} \right) + \left( \frac{x}{13} \right) = 30$$

$$\Rightarrow \frac{x}{17} + \frac{x}{13} = 30$$

$$\Rightarrow \frac{26x + 17x}{442} = 30$$

$$\Rightarrow 43x = 30 \times 442$$

$$\Rightarrow x = \frac{30 \times 442}{43} = 308 \text{ km.}$$

15. (4) Time taken by B =  $x$  minutes.

Time taken by A =  $(x + 10)$  minutes.

$$= \left( 60 + \frac{5}{10} \right) \text{ m/sec.}$$

$$= 2\frac{1}{2} \text{ m/sec}$$

$\therefore$  Relative speed of A

Length of train A and B

$\times$  Time taken in crossing

$$\Rightarrow 35 + 25 = \frac{x + \frac{x}{2}}{2}$$

$$\Rightarrow 60 \times 2 = \frac{x + \frac{x}{2}}{2}$$

$$\Rightarrow 3x = 90 \times 2 \Rightarrow x = \frac{90 \times 12}{3}$$

$$= 360 \text{ metre}$$

$\therefore$  Time taken in crossing the platform

$$= \frac{\text{Length of train A and platform}}{\text{Speed of train A}}$$

$$= \left( \frac{360 + 690}{35} \right) \text{ seconds}$$

$$= \frac{1050}{35} \text{ seconds} = 30 \text{ seconds}$$

$$17. (4) C = k \cdot x^2$$

$$\text{When } x = 16 \text{ kmph}$$

Rate downstream of boat A

$$= 10 + x + y = 10 \text{ kmph}$$

$$\Rightarrow \frac{40}{10 + x + y} = 3$$

$$\Rightarrow \frac{40}{10 + x} = 3$$

$$\Rightarrow 30 - 5x = 40$$

$$\Rightarrow 5x = 50 - 40 = 10$$

$$\Rightarrow x = \frac{10}{5} = 2 \text{ kmph}$$

$\therefore$  Speed of boat A in still water

$$= 10 - 2 = 8 \text{ kmph}$$

20. (4) Speed of boat A in still water =  $x$  kmph

Speed of boat B in still water =  $3x$  kmph

Speed of current =  $y$  kmph

According to the question,

$$\frac{80}{x + y} + \frac{80}{x - y} = 30$$

$$\Rightarrow 80 \left( \frac{x + y + x - y}{x^2 - y^2} \right) = 30$$

$$\Rightarrow 16x = 3(x^2 - y^2)$$

Again,

$$80 = 30 \dots (i)$$

$$\Rightarrow 16x = 3 \times \frac{8x^2}{9}$$

$$\Rightarrow 6x = x^2 \Rightarrow x^2 - 6x = 0$$

$$\Rightarrow x(x - 6) = 0$$

$$\Rightarrow x = 6$$

$\therefore$  Speed of boat B in still water =  $6 \times 3 = 18$  kmph

**21. (4)** Distance covered by car in 2 hours

$$= \frac{300 \times 40}{100} = 120 \text{ km}$$

Remaining distance

$$= 300 - 120 = 180 \text{ km}$$

$$\text{Remaining time} = 4 - 2$$

$$= 2 \text{ hours}$$

$$\therefore \text{Required speed} = \frac{180}{2}$$

$$= 90 \text{ kmph}$$

$$\text{Original speed of car} = \frac{120}{2}$$

$$= 60 \text{ kmph}$$

$\therefore$  Required increase in speed

$$= 90 - 60 = 30 \text{ kmph}$$

**22. (1)** Speed of flight A =  $x$  kmph.

Speed of flight B =  $y$  kmph.

Case 1.

Distance

$$\Rightarrow \frac{1440}{y} = \frac{8}{5}$$

$$\Rightarrow 8y = 1440 \times 5$$

$$\Rightarrow y = \frac{1440 \times 5}{8} = 900 \text{ kmph.}$$

From equation (ii).

$$\frac{7200}{x} - \frac{7200}{900} = 1$$

$$\Rightarrow \frac{7200}{x} - 8 = 1$$

$$\Rightarrow \frac{7200}{x} = 9$$

$$\Rightarrow x = \frac{7200}{9} = 800 \text{ kmph.}$$

OR

Let consider flight B

To cover 7200 km distance it takes 1 hour 36

minutes =  $\frac{8}{5}$  hour more when its speed de-

creases by  $\frac{1}{6}$  factor.

Let initial speed of flight B =  $6x$

$$\text{Speed of A} = \frac{7200}{9}$$

$$= 800 \text{ km/hr.}$$

**23. (1)** Speed of train Y =  $x$  kmph.

$\therefore$  Speed of train X =  $(x - 10)$  kmph.

Relative speed =  $(2x - 10)$  kmph.

$\therefore$  Relative speed  $\times$  Time = Distance

$$\Rightarrow (2x - 10) \times \frac{11}{2} = 396$$

$$\Rightarrow 2x - 10 = \frac{396 \times 2}{11} = 72$$

$$\Rightarrow 2x = 72 + 10 = 82$$

$$\Rightarrow x = \frac{82}{2} = 41 \text{ kmph.}$$

$$\xrightarrow{\text{A}} \text{X} \xrightarrow{\text{A}} \text{P} \xrightarrow{\text{A}} \text{B} \xrightarrow{\text{A}} \text{Y}$$

**24. (2)**

Let speed of B =  $10x$  km/hr

Speed of A =  $12x$  km/hr

Let Distance (XY) =  $60$  D km

L.C.M. of 10 & 12

Assume that they meet after  $t$  hours.

$$\text{Time taken by B} = \frac{60D}{10x} = \frac{6D}{x}$$

$$= 12x$$

$$\text{Also distance travelled by A}$$

$$= 12x$$

Dist

$$\Rightarrow \frac{36}{5} = \frac{6D}{x} = \frac{36D}{11x}$$

$$\Rightarrow \frac{36}{5} = \frac{6D}{x} = \frac{36D}{11x}$$

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$$\text{25. (3)} \quad \text{X} \xrightarrow{\text{A}} \text{P} \xrightarrow{\text{A}} \text{B} \xrightarrow{\text{A}} \text{Y}$$

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$$\text{X} \xrightarrow{\text{A}} \text{P} \xrightarrow{\text{A}} \text{B} \xrightarrow{\text{A}} \text{Y}$$

Let the distance between X and Y = D km

Speed of A = 33 km/hr

Let speed of B =  $x$  km/hr

Suppose they meet after T hours

$\Rightarrow$  Distance travelled by A

$$\Rightarrow x = \sqrt{\frac{49 \times 11 \times 11}{9}}$$

$$x = \frac{77}{3}; x = 25\frac{2}{3}$$

26. (3) Total time taken by delivery boy  
= 1 hr 25 min - 15 min  
= 1 hr 10 min

$$= 1\frac{10}{60}$$

$$= \frac{7}{6} \text{ hrs}$$

$$A \xrightarrow{t_1} \text{---} \text{---} \text{---} B$$

$$32 \text{ km/hr} \quad \text{---} \quad 24 \text{ km/hr}$$

Now it is very clear,

$$t_1 + t_2 = \frac{7}{6} \quad \dots (1)$$

Distance travelled from A to B = Distance  
travelled from B to A

$$\Rightarrow 32 \times t_1 = 24 \times t_2$$

$$4t_1 = 3t_2$$

If  $\frac{40}{V_1} = \frac{50}{V_2}$  then they will collide i.e. cars will  
reach at the same time.

$$\therefore \frac{V_1}{V_2} = \frac{40}{50} = \frac{4}{5} = 4 : 5$$

28. (2) Relative speed =  $50 + 50 = 100 \text{ kmph}$

Time taken by cars in meeting =  $\frac{100}{100} = 1 \text{ hour}$

$\therefore$  Distance covered by the bird =  $100 \times 1$   
= 100 km

29. (1) A's speed =  $\frac{1000}{5} = 200 \text{ m/minute}$

B's speed =  $\frac{1000}{8} = 125 \text{ m/minute}$

C's speed =  $\frac{1000}{10} = 100 \text{ m/minute}$

Distance covered by C in 2 minutes =  $200$   
metre/minute

Distance covered by B in 1 minute =  $125 \text{ m/min}$

Relative speed of A with respect to C =  $100$   
metre/minute

$\therefore$  Time =  $\frac{200}{100} = 2 \text{ minutes}$

If means both of them together cover a  
distance of 4000 miles between them in 60  
min or 1 hour miles 1 min

therefore, they should be 1000 miles apart, 1 min  
before impact

3. (3) According to question, relative speed = 18.  
Distance = 3600 m

Time taken to cover this distance =  
 $\frac{3600}{18} = 200 \text{ s}$

4. (3) Let the speed of the car be  $x \text{ km/h}$ .

$\therefore$  Speed of train =  $x \times 1.2x$   
 $= \frac{6x}{5} \text{ km/h}$

$$\therefore \frac{5}{6x} \times 75 + \frac{25}{2 \times 60} = \frac{75}{x}$$

$$\Rightarrow x = \frac{75 \times 2 \times 60}{6 \times 25} = 60 \text{ km/h}$$

5. (1) Let second car overtake the first car after  
 $n \text{ hr}$ .

$\therefore$  Distance covered by first car = Distance  
covered by second car

$$8x = 8 \left( 8 + \frac{1}{n} \right) + \left( 8 + \frac{2}{n} \right) \times (n-1)$$

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}}$$

$$550 = \frac{2x \cdot 770}{x + \frac{x-770}{660}}$$

$$\Rightarrow \frac{2x \cdot 770}{660} = \frac{x \cdot x - 770}{660}$$

$$\Rightarrow x = 1760$$

$$\therefore \text{Total distance covered}$$

$$= 2x \cdot 770 = 2 \times 1760 \cdot 770$$

$$= 2750 \text{ km}$$

$$\therefore \text{Time taken by A to complete one round} =$$

$$\frac{144}{6} = 24 \text{ minutes}$$

$$\text{Time taken by B to complete one round} =$$

$$\frac{144}{8} = 18 \text{ minutes}$$

$$\text{Time taken by C to complete one round}$$

$$= \frac{144}{12} = 12 \text{ minutes}$$

$$\text{LCM of 24, 18 and 12}$$

$$= 72 \text{ minutes} = 1 \text{ hr}$$

# TIME AND DISTANCE

Since, Barun & Kiranmala overtake Arun at the same instant, Kiranmala has to cover a distance of 240 km such that she reaches the point at the same instant when Arun and Barun are there. Kiranmala will take  $\frac{240}{60} = 4$  hrs to be at that point.

**Method-2 :**

|            |      |      |     |
|------------|------|------|-----|
|            | A :  | B :  | K : |
| Speed      | 30 : | 40 : | 60  |
| or,        | 3 :  | 4 :  | 6   |
| Time taken | 1 :  | 1 :  | 1 : |
|            | 3 :  | 4 :  | 6 : |
| or,        | 4 :  | 3 :  | 2 : |

∴ (Since distance travelled in same ratio of time is reciprocal of ratio of speed)  
Barun starts 2 hours after Arun.

Let, distance travelled by Ram be  $x$ , when he has overtaken by Shyam. Time taken by Ram to travel  $x$  km (from B) is same as time taken by Shyam to travel  $(x + 2.5)$  km.  
Hence,  $\frac{x}{5} = \frac{2.5 + x}{10} \Rightarrow x = 2.5$  km.

2.5 km would be covered by Ram in  $\frac{1}{2}$  an hour or 30 minutes.  
At 10 a.m., Ram is at point B, 30 minutes after 10 a.m. is 10.30 a.m. (You can also check by options.)

14. (3) Since both the runners are starting from the same point, at the same time, in the same direction (since it is a race), hence the fastest runner has completed one round more than the slowest runner at end of the 5th minute.

# TIME AND DISTANCE

17. (1) Distance covered on foot, cycle, scooter & car is same i.e.  $\frac{4800}{4} = 1200$  km.  
Let speed of scooter be  $s$ , then speed of car is  $2s$ , that of cycle is  $(2s - 20)$  and speed on foot is  $(2s - 40)$   
Since it takes 4 hours more, when on foot as compared when in car, so

$$\frac{1200}{2s - 40} - \frac{1200}{2s} = 4$$

On solving we get,

$$4s^2 - 80s - 1200 = 0 \text{ or } s^2 - 20s - 300 = 0$$

$$\Rightarrow s^2 - 30s + 10s - 300 = 0$$

$$\Rightarrow s(s - 30) + 10(s - 30) = 0$$

$$\Rightarrow [s + 10](s - 30) = 0 \Rightarrow s = -10, 30$$

where  $s$  (speed of scooter) = 30, speed of car = 60.

$$\text{speed of cycle} = 60 - 20 = 40$$

$$60 - 40 = 20$$

21. (5) The lions are running in the opposite direction. Time when they would meet each other is  $\frac{250}{(18 + 36)} = \frac{250}{54} = \frac{25}{5.4} = \frac{250}{54} \times \frac{60}{60} = \frac{250}{9} \times \frac{10}{3} = \frac{2500}{27} \text{ sec.}$

Distance travelled by deer in  $40/3$  seconds

$$\text{at speed of } \left( \frac{215}{5} \times \frac{5}{18} \right) \text{ m/s} = \frac{40}{3} \times 12 = \frac{480}{3} \text{ m} = 160 \text{ m}$$

22. (4) Neil & Bipasha would meet in  $\frac{240}{12 - 8} = 60$  seconds



## TIME AND DISTANCE

25. (3) N is beaten by half a round or 5 sec, i.e., one round is covered by N in 10 seconds, hence 2 rounds would take 20 seconds. M has covered 2 rounds in 5 seconds less than N i.e., 15 seconds.

2 rounds are covered by M in 15 sec.

$\therefore$  3 rounds would take 22.5 sec.  $(15 + 7.5)$

26. (2) Let Shyama take 1 minute for every 3 steps

he takes. This means he requires  $\frac{1}{3}$  minute

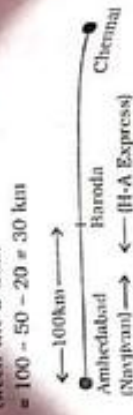
to take one step. However, since Vyom is taking only 2 steps in time when Shyama is taking 3 steps, Vyom will take  $\frac{1}{2}$  minute per step.

In 1 minute when Shyama takes 3 steps

The other train starts at 7:00 am, hence has travelled for 30 minutes and will cover a distance of  $\frac{40}{2} = 20$  km. Remaining distance between the 2 train

$= 100 - 50 - 20 = 30$  km

$\leftarrow 100 \text{ km} \rightarrow$



This distance of 30 km would be covered at relative speed of  $50 + 40 = 90$  km/hr, since both are travelling in opposite direction.

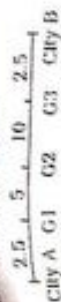
Remaining Time =  $\frac{30}{90}$

## TIME AND DISTANCE

$$1. G2G3 + G2G3 = 15$$

$$G2G3 = 10 \text{ km Hence } G1G2 = 5 \text{ km}$$

We get the following diagram



total time taken to go from City A to G3 and then return to City A

$\therefore$  time taken to cover  $[AG1 + G1G2 + G3A]$

$$= \frac{5}{60} + \frac{15}{60} + \frac{17.5}{60}$$

$$= \frac{37.5}{60} \times 60 \text{ minutes} = 37.5 \text{ minutes}$$

$$\Rightarrow 30 = \frac{x + 365}{21}$$

$$\Rightarrow 20 \times 21 = x + 365$$

$$\Rightarrow 420 = x + 365$$

$$\Rightarrow x = 420 - 365 = 255 \text{ metre}$$

Quantity II : Average speed of car

$$= \frac{\text{Distance covered}}{\text{Time taken}}$$

$$= \frac{232}{4} = 58 \text{ kmph}$$

$\therefore$  Average speed of bike

$$= \frac{58 \times 150}{100} = 87 \text{ kmph}$$

$$\Rightarrow 42x = 36x + 48 \times 36$$

$$\Rightarrow 6x = 48 \times 36$$

$$\Rightarrow x = \frac{48 \times 36}{6} = 48 \times 6 = 288.$$

$$\text{Distance} = 2 \times 288 + 48 = 624 \text{ km}$$

Quantity I > Quantity II

5. (3) **Quantity I** : Length of both trains = 250 metre

Speed of second train =  $x$  kmph

Relative speed

$$= (65 + x) \text{ kmph}$$

$$= (65 + x) \times \frac{5}{18} \text{ m/sec}$$

$\therefore$  Time

$$= \frac{\text{Sum of lengths of trains}}{\text{Relative speed}}$$

$$\Rightarrow 6 = \frac{250}{(65 + x) \times \frac{5}{18}}$$

$$\Rightarrow 6 \times \frac{5}{18} \times (65 + x) = 250$$

$$\Rightarrow 65 + x = \frac{250 \times 3}{5}$$

$$\Rightarrow 65 + x = 150$$

$$\Rightarrow x = 150 - 65 = 85 \text{ kmph}$$

**Quantity II** : Speed of car =  $x$  kmph

$$\text{Speed of train} = \frac{3x}{2} \text{ kmph}$$

$$\therefore \frac{75}{x} - \frac{75}{\frac{3x}{2}} = \frac{125}{60}$$

$$\Rightarrow 75 \left( \frac{1}{x} - \frac{2}{3x} \right) = \frac{5}{24}$$

$$\Rightarrow 75 \left( \frac{3-2}{3x} \right) = \frac{5}{24}$$

$$\Rightarrow \frac{75}{3x} = \frac{5}{24}$$

$$\Rightarrow \frac{15}{x} = \frac{1}{8}$$

$$\Rightarrow x = 15 \times 8 = 120 \text{ kmph}$$

Quantity I < Quantity II

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# LEVEL

1. What amount a man w  
a principal of Rs. 4,00  
simple interest at the  
annum ?  
(1) Rs. 4,161  
(3) Rs. 4,400  
(5) Rs. 4,190
2. A sum of Rs. 3200 beco  
years at a certain rate of  
is the rate of interest pe  
(1) 5.5% (2) 6%  
(5) None of these
3. Simple interest on a cert  
annual rate of interest is  
the numbers representin  
time in years be equal. th  
est is  
(1) 4% (2) 6% (3)  
(5) None of these
4. What will be the differ  
terest accrued on a sum  
per annum for 2 years an  
Rs. 5600 at 9% per annu  
(1) ₹ 75 (2) ₹ 72 (3)  
(5) None of these
5. A sum of money at simpl  
to Rs. 14,160 in 3 years. If  
is increased to 25%, the s  
to Rs. 14,700 in the same t  
terest is  
(1) 5% (2)  $5\frac{1}{2}\%$  (3)  
(5) None of these
6. Veena obtained an amoun  
simple interest on a cert  
p.c.p.a. after 6 years. What  
vested by Veena ?  
(1) Rs. 17,180 (2) F  
(3) Rs. 16,660 (4) F  
(5) None of these
7. A certain amount earns simple  
1,750 after 7 years. Had the in  
more, how much more interest  
earned ?  
(1) Rs. 35 (2) Rs. 350 (3) Rs. 2

the simple interest for 7 years  
the amount deposited by:  
(1) ₹ 9000 (2)  
(3) ₹ 9200 (4)  
(5) 12%

28. Find the value of  $r$  if rate  
and SI for 2 years for a sum  
of ₹ is 2684.  
(1) 10% (2) 20% (3)  
(4) None of these (5)

29. Two equal sums were lent  
S.I. respectively. The interest  
two loans add up to ₹ 9  
total sum lent  
out is  
(1) ₹ 3500 (2) ₹ 2500 (3)  
(5) ₹ 4000

30. The simple interest accrued  
principal of ₹ 24000 is on  
principal. What is the rate  
p.c.p.a. ?  
(1) 5 (2) 4.5 (3)  
(5) None of these

31. How much will a sum of  
a rate of 9% per annum  
13 years amount to ?  
(1) ₹ 14.040  
(3) ₹ 13.404  
(5) ₹ 26.040

32. The sum invested in scheme  
sum invested in scheme  
scheme A is made for 3 years  
interest and in Scheme B  
p.a. simple interest. The interest  
from both the schemes is ₹ 1400  
was invested in Scheme A  
(1) Rs. 4000 (2)  
(3) Rs. 3000 (4)  
(5) Rs. 4500

33. Ravi invested Rs. P in a scheme  
simple interest at 10% p.a. for 2 years  
vested the whole amount in  
scheme A. In another scheme B  
simple interest at 12% p.a. for 3 years  
difference between the interest  
schemes A and B was ₹ 2000  
value of P?  
(1) 2500 (2) 2000 (3)  
(5) 3200



paid amount as well as  
as simple interest for an  
same rate of interest. What  
interest will he obtain after  
years?

(1) ₹ 4,800

(3) ₹ 4,801.28

(5) None of these

16. Simple interest on a certain  
annual rate of interest for  
the numbers represent  
time in years be equal. The  
rate is

(1) 4% (2) 6%

(4) 6.5% (5) None of these

17. The rate of interest on  
the first two years is 6%  
years it is 7% p.a. and 8%  
exceeding four years; all  
a person earns an interest  
end of the seven years.  
at the end of the period

(1) ₹ 15.072

(3) ₹ 22.608

(5) ₹ 20.000

18. A sum of ₹ 5,000 lent  
amounts to ₹ 5,700 in 4 years.  
interest is increased by 2%  
what would the sum amount  
period?

(1) ₹ 4200

(3) ₹ 5,000

(5) Cannot be determined

19. Neil invests an amount of  
interest equal to 'r%' for a  
Had he invested at  $(r + 5)\%$   
earned ₹ 750 more. Find the

(1) ₹ 4000

(2)

(3) ₹ 5000

(4)

(5) ₹ 2500

20. A certain amount is lent  
interest for two years. Instead  
was lent at  $2x\%$  p.a. simple  
more years, then the interest  
been five times the earlier  
the value of  $y$ ?

(1) 2 years

(3) 4 years

(5) 6 years

## SIMPLE INTEREST

8. Quantity 1: ₹ 55,720 amounts to ₹ 79,000 in four years at simple interest. What is the rate of interest paid?

Quantity 2: If the interest paid for Sumit's sum ₹ 14,000 is ₹ 1260 for a period of 3 years, then find the rate of simple interest charged.

## DATA SUFFICIENCY QUESTIONS

Directions (1-5): Each question consists of a question and two statements, numbered 1 and 2 given below it. You have to decide whether the data provided in the statements is sufficient to answer the question. Read the question and both the statements and—

**Give answer (1)** If the data in statement 1 alone is sufficient to answer the question while the data in statement 2 alone is not sufficient to answer the question.

**Give answer (2)** If the data in statement 2 alone is sufficient to answer the question, while the data in statement 1 alone is not sufficient to answer the question.

**Give answer (3)** If the data either in

8. What is the principal?

1. The simple interest accrued on this sum at the rate of 12% per annum in 2 years is ₹ 288 less than the compounded interest on the same sum at 12% per annum in 2 years.

2. The sum doubles itself in 10 years at 10% per annum rate of simple interest.

## SHORT ANSWERS

## LEVEL-1

|         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|
| 1. (3)  | 2. (3)  | 3. (1)  | 4. (2)  | 5. (3)  | 6. (1)  |
| 7. (4)  | 8. (3)  | 9. (1)  | 10. (1) | 11. (3) | 12. (4) |
| 13. (2) | 14. (1) | 15. (2) | 16. (2) | 17. (2) | 18. (1) |
| 19. (2) | 20. (2) | 21. (4) | 22. (2) | 23. (4) | 24. (3) |
| 25. (1) | 26. (2) | 27. (1) | 28. (2) | 29. (3) | 30. (3) |
| 31. (5) | 32. (3) | 33. (1) | 34. (2) | 35. (2) |         |

## LEVEL-2

## SIMPLE INTEREST

6. (1) Let the annual interest be Rs.  $x$ .

$$\begin{aligned} \text{Principal} &= \frac{\text{Interest} \times 100}{\text{Rate} \times \text{Time}} \\ &= \frac{1000 \times 100}{10 \times 10} = 1000 \end{aligned}$$

7. (1) Let the principal be Rs.  $x$  and rate of interest be  $R$ .

$$\text{Case I: } \frac{x \times R \times 7}{100} = 1790$$

$$\Rightarrow xR = \frac{1790 \times 100}{7}$$

$$= \text{Rs. } 25571.43$$

Case II:

$$\frac{x \times R \times 3}{100} = 1790$$

$$\Rightarrow xR = \frac{1790 \times 100}{3}$$

$$= \text{Rs. } 59666.67$$

Which cannot be determined.

$$\text{Rs. } \left[ \frac{1000 \times 2 \times 7}{100} \right] = \text{Rs. } 140$$

Amount received

$$= \text{Rs. } (1000 + 140) = \text{Rs. } 1140$$

$$\therefore \text{Interest} = 1140 - 1000 = 140$$

$$\therefore \text{Rs. } 140$$

$$\therefore \text{Rate} = \frac{\text{Interest} \times 100}{\text{Principal} \times \text{Time}}$$

$$= \frac{140 \times 100}{1000 \times 2} = 7\%$$

$$\therefore \text{Rate} = 7\% \text{ per annum}$$

$$\therefore \text{Rate} = 7\% \text{ per annum}$$

$$\therefore \text{Rate} = 7\% \text{ per annum}$$

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$$\therefore \text{Rate} = 7\% \text{ per annum}$$

## SIMPLE INTEREST

11. (3) Increase in interest in 3 years due to increase in rate by 4%

$$= \frac{800 \times 3 \times 4}{100} = \text{Rs. } 96$$

Total amount at the end of 3 years

$$= \text{Rs. } 956 + \text{Rs. } 96$$

$$= \text{Rs. } 1052$$

12. (4) S.I. on Rs. 12000

$$= \frac{12000 \times 8 \times 1}{100} = \text{Rs. } 960$$

Desired gain on Rs. 20000

$$= 20000 \times \frac{10}{100} = \text{Rs. } 2000$$

$$\therefore \text{S.I. on Rs. } 8000 = 2000 - 960$$

$$= \text{Rs. } 1040$$

$$\therefore \text{Rate} = \frac{\text{S.I.} \times 100}{\text{Principal} \times \text{Time}}$$

$$= \frac{1040 \times 100}{8000}$$

$$= 13\% \text{ per annum}$$

$$\text{Principal} \times \text{Time} \times \text{Rate}$$

$$15. (2) \text{Principal} = \frac{\text{S.I.} \times 100}{\text{Time} \times \text{Rate}}$$

$$= \frac{1 \times 100}{\frac{1}{365} \times 5} = 365 \times 100$$

$$= \text{Rs. } 7300$$

16. (2) Amount after 10 years

$$= P \left( 1 + \frac{R \times T}{100} \right) = P \left( 1 + \frac{R \times 10}{100} \right)$$

$$= \text{Rs. } P \left( 1 + \frac{R}{10} \right)$$

$$\therefore \text{Interest} = \text{Rs. } P \left( 1 + \frac{R}{10} \right) \times \frac{2}{5}$$

$$\therefore \text{Rate} = \frac{\text{S.I.} \times 100}{\text{Principal} \times \text{Time}}$$

$$\Rightarrow R = \frac{P \left( 1 + \frac{R}{10} \right) \times \frac{2}{5} \times 100}{P \times 10}$$

$$\Rightarrow R = 4 \left( 1 + \frac{R}{10} \right)$$

$$R = R$$

## SIMPLE INTEREST

$$\Rightarrow \frac{22x}{100} = \frac{23y}{100} = \frac{24z}{100}$$

$$\text{H.C.M. of } 22, 23 \text{ and } 24 = 10772$$

$$\Rightarrow \frac{x}{276} = \frac{y}{264} = \frac{z}{253}$$

$$\therefore x : y : z = 276 : 264 : 253$$

Sum of terms of ratio

$$= 276 + 264 + 253 = 793$$

$$\therefore \text{A's loan} = \frac{276}{793} \times 7930$$

$$= \text{Rs. } 2760$$

18. (1) According to the question,

$$\text{S.I. for 2 years 6 months}$$

$$= \text{Rs. } (5500 - 4000)$$

$$\Rightarrow \text{S.I. for } \frac{5}{2} \text{ years} = \text{Rs. } 1500$$

$$\therefore \text{S.I. for 1 year} = \frac{1500 \times 2}{5}$$

$$= \text{Rs. } 600$$

$$\therefore \text{S.I. for 2 years} = \text{Rs. } 1200$$

$$\therefore \text{Principal} = \text{Rs. } (4000 - 1200) = \text{Rs. } 2800$$

$$\therefore \text{Rate} = \frac{\text{S.I.} \times 100}{\text{Principal} \times \text{Time}}$$

$$\text{Let sum of Rs. } 225000 \times \frac{1}{4}$$

$$= \text{Rs. } 56250 \text{ is invested at the rate of } r\%$$

$$\therefore \text{S.I.} = \frac{\text{Principal} \times \text{Time} \times \text{Rate}}{100}$$

$$\frac{56250 \times r \times 2}{100} = \frac{18750 \times (r - 5) \times 2}{100}$$

$$= 4125$$

$$\Rightarrow \frac{12500r}{100} = \frac{37500r - 18750 \times 2}{100}$$

$$= 4125$$

$$= 125r = 375r - 1500 = 4125$$

$$\Rightarrow 500r = 4125 - 1500 = 2625$$

$$\Rightarrow r = \frac{2625}{500} = 5.25$$

$$\therefore \text{Required answer}$$

$$= 5.25 = 4\frac{1}{4}\% \text{ per annum}$$

$$= 9.25\% \text{ per annum}$$

21. (4) Let the sum = Rs. x

According to the question

$$\frac{x \times 4 \times 3}{100} = \frac{x \times 8 \times 4}{100} - \frac{x \times 12 \times 4}{100} = 27500$$

$$\begin{aligned}
 &= \left( x + x \times \frac{2}{12} \times \frac{10}{100} \right) + \left( x + x \times \frac{1}{12} \times \frac{10}{100} \right) + x \\
 &\Rightarrow 500 + 12.5 = 3x + \frac{10}{1200}x \\
 &\Rightarrow 512.5 = 3x + \frac{x}{120} \Rightarrow x = \frac{512.5 \times 40}{121} = ₹ 169.42 \\
 \text{24. (3) Let amount of each instalment} &= ₹ x, \text{ then} \\
 10,500 &= \text{Present worth of instalment due after} \\
 1 \text{ year} + \text{Present worth of instalment due after} & \\
 2 \text{nd year.} & \\
 10,500 &= \frac{x}{\left(1 + \frac{15}{100}\right)} + \frac{x}{\left(1 + \frac{15}{100}\right)^2} \\
 &\Rightarrow 10,500 = \frac{100}{115}x + \frac{100}{115} \times \left(\frac{100}{115}\right)x \\
 &\Rightarrow 10,500 = \frac{100}{115}x \left(1 + \frac{100}{115}\right) \\
 &\Rightarrow 10,500 = \frac{100 \times 215}{115 \times 115}x \\
 &\Rightarrow x = \frac{10,500 \times 115 \times 115}{100 \times 215} = 6458.7 = ₹ 6458 \\
 & \quad \text{(approx.)}
 \end{aligned}$$

$$\begin{aligned}
 &\Rightarrow \frac{54}{100}P = 4800 \\
 &P = \frac{480000}{54} = ₹ 8900
 \end{aligned}$$

28. (2) Let 'P' be the fixed amount,  $r$  is the rate per cent given to us.

$$C.I. \text{ for 4 years} = P \left[ \left( 1 + \frac{r}{100} \right)^4 - 1 \right] = P \times 8$$

$$S.I. \text{ for 2 years} = \frac{P \times r \times 2}{100} = 50$$

$$\text{We are given that } \frac{P \left( 1 + \frac{r}{100} \right)^4 - P}{\frac{P \times r}{50}} = 2.684$$

$$\Rightarrow P \left[ \left( 1 + \frac{r}{100} \right)^4 - 1 \right] = \frac{Pr \times 2.684}{50}$$

$$\Rightarrow \left( 1 + \frac{r}{100} \right)^4 - 1 = \frac{2.684}{50}r$$

Checking options, (1) When  $r = 10\%$ , then

$$(1.1)^4 - 1 = \frac{2.684 \times 10}{50} \Rightarrow 0.4641 \neq 0.5366$$

## SIMPLE INTEREST

Method-2 :

$$\frac{S.I.}{\text{Principal}} = \frac{1}{R}$$

$$\therefore \text{Rate} = \frac{S.I.}{\text{Principal}} \times \frac{100}{\text{Time}}$$

$$= \frac{1}{8} \times \frac{100}{2} = \frac{25}{4}$$

$$= 6.25\% \text{ per annum.}$$

$$\therefore S.I. = \frac{12000 \times 9 \times 13}{100} = ₹ 14040$$

$$\therefore \text{Amount} = 12000 + 14040 = ₹ 26040$$

32. (3) Amount invested in scheme A =  $P_A \times 0.01$

$\therefore$  Amount invested in scheme B =  $P_B \times 2x$

$$S.I. = \frac{\text{Principal} \times \text{Time} \times \text{Rate}}{100}$$

According to the question,

$$\frac{x \times 3 \times 8}{100} + \frac{2x \times 2 \times 9}{100} = 1800$$

$$\Rightarrow \frac{24x + 36x}{100} = 1800$$

$$\Rightarrow x = \frac{1800 \times 100}{60} = ₹ 3000$$

34. (2) Let the principal be  $P_A \times x$ .

According to the question,

$$x = 5.1 \text{ for 2 years} = ₹ 1008$$

$$x = 3.1 \text{ for } \frac{7}{2} \text{ years} = ₹ 1164$$

By equation (i) - (ii),

$$5.1 \text{ for } \frac{7}{2} \text{ years}$$

$$= P_A (1164 - 1008) = P_A \times 156$$

$$\Rightarrow 5.1 \text{ for 2 years}$$

$$= \frac{156 \times 2 \times 2}{3} = P_A \times 208$$

From equation (iii),

$$x = 258 = 1008$$

$$\therefore x = P_A (1008 - 258)$$

$$= P_A \times 750$$

$$\therefore \text{Rate} = \frac{5.1 \times 100}{\text{Principal} \times \text{Time}}$$

$$= \frac{258 \times 100}{750 \times 2} = 17\% \text{ per annum}$$



# SIMPLE INTEREST

$$= ₹ 31590$$

Remaining amount  
= ₹ (55500 - 31590)

$$= ₹ 26910$$

Investment in mutual fund

$$= 26910 \times \frac{2}{3} = ₹ 17940$$

∴ Required amount

$$= ₹ (26910 - 17940)$$

$$= ₹ 8970$$

3. (2) Required sum = ₹ P

Principal × Difference of rates

$$\frac{\text{Principal} \times \text{Time}}{100} = 72$$

$$= \frac{P \times 3 \times 2}{100} = 72$$

$$\Rightarrow P = \frac{72 \times 100}{3 \times 2} = ₹ 1200$$

4. (1) When T = 15

$$S.I. = \frac{6000 \times 15 \times 3}{100}$$

$$= ₹ 2700$$

$$[C, R, T]$$

$$\Rightarrow P_1 = \frac{2880 \times 100}{3 \times R}$$

$$= ₹ \frac{96000}{R}$$

Case II.

$$P_2 = \frac{4000 \times 100}{R \times 5} = ₹ \frac{80000}{R}$$

∴ Required percent

$$= \left( \frac{96000 - 80000}{80000} \right) \times 100$$

$$= \frac{16000}{80000} \times 100 = 20\%$$

6. (2) A's investment = Rs. x

B's investment

$$= ₹ (4000 - x)$$

$$S.I. = \frac{\text{Principal} \times \text{Time} \times \text{Rate}}{100}$$

According to the question,

$$\frac{x \times 5 \times 2}{100}$$

# SIMPLE INTEREST

8. (1) Amount invested in scheme B = Rs. x

Amount invested in scheme A = Rs. (13000 - x)

∴

$$S.I. = \frac{\text{Principal} \times \text{Time} \times \text{Rate}}{100}$$

$$\therefore \frac{x \times 11 \times 2}{100} + \frac{(13000 - x) \times 14 \times 2}{100} = 3508$$

$$\Rightarrow \frac{22x}{100} + \frac{36400 - 28x}{100} = 3508$$

$$\Rightarrow \frac{28x}{100} - \frac{22x}{100} = 36400 - 3508$$

$$\Rightarrow \frac{6x}{100} = 384$$

$$\Rightarrow x = \frac{384 \times 100}{6} = ₹ 6400$$

$$\Rightarrow x = \frac{384 \times 100}{6} = ₹ 6400$$

9. (5) Case I.

$$S.I. = \frac{\text{Principal} \times \text{Time} \times \text{Rate}}{100}$$

$$\frac{P \times 2 \times 12}{100} = ₹ \frac{24P}{100}$$

$$\therefore \text{Amount} = ₹ \left( P + \frac{24P}{100} \right)$$

$$= \frac{100 \times 23 \times 2}{100} = ₹ 46$$

Now, invested in scheme B

$$= ₹ (100 - 24) = ₹ 76$$

$$\therefore \text{Interest from scheme B}$$

$$= \frac{76 \times 15 \times 2}{100} = ₹ 22.8$$

Now, According to question

$$22.8 - 24 = 264$$

$$\Rightarrow 13.2 = 264$$

$$\Rightarrow 1 = \frac{264 \times 100}{132} = 200$$

$$\therefore P = 100 \text{ unit} = 100 \times 20$$

$$= ₹ 2000$$

$$10. (4) S.I. = \frac{P \times R \times T}{100}$$

$$\therefore y = \frac{x \times T \times R}{100}$$

$$\therefore z = \frac{y \times T \times R}{100}$$

$$\therefore \frac{y}{x} = \frac{z}{y} \Rightarrow y^2 = xz$$

$$1.$$

# SIMPLE INTEREST

$$= 20 \times 120000 - 20x + 480000 - 4x$$

$$\Rightarrow 50x = 2400000 + 480000$$

$$\Rightarrow 50x = 2880000$$

$$\Rightarrow x = \frac{2880000}{50} = \text{Rs } 57600$$

14. (1) Let amounts be equal in  $T$  years.

$$\text{S.I.} = \frac{\text{Principal} \times \text{Time} \times \text{Rate}}{100}$$

$$\therefore P + \frac{P \times x \times T}{100}$$

$$= Q + \frac{Q \times y \times T}{100}$$

$$= \frac{P \times T}{100} - \frac{Q \times y \times T}{100} = Q - P$$

$$= T \left( \frac{Px - Qy}{100} \right) = Q - P$$

$$\Rightarrow T = 100 \left( \frac{Q - P}{Px - Qy} \right)$$

# SIMPLE INTEREST

$$\Rightarrow x = 2 \times 7536 = \text{₹ } 15072$$

$$\therefore \text{Amount} = 15072 + 7536 = \text{₹ } 22608$$

18. (2) S.I. = 5700 - 5000 = ₹ 700

$$\therefore R = \frac{\text{S.I.} \times 100}{\text{Principal} \times \text{Time}}$$

$$= \frac{700 \times 100}{5000 \times 2} = 7\% \text{ per annum}$$

New rate = 11%

$$\therefore \text{S.I.} = \frac{5000 \times 11 \times 2}{100} = \text{₹ } 1100$$

$$\therefore \text{Amount} = \text{₹ } (5000 + 1100) = \text{₹ } 6100$$

19. (3) We are given that,

$$x \times \frac{5}{100} \times 3 = 750$$

$$\Rightarrow x = \frac{75000}{15} = \text{₹ } 5000$$

20. (2) Let the sum be ₹ 100

$$100 \times 2x \times (y+2)$$

# SIMPLE INTEREST

9. (3) According to question,

$$P \left( 1 + \frac{r}{100} \right)^2 = \frac{6}{5}$$

$$\Rightarrow \left( 1 + \frac{r}{100} \right)^2 = \frac{6}{5}$$

$$\Rightarrow \left( 1 + \frac{r}{100} \right)^2 = \frac{6}{5} \Rightarrow r = 20\%$$

9. (3) Let the rate of interest be  $r\%$ .

$$\text{Then, } A = 1200 + \frac{1200 \times r \times 7}{12 \times 100}$$

$$\Rightarrow A = 1200 + 7r$$

$$\text{Again } 1200 + 7r = 1016 + \frac{1016 \times r \times 2.5}{100}$$

$$\Rightarrow 1200 + 7r = 1016 + 25.4r$$

$$\Rightarrow 18.4r = 184$$

$$\therefore r = \frac{184}{18.4} = 10\%$$

# SIMPLE INTEREST

7. (2) Let the present value of money be  $x$

$$\therefore 1346 = x + 10 \times \frac{12 \times 19 \times 1}{100}$$

$$\Rightarrow 1346 = 10x + 1219$$

$$\Rightarrow x = \text{₹ } 771$$

So, amount invested at 12% is ₹ 500

7. (2) Let the present value of money be  $x$

$$\text{Then, } \frac{x \times 12}{100} = x = 10128$$

$$\Rightarrow 0.12x = x = 10128$$

$$\Rightarrow x = \frac{10128}{1.12}$$

This amount after 3 months

$$= \frac{10128 \times 12 \times 3}{12 \times 100} = \frac{10128}{1.12}$$

$$= \frac{10128 \times 3}{112 \times 100} = \frac{10128}{1.12}$$

Rate of 8% per annum payable half-yearly

Principal for next six months = ₹ 105

$$\therefore \text{S.I.} = \frac{105 \times 10 \times 6}{100 \times 12}$$

$$= ₹ 5.25$$

$$\therefore \text{Total interest} = ₹ 10.25$$

$$10.25 = \frac{100 \times 1 \times 1 \times r}{100}$$

$$\Rightarrow r = 10.25\%$$

11. (1) For first investment.

$$\text{S.I.} = \frac{\text{Principal} \times \text{Time} \times \text{Rate}}{100}$$

$$= \frac{x \times 5 \times 12}{100} = \text{Rs. } \frac{3x}{5}$$

$$\therefore \text{Total amount} = \text{Rs. } \left( x + \frac{3x}{5} \right)$$

$$= \text{Rs. } \frac{8x}{5}$$

Second investment.

$$\text{S.I.} = \frac{8x}{5} \times \frac{5 \times 12}{100}$$

$$= \text{Rs. } \frac{24x}{25}$$

$$\therefore \text{Total amount} = \frac{8x}{5} + \frac{24x}{25}$$

$$= \frac{40x + 24x}{25}$$

$$= \text{Rs. } \frac{64x}{25}$$

According to the question.

$$\frac{64x}{25} - x = 780$$

$$\Rightarrow \frac{64x - 25x}{25} = 780$$

$$\Rightarrow \frac{39x}{25} = 780$$

$$\Rightarrow x = \frac{780 \times 25}{39} = \text{Rs. } 500$$

12. (1) Let the annual payment be ₹  $x$ . Since the first instalment is to be paid after 1 year, the sum of amounts of ₹  $x$  for 3 years, of ₹  $x$  for 2 years, of ₹  $x$  for 1 year and ₹  $x$ , gives the amount of debt.

Now, amount of ₹  $x$  for three years

$$= ₹ \left[ x + \frac{x \times 8 \times 3}{100} \right] = ₹ \left( \frac{31x}{25} \right)$$

## SIMPLE INTEREST

$$P + S.I. = 2P$$

S.I. for 5 years = P

$$\frac{P \times R \times 5}{100} = P \Rightarrow R = 20\%$$

Now Put  $R = 20\%$  in (i)

$$\frac{P \times 20 \times 7}{100} = 7000$$

$$\Rightarrow \frac{P \times 20 \times 7}{100} = 7000$$

$$P = ₹ 5000$$

Hence, both are required.

3. (2) The data in statement II alone is sufficient to answer the question, while the data in statement I alone is not sufficient to answer the question.

4. (3) From statement II,

If principal = ₹ x, then interest after 10 years = ₹ x

$$x \times \text{Rate} \times 10$$

$$\Rightarrow r = \frac{10}{4} = 2\frac{1}{2}\%$$

Quantity I > Quantity II

Quantity I : S.I. = 70900 - 68750 = ₹ 21150

5. (1) Quantity I : S.I. = 70900 - 68750 = ₹ 21150

We know S.I. =  $\frac{P \times R \times T}{100}$

$$\text{Rate} = \frac{S.I. \times 100}{\text{Principal} \times \text{Time}}$$

$$= \frac{21150 \times 100}{68750 \times 4} = 9\% \text{ per annum}$$

Quantity II : 1260 =  $\frac{14000 \times 3 \times r}{100}$

$$1260 = 3\% \text{ per annum}$$

$$r = \frac{140 \times 3}{1260} \times 100 = 3\%$$

Method-2 : Interest for 3 years = ₹ 1260

$$1260 = ₹ 420$$

# 13 Compound Interest

## IMPORTANT FACTS AND FORMULAE

1. We have learnt earlier that if Principal = Rs. P, Rate = R% per annum and Time = T years then the simple interest is given by the formula

$$S.I. = \frac{P \times R \times T}{100}$$

Clearly, when money is borrowed on simple interest then the interest is calculated uniformly on the original principal throughout the loan period.

However, in post offices, banks, insurance corporations and the other money lending and deposit taking companies, the method of calculation

Now, Interest on Rs. 1100 at 10% for 1 year

$$= \frac{1100 \times 10 \times 1}{100} = \text{Rs. } 110$$

Hence, amount after 2 years

$$= \text{Rs. } 1100 + \text{Rs. } 110 = \text{Rs. } 1210$$

It is the final amount.

$$\therefore \text{C.I.} = \text{Final amount} - \text{Original amount}$$

$$= \text{Rs. } 1210 - 1000 = \text{Rs. } 210$$

4. Formulae.

(i) If P is the original Principal, R is the rate of interest per annum, T is the number of years, for



# COMPOUND INTEREST

10. The compound interest is calculated annually in general unless some other period is clearly otherwise stated.

11. The compound interest is always greater than the simple interest for the same period and at the same rate. However, the C.I. for one year is equal to the S.I. for one year if calculated annually.

## An important table

| Time →         | 1 year                           | 2 years                            | 3 years                            |
|----------------|----------------------------------|------------------------------------|------------------------------------|
| $R \downarrow$ | $\left(1 + \frac{R}{100}\right)$ | $\left(1 + \frac{R}{100}\right)^2$ | $\left(1 + \frac{R}{100}\right)^3$ |
| 10%            | $\frac{11}{10}$                  | $\frac{121}{100}$                  | $\frac{1331}{1000}$                |
|                | 21                               | 441                                | 9261                               |
|                | 100                              | 10000                              | 8000000                            |

6. If a certain sum becomes  $n$  times in  $T$  years, the rate of compound interest  $R$  is equal to  $100 \left[ \left( \frac{n}{1} \right)^{1/T} - 1 \right]$

7. If the difference between SI and CI on a certain sum of money for 2 years at  $R\%$  per annum is  $D$ . Then the sum (Principal)

$$P = \left( \frac{100}{R} \right)^2 \times D$$

8. If the difference between SI and CI for a certain sum of money for 3 years at  $R\%$  per annum is  $D$ . Then the sum (Principal)

$$P = \left( \frac{100}{R} \right)^3 \times D$$

# COMPOUND INTEREST

9. The difference between compound interest and simple interest on a certain sum at  $12\%$  per annum for two years is Rs. 90. What will be the value of the amount at the end of three years at compound interest?

- (1) Rs. 9000
- (2) Rs. 6250
- (3) Rs. 8530.80
- (4) Rs. 8780.60
- (5) Rs. 8800

10. Sudhanshu invested Rs. 15,000 at interest @ 10 p.c. p.a. for one year. If the interest is compounded every six months, what amount will Sudhanshu get at the end of the year?

- (1) Rs. 16,537.50
- (2) Rs. 16,500
- (3) Rs. 16,525.50
- (4) Rs. 18,150
- (5) None of these

11. The simple interest accrued on a sum of certain principal is Rs. 6,400 in four years at the rate of 8 p.c.p.a.

16. A person invested equal amounts in two schemes A and B at the same rate of interest. Scheme A offers simple interest, while scheme B offers compounded interest. After two years he got Rs. 1520 from scheme A as interest and Rs. 2112 from scheme B. If the rate of interest is increased by 4%, what will be the total interest after two years from both schemes?

- (1) Rs. 4254.48
- (2) Rs. 4254.48
- (3) Rs. 4254.94
- (4) Rs. 4254.48
- (5) None of these

17. A invests a certain sum in scheme A at compound interest (compounded annually) of 10% per annum for 2 years. In scheme B he invests at simple interest of 8% per annum

# COMPOUND INTEREST

interest at 10.5% per annum. what should be the period of his investment at simple interest so that the compound interest earned is only 75% of the simple interest earned?

- (1) 3 years  
(2) 4 years  
(3) 5 years  
(4) 6 years

(5) None of these

21. Aman invests a certain sum - Rs. 'X' for two years in scheme A offering compound interest at 20% p.a. (compounded annually). He also invests Rs. (x-1600) in scheme B for three years offering simple interest at 10% p.a. The interest he received from scheme A was twice the interest he received from scheme B. What is the sum of money that

interest at 10% in  $1\frac{1}{2}$  years compounded half yearly?

- (1) ₹ 40,000  
(2) ₹ 35,000  
(3) ₹ 32,000  
(4) ₹ 28,000  
(5) ₹ 30,000

26. If the compound interest on a sum for 2 years at  $12\frac{1}{2}$  p.a is ₹ 510, the simple interest on the same sum at the same rate for the same period of time is

- (1) ₹ 400 (2) ₹ 450 (3) ₹ 460 (4) ₹ 480  
(5) ₹ 500

27. A man borrows money at 3% per annum interest payable yearly and lend it at 10% per annum

# COMPOUND INTEREST

32. Rs. 260200 is divided between Ram and Shyam so that the amount that Ram receives in 4 years is the same as that Shyam receives in 6 years. If the interest is compounded annually at the rate of 4% per annum then Ram's share is

- (1) Rs. 125000  
(2) Rs. 135200  
(3) Rs. 152000  
(4) Rs. 165200  
(5) Rs. 120000

33. If the amount on a certain principal in 3 years at 12% rate of interest compounded annually is Rs. 12,000, what will be the amount in Rs. after the 4th year?

- (1) 14330 (2) 15440 (3) 13440 (4) 14550  
(5) 13000

40. A sum of ₹ 16,764 is paid back, along with interest, in 4 equal quarterly instalments. Find the amount of each instalment if the rate of interest charged is 17% p.a. compounded annually.

- (1) ₹ 5458.92  
(2) ₹ 4598  
(3) ₹ 5424  
(4) ₹ 3945  
(5) ₹ 3764

41. An amount of Rs. is distributed by paying 3 equal quarterly instalments of ₹ 9251 each. If the rate of interest charged was 25% per annum, compounded quarterly, then find the amount of loan borrowed.

- (1) ₹ 27220  
(2) ₹ 25239  
(3) ₹ 22720  
(4) None of these  
(5) Data insufficient

# COMPOUND INTEREST

invested by Rajni in each of the two banks, if the rate of interest is same in both banks.  
(1) ₹ 5000 (2) ₹ 3000 (3) ₹ 2000 (4) ₹ 4000  
(5) ₹ 1000

## LEVEL-II

- What would be the compound interest obtained on an amount of ₹ 7,790 at the rate of 10 p.c.p.a. after two years?  
(1) ₹ 1532.60 (2) ₹ 1495.90  
(3) ₹ 1653.50 (4) ₹ 1635.90
- None of these
- What is the difference between the simple and compound interest on ₹ 7500 at the rate of 6 p.c.p.a. in 2 years?  
(1) ₹ 29.37 (2) ₹ 26.28  
(3) ₹ 25.22 (4) ₹ 23.22

- The simple interest accrued on an amount of ₹ 22,500 at the end of four years is ₹ 10,500. What would be the compound interest accrued on the same amount at the same rate at the end of two years?  
(1) ₹ 14,908 (2) ₹ 15,724  
(3) ₹ 26,234 (4) ₹ 8,568
- None of these

- The difference between the amount of compound interest and simple interest accrued on an amount of ₹ 26000 at the end of 3 years is ₹ 2994.134. What is the rate of interest p.c.p.a.?  
(1) 22% (2) 17% (3) 19%  
(4) Cannot be determined
- None of these

- A sum of money was invested at 10% per annum compound interest compounded annually for 2 years. If the interest

# COMPOUND INTEREST

- ₹ 7350 was partly invested in scheme A at 10% p.a. compound interest compounded annually for 2 years and partly in scheme B at 7% p.a. simple interest for 4 years. Both the schemes earn equal interests. How much was invested in scheme A?  
(1) ₹ 5600 (2) ₹ 4200  
(3) ₹ 4400 (4) ₹ 6600
- None of these

- An amount of Rs. 6000 becomes Rs. 7200 in 3 years at a certain rate of compound interest. What will be the amount received after 9 years at the same rate of interest?  
(1) Rs. 10,614 (2) Rs. 11,298  
(3) Rs. 10,368 (4) Rs. 11,612  
(5) Rs. 19,998

- All deposited Rs. P in scheme A for 3 years at a rate of 10% per annum simple interest.

- A certain sum is deposited for 2 years in scheme A at 10% p.a. compound interest compounded annually. The same sum is deposited for 3 years in scheme B at 7% p.a. simple interest. The interest earned from scheme A is half to that of earned from scheme B. What is the value of P?  
(1) 27 (2) 9 (3) 12 (4) 18

- Parmanu deposited Rs. P for 2 years in scheme A which offered 20% per annum compounded interest compounded annually. He got the interest earned from scheme A to Shashi, at the rate of 7.5% per annum simple interest. If at the end of 2 years Shashi gave Rs. 3756 to Parmanu and thereby repaid the whole interest earned from scheme A.



## COMPOUND INTEREST

25. Ram invested a certain sum for 2 years in scheme X at 50% p.a. compound interest (compounded annually). He also invested an equal sum in scheme Y at 15% p.a. simple interest for 2 years. If the difference between the interest earned was the sum?

- Y is Rs. 2400  
(1) Rs. 2400  
(2) Rs. 1200  
(3) Rs. 4000  
(4) Rs. 4000

### LEVEL-III

1. ₹ 5000 is divided between Shyam and Ram. Shyam's share at the end of 9 yr is equal to Ram's share at the end of 11 yr, compounded at the rate of 5%. The share

a compound interest of 20%, compounded annually. Find the least number of complete years after which his sum will be more than double.

- (1) 4 years  
(2) 2 years  
(3) 6 years  
(4) 8 years

(5) None of these

7. A sum of money lent at compound interest for 2 years at 20% per annum would fetch ₹ 482 more if the interest was payable half yearly than if it was payable annually. The sum is

- (1) ₹ 10,000  
(2) ₹ 20,000  
(3) ₹ 40,000  
(4) ₹ 50,000

(5) None of these

8. A sum of ₹ 3000

## COMPOUND INTEREST

12. Sanju puts equal amounts of money now at half yearly and the second at a certain per cent per annum compound interest payable yearly. If he gets equal amount after 3 years what is the value of the second per cent?

- (1)  $10\frac{1}{4}\%$  (2) 10% (3)  $9\frac{1}{2}\%$  (4)  $9\frac{1}{4}\%$   
(5) None of these

13. The simple interest accrued on an amount of ₹ 25,000 at the end of four years is ₹ 5000. What would be the compound interest accrued on the same amount at the same rate in the same period?

2. **Quantity I :** The rate of the simple interest for 3 years under ₹ 1 annually and for the same rate in 3 years under ₹ 1.25 if the same rate of simple interest is charged then find the rate.  
**Quantity II :** A sum of money between 1200 and 1300 is compounded annually. The rate of interest is 10% per annum. Find the principal amount.  
**Quantity I :** A certain amount of money is invested at 10% p.a. for 3 years and 12% p.a. for 2 years. Find the principal amount.  
**Quantity II :** A sum of money between 1200 and 1300 is compounded annually. The rate of interest is 10% per annum. Find the principal amount.



Give answer ( )  
statements I and II  
answer the question

1. What was the sum after three years?  
I. The interest on the sum was ₹ 6,000.  
II. The difference between the compound interest and the simple interest at the end of 2 years was ₹ 600.
2. What is the rate of interest?  
I. The simple interest on ₹ 6,000 deposited for 2 years was ₹ 1,200.  
II. The difference between the compound interest and the simple interest on ₹ 6,000 at the end of 2 years was ₹ 600.
3. Whose income is more?  
I. Income of X grows at 10% p.a. C.I. annually at a rate of ₹ 10,000.  
II. Two years ago, the income of Y was ₹ 10,000.
4. What is the rate of interest per annum?  
I. Mr. A invests a sum of ₹ 10,000 at 10% p.a. C.I. annually.  
II. The sum becomes ₹ 12,100 at the end of 2 years when compounded annually.
5. What is the rate of interest?  
I. An amount doubles in 10 years at simple interest.  
II. Difference between the compound interest and the simple interest on ₹ 10,000 at the end of 2 years is ₹ 600.

### SHORT ANSWERS

#### LEVEL-I

|         |         |         |         |
|---------|---------|---------|---------|
| 1. (1)  | 2. (3)  | 3. (4)  | 4. (1)  |
| 7. (4)  | 8. (4)  | 9. (4)  | 10. (1) |
| 13. (3) | 14. (1) | 15. (3) | 16. (1) |
| 19. (3) | 20. (2) | 21. (1) | 22. (3) |
| 25. (3) | 26. (4) | 27. (4) | 28. (1) |
| 31. (3) | 32. (2) | 33. (3) | 34. (1) |
| 37. (2) | 38. (1) | 39. (4) | 40. (3) |
| 43. (3) | 44. (2) | 45. (1) |         |

$$= 10000 \left( \frac{441}{400} - 1 \right)$$

$$= \frac{10000 \times 41}{400} = \text{Rs. } 1025$$

$$8. (4) \text{ C.I.} = P \left[ \left( 1 + \frac{R}{100} \right)^T - 1 \right]$$

$$= 7400 \left[ \left( 1 + \frac{13.5}{100} \right)^2 - 1 \right]$$

$$= 7400 [(1.135)^2 - 1]$$

$$= 7400 \times 0.288225 = \text{Rs. } 2132.87$$

9. (4) For two years,

$$\text{C.I.} - \text{S.I.} = \frac{PR^2}{10000}$$

$$\Rightarrow 90 = \frac{P \times 12 \times 12}{10000}$$

$$\Rightarrow P = \frac{10000 \times 90}{12 \times 12} = \text{Rs. } 6250$$

$$\text{Amount} = P \left( 1 + \frac{R}{100} \right)^T$$

$$= 6250 \left( 1 + \frac{12}{100} \right)^3$$

$$= 6250 \left( 1 + \frac{3}{25} \right)^3 = 6250 \left( \frac{28}{25} \right)^3$$

$$= \text{Rs. } \left( \frac{6250 \times 28 \times 28 \times 28}{25 \times 25 \times 25} \right)$$

$$= \text{Rs. } 8780.80$$

10. (1) Interest is compounded half yearly.

$$\text{Hence rate of interest} = \frac{10}{2} = 5\%$$

No. of periods = 2

$\therefore$  Amount

$$= 15000 \left( 1 + \frac{5}{100} \right)^2$$

$$= 15000 \times \frac{21}{20} \times \frac{21}{20}$$

$$= \text{Rs. } 16537.50$$

11. (2) P =

= Rs.

CI =

= 2000

= 2000

= Rs. 8

12. (2) Ra

$$= \frac{1080}{225}$$

$\therefore$  CI =

= 2250

= 2250

= 22500

$$= \frac{22500 \times 1}{625}$$

13. (3) Rate =

$$= \frac{6800 \times 10}{17000 \times 4}$$

$$\therefore \text{C.I.} = P \left[ \left( 1 + \frac{R}{100} \right)^T - 1 \right]$$

$$= 17000 \left[ \left( 1 + \frac{10}{100} \right)^2 - 1 \right]$$

$$= 17000 \left[ \left( \frac{11}{10} \right)^2 - 1 \right]$$

$$= 17000 \times \left( \frac{121}{100} - 1 \right)$$

# COMPOUND INTEREST

$$\Rightarrow \frac{32x}{100} - x \left( \frac{121}{100} - 1 \right) = 990$$

$$\Rightarrow \frac{32x}{100} - \frac{21x}{100} = 990$$

$$\Rightarrow \frac{11x}{100} = 990$$

$$\Rightarrow x = \frac{990 \times 100}{11} = \text{Rs. } 9000$$

18. (4) Let the rate of interest be R% per annum.  
Difference between C.I. and S.I. for 2 years

$$= \frac{PR^2}{10000} \dots (i)$$

Again, C.I. - S.I.

$$= \text{Rs. } (2688 - 2560) = \text{Rs. } 128$$

$$\text{S.I. for 1 year} = \frac{2560}{2}$$

$$= \text{Rs. } 1280$$

$$\therefore \text{Rate} = \frac{128}{1280} \times 100 = 10\% \text{ per annum}$$

$$\text{New rate} = (10 + 6\%) = 16\% \text{ per annum}$$

From equation (i),

$$128 = \frac{P \times 10 \times 10}{10000}$$

$$\Rightarrow P = \text{Rs. } 1280$$

$$\text{C.I.} = x \left[ \left( 1 + \frac{10}{100} \right)^2 - 1 \right] = \frac{21}{100} x$$

$$\text{S.I.} = \left( \frac{2x}{3} \right) \times 10.5 \times t = \frac{7xt}{100}$$

According to the question,

$$\frac{21x}{100} = \frac{75}{100} \left( \frac{7xt}{100} \right)$$

$$\Rightarrow t = \frac{21 \times 100}{7 \times 75} = 4 \text{ years}$$

$$21. (1) \text{ C.I.} = P \left[ \left( 1 + \frac{R}{100} \right)^T - 1 \right]$$

$$\text{S.I.} = \frac{P \times R \times T}{100}$$

According to the question,

$$x \left[ \left( 1 + \frac{20}{100} \right)^2 - 1 \right] = \frac{2(x - 1600) \times 10 \times 3}{100}$$

$$\Rightarrow x \left[ \left( 1 + \frac{1}{5} \right)^2 - 1 \right] = \frac{6x - 9600}{10}$$

# COMPOUND INTEREST

Difference

$$= P \left[ \frac{r^3}{10000} - \frac{r^2}{1000} \right]$$

$$\Rightarrow 244 = P \left[ \frac{125}{10000} - \frac{75}{1000} \right]$$

$$\Rightarrow 244 = P \left[ \frac{7525}{10000} \right]$$

$$\Rightarrow P = \frac{244 \times 10000}{7525}$$

$$= \text{Rs. } 32000$$

26. (4) Principal = Rs. P (say)

$$\therefore \text{C.I.} = P \left[ \left( 1 + \frac{R}{100} \right)^T - 1 \right]$$

$$\Rightarrow 510 = P \left[ \left( 1 + \frac{25}{100} \right)^2 - 1 \right]$$

$$\Rightarrow 510 = P \left[ \left( 1 + \frac{1}{4} \right)^2 - 1 \right]$$

$$\Rightarrow 510 = P \left[ \frac{9}{16} - 1 \right]$$

Time = 2 half years

$$\therefore \text{C.I.} = P \left[ \left( 1 + \frac{R}{100} \right)^T - 1 \right]$$

$$= x \left[ \left( 1 + \frac{5}{200} \right)^2 - 1 \right]$$

$$= x \left[ \left( 1 + \frac{1}{40} \right)^2 - 1 \right]$$

$$= x \left[ \left( \frac{41}{40} \right)^2 - 1 \right]$$

$$= x \left( \frac{1681}{1600} - 1 \right)$$

$$= x \left( \frac{1681 - 1600}{1600} \right) = \text{Rs. } \frac{81x}{1600}$$

$$\text{Difference} = \frac{81x}{1600} - \frac{3x}{100}$$

$$= \frac{81x - 48x}{1600}$$

$$= \text{Rs. } \frac{33x}{1600}$$

$$\therefore \frac{33x}{1600} = 330$$

$$\Rightarrow x = \frac{1600 \times 330}{33} = \text{Rs. } 16000$$

28. (1) Let the sum of the loan taken be Rs. P.  
Amount due in 1 year

$$= P \left( 1 + \frac{R}{100} \right) = P \left( 1 + \frac{8}{100} \right)$$

$$= P \left( 1 + \frac{2}{25} \right)$$

$$= \text{Rs. } \frac{27P}{25}$$

Remaining principal for second year

$$= \text{Rs. } \left( \frac{27P}{25} - 50000 \right)$$

Amount due at the end of the second year

$$= \left( \frac{27P}{25} - 50000 \right) \left( 1 + \frac{8}{100} \right)$$

$$= \text{Rs. } \left( \frac{27P}{25} - 50000 \right) \times \frac{27}{25}$$

According to the question,

$$\therefore$$

$$=$$

$$\Rightarrow$$

$$\Rightarrow P$$

$$29. (3)$$

$$\therefore S$$

$$A$$

$$A$$

$$x$$

$$=$$

$$\Rightarrow x$$

$$\Rightarrow x$$

$$\Rightarrow \left( \frac{2}{2} \right)$$

$$\Rightarrow \frac{40}{4}$$

$$\Rightarrow \frac{40}{4}$$

$$\Rightarrow 84$$

$$\Rightarrow x =$$

$$30. (2)$$

$$= P$$

$$= P$$

$$A = P$$

$$\Rightarrow 34$$



35. (3) Principal increases by 15% in 3 years at simple interest.

∴ Rate = 5% per annum

$$\therefore C.I. = P \left[ \left( 1 + \frac{R}{100} \right)^T - 1 \right]$$

$$= \text{Rs. } 15 \left[ \left( 1 + \frac{5}{100} \right)^3 - 1 \right] \text{ lakhs}$$

$$= \text{Rs. } 15 \left[ \left( 1 + \frac{1}{20} \right)^3 - 1 \right] \text{ lakhs}$$

$$= \text{Rs. } 15 \left[ \left( \frac{21}{20} \right)^3 - 1 \right] \text{ lakhs}$$

$$= \text{Rs. } 15 \left( \frac{9261}{8000} - 1 \right) \text{ lakhs}$$

$$= \text{Rs. } \frac{15 \times 1261}{8000} \text{ lakhs}$$

$$= \text{Rs. } 2.36 \text{ lakhs}$$

36. (1)  $A = P \left( 1 + \frac{R}{100} \right)^T$

$$\therefore 9600 = P \left( 1 + \frac{R}{100} \right)^2 \dots (i)$$

$$10272 = P \left( 1 + \frac{R}{100} \right)^3 \dots (ii)$$

Dividing equation (ii) by (i),

$$1 + \frac{R}{100} = \frac{10272}{9600} = \frac{107}{100}$$

$$\Rightarrow \frac{R}{100} = \frac{107}{100} - 1 = \frac{7}{100}$$

$$\Rightarrow R = 7\% \text{ per annum}$$

OR

C.I

2nd year

9,600

3rd year

10,272

$$S.I \text{ for 1 year} = 10,272 - 9,600$$

37. (2)

sin

∴

=

∴

= Rs

= Rs

= Rs

= Rs

= Rs

= Rs. 8

38. (1) Let in

$$3x = x(1 +$$

quired to

$$3 = (1.1)^t$$

t = almost

of 2 minu

minutes

39. (4) Let m

loan = ₹ 60

So,

$$500 + (500$$

$$= (x + x \times \frac{1}{100})$$

## LEVEL-II

$$1. (4) \text{ C.I.} = P \left[ \left( 1 + \frac{R}{100} \right)^T - 1 \right]$$

$$= 7790 \left[ \left( 1 + \frac{10}{100} \right)^2 - 1 \right]$$

$$= 7790 \left( \frac{121}{100} - 1 \right)$$

$$= \frac{7790 \times 21}{100} = ₹ 1635.9$$

$$2. (2) \text{ Difference} = \frac{P \times R^2}{(100)^2}$$

$$= \frac{7300 \times 36}{10000} = ₹ 26.28$$

3. (1) Present price of machine = ₹  $x$  (let)  
Price of machine after two years

$$= P \left( 1 - \frac{R}{100} \right)^T = x \left( 1 - \frac{25}{100} \right)^2$$

$$= x \times \left( \frac{3}{4} \right)^2 = \text{Rs. } \frac{9x}{16}$$

$$\therefore \text{Required ratio} = x : \frac{9x}{16} = 16 : 9$$

4. (3) Rate of growth of population =  $R\%$  per annum (let)

$$\therefore P = P_0 \left( 1 + \frac{R}{100} \right)^T$$

$$\Rightarrow 49972 = 32500 \left( 1 + \frac{R}{100} \right)^2$$

$$\Rightarrow \frac{49972}{32500} = \left( 1 + \frac{R}{100} \right)^2$$

$$\Rightarrow \frac{961}{625} = \left( 1 + \frac{R}{100} \right)^2$$

$$\Rightarrow \left( \frac{31}{25} \right)^2 = \left( 1 + \frac{R}{100} \right)^2$$

$$\Rightarrow 1 + \frac{R}{100} = \frac{31}{25}$$

$$\Rightarrow \frac{R}{100} = \frac{31}{25} - 1 = \frac{6}{25}$$

$$\Rightarrow R = \frac{6}{25} \times 100 = 24\%$$

5. (1)

=

(

=

S

(

R

=

6. (2)

.

 $\Rightarrow$ 

=

 $\Rightarrow$  $\Rightarrow$ 

=

7. (4) P

$$= \frac{80 \times}{2 \times}$$

Case I

Amount

$$= 1000$$

$$= 1000$$

$$= 1000$$

$$= \text{Rs. } 3$$

$$\therefore \text{Com}$$

$$= \text{Rs. } ($$

$$= \text{Rs. } 8$$

$$= \text{Rs. } x \left[ \left( 1 + \frac{1}{5} \right)^2 - 1 \right]$$

$$= \text{Rs. } x \left[ \left( \frac{6}{5} \right)^2 - 1 \right]$$

$$= \text{Rs. } x \left( \frac{36}{25} - 1 \right) = \text{Rs. } \left( \frac{11x}{25} \right)$$

S.I. from scheme B

$$= \frac{\text{Principal} \times \text{Time} \times \text{Rate}}{100}$$

$$= \frac{3x \times 2 \times 9}{100} = \text{Rs. } \frac{54x}{100}$$

$$\therefore \frac{54x}{100} - \frac{11x}{25} = 1200$$

$$\Rightarrow \frac{54x - 44x}{100} = 1200$$

$$\Rightarrow \frac{10x}{100} = 1200$$

$$\Rightarrow x = 1200 \times 10 = \text{Rs. } 12000$$

$$13. (3) \text{ Rate} = \frac{\text{S.I.} \times 100}{\text{Principal} \times \text{Time}}$$

$$= \frac{6800 \times 100}{17000 \times 4}$$

$$= 10\% \text{ per annum}$$

$$\therefore \text{C.I.} = P \left[ \left( 1 + \frac{R}{100} \right)^T - 1 \right]$$

$$= 17000 \left[ \left( 1 + \frac{10}{100} \right)^2 - 1 \right]$$

$$= 17000 \left[ \left( \frac{11}{10} \right)^2 - 1 \right]$$

$$= 17000 \times \left( \frac{121}{100} - 1 \right)$$

$$= 17000 \times \frac{21}{100} = \text{Rs. } 3570$$

14. (2) Investment in scheme A = Rs.  $x$

$\therefore$  Investment in scheme B = Rs.  $(7350 - x)$

According to the question,

$$\frac{P_1 \times R_1 \times T_1}{100} = P_2 \left[ \left( 1 + \frac{R_2}{100} \right)^{T_2} - 1 \right]$$

$$\Rightarrow (7350 - x)$$

$$= x \left[ \left( 1 + \frac{10}{100} \right)^2 - 1 \right]$$

$$\Rightarrow (7350 - x)$$

$$\Rightarrow (7350 - x)$$

$$\Rightarrow (7350 - x)$$

$$\Rightarrow 7350$$

$$\Rightarrow 7x =$$

$$\Rightarrow x = \frac{7}{10}$$

15. (3)  $\therefore A$

$$\Rightarrow 7200$$

$$\Rightarrow \left( 1 + \frac{10}{100} \right)^2$$

$$= \frac{6}{5}$$

$\therefore$  Amount

$$= 6000 \left( 1 + \frac{10}{100} \right)^2$$

$$= 6000 \times \frac{6}{5}$$

$$= \text{Rs. } 10360$$

16. (2) S.I. =  $\frac{P \times R \times T}{100}$

$$= \frac{P \times 3 \times 10}{100}$$

$\therefore$  Amount = Rs

$$\text{C.I.} = \frac{13P}{10}$$

$$\Rightarrow 819 = \frac{13P}{10}$$

$$\Rightarrow \frac{8190}{13} =$$

$$\therefore \frac{21x}{100} = \frac{(6100 - x) \times 40}{100}$$

$$\Rightarrow 21x = 244000 - 40x$$

$$\Rightarrow 61x = 244000$$

$$\therefore x = \frac{244000}{61} = \text{Rs. } 4000$$

$$20. (5) \text{ C.I.} = P \left[ \left( 1 + \frac{R}{100} \right)^T - 1 \right]$$

$$\text{S.I.} = \frac{P \times R \times T}{100}$$

According to the question.

$$2P \left[ \left( 1 + \frac{10}{100} \right)^2 - 1 \right] = \frac{Px \times 3}{100}$$

$$\Rightarrow 2 \left[ \left( 1 + \frac{1}{10} \right)^2 - 1 \right] = \frac{3x}{100}$$

$$\Rightarrow 2 \left[ \left( \frac{11}{10} \right)^2 - 1 \right] = \frac{3x}{100}$$

$$\Rightarrow 2 \left( \frac{121}{100} - 1 \right) = \frac{3x}{100}$$

$$\Rightarrow \frac{2 \times 21}{100} = \frac{3x}{100}$$

$$\Rightarrow 3x = 2 \times 21$$

$$\Rightarrow x = \frac{2 \times 21}{3}$$

$$= 7 \times 2 = 14$$

$$21. (1) \text{ C.I.} = P \left[ \left( 1 + \frac{R}{100} \right)^T - 1 \right]$$

$$= P \left[ \left( 1 + \frac{20}{100} \right)^2 - 1 \right] = P \left[ \left( \frac{6}{5} \right)^2 - 1 \right]$$

$$= P \left[ \frac{36}{25} - 1 \right] = \text{Rs. } \frac{11P}{25}$$

$$\text{S.I.} = \frac{\text{Principal} \times \text{Time} \times \text{Rate}}{100}$$

$$= \frac{11P}{25} \times \frac{2 \times 7.5}{100} = \text{Rs. } \frac{33P}{500}$$

Accor

$$\frac{33P}{500}$$

$$\Rightarrow \frac{33}{500}$$

$$\Rightarrow 25$$

$$\Rightarrow P =$$

$$22. (2) \text{ S.I.}$$

$$= \frac{1500}{1}$$

$$= \text{Rs. } 1$$

Amoun

$$= \text{Rs. } 2$$

If addit

$$\text{C.I.} = P$$

$$\Rightarrow 1408$$

$$\left[ \left( 1 + \frac{20}{100} \right)^2 - 1 \right]$$

$$\Rightarrow 1408 =$$

$$\Rightarrow 1408 =$$

$$\Rightarrow (2550 +$$

$$\Rightarrow x = 3200$$

$$23. (2) \text{ Let } X's$$

$$\therefore Y's \text{ sha}$$

$$\therefore A = P \left( \right.$$

$$\Rightarrow x \left( 1 + \frac{10}{100} \right)$$

$$\Rightarrow x \times \frac{11}{10} =$$

$$\Rightarrow 11x = 7$$

$$\Rightarrow 21x = 7$$

$$\Rightarrow x = \frac{7140}{21}$$

$$24. (1) \text{ Case-I.}$$

$$\text{S.I.} = \frac{\text{Princ}}$$



$$\Rightarrow 16 = 2^n \Rightarrow 2^4 = 2^n$$

$$n = 4 \text{ yr}$$

5. (2) Let the principal be P and rate of interest be r%. Then, principal (i) when difference between C.I. and SI is for 2 yr is given by

$$P = \frac{20 \times (100)^2}{r^2}$$

(ii) When difference between CI & SI in for 3 yrs is given by

$$P = \frac{61 \times 10^6}{r^2(300 + r)}$$

From eqs. (i) and (ii),

$$\frac{20 \times 10^4}{r^2} = \frac{61 \times 10^6}{r^2(300 + r)}$$

$$\Rightarrow r = 305 - 300 = 5\%$$

From Eq. (i),  $P = \frac{20 \times 10^4}{25} = ₹ 8000$

6. (1) Let the sum of money be P.

$$\therefore P \left( 1 + \frac{20}{100} \right)^T \geq 2P$$

$$\Rightarrow \left( \frac{6}{5} \right)^T \geq 2 \Rightarrow (1.2)^T \geq 2$$

$$\text{If } T = 4$$

$$\text{then } 1.2^4 = 2.07 \geq 2$$

Hence, least value of T is 4 yrs.

7. (2) Let the sum be x.

$$x \left( 1 + \frac{10}{100} \right)^4 - x \left( 1 + \frac{20}{100} \right)^2 = 482$$

$$\Rightarrow x \left( \frac{11}{10} \right)^4 - x \left( \frac{6}{5} \right)^2 = 482$$

$$\Rightarrow x(1.4641 - 1.44) = 482$$

$$\Rightarrow 0.0241x = 482$$

$$\Rightarrow x = \frac{482}{0.0241} = ₹ 20000$$

8. (2) Let the share of Anjali be x.

Then, the share of Arun is (30600 - x).

$$x \times \left( 1 + \frac{4}{100} \right)^3$$

$$\left(1 + \frac{r}{100}\right)^2 = \frac{36}{25} \Rightarrow 1 + \frac{r}{100} = \frac{6}{5}$$

$$1 + \frac{r}{100} = 1 + \frac{1}{5} \Rightarrow \frac{r}{100} = \frac{1}{5} \text{ and } r = 20\%$$

**Quantity II :** If principal = ₹ 1000, amount = ₹ 1331

$$\therefore A = P \left(1 + \frac{R}{100}\right)^T$$

$$\Rightarrow \frac{1331}{1000} = \left(1 + \frac{R}{100}\right)^3$$

$$\Rightarrow \left(\frac{11}{10}\right)^3 = \left(1 + \frac{R}{100}\right)^3$$

$$\Rightarrow 1 + \frac{R}{100} = \frac{11}{10}$$

$$\Rightarrow \frac{R}{100} = \frac{1}{10}$$

$$\Rightarrow R = \frac{1}{10} \times 100 = 10\%$$

Quantity I < Quantity II

3. (1) **Quantity I :** We are given,

$$P \left(1 + \frac{r}{100}\right)^2 = 771.75 \text{ and}$$

$$P \left(1 + \frac{r}{100}\right)^3 = 810.338$$

On Dividing, we get

$$\frac{P \left(1 + \frac{r}{100}\right)^3}{P \left(1 + \frac{r}{100}\right)^2} = \frac{810.338}{771.75}$$

$$1 + \frac{r}{100} = 1.05 \Rightarrow r = 0.05 \times 100 = 5\%$$

$$\text{To find } P \rightarrow P \left(1 + \frac{5}{100}\right)^2 = 771.75$$

$$\Rightarrow P = \frac{771.75}{1.05 \times 1.05} = ₹ 700$$

$$\text{Quantity II : } A = P \left(1 + \frac{R}{100}\right)^T$$

$$\Rightarrow 291$$

$$\Rightarrow 291$$

$$\Rightarrow x =$$

$$\therefore \text{S.I.}$$

$$= \frac{2500}{100}$$

4. (1) **Quantity I :**

$$\Rightarrow 700$$

$$10000$$

By eq

$$\frac{10000}{7000}$$

$$\Rightarrow \frac{10}{7}$$

From

$$7000 =$$

$$\Rightarrow P =$$

**Quantity II :**

$$\text{C.I.} =$$

$$= 13033$$

Difference

5. (3) **Quantity I :**

Present value

$$= \frac{x}{1 + \frac{10}{100}}$$

In this type of problem, we have to find out the value after rounding. The answer can be found by rounding a number to a desired multiple of the place which is closest. Thus, rounding to the nearest hundred means finding the number which is closest to the nearest hundred.

of  $\frac{1}{10}$  which is closest to the nearest ten.

After a number is rounded to a particular decimal place, the value of that particular decimal place becomes zero.

**Example :** Round 7.35 to the nearest hundred.

**Solution :** You need to round to the nearest hundred which is closest to the number.

The answer is 7.35.

To round a number to the nearest hundred, look at the tens digit. If the tens digit is 5 or greater, then change the hundreds digit and ones digit to zero. If the tens digit is less than 5, then leave the hundreds digit unchanged and change the ones digit to zero.

To round a number to the nearest hundred, consider the tens digit. If the tens digit is 5 or greater, then change the hundreds digit and ones digit to zero. If the tens digit is less than 5, then leave the hundreds digit unchanged and change the ones digit to zero.

To round a number to the nearest hundred, consider the hundreds digit. If the hundreds digit is 5 or greater, then change the thousands digit and the hundreds digit to the next higher digit. If the hundreds digit is less than 5, then leave the thousands digit unchanged but change all the digits before that to zero.

Study the following

| Number | Rounded to nearest |
|--------|--------------------|
| 395    | ten                |
| 625    | hundred            |
| 875623 | thousand           |
| 531924 | ten-thousand       |
| 472804 | lakh               |

28.  $127.001 \times 7.998 + 6.05 \times$   
(1) 1440 (2) 1400  
(5) 1140
29.  $\sqrt{1000} + \frac{3.001}{4.987}$  of 1891.  
(1) 2500 (2) 1230  
(5) 2130
30.  $2439.97 - 1234.01 + 4$   
(1) 620 (2) 650  
(5) 600
31.  $\frac{340}{20.002} + \frac{29.997}{510} \times \frac{179.909}{59.919}$   
(1) 760 (2) 800  
(5) 780
32. 34% of 576 + 18% of 84  
(1) 75 (2) 72  
(5) 66
33.  $331.8 \div 23.7 + [-21]^2 - 5$   
(1) 15 (2) 16 (3)  
(5) 17
34.  $10^3 \times 100^3 + 999999999 =$   
(1) 6 (2) 9 (3)  
(5) 12
35. 24.9% of 5679 + 44.9% of 4  
(1) 3455 (2) 3355 (3)  
(5) None of these
36.  $248.251 \div 12.62 \times 20.52 =$   
(1) 400 (2) 450 (3) 6  
(5) 375
37.  $33\frac{1}{3}\%$  of 768.9 + 25% of 16  
(1) 230 (2) 225 (3) 2  
(5) 240
38.  $39.05 \times 14.95 - 27.99 \times 10.$   
(1) 22 (2) 29 (3) 3  
(5) 25
39.  $561 \div 35.05 \times 19.99 = ?$   
(1) 320 (2) 330 (3) 3  
(5) 335
40. 105.01% of 8451 -  $\frac{3}{7}\%$  of 50  
(1) 8879 (2) 8860 (3) 88  
(5) None of these

#### LEVEL-II

**Directions (1-5) :** What app will come in place of the question following questions ? (You are not calculate the exact value.)



7. 40.02% of  $224.9 + 4$

- (1) 200 (2) 245  
(5) 165

8.  $(859.98 - 379.88) \div 11$

- (1) 30 (2) 60  
(5) 50

9.  $295 - 14 \times 3.99 + 99.95$

- (1) 19 (2) 11  
(5) 18

10.  $15.99^2 + 4.98^2 \times 8.03 + 6$

- (1) 575 (2) 425  
(5) 480

**Directions (11-20) :**

value will come in place of the  
in each of the following ques  
expected to calculate the exa

11.  $\left(\frac{18}{4}\right)^2 \times \frac{455}{19} \div \frac{61}{799} = ?$

- (1) 1840 (2) 1860 (3)  
(5) 1900

12. 73.99% of  $1299 + 9.98\%$

- (1) 1142 (2) 1152 (3)  
(5) None of these

13.  $(9.979)^3 - (23.99)^2 + (1.99)$

- (1) 446 (2) 436 (3)  
(5) None of these

14.  $5891 \div 14.99 + 589.01 - 11$

- (1) 820 (2) 830 (3)  
(5) None of these

15.  $2439.97 - 1234.01 + 401.8$

- (1) 620 (2) 640 (3)  
(5) None of these

16. 149.9% of  $149.9 + 149.9 =$

- (1) 375 (2) 400 (3)  
(5) 450

17.  $(\sqrt{7921} - \sqrt{2070.25}) \times \frac{1}{4} = ?$

- (1) 11 (2) 14 (3) 1  
(5) 13

18.  $(21.98)^2 - (25.02)^2 + (13.03)^2$

- (1) 25 (2) 120 (3) 1  
(5) 140

19.  $441.01 - 232.99 + 1649.99 =$

- (1) 600 (2) 630 (3) 6  
(5) 720

20.  $(49.99)^2 - (8.9)^2 - (15.9)^2 = ?$

- (1) 2165 (2) 2000 (3) 19  
(5) 1885

$$31. (4) ? = \frac{340}{20} + \frac{30}{510} \times \frac{180}{60}$$

$$= \frac{340}{20} + \frac{510}{30} \times \frac{180}{60}$$

$$= 867$$

∴ Required answer = 870

$$32. (5) \frac{576 \times 34}{100} + \frac{842 \times 18}{100}$$

$$= \frac{400 \times ?}{100} + 83.4$$

$$\Rightarrow 195.84 + 151.56 = 4 \times ? + 83.4$$

$$\Rightarrow 347.4 = 4 \times ? + 83.4$$

$$\Rightarrow 4 \times ? = 347.4 - 83.4 = 264$$

$$\Rightarrow ? = \frac{264}{4} = 66$$

$$33. (4) (?)^2 = 331.8 + 23.7 + (-21)^2 - 94$$

$$= 14 + 441 - 94 = 361$$

$$\Rightarrow ? = \sqrt{361} = 19$$

$$34. (2) 10^3 \times 10^6 + 10^9 = 10^7 + 10^7$$

$$= 10^9 + 10^9 = 2 \times 10^9$$

$$= ? = 9$$

$$35. (2) ? = \frac{25}{100} \times 5680 + \frac{45}{100} \times 43$$

$$= 1420 + 1935 = 3355$$

$$36. (1) 248.251 \div 12.62 \times 20.52$$

its approximate value may be w

$$= 250 \div 12.5 \times 20.5$$

$$= 20 \times 20.5 = 410$$

The answer choice nearest to 410

$$37. (5) 33\frac{1}{3}\% \text{ of } 768.9 + 25\% \text{ of } 161.2 =$$

$$= \frac{100}{3 \times 100} \times 768.9 + \frac{25}{100} \times 161.2 = 58.12$$

$$= 256.3 + 40.3 = 58.12$$

$$= 238.48 = 240 \text{ (approx.)}$$

$$38. (5) 39.05 \times 14.95 - 27.99 \times 10.12 =$$

$$583.7975 - 283.2588$$

$$= (36 + ?) \times 5$$

$$300.5387 = (36 + ?) \times 5$$

$$\frac{300.5387}{5} = 36 + ?$$

$$60.10774 = 36 + ?$$

$$60.10774 - 36 = ?$$

∴ ? = 24.10774 = 24 (approx.) or 25  
as per the answer choice.

**Approximate calculation :**

$$39.05 \times 14.95 - 27.99 \times 10.12 = (36 + ?) \times 5$$

This can be approximated as

$$39 \times 15 - 28 \times 10 = (36 + ?) \times 5$$

$$585 - 280 = (36 + ?) \times 5$$

$$\text{or. } 85.147 + 216.4544 + ?$$

$$= 802.293$$

$$301.6014 + ? = 802.293$$

$$? = 500.6916 \approx 500$$

**Approximate calculation :**

$$85.147 + 34.912 \times 6.2 + ?$$

$$= 802.293$$

Approximate equation can be written as

$$85 + 35 \times 6 + ? = 802$$

$$85 + 210 + ? = 802$$

$$295 + ? = 802$$

$$? = 507$$

The answer choice nearest to 507 is 500.

$$30. (1) ? = \frac{700 \times 69}{100} + \frac{400 \times 33}{100}$$

$$= 483 + 132 = 615$$

### LEVEL-III

$$1. (1) 600 \times \sqrt{?} + 60 \times \sqrt{64} = 2280$$

$$\Rightarrow 600 \times \sqrt{?} + 60 \times 8 = 2280$$

$$\Rightarrow 600 \times \sqrt{?} = 2280 - 480$$

$$\Rightarrow 600 \times \sqrt{?} = 1800$$

$$\Rightarrow \sqrt{?} = \frac{1800}{600} = 3$$

$$\Rightarrow ? = 9$$

$$2. (5) ?^2 = \frac{20 \times 500}{100} + 61$$

$$= 100 + 61 = 161$$

$$\therefore ? = \sqrt{161} = 13$$

$$3. (5) \frac{18^3}{?} \times 20 \times 3 = 360$$

$$\Rightarrow ? = \frac{18^3 \times 20 \times 3}{360} = 972$$

$$4. (1) ? = \frac{4500}{15 \times \sqrt{144}} \times 5$$

$$= \frac{4500 \times 5}{15 \times 12} = 125$$

$$5. (4) ? = (1280 + 520) \div 12$$

$$= \frac{1800}{12} = 150$$

$$6. (5) 180.15 \times 17.98 - ?^2 = 2984$$

$$\Rightarrow 180 \times 18 - ?^2 = 2984$$

$$\Rightarrow 3240 - ?^2 = 2984$$

$$\Rightarrow ?^2 = 3240 - 2984 = 256$$

$$\therefore ? = \sqrt{256} = 16$$

P

**Definition :** Each which can be made ber of given things permutation.

**Note :** It should be the order of arrangement when the order is considered is obtained.

**The symbol  ${}^n P_r$  :** number of permutations taken  $r$  at a time. The

tions.  
Thus the symbol  ${}^5 P_2$  permutations or arrangements taken 2 at a time.

COM

**Definition :** Each collection which can be of a number of given things called a combination.

**The symbol  ${}^n C_r$  :** The number of combinations taken  $r$  at a time. The

nation.  
Thus, the symbol  ${}^5 C_2$  combinations or selections taken 2 at a time.

**Remark :** In combinations things is not taken into account pointed out that

Number of arrangements

$= \frac{n!}{(n-r)!}$  (number of selections

where  $r$  is the no. of objects

Thus  ${}^n P_r = \frac{n!}{(n-r)!}$ .

#### Difference between Permutation and Combination

(i) In a combination only the selection of objects is considered whereas in a permutation both the selection and the arrangement are made but also an arrangement is taken into consideration.

(ii) In a combination, the order of objects is immaterial whereas in a permutation the ordering is essential. For example, the permutations of 1, 2, 3 are 123, 132, 213, 231, 312, 321. The combinations of 1, 2, 3 are 123, 132, 213, 231, 312, 321. They are same as combinations.

(iii) Practically to find the permutations of  $n$  different items or objects, taken  $r$  at a time, we select  $r$  items from  $n$  items and then arrange them. So, usually the number of permutations is the number of combinations.



# LEVEL

1. In how many different ways can the word HARYANA be arranged?  
(1) 820 (2) 840  
(3) 750
2. In how many different ways can the word RUMOUR be arranged?  
(1) 180 (2) 720  
(3) None of these
3. In how many different ways can the word RECTITUDE can be arranged so that vowels come together?  
(1) 4302 (2) 7200  
(3) None of these
4. In how many different ways can the word DRASTIC be arranged so that the vowels always come together?  
(1) 720 (2) 360  
(3) None of these
5. In how many different ways can the word THERAPY be arranged so that vowels never come together?  
(1) 720 (2) 1440  
(3) 4800
6. In how many ways the letters of the word SITING can be rearranged so that the vowels always come together?  
(1) 6720 (2) 5720  
(3) None of these
7. In how many different ways can the word CANDIDATE be arranged so that the vowels always come together?  
(1) 4320 (2) 1440  
(3) 1560
8. In how many different ways can the word JUDGE be arranged so that the vowels always come together?  
(1) 48 (2) 124 (3) 144  
(3) None of these
9. In how many different ways can the word REPLACE be arranged so that the vowels always come together?  
(1) 2630 (2) 5040 (3) 1440  
(3) None of these
10. Out of 5 women and 4 men a committee of 5 members is to be formed in such a way that at least one member is a woman. In how many different ways can it be done?  
(1) 80 (2) 84 (3) 76  
(3) None of these

**LEVEL**

1. Out of 7 consonants and 4 vowels, words of 3 consonants and 2 vowels can be formed ?

(1) 24400 (2) 21300  
(5) None of these

2. There are 6 men and 4 women. In how many ways, a committee of 3 men and 2 women can be formed such that at least 1 woman is included?

(1) 220 (2) 170  
(5) 180

3. From a group of 7 men and 6 women, a committee of 5 people is to be selected such that at least 3 men are included. In how many ways can the committee be formed ?

(1) 756 (2) 765  
(5) 760

4. There is a group of 10 children, 1 man, 1 woman, and 1 child going to be selected. In how many ways can the group be selected?

(1) 480 (2) 240  
(5) 480

**Directions (5-6) :** Find the number of ways in which the flag can be made by using six flags of different colors, 3 blue flags and 1 red flag.

5. In how many ways can the flag be made?

(1) 108 (2) 185  
(5) 110

6. In how many ways can the flag be made?

(1) 21 (2) 90  
(5) 125

7. In how many different ways can the word 'OPTICAL' be formed?

(1) 600 (2) 540  
(5) 720

8. A dinner party is to be given for 10 persons. In this party, 60 prefer chicken, 60 prefer fish, 60 prefer both, and 60 prefer neither. In how many ways can the party be given?

(1) 20 (2) 30  
(5) 15

9. In how many different ways can the number '256974' be arranged such that its 6 and 5 are at the ends?

(1) 48 (2) 720  
(5) None of these



- (1) 15 (2) 56 (3)  
(5) None of these
20. One red flag, three white flags are arranged in a line.
- I. No two adjacent flags colour
- II. The flags at the two ends different colours.
- In how many different ways arranged ?
- (1) 6 (2) 4 (3)  
(5) None of these
21. For a scholarship, at most of  $2n + 1$  can be selected. In ways of selection of at least 63 the maximum no. of can be selected for the scholarship
- (1) 3 (2) 4 (3)  
(5) None of these
22. Four volunteers represent school. Each volunteer represents the school. Each volunteer shakes hands with from school other than himself. Each volunteer shakes hands exactly the maximum possible shakes ?
- (1) 153 (2) 141 (3)  
(5) 115
23. Six identical cards are arranged in a row. Each card has number '1' on one side and number '2' on the other side. All the six cards are placed on a table. If number '1' is on the upper side of exactly four (neither more nor less) cards are turned upside down. In how many ways can the cards be turned so that all the six cards show '1' on the upper side ?
- (1) 3 (2) 5 (3) 7  
(5) Data is insufficient

**EXPLANA****LEVEL**

1. (2) The word HARYAN  
which A comes thrice.  
∴ Number of arrangements  
$$= \frac{7!}{3!} = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$$
$$= 7 \times 6 \times 5 \times 4$$
$$= 840$$
2. (1) The word RUMOUR  
in which each of R and U  
∴ Number of arrangements  
$$= \frac{6!}{2!2!} = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 2}$$
$$= 180$$
3. (3) The word RECTITUDE  
which RCTTD are consonants  
vowels and T and E come  
We have to arrange  
RCTTD (EEIU)  
∴ Number of arrangements  
$$= \frac{6! \times 4!}{2! \times 2!}$$
$$= 6 \times 5 \times 4 \times 3 \times 2 \times 3 \times 2$$
$$= 4320$$
4. (3) There are 7 letters in the word  
including 2 vowels (A, I) and 5 consonants  
(D, R, S, T, C.). Considering the first letter,  
we have 6 letters which can be arranged  
in 6! ways. But each of these arrangements  
can be put together in 2! ways.  
∴ Total arrangements =  $6! \times 2!$
5. (4) The word THERAPY contains 7 letters  
in which E, A are two vowels and T, H, R, P, Y are  
consonants. We get THRPY (EA) keeping E and A as  
a single entity.  
Number of permutations with 6 entities together  
=  $6! \times 2! = 1440$   
∴ Required number of arrangements  
=  $7! - 1440 = 5040 - 1440 = 3600$
6. (1) The word VISITING has 8 letters  
in which I comes thrice.  
∴ Number of arrangements  
$$= \frac{8!}{3!} = 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$$
$$= 1680$$



ent ways that w  
committee are :

(i) 2m. 2w in  ${}^6C_2$

$$= \frac{6 \times 5}{2 \times 1} \times \frac{4 \times 3}{2 \times 1} = 90$$

(ii) 3m. 1w in  ${}^6C_3$  ×

$$= \frac{6 \times 5 \times 4}{3 \times 2 \times 1} \times 4 = 80$$

$$(iii) 4m \text{ in } {}^6C_4 = \frac{6 \times 5}{2 \times 1}$$

∴ Total no. of differ  
mittee of 4 persons c  
has at least 2 men

$$= 90 + 80 + 15 = 185$$

**23.** (3) There are 9 women  
tee of 12, consisting of  
be formed by choosing

(i) 5 women and 7 men

(ii) 6 women and 6 men

(iii) 7 women and 5 men

(iv) 8 women and 4 men

(v) 9 women and 3 men

∴ Total number of ways  
mittee

$$= {}^9C_5 \times {}^8C_7 + {}^9C_6 \times {}^6C_6 +$$

$${}^9C_9 \times {}^4C_3$$

$$= 126 \times 8 + 84 \times 28 + 36 \times$$

$$= 6062$$

**24.** (4) Women are in majorit  
cases as discussed in qu

∴ Total number of such

$$= {}^9C_7 \times {}^6C_5 + {}^9C_8 \times {}^5C_4 + {}^9C_9$$

$$= 36 \times 56 + 9 \times 70 + 1 \times 5$$

**25.** (1) Men are in majority in  
discussed in question 37.

∴ Total number of such

$${}^6C_7 = 126 \times 8 = 1008$$

**26.** (1) Number of combination

$$= ({}^4C_4 \times {}^6C_1 + {}^3C_3 \times {}^4C_2)$$

$$= 1 \times 6 + 1 \times 6 = 12$$

**27.** (3) Number of combinations

= Selecting 2 trainees out of  
3 research associates out of

$$= 3 \times \frac{6 \times 5 \times 4}{1 \times 2 \times 3} = 60$$

**28.** (2) Required number of comm

$$= 4C_2 \times 5C_2 \times 3C_2$$

$$\begin{aligned}
 &+4c_2 \times 7c_1 + 4c_1 \times 7c_2 + 4c_3 \times \\
 &= 4 \times 21 + 4 \times 35 + 6 \times 2 \\
 &+ 6 \times 7 + 6 \times 21 + 4 \times \\
 &= 126 + 266 = 266 \\
 \text{Total no. of ways} \\
 &= 126 + 266 + 266 = 658
 \end{aligned}$$

### LEVEL-II

1. (3) From question, Required

$$= {}^{9-2}C_{5-2} = {}^7C_3$$

$$= \frac{7 \times 6 \times 5}{3 \times 2} = 35 \text{ ways}$$

2. (4) According to question

$$\text{ways} = 4^3 \times 5^3$$

$$= 64 \times 125$$

$$= 8000$$

3. (2) According to question,

sons in tennis and rest in ways

$$= {}^{11}C_4 \times {}^7C_7$$

$$= \frac{11 \times 10 \times 9 \times 8}{4 \times 3 \times 2} \times 1 = 330$$

Case II When 5 persons in theatre, Number of ways =

$$= \frac{11 \times 10 \times 9 \times 8 \times 7}{5 \times 4 \times 3 \times 2 \times 1} \times 1 = 462$$

Case III When 6 persons in theatre,

$$\text{Number of ways} = {}^{11}C_6 \times {}^5C_5 =$$

Case IV When 7 persons in theatre,

$$\text{Number of ways} = {}^{11}C_7 \times {}^4C_4 =$$

$$\therefore \text{Total Number of ways}$$

$$= 2(330 + 462) = 2(792)$$

$$= 1584$$

4. (4) According to question, Total seats

$$= 1 \text{ grandfather} + 5 \text{ sons and grandchildren} = 14$$

The grandchildren occupy the 4 other side of the table in  $8P_4 \times 4! =$

The grandfather can occupy a seat (ie,  $S_6, S_7, S_8, S_9$ ).

And, the remaining seats can be  $5! = 5 \times 4 \times 3 \times 2 = 120$  ways (5 sons and daughters)

Hence, the total number of required arrangements

$$= 8! \times 480 = 19353600$$

# PERMUTATIONS

$$\begin{aligned}\text{No. of ways} &= {}^6C_3 \times {}^3C_3 = \frac{8!}{5!3!} \times 1 \\ &= \frac{8 \times 7 \times 6}{6} = 56 \text{ ways.}\end{aligned}$$

**Method-2 :** No. of roads going west are 4 & no. of roads going north are 6, no. of ways that a person will need to travel =  $(4-1) + (6-1)$  or  $4 + 6 - 2 = 8$

$$\text{No. of ways} = {}^8C_3 \times {}^5C_5 = [{}^8C_5 \times {}^3C_3]$$

i.e. if there are 'm' & 'n' roads, no. of ways

$$= \left[ {}^{m+n-2}C_{m-1} \times {}^{n-1}C_{n-1} \right]$$

20. (1) The possible arrangements are :

W \_ W \_ W \_ or \_ W \_ W \_ W

The white flags can be arranged in 3! ways

are same or  $\frac{3!}{3!}$  ways] & remaining 2 blue flags

red flag can be arranged in 3 places i.e. 3! ways

[Since, 2 blue flags are same].

Since there are two possible arrangements of white flags

$$\text{Hence, no. of ways} = \frac{2 \times 3!}{2!} = 6 \text{ ways.}$$

21. (1) Clearly, we have to find when 1 or more than 1 or n candidates are selected out of n candidates.

$$\text{i.e. } {}^{2n+1}C_1 + {}^{2n+1}C_2 + {}^{2n+1}C_3 + \dots + {}^{2n+1}C_{2n+1}$$

Check through options

(1) If  $n = 3$ , then  $2n + 1 = 7$

$${}^7C_1 + {}^7C_2 + {}^7C_3 = 7 + \frac{7 \times 6}{2} + \frac{7 \times 6 \times 5}{6}$$

$$= 7 + 21 + 35 = 63$$

Hence choice (1) is correct.

22. (3)

|   |   |   |   |
|---|---|---|---|
| A | B | C | D |
| 4 | 5 | 6 | 3 |

A volunteer of school A would shake hands with volunteers from school B, C and D. So 4 volunteers from A will shake hands with 5+6+3 = 14 volunteers. So 4 volunteers from A will shake hands  $14 \times 4 = 56$  times. Similarly, volunteers from B will shake hands  $5 \times (4+6+3) = 55$  times. [We haven't added handshake as it has been counted already in 56 times]. Similarly, volunteers from C will shake hands  $6 \times (4+5+3) = 66$  times. Similarly, volunteers from D will shake hands  $3 \times (4+5+6) = 39$  times.

Total no. of handshakes

$$= 56 + 55 + 66 + 39 = 216 \text{ times.}$$

## QUANTITY BASI

### 1. (1) Quantity I : We

& 3 digit numbers. (fr

When 7 comes at one p

$$({}^3C_1 \times 1) \times 9 \times 9 = 81 \times$$

When 7 comes at two p

$$({}^3C_2 \times 1) \times 9 = 3 \times 9 = 2$$

When 7 comes in 3 plac

there is only one 3 digit

No. of times 7 is written

ber, + twice in 27 num

number.

$$= 1 \times 243 + 2 \times 27 + 3$$

$$= 243 + 54 + 3 = 300$$

**Quantity II** : In a 5-dig

be filled by the 3 given

ing 2 places, either one

[one digit appears for a t

two digits may appear

appear twice in the num

In case 1 : A digit appea

no.,

$$\text{Total numbers} = \frac{5!}{3!} \times {}^3C_1$$

In case 2 : Any two digit

same no.

$$\text{Total numbers} = \frac{5!}{2!2!} \times {}^3C_2$$

$$\text{Total numbers} = 60 + 90 =$$

$$\text{Quantity I} > \text{Q}$$

$$\text{Quantity I} > \text{Q}$$

### 2. (3) Quantity I :

First arrange 4 Americans

lar table, which can be do

we have 4 seats available

Chinese can be arranged i

$$\text{Total no. of ways} = 3! \times {}^4P_4$$

**Quantity II** : 1st friend can

in 6 ways, 2nd can select

two go to the same institut

lect in 4 and 3 ways respe

no. of ways

$$= 6 \times 5 \times 4 \times 3 = 360.$$

$$\text{Quantity I} < \text{Qu}$$

### 3. (3) Quantity I : 3 number

from 1, 2, 3, ....., 18, such t

in the following ways

→ when common difference

{1, 2, 3}, {2, 3, 4}, {3, 4, 5} ...

ways



**Mathematical Definition** c  
are n elementary events asso  
experiment and m of them  
event E. then the probability  
denoted by P (E) and is defin

$$\text{or, } P(E) = \frac{\text{Number of cases favourable to event}}{\text{Total no. of cases}}$$

$$\text{or, } P(E) = \frac{\text{No. of desired events}}{\text{Total no. of events}}$$

$$= \frac{\text{Favourable number of cases}}{\text{Exhaustive number of cases}}$$

**Note :** 1. Probability of occur  
a number lying between 0 and  
2. Probability of a certain even

### LEVEL-I

1. Naman rolled a pair of die  
the probability that first die  
ple of 3 and the second die  
multiple of 3 or 2?

- (1)  $\frac{5}{18}$  (2)  $\frac{1}{8}$  (3)  $\frac{2}{9}$   
(5)  $\frac{7}{18}$

2. Abhay rolled a pair of dice t  
the probability that one dice  
ple of 2 and the second dice  
a multiple of 3 nor 2 ?

- (1)  $\frac{1}{3}$  (2)  $\frac{1}{9}$  (3)  $\frac{1}{12}$   
(5)  $\frac{1}{6}$

3. When two dice are rolled, find  
getting a greater number on  
than the one on the second,  
sum should equal 6.

- (1)  $\frac{5}{18}$  (2)  $\frac{2}{36}$  (3)  $\frac{5}{36}$   
(5) None of these

4. There are 27 cards having num  
to 27. Two cards are picked at  
by one without replacement.

20. An urn contains 3 red and 2 white marbles. If three marbles are picked at random, the probability that two are red is

(1)  $\frac{3}{7}$  (2)  $\frac{18}{35}$  (3)  $\frac{1}{3}$

(5) None of these

**Directions (21-25) :** Study the information carefully to answer the questions that follow.

A basket contains 6 blue, 2 white and 3 yellow balls.

21. If 2 balls are picked at random, the probability that either both are blue or both are yellow is

(1)  $\frac{2}{5}$  (2)  $\frac{3}{35}$  (3)  $\frac{1}{3}$

(5) None of these

22. If 5 balls are picked at random, the probability that at least one is blue is

(1)  $\frac{137}{143}$  (2)  $\frac{9}{91}$  (3)  $\frac{18}{45}$

(5) None of these

23. If 2 balls are picked at random, the probability that both are blue is

(1)  $\frac{1}{5}$  (2)  $\frac{8}{91}$  (3)  $\frac{2}{15}$

(5) None of these

24. If 4 balls are picked at random, the probability that 2 are red and 2 are white is

(1)  $\frac{4}{15}$  (2)  $\frac{5}{27}$  (3)  $\frac{1}{3}$

(5) None of these

25. If 3 balls are picked at random, the probability that none is yellow is

(1)  $\frac{3}{455}$  (2)  $\frac{1}{5}$  (3)  $\frac{44}{91}$

(5) None of these

**Directions (26-30) :** Study the information carefully and answer the questions that follow:

An urn contains 3 red, 6 blue, 2 white and 3 yellow marbles.

26. If two marbles are picked at random, the probability that both are green is

(1)  $\frac{2}{15}$  (2)  $\frac{1}{15}$  (3)  $\frac{2}{7}$

(5) None of these

46. If two cards are drawn at random from a pack of 52 cards, find the probability that one is a club and other is a diamond.

- (1)  $\frac{15}{103}$  (2)  $\frac{13}{102}$  (3)  $\frac{14}{101}$

(4) None of these

(5) Data is insufficient

**Directions (47- 49) :** The probability that Dilip will hit a target is  $\frac{1}{3}$  and the probability that Raj will hit a target is  $\frac{1}{2}$ . Find the probability that

(47) Both Dilip and Raj will hit the target

(1)  $\frac{1}{3}$  (2)  $\frac{1}{4}$  (3)  $\frac{1}{6}$  (4)  $\frac{1}{5}$

(5)  $\frac{1}{5}$

48. Exactly one person hits the target ?

- (1)  $\frac{1}{6}$  (2)  $\frac{1}{2}$  (3)  $\frac{1}{3}$  (4)  $\frac{2}{3}$

(5)  $\frac{2}{3}$

49. At least one person hits the target ?

- (1)  $\frac{1}{3}$  (2)  $\frac{2}{3}$  (3)  $\frac{1}{4}$  (4)  $\frac{1}{2}$

(5)  $\frac{1}{5}$

50. The probabilities of P, Q and R solving a problem are  $\frac{2}{3}$ ,  $\frac{5}{7}$  and  $\frac{5}{8}$  respectively. Find the probability that exactly one of them solves the problem.

- (1)  $\frac{33}{167}$  (2)  $\frac{35}{168}$  (3)  $\frac{37}{168}$

(4) None of these

(5) Data is insufficient

## LEVEL-II

1. Find the probability that a number from 1 to 300 is divisible by either 3 or 7?

- (1)  $\frac{37}{75}$  (2)  $\frac{32}{75}$  (3)  $\frac{36}{75}$  (4)  $\frac{28}{75}$

(5)  $\frac{26}{75}$

$$(1) \frac{1}{25} \quad (2) \frac{2}{25} \quad (3)$$

19. In his wardrobe, Timothy has 10 shirts. Two of them are black, the second is brown. In his wardrobe, there are 10 shirts. One of them is black, the other is white. He opens his wardrobe and picks out one shirt, then examines the colour. What is the probability that neither the shirt is black?

(1)  $\frac{1}{12}$       (2)  $\frac{1}{6}$       (3)

20. A programmer noted the following results in running 20 programs. The first 2 programs ran correctly on the first attempt, 7 ran correctly in the second attempt, 5 ran correctly in the third attempt, 3 ran correctly in the fourth attempt, and 3 ran correctly in the fifth attempt.

(1)  $\frac{1}{4}$       (2)  $\frac{1}{3}$       (3)  $\frac{1}{6}$

21. Course materials are sent to distance teaching institution with the understanding that they will send a written report to the Commission.

$\frac{3}{4}$  that the package is damaged.

(1)  $\frac{4}{5}$       (2)  $\frac{8}{60}$       (3)  $\frac{8}{15}$

(5) None of these



11. There are 200 balls numbered 1 to 200. Find the probability that a ball which bears a perfect square and the number is a multiple of 3 or multiple of 4.

(1)  $\frac{33}{199}$  (2)  $\frac{11}{200}$   
(3)  $\frac{9}{197}$  (4)  $\frac{11}{199}$

12. A bag has 4 red and 3 black balls. Another bag has 3 red and 7 black balls. A ball is drawn from the first bag and then a ball is drawn from the second. The probability that the balls drawn are of the same color is

(1)  $\frac{14}{45}$  (2)  $\frac{11}{45}$   
(3)  $\frac{14}{90}$  (4)  $\frac{11}{90}$

13. A class consists of 100 students. 40 are girls and 75 boys are of fair complexion and the remaining are of dark complexion. The probability that a student chosen at random is a fair complexioned rich girl is

(1) 0.05 (2) 0.04  
(3) 0.02 (4) 0.01

14. An anti-aircraft gun can fire four shots at an enemy plane. The probability of hitting the plane on the first, second, third, fourth, and fifth shots are 0.4, 0.3, 0.2 and 0.1 respectively. The probability that the plane is hit by at least one of the four shots is

(1) 0.6872 (2) 0.4379  
(3) 0.6872 (4) None of these

15. Three groups A, B and C are competing for a position on the Board of Directors of a company. The probabilities of A, B and C winning are 0.5, 0.3 and 0.2 respectively. If A wins, the probability that a new product is introduced is 0.7. If B wins, the probability is 0.5. If C wins, the probability is 0.5. The probability that a new product will be introduced is

(1) 0.52 (2) 0.74  
(3) 0.52 (4) None of these

16. An article manufactured by a company consists of two parts A and B. In a day, 6000 articles of part A are manufactured and 5000 articles of part B are manufactured. The probability that an article of part A is defective is 0.1 and the probability that an article of part B is defective is 0.2. The probability that an article chosen at random is a defective article is

(1) 0.8645  
(2) 0.6243  
(3) 0.6243  
(4) None of these

## QUANTITY BASED

**Directions (1-5) :** In each question, two quantities are given. Choose the correct option.

- (1) Quantity I > Quantity II
- (2) Quantity I  $\geq$  Quantity II
- (3) Quantity I < Quantity II
- (4) Quantity II  $\geq$  Quantity I
- (5) Quantity I = Quantity II or the relationship cannot be established

1. **Quantity I :** An urn contains 4 red balls and 6 blue balls of different sizes. Then find the probability of selecting a green ball being selected ball.

**Quantity II :** Two dice are thrown together simultaneously. Find the probability that there is a sum of at least two tails on the

2. **Quantity I :** A fruit basket contains 10 apples and 12 oranges. If 2 mangoes are selected at random, find the probability that either both are mangoes or both are apples.

**Quantity II :** A bag contains 4 red balls and 6 blue balls. If the balls are drawn one by one, find the probability that the 1st ball is red.

3. **Quantity I :** From a well-shuffled deck of 52 playing cards, one card is drawn. What is the probability that the card drawn will be a black king?

**Quantity II :** A bag contains 4 red balls and 6 blue balls. Two balls are drawn simultaneously. What is the probability that both balls are of the same colour?

4. **Quantity I :** The probability of getting a sum of 2 on the first dice and a sum of 12 on the other dice, when two dice are thrown simultaneously is :

**Quantity II :** When two dice are thrown together, the probability that the sum of the faces is 7 is :

5. **Quantity I :** There are two defective bulbs in a box of 10 bulbs. One bulb is selected at random and tested. Find the probability that the other bulb is also defective.

**Quantity II :** A box contains 10 bulbs, 2 of which are defective. Two bulbs are selected at random and tested. Find the probability that both bulbs are defective.

10. (5) Probability of the first ball to be red

$$= \frac{{}^5C_1}{{}^{15}C_1} = \frac{5}{15} = \frac{1}{3}$$

Probability of the second ball to be yellow

$$= \frac{{}^7C_1}{{}^{14}C_1} = \frac{7}{14} = \frac{1}{2}$$

$$\therefore \text{Required probability} = \frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$$

11. (2) Total outcomes = 8

Favourable outcomes = 3

$$\text{Required probability} = \frac{3}{8}$$

12. (1) Total possible outcomes

= Selection of 2 eggs out of 28

$$= {}^{28}C_2 = \frac{28 \times 27}{1 \times 2} = 14 \times 27$$

Favourable outcomes.

= Selecting 1 unrotten egg out of 20 and 1 rotten egg out of 8 + selecting 2 rotten eggs out of 8

$$= {}^{20}C_1 \times {}^8C_1 + {}^8C_2$$

$$= 160 + \frac{8 \times 7}{2} = 188$$

$\therefore$  Required probability

$$= \frac{188}{14 \times 27} = \frac{94}{7 \times 27}$$

$$= \frac{94}{189}$$

13. (3) Total possible outcomes

$$= {}^{17}C_5$$

$$= \frac{17 \times 16 \times 15 \times 14 \times 13}{1 \times 2 \times 3 \times 4 \times 5}$$

$$= 6188$$

Total favourable outcomes

$$= {}^8C_2 \times {}^4C_1 \times {}^5C_2$$

$$= \frac{8 \times 7}{1 \times 2} \times 4 \times \frac{5 \times 4}{1 \times 2}$$

$$= 28 \times 4 \times 10 = 1120$$

$$\text{Required probability} = \frac{1120}{6188} = \frac{280}{1547}$$

14

15.

16.(3)

17.(4)

18.(3)

$$= {}^6C_2 \times {}^4C_1 = \frac{6 \times 5}{1 \times 2} \times 4 = 60$$

$$\text{Required probability} = \frac{P(E)}{P(S)} = \frac{60}{455} = \frac{12}{91}$$

$$28. (2) P(S) = {}^{15}C_4 = \frac{15 \times 14 \times 13 \times 12}{1 \times 2 \times 3 \times 4} = 1365$$

Let no yellow marble is selected.

$\therefore P(E)$  = Selection of 4 marbles out of blues

$$= {}^{11}C_4 = \frac{11 \times 10 \times 9 \times 8}{1 \times 2 \times 3 \times 4} = 330$$

$$\therefore \text{Required probability} = 1 - \frac{330}{1365}$$

$$= 1 - \frac{22}{91} = \frac{91 - 22}{91} = \frac{69}{91}$$

$$29. (2) P(S) = {}^{15}C_2 = 105$$

$$P(E) = {}^3C_2 + {}^2C_2$$

$$= \frac{3 \times 2}{1 \times 2} + 1 = 4$$

$$\therefore \text{Required probability} = \frac{4}{105}$$

$$30. (3) P(S) = {}^{15}C_4 = 1365$$

$$P(E) = {}^2C_1 \times {}^6C_2 \times {}^3C_1$$

$$= 2 \times 15 \times 3 = 90$$

$$\therefore \text{Required probability} = \frac{P(E)}{P(S)} = \frac{90}{1365} = \frac{6}{91}$$

31. (3) Probability of a student being not a swimmer

$$= \frac{1}{5}$$

$\therefore$  Probability of a student being a swimmer

$$\left(1 - \frac{1}{5}\right) = \frac{4}{5}$$

$$\therefore \text{Required probability} = {}^5C_4 \times \left(\frac{1}{5}\right) \left(\frac{4}{5}\right)^4$$

32. (3) According to question, required probability

= probability that ball from bag A is red and both the balls from bag B are black

Or

probability that ball from bag A is black and one black and one red is drawn from bag B

$$= \frac{{}^4C_1 \times {}^7C_2}{{}^9C_1 \times {}^{10}C_2} + \frac{{}^5C_1 \times {}^3C_1 \times {}^7C_2}{{}^9C_1 \times {}^{10}C_2}$$

$$= \frac{4}{9} \times \frac{7}{15} + \frac{5}{9} \times \frac{7}{15} = \frac{7}{15}$$



$$\text{Required probability} = \frac{{}^7C_3 + {}^7C_5}{{}^9C_5}$$

$$= \frac{\frac{7 \times 6 \times 5}{3 \times 2} + \frac{7 \times 6}{2}}{\frac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2}} = \frac{35 + 21}{9 \times 14} = \frac{56}{126} = \frac{4}{9}$$

46. (2) Required probability =  $\frac{C(1)}{C(1)}$

$$= \frac{13 \times 13 \times 2}{52 \times 51} = \frac{13 \times 13}{26 \times 51} = \frac{13}{102}$$

47. (3) A  $\rightarrow$  Dilip hits the target  
B  $\rightarrow$  Raj hits the target

$$\text{Required Probability} = P(A) \cdot P(B)$$

48. (2) Required Probability =  $P(A) \cdot P(B)$

$$= \frac{1}{3} \times \frac{1}{2} + \frac{2}{3} \times \frac{1}{2} = \frac{1}{6} + \frac{2}{6} = \frac{1}{2}$$

49. (2) Required Probability =  $1 -$

$$\text{by any one} = 1 - \left( \frac{2}{3} \times \frac{1}{2} \right) = 1 - \frac{1}{3}$$

**Method-2 :**

Required Probability = P (target is hit by one) + P (target is hit by two)

$$= \left[ \left( \frac{1}{3} \times \frac{1}{2} \right) + \left( \frac{2}{3} \times \frac{1}{2} \right) \right] + \left( \frac{1}{3} \times \frac{1}{2} \right)$$

$$= \frac{4}{6} = \frac{2}{3}$$

50. (3) Required probability =  $P(A) \cdot P(B) \cdot P(C)$

$$= P(\bar{A}) \cdot P(\bar{B}) \cdot P(\bar{C}) + P(\bar{A}) \cdot P(B) \cdot P(\bar{C}) + P(A) \cdot P(\bar{B}) \cdot P(\bar{C}) + P(A) \cdot P(B) \cdot P(\bar{C})$$

$$= \left( \frac{2}{3} \times \frac{2}{7} \times \frac{3}{8} \right) + \left( \frac{1}{3} \times \frac{5}{7} \times \frac{3}{8} \right) + \left( \frac{1}{3} \times \frac{2}{7} \times \frac{3}{8} \right) + \left( \frac{1}{3} \times \frac{2}{7} \times \frac{3}{8} \right)$$

$$= \frac{12 + 15 + 10}{3 \times 7 \times 8} = \frac{37}{3 \times 56} = \frac{37}{168}$$

## LEVEL-II

1. (2) Number of numbers from 1 to 100 divisible by 3 = 100

Number of numbers divisible by 4 = 25

Number of numbers divisible by 14 = 7

14

$$\therefore \frac{x}{76} = \frac{2}{19}$$

$$\Rightarrow x = \frac{2 \times 76}{19} = 8$$

$$\therefore \text{Blue balls} = 15 - 8 = 7$$

$$\therefore \text{Required difference} \\ = 8 - 7 = 1$$

14. (3) There are 30 balls in the box as probabil-

$$\text{ity of choosing 1 blue ball} = \frac{1}{3}$$

Multiples of 2 up to 30

$$= \frac{30}{2} = 15$$

Multiples of 3 up to 30

$$= \frac{30}{3} = 10$$

Multiples of 6 ( $2 \times 3$ ) up to 30

$$= \frac{30}{6} = 5$$

So, total such numbers are

$$= 15 + 10 - 5 = 20$$

So, required probability

$$= \frac{20}{30} = \frac{2}{3}$$

15. (3) The required probability = X is selected and Y is rejected and Y is selected and X is rejected

$$= \frac{1}{5} \times \left(1 - \frac{2}{7}\right) + \left(1 - \frac{1}{5}\right) \times \frac{2}{7}$$

$$= \left[\left(\frac{1}{5}\right) \times \left(\frac{5}{7}\right)\right] + \left[\left(\frac{4}{5}\right) \times \left(\frac{2}{7}\right)\right]$$

$$= \frac{5}{35} + \frac{8}{35} = \frac{13}{35}$$

16. (2) Total number of balls

$$= 4 + 5 + 6 = 15$$

Total possible outcomes

$$= \text{Selection of 1 ball out of 15 balls} = {}^{15}C_1 = 15$$

Let  $E_1$  = Event of drawing 1 red ball

$E_2$  = Event of drawing 1 green ball.

Then,  $E_1 \cap E_2 = \phi$

$$P(E_1 \text{ or } E_2) = P(E_1) + P(E_2)$$

$$= \frac{4}{15} + \frac{5}{15} = \frac{9}{15} = \frac{3}{5}$$

(ii) 0 0 6 5  
1 1 5 4 net effect  $\rightarrow$  1  
ward

2 2 4 3  
3 3 3 2  
4 4 2 1  
5 5 1 0

(iii) 6 5 0 0 Backward an  
steps

5 4 1 1 are nullified ne  
4 3 2 2 1 step towards  
3 2 3 3  
2 1 4 4  
1 0 5 5

(iv) 5 6 0 0  
4 5 1 1 net effect  $\rightarrow$  1 st  
3 4 2 2 right

2 3 3 3  
1 2 4 4  
0 1 5 5

After listing the exhaustive no. of  
shown above, let us now find the  
probability.

When, out of 11 steps, 6 steps are  
steps are same, no. of arrangement  
there are a Total of 4 such cases her

$$\frac{11!}{6!5!}$$

Similarly, all the cases, can be listed

$$4 \left[ \frac{11!}{6!5!} + \frac{11!}{4!5!1!1!} + \frac{11!}{3!4!2!2!} + \frac{11!}{2!3!3!3!} + \frac{11!}{1!2!4!4!} \right]$$

$$= 4 \left[ \frac{11 \times 10 \times 9 \times 8 \times 7}{120} + \frac{11 \times 10 \times 9 \times 8 \times 7 \times 6}{24} + \frac{11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5}{6 \times 2 \times 2} + \frac{11!}{2 \times 6 \times 6 \times 6} + \frac{11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5}{24 \times 2} + \frac{11 \times 10 \times 9 \times 8 \times 7 \times 6}{120} \right]$$

$$= 4[462 + 13860 + 69300 + 92400 + 34650]$$

$$= 4(213444)$$

Total no. of arrangements of 11 steps =

$$\text{Required probability} = \frac{4 \times 213444}{4^{11}} = \frac{5336}{4^9}$$

**Method-2 :** After listing the above cases  
can calculate as follows :

Required probability

$$= \frac{12 \times 11 \times 10 \times 9}{36 \times 35 \times 34 \times 33} = \frac{1}{119}$$

Required probability

$$= \left( \frac{352}{3927} \right) + \left( \frac{1}{119} \right)$$

$$= \frac{352 + 33}{3927} = \frac{385}{3927} = \frac{5}{51}$$

2. (4) The team should not comprise of all men or all women

Total number of persons = 6 + 6 = 12

Number of ways of selecting 5 persons out of 12

$$= {}^{12}C_5 = \frac{12!}{7!5!}$$

$$= \frac{12 \times 11 \times 10 \times 9 \times 8}{5 \times 4 \times 3 \times 2 \times 1}$$

$$= 99 \times 8 = 792$$

Number of ways of selecting 5 men out of 6

$$\text{men} = {}^6C_5 = 6$$

Number of ways of selecting 5 women out of 6

$$\text{women} = {}^6C_5 = 6$$

Number of ways of selecting all 5 men or all 5 women team = 6 + 6 = 12

Number of ways of selecting a team of 5 persons with atleast 1 man and 1 woman

$$= 792 - 12 = 780$$

$$\therefore \text{Required probability} = \frac{780}{792} = \frac{65}{66}$$

3. (1) Total words that can be formed using B, R, I, N, G = 5!

$$\text{Probability of getting BRING} = \frac{1}{5!} = \frac{1}{120}$$

4. (1)  $\therefore$  Required probability

$$= P(A) + P(A)P(B)P(A) + \dots$$

$$= \frac{1}{2} + \left( \frac{1}{2} \right)^3 + \left( \frac{1}{2} \right)^5 + \dots$$

$$\text{Sum of G.P. upto infinity} = \frac{\frac{1}{2}}{1 - \frac{1}{4}} = \frac{2}{3}$$

5. (3) According to question, the product of four numbers will be positive in the following ways.



$$= 4 \times 3 \times 2 \times 1 = 24$$

number of favourable

Required probability

[ **Note :** If we look at all end in either 25 or 75, then only possible should be 25 only.  
No. of favourable cases

| Th | H |
|----|---|
| 2  | 1 |

Ten's & units place  
Thousands place by  
its and hundred's place

So favourable cases =  
total no of cases  
 $= 4 \times 3 \times 2 \times 1 = 24$

$$\frac{2}{24} = \frac{1}{12}$$

19. (4)  $px^2 + qx + r = 0$  will  
discriminant is greater than  
i.e.  $D \geq 0$  or  $b^2 - 4ac \geq 0$   
 $q^2 > 4pr$   
→ when,  $q = 1$ , no value  
for which  $q^2 > 4pr$ .  
→ when,  $q = 2$ ,  $q^2 = 4$ .  
→ when,  $q = 3$ ,  $q^2 = 9$ .  
→ when,  $q = 4$ ,  $q^2 = 16$ .  
 $= (1, 1) (1, 2) (1, 3) (1, 4)$   
→ when,  $q = 5$ ,  $q^2 = 25$ .  
 $(1, 3) (1, 4) (1, 5) (2, 1) (2, 2)$   
 $(5, 1)$

Favourable no. of cases =  
Total no. of cases =  $5 \times 5$

$$\text{Required probability} = \frac{24}{125}$$

20. (3) Out of 20 coins, 10 always  
maximum we can have 10  
definite with probability =  
 $P(\text{at least 9 tails}) = P(9 \text{ tails})$   
 $P(9 \text{ tails}) = P(4 \text{ biased tails})$   
 $\times P(5 \text{ tails and 1 head})$   
unbiased coins)

$$= 1 \times 1 \times \left[ \left( \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \right) \times \left( \frac{1}{2} \right) \right]$$

28. (4) We are given Mayank has passed in Q and E only or Q and D only or Q and D and E.

$$\text{i.e. } P(Q)P(E)P(\bar{D}) + P(Q)P(D)P(\bar{E}) + P(Q)P(D)P(E)$$

$$= (a)(b)\left(\frac{2}{3}\right) + (a)\left(\frac{1}{3}\right)(1-b) + (a)\left(\frac{1}{3}\right)(b)$$

$$= \frac{2ab + a(1-b) + ab}{3}$$

$$= \frac{2ab + a - ab + ab}{3} = \frac{2ab + a}{3}$$

$$\text{Now, } \frac{2ab + a}{3} \text{ is equal to } \frac{1}{3}$$

$$\Rightarrow 2ab + a = 1 \Rightarrow a = \frac{1}{1+2b}$$

$$\text{when, } b=1, a = \frac{1}{3}$$

$$\text{when, } a=1, b=0$$

$$\text{when, } b = \frac{1}{2}, a = \frac{1}{2}$$

$$\text{when, } b = \frac{1}{4}, a = \frac{2}{3} \text{ \& so on.}$$

Hence, there are infinite solutions.

29. (2) Maximum value of  $xy = 2500$  i.e. when  $x = y$

$$= 50 \text{ and } \frac{16}{25}(2500) = 1600$$

i.e.  $xy$  should not be less than 1600

$$xy \geq 1600$$

When,  $x = 20$ ,  $y = 80$ , then  $xy = 1600$

$$\text{i.e. } 20 \times 80 = 1600$$

$$21 \times 79 = 1659$$

$$22 \times 78 = 1716$$

$$30 \times 70 = 2100 \text{ still increasing}$$

$$40 \times 60 = 2400$$

$$50 \times 50 = 2500$$

$$51 \times 49 = 2499$$

$$80 \times 20 = 1600$$

30.

QUA

1. (1) Qu

 $B \rightarrow gr$ 

$$P\left(\frac{A}{B}\right) =$$

$$A \cap B =$$

$$\therefore P(A \cap$$

$$\therefore P\left(\frac{A}{B}\right) =$$

Quantity

$$n(S) = 3$$

$$n(A) = \{(1$$

$$P(A) = \frac{5}{36}$$

Some useful terms

**Constants** : A symbol

value is called a constant.

**Variables** : A symbol for numerical values is called a variable.

We know that the circumference of a circle is given by the formula  $C = 2\pi r$ , where  $C$  is the circumference,  $r$  is the radius of the circle. Here  $C$  and  $r$  are variables.

**Algebraic Expression** : An expression containing constants and variables, connected by fundamental operations (addition, subtraction, multiplication, division) is called an algebraic expression.

The several parts of the expression connected by + or - sign are called terms.

**Polynomials** : An algebraic expression in which the variables involved have only integral powers, is called a polynomial.

**Degree of Polynomial in One Variable** : The highest power of a polynomial in one variable is called the degree of the polynomial.

**Degree of Polynomial in Two Variables** : In case of polynomials in two variables, the sum of the powers of the variables in each term is taken up and the highest value obtained is known as the degree of the polynomial.

**Polynomials of Various Degrees**

(i) A polynomial of degree 1 is called a linear polynomial, e.g.,  $3 + 5x$ ,  $5 - 3y$

(ii) A polynomial of degree 2 is called a quadratic polynomial.

(iii) A polynomial of degree 3 is called a cubic polynomial.

(iv) A polynomial of degree 4 is called a quartic polynomial.

**Some useful terms**

**Constants** : A symbol whose value is called a constant.

**Variables** : A symbol whose numerical values is known.

We know that the circumference of a circle is given by the formula  $C = 2\pi r$ , where  $C$  is the circumference,  $r$  is the radius of the circle. Here  $C$  is a constant, whereas  $C$  and  $r$  are variables.

**Algebraic Expressions** : An expression containing constants and variables, connected by the fundamental operations (addition, subtraction, multiplication, division) is called an algebraic expression.

The several parts of the expression separated by  $+$  or  $-$  sign are called terms.

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**Degree of Polynomial in one variable** : The highest power of the variable in one variable polynomial is called the degree of the polynomial.

**Degree of Polynomial in two or more variables** : In case of polynomials in two or more variables, the sum of the powers of the variables in each term is taken up and the highest value obtained is known as the degree of the polynomial.

**Polynomials of Various Degrees**

(i) A polynomial of degree 1 is called a linear polynomial, e.g.,  $3 + 5x$ ,  $5 - 3y$ .

(ii) A polynomial of degree 2 is called a quadratic polynomial.

(iii) A polynomial of degree 3 is called a cubic polynomial.

(iv) A polynomial of degree 4 is called a quartic polynomial.



(13) If  $a + b + c = 0$ ,  
then  $a^3 + b^3 + c^3 = 3abc$ .

(14)  $(a + b + c)^3 = a^3 + b^3 + c^3 + 3(a+b)(b+c)(c+a)$ .

(15)  $a^2 + b^2 = (a+b)^2 - 2ab$ .

(16)  $a^2 + b^2 = (a-b)^2 + 2ab$ .

(17)  $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$ .

### Method of Sp

Factors of the p

Examine the f

$(x + 2)(x + 5) = x^2 + 7x + 10$ .

Note that, in the above example, the factors are obtained by multiplying the constant term (10) and adding together 2 and 5 to get 7.

Hence, in order to factorise a quadratic equation, we must find two numbers whose product is the constant term and sum is 7.

Since  $(x + a)(x + b) = x^2 + (a + b)x + ab$ ,

$= x^2 + (a + b)x + ab$ .

$\therefore x^2 + (a + b)x + ab = (x + a)(x + b)$ .

### QUADRATIC EQUATION

#### Quadratic Equation

An equation of the form  $ax^2 + bx + c = 0$ , where  $a, b, c$  are given numbers and  $x$  is an unknown quantity, is called a quadratic equation.

The numbers  $a, b, c$  are called the coefficients of the quadratic equation.  $a$  is the coefficient of  $x^2$ ,  $b$  is the coefficient of  $x$ , and  $c$  is the constant term.

**Ex.**  $3x^2 + 2x - 1 = 0$ . Here,  $a = 3, b = 2, c = -1$ .

The values of the variable ( $x$ ) which satisfy the equation are called its roots.

**For example,**  $x = 1$  is a root of  $3x^2 + 2x - 1 = 0$  because  $3(1)^2 + 2(1) - 1 = 0$ .

**Remember :**

● A quadratic equation has at most two roots.

● Solving a quadratic equation means finding its all roots.

### SOLVING A QUADRATIC EQUATION

A quadratic equation can be solved by the factor method or by the formula method.

**(i) Factor Method :** It is the method in which the quadratic equation is written in the form of two linear factors.

We take an example :

$x^2 - 5x + 6 = 0$

**Directions**  
question  
have to s

Give a

(1)  $x$

(3)  $x$

(5)  $x$

no

21. I. (17

II. (26

22. I.  $\sqrt{25}$

II.  $\sqrt{36}$

23. I.  $9x -$

II.  $\sqrt{y + 1}$

24. I.  $x^2 + 1$

II.  $y^2 + 7$

25. I.  $x^2 \times 13$

II.  $y^{1/3} \times 14$

**Directions (26-30)**  
one or more equa  
basis of these yo  
tween p and q.

**Give answer (1)**  
be established.

**Give answer (2)**

**Give answer (3)**

**Give answer (4)**

**Give answer (5)**

26. I.  $4q^2 + 8q = 4q$

II.  $p^2 + 9p = 2p$

27. I.  $2p^2 + 40 = 18$

II.  $q^2 = 13q - 42$

28. I.  $6q^2 + \frac{1}{2} = \frac{7}{2}q$

II.  $12p^2 + 2 = 10p$

**Directions (29-33)**  
questions two equati  
solve them and

**Give answer (1)** if p

**Give answer (2)** if p

**Give answer (3)** if p

**Give answer (4)** if p

**Give answer (5)** if p  
be established.

29. I.  $p^2 - 7p = -12$

II.  $q^2 - 3q + 2 = 0$

30. I.  $12p^2 - 7p = -1$

II.  $6q^2 - 7q + 2 = 0$

31. I.  $p^2 + 12p + 35 = 0$

II.  $2q^2 + 22q + 56 = 0$

17. I.  $2x^2 - 9x + 10 = 0$

II.  $2y^2 - 13y + 20 = 0$

(1)  $x \leq y$  (2)  $x \geq y$

(3) relationship between  $x$  and  $y$  cannot be determined

(4)  $x < y$  (5)  $x > y$

18. I.  $2x^2 + 15x + 27 = 0$

II.  $2y^2 + 7y + 6 = 0$

(1)  $x \leq y$  (2)  $x \geq y$

(3) relationship between  $x$  and  $y$  cannot be determined

(4)  $x < y$  (5)  $x > y$

19. I.  $3x^2 - 13x + 12 = 0$

II.  $3y^2 - 13y + 14 = 0$

(1)  $x \leq y$  (2)  $x \geq y$

(3) relationship between  $x$  and  $y$  cannot be determined

(4)  $x < y$  (5)  $x > y$

20. I.  $5x^2 + 8x + 3 = 0$

II.  $3y^2 + 7y + 4 = 0$

(1)  $x \leq y$  (2)  $x \geq y$

(3) relationship between  $x$  and  $y$  cannot be determined

(4)  $x < y$  (5)  $x > y$

**Directions (21-25) :** In each of the following questions, two equations I and II have been given. Solve these questions and answer the following questions.

(1) If  $x < y$

(2) If  $x \leq y$

(3) If  $x = y$  or the relation cannot be established

(4) If  $x \geq y$

(5) If  $x > y$

21. I.  $30x^2 + 11x + 1 = 0$

II.  $42y^2 + 13y + 1 = 0$

22. I.  $x^2 - x - \sqrt{2}x + \sqrt{2} = 0$

II.  $y^2 - 3y + 2 = 0$

23. I.  $x^2 - 2x - \sqrt{5}x + 2\sqrt{5} = 0$

II.  $y^2 - \sqrt{3}y - \sqrt{2}y + \sqrt{6} = 0$

24. I.  $x^2 + 12x + 36 = 0$

II.  $y^2 = 16$

25. I.  $9x^2 + 3x - 2 = 0$

II.  $8y^2 + 6y + 1 = 0$

## SHORT ANSWER

## LEVEL-I

|         |         |         |         |   |
|---------|---------|---------|---------|---|
| 1. (3)  | 2. (2)  | 3. (1)  | 4. (4)  |   |
| 7. (3)  | 8. (4)  | 9. (2)  | 10. (5) | 1 |
| 13. (3) | 14. (1) | 15. (4) | 16. (5) | 1 |
| 19. (3) | 20. (3) | 21. (3) | 22. (5) | 2 |
| 25. (3) | 26. (3) | 27. (3) | 28. (4) | 2 |
| 31. (5) | 32. (4) | 33. (3) | 34. (3) | 3 |
| 37. (1) | 38. (5) | 39. (3) | 40. (3) | 4 |
| 43. (4) | 44. (2) | 45. (5) | 46. (3) | 4 |
| 49. (1) | 50. (5) |         |         |   |

## LEVEL-II

|         |         |         |         |   |
|---------|---------|---------|---------|---|
| 1. (5)  | 2. (3)  | 3. (1)  | 4. (5)  |   |
| 7. (3)  | 8. (2)  | 9. (4)  | 10. (1) | 1 |
| 13. (1) | 14. (4) | 15. (2) | 16. (1) | 1 |
| 19. (3) | 20. (2) | 21. (2) | 22. (3) | 2 |
| 25. (3) |         |         |         |   |

## LEVEL-III

|         |         |         |         |     |
|---------|---------|---------|---------|-----|
| 1. (5)  | 2. (2)  | 3. (3)  | 4. (1)  | 5.  |
| 7. (1)  | 8. (5)  | 9. (1)  | 10. (1) | 11. |
| 13. (2) | 14. (2) | 15. (2) | 16. (3) | 17. |
| 19. (2) | 20. (3) |         |         |     |

## EXPLANATIONS

## LEVEL-I

1. (3) I.  $2x^2 + 11x + 14 = 0$   
 $\Rightarrow 2x^2 + 4x + 7x + 14 = 0$   
 $\Rightarrow 2x(x+2) + 7(x+2) = 0$   
 $\Rightarrow (x+2)(2x+7) = 0$

$\Rightarrow x = -2$  or,  $-\frac{7}{2}$

II.  $4y^2 + 12y + 9 = 0$   
 $\Rightarrow (2y)^2 + 2 \times 2y \times 3 + (3)^2 = 0$   
 $\Rightarrow (2y+3)^2 = 0$

$\Rightarrow 2y+3 = 0 \Rightarrow y = -\frac{3}{2}$

Clearly,  $x < y$



# SHORT ANSWERS

## LEVEL-I

|         |         |         |     |
|---------|---------|---------|-----|
| 1. (3)  | 2. (2)  | 3. (1)  | 4.  |
| 7. (3)  | 8. (4)  | 9. (2)  | 10. |
| 13. (3) | 14. (1) | 15. (4) | 16. |
| 19. (3) | 20. (3) | 21. (3) | 22. |
| 25. (3) | 26. (3) | 27. (3) | 28. |
| 31. (5) | 32. (4) | 33. (3) | 34. |
| 37. (1) | 38. (5) | 39. (3) | 40. |
| 43. (4) | 44. (2) | 45. (5) | 46. |
| 49. (1) | 50. (5) |         |     |

## LEVEL-II

|         |         |         |     |
|---------|---------|---------|-----|
| 1. (5)  | 2. (3)  | 3. (1)  | 4.  |
| 7. (3)  | 8. (2)  | 9. (4)  | 10. |
| 13. (1) | 14. (4) | 15. (2) | 16. |
| 19. (3) | 20. (2) | 21. (2) | 22. |
| 25. (3) |         |         |     |

## LEVEL-III

|         |         |         |         |
|---------|---------|---------|---------|
| 1. (5)  | 2. (2)  | 3. (3)  | 4. (1)  |
| 7. (1)  | 8. (5)  | 9. (1)  | 10. (1) |
| 13. (2) | 14. (2) | 15. (2) | 16. (3) |
| 19. (2) | 20. (3) |         |         |

# EXPLANATION

## LEVEL-I

1. (3) I.  $2x^2 + 11x + 14 = 0$   
 $\Rightarrow 2x^2 + 4x + 7x + 14 = 0$   
 $\Rightarrow 2x(x+2) + 7(x+2) = 0$   
 $\Rightarrow (x+2)(2x+7) = 0$   
 $\Rightarrow x = -2 \text{ or } -\frac{7}{2}$   
 II.  $4y^2 + 12y + 9 = 0$   
 $\Rightarrow (2y)^2 + 2 \times 2y \times 3 + (3)^2 = 0$   
 $\Rightarrow (2y+3)^2 = 0$   
 $\Rightarrow 2y+3 = 0 \Rightarrow y = -\frac{3}{2}$   
 Clearly,  $x < y$

$$\Rightarrow y^3 - \frac{(14)^2}{\sqrt{y}}$$

$$\Rightarrow y^3 \times \sqrt{y} = (14)^2$$

$$\Rightarrow y^{\frac{7}{2}} = (14)^2 \Rightarrow y = 14$$

$$17. (3) \text{ I. } \frac{5}{7} - \frac{5}{21} = \frac{\sqrt{x}}{42}$$

$$\Rightarrow \frac{15-5}{21} = \frac{\sqrt{x}}{42}$$

$$\Rightarrow \sqrt{x} = \frac{10}{21} \times 42 = 20$$

$$\therefore x = 20 \times 20 = 400$$

$$\text{II. } \frac{\sqrt{y}}{4} + \frac{\sqrt{y}}{16} = \frac{250}{\sqrt{y}}$$

$$\Rightarrow \frac{4\sqrt{y} + \sqrt{y}}{16} = \frac{250}{\sqrt{y}}$$

$$\Rightarrow 5\sqrt{y} \times \sqrt{y} = 250 \times 16$$

$$\Rightarrow 5y = 250 \times 16$$

$$\Rightarrow y = \frac{250 \times 16}{5} = 800$$

$$18. (1) \text{ I. } (625)^{\frac{1}{4}} x + \sqrt{1225} = 155$$

$$\Rightarrow (5^4)^{\frac{1}{4}} x + 35 = 155$$

$$\Rightarrow 5x = 155 - 35 \Rightarrow 5x = 120$$

$$\Rightarrow x = \frac{120}{5} = 24$$

$$\text{II. } \sqrt{196y} + 13 = 279$$

$$\Rightarrow 14y = 279 - 13 = 266$$

$$\Rightarrow y = \frac{266}{14} = 19$$

$$19. (3) \text{ I. } \frac{13}{\sqrt{x}} + \frac{9}{\sqrt{x}} = \sqrt{x}$$

$$\Rightarrow 13 + 9 = \sqrt{x} \times \sqrt{x} = x$$

$$\Rightarrow x = 22$$

$$\text{II. } y^4 - \frac{(26)^2}{\sqrt{y}} = 0$$

$$\Rightarrow (y)^2 = (26)^2$$

$$20. (3) \text{ I. } \sqrt{500x} = -\sqrt{402}$$

$$\Rightarrow x = -\sqrt{\frac{402}{500}} = -\sqrt{\frac{400}{500}} = -0.9$$

21. (3)

II

C

22. (5)

 $\Rightarrow$  $\Rightarrow$  $\Rightarrow$  $\therefore$ 

II.

 $\Rightarrow$  $\Rightarrow$ 

23. (5)

 $\Rightarrow$ II.  $\sqrt{y}$  $\Rightarrow y +$  $\Rightarrow y =$ 

24. (3) I. x

 $\Rightarrow x^2 +$  $\Rightarrow x(x$  $\Rightarrow (x +$  $\therefore x = -$ II.  $y^2 +$  $\Rightarrow y^2 +$  $\Rightarrow y(y +$  $\Rightarrow (y +$  $\therefore y = -$ 

Clearly

25. (3) I.  $x^3$  $\Rightarrow \frac{x^3}{x^2} =$ II.  $y^3 \times$  $\Rightarrow \frac{1}{y^3} \times$  $\Rightarrow \frac{1}{y^3} \times$  $\Rightarrow \frac{1}{y^3} \times$  $\Rightarrow \frac{1}{y^3} \times$  $\Rightarrow \frac{1}{y^3} \times$  $\Rightarrow \frac{1}{y^3} \times$  $\Rightarrow \frac{1}{y^3} \times$  $\Rightarrow \frac{1}{y^3} \times$  $\Rightarrow \frac{1}{y^3} \times$  $\Rightarrow \frac{1}{y^3} \times$  $\Rightarrow \frac{1}{y^3} \times$  $\Rightarrow \frac{1}{y^3} \times$  $\Rightarrow \frac{1}{y^3} \times$  $\Rightarrow \frac{1}{y^3} \times$

$$\begin{array}{r}
 x^2 + 5x + 6 \\
 x+1 \overline{) x^2 + 6x^2 + 11x + 6} \\
 \underline{x^2 + x^2} \phantom{+ 6x + 6} \\
 5x^2 + 11x \phantom{+ 6} \\
 \underline{5x^2 + 5x} \phantom{+ 6} \\
 6x + 6 \\
 \underline{6x + 6} \\
 0
 \end{array}$$

$$\begin{aligned}
 \therefore x^2 + 5x + 6 &= 0 \\
 \Rightarrow x^2 + 3x + 2x + 6 &= 0 \\
 \Rightarrow x(x+3) + 2(x+3) &= 0 \\
 \Rightarrow (x+3)(x+2) &= 0 \\
 \Rightarrow x &= -3 \text{ or } -2
 \end{aligned}$$

$$\text{II. } \frac{y^3 + 15y^2 + 74y + 120}{y+4} = 0$$

$$\begin{array}{r}
 y^2 + 11y + 30 \\
 y+4 \overline{) y^3 + 15y^2 + 74y + 120} \\
 \underline{y^3 + 4y^2} \phantom{+ 74y + 120} \\
 11y^2 + 74y \phantom{+ 120} \\
 \underline{11y^2 + 44y} \phantom{+ 120} \\
 30y + 120 \\
 \underline{30y + 120} \\
 0
 \end{array}$$

$$\begin{aligned}
 \therefore y^2 + 11y + 30 &= 0 \\
 \Rightarrow y^2 + 5y + 6y + 30 &= 0 \\
 \Rightarrow y(y+5) + 6(y+5) &= 0 \\
 \Rightarrow (y+5)(y+6) &= 0 \\
 \Rightarrow y &= -5 \text{ or } -6
 \end{aligned}$$

Clearly,  $x > y$

$$35. (3) \text{ I. } x(x+2) - 6\left(\frac{4x}{3} - \frac{3}{2}\right) = 0$$

$$\begin{aligned}
 \Rightarrow x^2 + 2x - 8x + 9 &= 0 \\
 \Rightarrow x^2 - 6x + 9 &= 0 \\
 \Rightarrow (x-3)^2 &= 0 \\
 \Rightarrow x-3 &= 0 \\
 \Rightarrow x &= 3
 \end{aligned}$$

$$\text{II. } 5\left(y - \frac{2}{3}\right) - y(1+y) = 0$$

$$\Rightarrow 5y - \frac{10}{3} - y - y^2 = 0$$

$$\Rightarrow y^2 - 4y + \frac{10}{3} = 0$$

$$\Rightarrow 3y^2 - 12y + 10 = 0$$

$$\Rightarrow y = \frac{+12 \pm \sqrt{144 - 4 \times 3 \times 10}}{6}$$

36.

I.

II. y

37. (1)

II.

⇒

∴

⇒

⇒

Fro

x =

=

=

Clea

38. (5) I

⇒ 5.

⇒ 5.

⇒ 5.

⇒ 5.

⇒ x.

II. 6y

⇒ 6y

⇒ 6y

⇒ (y

⇒ y =

39. (3)

I. 2x

⇒ 2x

⇒ 2x

⇒ (2x

⇒ x =

II. 2y

⇒ 2y

⇒ 2y

$$\Rightarrow x = \frac{4}{3} \text{ or } \frac{5}{3}$$

$$\text{II. } 6y^2 - 5y + 1 = 0$$

$$\Rightarrow 6y^2 - 3y - 2y + 1 = 0$$

$$\Rightarrow 3y(2y - 1) - 1(2y - 1) = 0$$

$$\Rightarrow (3y - 1)(2y - 1) = 0$$

$$\Rightarrow y = \frac{1}{3} \text{ or } \frac{1}{2}$$

Clearly,  $x > y$

$$48. (4) \text{ I. } x^2 - 6x + 9 = 0$$

$$\Rightarrow (x - 3)^2 = 0 \Rightarrow x - 3 = 0$$

$$\Rightarrow x = 3$$

$$\text{II. } y^2 - 11y + 24 = 0$$

$$\Rightarrow y^2 - 8y - 3y + 24 = 0$$

$$\Rightarrow y(y - 8) - 3(y - 8) = 0$$

$$\Rightarrow (y - 3)(y - 8) = 0$$

$$\Rightarrow y = 3 \text{ or } 8$$

$$49. (1) \text{ I. } 8x^2 - 10x - 63 = 0$$

$$\Rightarrow 8x^2 + 18x - 28x - 63 = 0$$

$$\Rightarrow 2x(4x + 9) - 7(4x + 9) = 0$$

$$\Rightarrow (2x - 7)(4x + 9) = 0$$

$$\Rightarrow x = \frac{7}{2} \text{ or } -\frac{9}{4}$$

$$\text{II. } 5y^2 + 33y + 54 = 0$$

$$\Rightarrow 5y^2 + 15y + 18y + 54 = 0$$

$$\Rightarrow 5y(y + 3) + 18(y + 3) = 0$$

$$\Rightarrow (y + 3)(5y + 18) = 0$$

$$\Rightarrow y = -3 \text{ or } -\frac{18}{5}$$

$$50. (5) \text{ I. } x^2 - 9x - 52 = 0$$

$$\Rightarrow x^2 - 13x + 4x - 52 = 0$$

$$\Rightarrow x(x - 13) + 4(x - 13) = 0$$

$$\Rightarrow (x - 13)(x + 4) = 0$$

$$\Rightarrow x = 13 \text{ or } -4$$

$$\text{II. } y^2 - 5y - 84 = 0$$

$$\Rightarrow y^2 - 12y + 7y - 84 = 0$$

$$\Rightarrow y(y - 12) + 7(y - 12) = 0$$

$$\Rightarrow (y + 7)(y - 12) = 0$$

$$\Rightarrow y = -7 \text{ or } 12$$

### LEVEL-II

$$1. (5) \text{ I. } x^2 + 2x + 4 = 0$$

$$\Rightarrow x = \frac{-2 \pm \sqrt{(2)^2 - 4 \times 4}}{2}$$

$$= \frac{-2 \pm \sqrt{4 - 16}}{2}$$

$$= \frac{-2 \pm 2\sqrt{-3}}{2} = -1 \pm \sqrt{-3}$$

2. (3)

3. (1)

4. (5)

5. (5)

6. (5)

I. 4x



$$\Rightarrow (y+4)(y+5) = 0$$

$$\Rightarrow y = -4 \text{ or } -5$$

Clearly,  $x > y$

$$14. (4) \text{ I. } x^2 - 7x + 10 = 0$$

$$\Rightarrow x^2 - 2x - 5x + 10 = 0$$

$$\Rightarrow x(x-2) - 5(x-2) = 0$$

$$\Rightarrow (x-5)(x-2) = 0$$

$$\Rightarrow x = 5 \text{ or } 2$$

$$\text{II. } y^2 - 12y + 35 = 0$$

$$\Rightarrow y^2 - 5y - 7y + 35 = 0$$

$$\Rightarrow y(y-5) - 7(y-5) = 0$$

$$\Rightarrow (y-5)(y-7) = 0$$

$$\Rightarrow y = 5 \text{ or } 7$$

Clearly,  $x \leq y$

$$15. (2) \text{ I. } (x-12)^2 = 0$$

$$\Rightarrow (x-12)(x-12) = 0$$

$$\Rightarrow x = 12, 12$$

$$\text{II. } y^2 = 144$$

$$\Rightarrow y = \sqrt{144} = \pm 12$$

Clearly,  $x \geq y$

$$16. (1) \text{ I. } x^2 = 144$$

$$\Rightarrow x = \sqrt{144} = \pm 12$$

$$\text{II. } y^2 - 24y + 144 = 0$$

$$\Rightarrow (y-12)^2 = 0$$

$$\Rightarrow y - 12 = 0 \Rightarrow y = 12$$

Clearly,  $x \leq y$

$$17. (1) \text{ I. } 2x^2 - 9x + 10 = 0$$

$$\Rightarrow 2x^2 - 4x - 5x + 10 = 0$$

$$\Rightarrow 2x(x-2) - 5(x-2) = 0$$

$$\Rightarrow (x-2)(2x-5) = 0$$

$$\Rightarrow x = 2 \text{ or } \frac{5}{2}$$

$$\text{II. } 2y^2 - 13y + 20 = 0$$

$$\Rightarrow 2y^2 - 8y - 5y + 20 = 0$$

$$\Rightarrow 2y(y-4) - 5(y-4) = 0$$

$$\Rightarrow (y-4)(2y-5) = 0$$

$$\Rightarrow y = 4 \text{ or } \frac{5}{2}$$

Clearly,  $x \leq y$

$$18. (4) \text{ I. } 2x^2 + 15x + 27 = 0$$

$$\Rightarrow 2x^2 + 6x + 9x + 27 = 0$$

$$\Rightarrow 2x(x+3) + 9(x+3) = 0$$

$$\Rightarrow (2x+9)(x+3) = 0$$

$$\Rightarrow x = -\frac{9}{2} \text{ or } -3$$

$$\text{II. } 2y^2 + 7y + 6 = 0$$

$$\Rightarrow 2y^2 + 4y + 3y + 6 = 0$$

$$\Rightarrow 2y(y+2) + 3(y+2) = 0$$

$$\Rightarrow (2y+3)(y+2) = 0$$

22.

$$\begin{aligned} &\Rightarrow 5x(3x - 1) - 2(3x - 1) = 0 \\ &\Rightarrow (3x - 1)(5x - 2) = 0 \\ &\Rightarrow x = \frac{1}{3} \text{ or } \frac{2}{5} = 0.33 \text{ or } 0.4 \end{aligned}$$

**Quantity - 2**

$$\begin{aligned} 10y^2 - 9y + 2 &= 0 \\ \Rightarrow 10y^2 - 5y - 4y + 2 &= 0 \\ \Rightarrow 5y(2y - 1) - 2(y - 1) &= 0 \\ \Rightarrow (2y - 1)(5y - 2) &= 0 \\ \Rightarrow y &= 0.5 \text{ or } 0.4 \end{aligned}$$

**7. (1) Quantity - 1**

$$\begin{aligned} 10x^2 - 7x + 1 &= 0 \\ \Rightarrow 10x^2 - 5x - 2x + 1 &= 0 \\ \Rightarrow 5x(2x - 1) - 1(2x - 1) &= 0 \\ \Rightarrow (5x - 1)(2x - 1) &= 0 \\ \Rightarrow x &= \frac{1}{5} \text{ or } \frac{1}{2} = 0.2 \text{ or } 0.5 \end{aligned}$$

**Quantity - 2**

$$\begin{aligned} 5y^2 + 33y + 40 &= 0 \\ \Rightarrow 5y^2 + 25y + 8y + 40 &= 0 \\ \Rightarrow 5y(y + 5) + 8(y + 5) &= 0 \\ \Rightarrow (5y + 8)(y + 5) &= 0 \\ \Rightarrow y &= -5 \text{ or } -1.6 \end{aligned}$$

**8. (5) Quantity - 1**

$$\begin{aligned} 9y^2 + 32y + 15 &= 0 \\ \Rightarrow 9y^2 + 27y + 5y + 15 &= 0 \\ \Rightarrow 9y(y + 3) + 5(y + 3) &= 0 \\ \Rightarrow (9y + 5)(y + 3) &= 0 \\ \Rightarrow y &= -3 \text{ or } -0.55 \end{aligned}$$

**Quantity - 2**

$$\begin{aligned} 6x^2 - 13x - 44 &= 0 \\ \Rightarrow 6x^2 - 24x + 11x - 44 &= 0 \\ \Rightarrow 6x(x - 4) + 11(x - 4) &= 0 \\ \Rightarrow (6x + 11)(x - 4) &= 0 \\ \Rightarrow x &= 4 \text{ or } -1.83 \end{aligned}$$

**9. (1) Quantity - 1**

The pattern is :

$$\begin{aligned} 4 \times 0.5 &= 2 \\ 2 \times 1 &= 2 \\ 2 \times 1.5 &= 3 \\ 3 \times 2 &= 6 \\ 6 \times 2.5 &= 15 = x \end{aligned}$$

**Quantity - 2**

The pattern is :

$$\begin{aligned} 2 \times 1 + 1 &= 3 \\ 3 \times 2 + 4 &= 10 = y \\ 10 \times 3 + 9 &= 39 \\ 39 \times 4 + 16 &= 172 \\ 172 \times 5 + 25 &= 885 \end{aligned}$$

**10. (1)**

77

43

54

70

10

15

**9**

77

89

91

94

99

10

**11. (1)**

Let

:

=

=

=

=

=

=

=

**Quan**

Let t

 $\Rightarrow 2y$  $\Rightarrow 2y$  $\Rightarrow (2$  $\Rightarrow y$  $= -9.$ **12. (5) 9**

Let t

 $\therefore x^2$  $\Rightarrow x^2$  $\Rightarrow x$  $\Rightarrow (x$  $\Rightarrow x$  $\Rightarrow x$ **Quan**

Let t

 $\therefore 7y$  $\Rightarrow 7y$  $\Rightarrow 7y$  $\Rightarrow (7$  $\Rightarrow y$  $= -0$

## IMPORTANT

## AREA AND PERIMETER

A figure bounded by three or more sides is called a plane figure. It is to be noted that a circle is also a plane figure. In this section, we shall discuss finding the perimeters and areas of plane figures. Let us discuss relevant facts.

**Perimeter :** Perimeter of a plane figure is the measure of the length of its boundary. The unit of perimeter is same as the unit of length, i.e., metre (cm), metre (m), kilometre (km).

**Area :** Area of a plane figure is the measure of the region enclosed by it. The various units for measuring area are : 1 square cm ( $\text{cm}^2$ ), 1 square metre ( $\text{m}^2$ ), 1 hectare etc.

**Perimeter and Area**

**1. Triangle :** Let  $a, b, c$  denote the sides of a triangle. Then

$$(i) \text{ Perimeter} = a + b + c$$

The perimeter of a triangle is generally denoted by  $2s$ .

$$\therefore 2s = a + b + c$$

$$\Rightarrow s = \frac{1}{2} (a + b + c)$$

$$(ii) \text{ Area} = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\text{Also, Area} = \frac{1}{2} \times \text{Base} \times \text{Height}$$

**2. Equilateral Triangle :** For an equilateral triangle, each of whose side is  $a$ , we have

$$(i) \text{ Perimeter} = 3a$$

$$(ii) \text{ Height} = \frac{\sqrt{3}}{2} a$$

$$(iii) \text{ Area} = \frac{1}{2} \times \text{Base} \times \text{Height}$$

$$= \frac{1}{2} \times a \times \frac{\sqrt{3}}{2} a = \frac{\sqrt{3}}{4} a^2$$

**3. Rectangle :** For a rectangle of length  $l$  and breadth  $b$  we have

$$(i) \text{ Perimeter} = 2(l + b)$$

$$(ii) \text{ Diagonal} = \sqrt{l^2 + b^2}$$

### 3. Right Circular Cylinder

Like we see solids like road roller etc. These solids have a curved surface with congruent circular ends called right circular cylinders. If a rectangle is revolved about one of its sides, the solid generated by it in this motion is called a right circular cylinder.

For a cylinder of base radius  $r$  and height  $h$ , we have

(i) Area of base =  $\pi r^2$

(ii) Curved surface area  
= Perimeter of base  $\times$  height  
 $= 2\pi r h$

(iii) Total surface area =  $2\pi r^2 + 2\pi r h$

(iv) Volume = Area of base  $\times$  height  
 $= \pi r^2 h$

**4. Hollow Cylinder :** A solid formed by two concentric cylinders of the same height is called a hollow cylinder. Pipes and tubes are its examples.

Let  $R$  and  $r$  be the external and internal radii of hollow cylinder and  $h$  be the height.

(i) Area of base =  $\pi (R^2 - r^2)$

(ii) Curved surface = External curved surface + Internal curved surface  
 $= 2\pi R h + 2\pi r h$

(iii) Total surface area  
 $= 2\pi R h + 2\pi r h + 2 (\pi R^2 - \pi r^2)$

(iv) Volume of material =  $\pi (R^2 - r^2) h$

**5. Cone :** A cone is formed by revolving a right-angled triangle around its height.

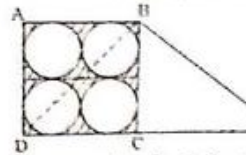
### LEVEL-I

- The cost of fencing a circular plot at the rate of Rs. 15 per metre is Rs. 3,300. Find the cost of flooring the plot at the rate of 100 per square metre ?  
(1) Rs. 3,85,000 (2) Rs. 3,50,000  
(3) Rs. 3,50,000 (4) Cannot be determined  
(5) None of these
- The circumference of two circles is 176 metres and 176 metres respectively. Find the difference between the area of the larger circle and the smaller circle ?  
(1) 1048 sq. metres (2) 1078 sq. metres  
(3) 1078 sq. metres (4) 1048 sq. metres  
(5) None of these



# MENSURAT

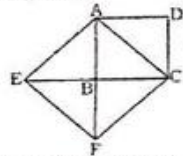
**Directions (18-19) :** Study the diagram to answer the question.



18. If the diameter of each circle is  $= CE$ , the area of  $\Delta BDE$  is  
 (1) 784 sq.cm (2) 748  
 (3) 874 sq.cm (4) 441  
 (5) None of these

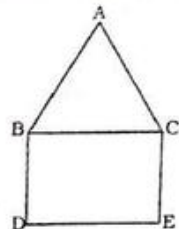
19. The area of the shaded region of s  
 is  
 (1) 186 sq. cm (2) 168 sq  
 (3) 188 sq. cm (4) 198 sq  
 (5) None of these

20. In the following figure, ABCD  
 whose each side is 10cm long. Tri  
 and  $\Delta EFC$  are congruent. Point B  
 point of side EC. Find the area  
 sq. cm.



- (1) 100 (2) 80 (3) 60  
 (5) None of these

21. In the following figure,  $\Delta ABC$  is  
 equilateral triangle and BCDE is a sq  
 each side is 8 cm long. Find the  
 polygon ABDEC in square cm.



- (1)  $18(4 + \sqrt{3})$  (2)  $16(4 +$   
 (3)  $8(4 + \sqrt{3})$  (4)  $16(2 +$   
 (5) None of these

22. There is a rectangular plot whos  
 36 metre and breadth is 28 metre  
 two paths parallel to length and  
 plot as is shown shaded in the fo

40. There is a semicircle whose area is equal to the area of a hemisphere of radius 7 cm. What is the radius of the semicircle?
- (1) 80 cm (2) 65 cm (3) 70 cm (4) 75 cm (5) 72 cm

41. The perimeter of a circle is 100 cm. The area of a square is 784 square cm. What is the radius of the circle? (1) 512 square cm. (2) 512 square cm. (3) 616 square cm. (4) 616 square cm. (5) 641 square cm.

42. The respective ratio of curved surface area and total surface area of a right circular cylinder is 3 : 5. If the curved surface area is 471 square cm, what is its height? (in cm.)
- (1) 24 (2) 14 (3) 21 (4) 18 (5) 18

43. A sphere of radius 3 cm is placed in a cylindrical vessel partly filled with water. The radius of the vessel is 6 cm. If the sphere is submerged completely then the water is raised by —
- (1)  $\frac{1}{4}$  cm (2)  $\frac{1}{2}$  cm (3)  $\frac{3}{4}$  cm (4)  $\frac{1}{2}$  cm (5) None of these

44. A water vessel is in the form of a hemispherical bowl on which a hollow hemisphere is placed. The diameter of the bowl is 14 cm. The total height of vessel is 15 cm. What is the approximate capacity of the vessel in litres?
- (1) 1910 (2) 1850 (3) 1760 (4) 1800 (5) 1760

45. The height of a room is 40 cm. The perimeter of the room is 260 m. It cost ₹ 260 to paper the walls of the room with paper 50 cm wide. The cost of paper is ₹ 2 per m allowing an area of 10% for doors and windows. The height of the room is —
- (1) 2.6 m (2) 3.9 m (3) 4 m (4) 2.6 m (5) None of these

46. The width of a rectangle is 10 cm. The area of the rectangle is 100 square cm. What is the length of the rectangle?
- (1) 10 cm (2) 20 cm (3) 30 cm (4) 40 cm (5) 50 cm

# MENSURATION : TWO AND THREE DIMENSIONS

7. 7480. The length of rectangular room is equal to the length of a square room whose area is 256 square feet. What is the breadth of the rectangular room?

- (1) 5.2 feet  
(2) 5.5 feet  
(3) 4.5 feet  
(4) 4.8 feet  
(5) None of these

16. A copper wire is bent in the form of an equilateral triangle and has area  $121\sqrt{3}$  cm<sup>2</sup>. If the same wire is bent into the form of a circle, the area (in cm<sup>2</sup>) enclosed by the wire is

(Take  $\pi = \frac{22}{7}$ )

- (1) 364.5 (2) 693.5 (3) 346.5 (4) 639.5

(5) None of these

21. M, N, O, P, Q, R, S are two rhombuses. Diagonals of ABCD are 6 cm and 8 cm. One of the angles of PQRS is 120 degree and diagonal bisecting that angle measures 15 cm. Side of ABCD = ML. Side of PQRS = JK. Find ON (median of trapezoid)



- (1) 10 cm (2) 12 cm (3) 8 cm (4) 15 cm  
(5) 9 cm

22. The base (9cm.) and the height of a right angled triangle (T) are equal to the breadth and

# MENSURATION : TWO AND THREE DIMENSIONS

## LEVEL-III

1. The tube of a fountain pen is cylindrical in shape which has radius of base as 0.7 cm and is 5 cm long. One such tube in the pen can be used to write 360 words. How many full of ink which has a capacity of 14 cubic cm?

- (1) 425 words (2) 590 words  
(3) 545 words (4) 529 words  
(5) 378 words

2. A square and a circle are inscribed inside a rectangle. The side

the square and breadth of the rectangle is 24m. If the length of the rectangle is breadth, what is the respective ratio between the area of the square and the area of the rectangle?

- (1) 9 : 2 (2) 7 : 4 (3) 6 : 5 (4) 9 : 8  
(5) 7 : 2

27. A conical cavity is drilled in a circular cylinder of height 15 cm and base radius 4 cm. The height and the base radius of the cone are also same. Then, the whole surface of the remaining solid is —

- (1)  $440\pi$  cm<sup>2</sup> (2)  $240\pi$  cm<sup>2</sup>  
(3)  $640\pi$  cm<sup>2</sup> (4)  $960\pi$  cm<sup>2</sup>  
(5) None of these

28. From a square piece of

the later rectangular  
2. What is the breadth of the rectangular field?

- (1) 11 metre
- (3) 22 metre
- (5) 16.8 metre

6. A semicircle is drawn with AB as its diameter. From C, a point on AB, a line segment CD is drawn meeting the semi-circle at D. If AC = 4 cm and CD = 6 cm, the area (in sq. cm) will be :

- (1)  $32\pi \text{ cm}^2$
- (3)  $40.5\pi \text{ cm}^2$
- (5) Cannot be determined

7. Rectangular tiles each measuring 12 cm by 10 cm must be laid horizontally on a rectangular floor of size 110 cm by 130 cm, such that tiles do not overlap. They are laid in any orientation so long as the edges are parallel to the edge of the floor. The number of tiles that can be laid on the floor is :

- (1) 4
- (2) 5
- (5) 8

8. A rectangular sheet of paper is folded in half by folding it at the mid-point of one of its longer sides. The resulting rectangle has shorter sides in the ratio 1:2. The longer and the shorter sides of the original rectangle are in the ratio of the sides of the resulting rectangle. If the area of the original rectangle is 2, what is the area of the smaller rectangle?

- (1)  $4\sqrt{2}$
- (2)  $2\sqrt{2}$
- (4) None of these
- (5) Data is insufficient

9. The length of the chord of a circle is equal to the radius of the circle. The area of the sector formed by the chord and also the perimeter of the sector are respectively  $c$ ,  $t$  and  $s$  respectively.

- (1)  $s > t > c$
- (2)  $t > s > c$
- (3)  $c > s > t$
- (4)  $s > c > t$
- (5) None of these

**Directions (10 – 11) :** Consider the following information and answer the questions that follow.

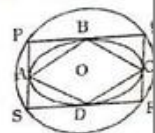
A cylinder of height  $h$  cm and radius  $r$  cm is shown in the figure (not drawn to scale).

The figure shows a line segment of length  $l$  cm, when wound around the surface, starting at point A and ending at point B, gives a maximum

# MENSUR

24. The length, breadth and height of a cuboid are in the ratio of 3 : 2 : 1. If the length and breadth are halved while the height remains the same, the total area of four walls will
- (1) remain the same
  - (2) decrease by 13.64%
  - (3) decrease by 15%
  - (4) decrease by 18.75%
  - (5) decrease by 30%

25. The figure below shows two concentric circles with centre O. PQRS is a square inscribed in the outer circle. It also circumscribes the inner circle, touching it at points A, B, C and D. What is the ratio of the perimeter of the inner circle, to that of polygon ABCD?



- (1)  $\frac{\pi}{4}$  (2)  $\frac{3\pi}{2}$  (3)  $\frac{\pi}{2}$   
(5) 35

26. Consider two different processes. In the first one,  $n$  circles are cut from a square cloth piece of side  $a$  in the following steps: the original square is divided into  $n$  smaller squares, not necessarily of the same size. The maximum possible area is cut from each of the smaller squares. In the second process, only one circle of maximum area is cut from the square of side  $a$ . The cloth pieces left after cutting the circles are scrap. The ratio of the total cloth generated in the first process to the latter is

- (1) 1 : 1 (2)  $\sqrt{2}$  : 1  
(3)  $\frac{n(4 - \pi)}{4n - \pi}$  (4)  $\frac{n\pi}{4n - \pi}$   
(5) None of these

27. Three horses are grazing within a semi-circular field. In the diagram given, AB is the diameter of the semi-circular field with centre O. Horses are tied up at points P and R such that PO and RO are the radii of the circles with centre P and R respectively. S is the centre of the circle touching the semi-circles with diameters AO and BO.



ME

## SHORT A

### LEVEL

|         |         |         |
|---------|---------|---------|
| 1. (1)  | 2. (3)  | 3. (5)  |
| 7. (5)  | 8. (1)  | 9. (2)  |
| 13. (4) | 14. (4) | 15. (4) |
| 19. (2) | 20. (1) | 21. (2) |
| 25. (1) | 26. (1) | 27. (3) |
| 31. (4) | 32. (1) | 33. (2) |
| 37. (1) | 38. (3) | 39. (1) |
| 43. (3) | 44. (3) | 45. (3) |
| 49. (1) | 50. (4) |         |

### LEVEL-I

|         |         |         |     |
|---------|---------|---------|-----|
| 1. (4)  | 2. (3)  | 3. (2)  | 4.  |
| 7. (5)  | 8. (1)  | 9. (3)  | 10. |
| 13. (4) | 14. (3) | 15. (2) | 16. |
| 19. (3) | 20. (3) | 21. (1) | 22. |
| 25. (2) | 26. (1) | 27. (1) | 28. |

### LEVEL-I

|         |         |         |     |
|---------|---------|---------|-----|
| 1. (3)  | 2. (4)  | 3. (4)  | 4.  |
| 7. (3)  | 8. (2)  | 9. (3)  | 10. |
| 13. (1) | 14. (2) | 15. (3) | 16. |
| 19. (4) | 20. (4) | 21. (2) | 22. |
| 25. (3) | 26. (1) | 27. (2) | 28. |

### QUANTITY B

|        |        |        |        |
|--------|--------|--------|--------|
| 1. (3) | 2. (1) | 3. (3) | 4. (1) |
|--------|--------|--------|--------|

### DATA SUFFIC

|        |        |        |        |
|--------|--------|--------|--------|
| 1. (3) | 2. (1) | 3. (4) | 4. (1) |
|--------|--------|--------|--------|

### EXPLANATIO

### LEVEL-I

1. (1) Total cost of fencing  
= Rs. 3300  
Rate = Rs. 15 /metre  
 $\therefore$  Circumference of circu  
$$= \frac{3300}{15} = 220 \text{ metre}$$

$$= \frac{22}{7} \times (28 + 7)(28 - 7) \text{ m}^2$$

$$= \frac{22}{7} \times 35 \times 21 \text{ m}^2 = 2310 \text{ m}^2$$

∴ Height of embankment

$$= \frac{\text{Volume}}{\text{Area}} = \frac{1540}{2310} \text{ m}$$

$$= \frac{2}{3} \text{ m}$$

12. (2) Tricky approach

$$\frac{\sqrt{3}}{4} \times \text{side}^2 = \text{Area of equilateral triangle}$$

$$= \frac{\sqrt{3}}{4} \times a^2 = 2$$

$$\Rightarrow a^2 = \frac{8}{\sqrt{3}}$$

Now, side of a regular hexagon

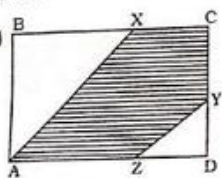
$$= \frac{3}{6} \times \text{side} = \frac{a}{2}$$

∴ Area of regular hexagon

$$= \frac{3\sqrt{3}}{2} \times \frac{a^2}{4} = \frac{3\sqrt{3}}{2} \times \frac{8}{\sqrt{3} \times 4}$$

$$= 3 \text{ cm}^2$$

13. (4)



$$BC = BX + XC = 3x + 2x$$

$$= 5x \text{ cm}$$

$$CD = CY + YD = 2y + y$$

$$= 3y \text{ cm}$$

$$\therefore 5x \times 3y = 120$$

$$\Rightarrow xy = 8 (= 2 \times 4)$$

$$\therefore BC = 20 \text{ cm}$$

$$CD = 6 \text{ cm}$$

$$BX = \frac{3}{5} \times 20 = 12 \text{ cm}$$

$$YD = \frac{1}{3} \times 6 = 2 \text{ cm}$$

$$ZD = \frac{1}{4} \times 20 = 5 \text{ cm}$$

14. (4)

Area

$$= 2\sqrt{3}$$

Area

$$= \frac{1}{2}(2x$$

$$= \frac{1}{2} \times 3x$$

$$= \frac{1}{2} \times 3x$$

Require

$$= \frac{3\sqrt{3}}{4} x$$

15. (4) Let

$$\therefore ab =$$

$$bc = 20$$

$$ac = 15$$

Volume

$$= \sqrt{a^2 b^2}$$

$$= \sqrt{3600}$$

16. (5) Quick

Effect on

$$= (-20 +$$

$$= -4\% \text{ l.}$$

17. (2)

$$\Rightarrow \pi r = 120$$

$$\Rightarrow 2\pi r = 240 \text{ metre}$$

26. (1) Volume of the silver used in the ball

$$= \frac{4}{3} \pi (r_2^3 - r_1^3) = \frac{4}{3} \pi (2^3 - 1^3)$$

$$= \frac{28\pi}{3} \text{ cu. cm}$$

27. (3) Surface area of the sphere =  $4\pi r^2$

$$= 4\pi \times 10^2 = 400\pi \text{ sq. cm.}$$

$$\text{Volume of sphere} = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \pi \times 10^3 = \frac{4000\pi}{3} \text{ cu. cm.}$$

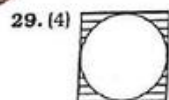
$\therefore$  Required percentage

$$= \frac{400\pi}{\frac{4000\pi}{3}} \times 100 = 30\%$$

28. (4) Volume of water in the pool

$$= 40 \times 15 \times \left( \frac{1.2 + 2.4}{2} \right)$$

$$= 1080 \text{ cubic metre}$$



29. (4)

Radius of the circular garden

$$= \frac{28}{2} = 14 \text{ metre}$$

$\therefore$  Its area =  $\pi r^2$

$$= \frac{22}{7} \times 14 \times 14 = 616 \text{ sq. metre}$$

Area of the square plot

$$= 28 \times 28$$

$$= 784 \text{ sq. metre}$$

$\therefore$  Area of the shaded region

$$= 784 - 616 = 168 \text{ sq. metre}$$

30. (1) According to the question,

$$\frac{2r}{h} = \frac{7}{5} \Rightarrow \frac{r}{h} = \frac{7}{10}$$

Let,  $r = 7x$  metre

$h = 10x$  metre

$\therefore$  Curved surface area =  $2\pi rh$

Volume of cylinder =  $\pi r^2 h$

$$\therefore \frac{2\pi \times 7x \times 10x}{\pi \times 7x \times 7x \times 10x} = \frac{1}{7}$$

$$\Rightarrow x = 2$$

31.

32. (

A  
=  
=  
R  
=  
Area

$$= \frac{1}{3} \times$$

Area

$$= 96$$

Area

No. o

33. (2) [

In b

$$\therefore \frac{4}{3}$$

$$\Rightarrow r = \sqrt{784} = 28 \text{ cm}$$

Here, the Diameter of the semicircle = radius of the hemisphere = 28 cm

Radius of the semicircle

$$= \frac{28}{2} = 14 \text{ cm}$$

Circumference of the semi-circle =  $\pi r + 2r =$

$$\left(\frac{22}{7}\right) 14 + 28 = 72 \text{ cm}$$

41. (3) Area of square = 784 square cm

Side of square =  $\sqrt{784}$

= 28 cm.

Perimeter of square =  $4 \times 28 \text{ cm}$

Circumference of the circle

$$= 4 \times 28 \times \frac{22}{7} \text{ cm}^2$$

$$\text{So, } 2\pi r = 4 \times 4 \times 22$$

$$\Rightarrow 2 \times \frac{22}{7} \times r = 4 \times 4 \times 22$$

$$\Rightarrow r = 16 \times 22 \times \frac{7}{2 \times 22}$$

= 56 cm.

$$\text{Radius of circle } Q = \frac{56}{4} = 14 \text{ cm.}$$

Area of circle Q

$$= \pi r^2 = \left(\frac{22}{7}\right) \times 14 \times 14$$

= 616 square cm

42. (3) Radius of cylinder =  $r \text{ cm}$  and height =  $h \text{ cm}$  (let)

According to the question,

Curved surface area =  $2\pi rh$

Total surface area =  $2\pi rh + 2\pi r^2$

$$\therefore \frac{2\pi rh}{2\pi r(h+r)} = \frac{3}{5}$$

$$\Rightarrow \frac{h}{h+r} = \frac{3}{5}$$

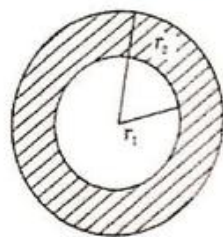
$$\Rightarrow 5h = 3h + 3r$$

$$\Rightarrow 2h = 3r \quad \dots(i)$$

Again,  $2\pi rh = 1848$

$$\Rightarrow 2\pi \times \frac{2}{3} h \times h = 1848$$

[From equation (i)]



$$= 2 \times \frac{22}{7} \times r_1 = 308$$

$$\Rightarrow r_1 = \frac{308 \times 7}{2 \times 22} = 49 \text{ metre}$$

$$\Rightarrow \text{New radius } (r_2) = 49 + 7$$

$$= 56 \text{ m}$$

$$\therefore \text{Area of the path} = \pi(r_2^2 - r_1^2) = \frac{22}{7}(56^2 - 49^2)$$

$$= \frac{22}{7}(56 + 49)(56 - 49)$$

$$= 22 \times 105 = 2310 \text{ sq. m.}$$

2. (3) Circumference of circle

$$= \pi \times \text{diameter}$$

$$= \frac{22}{7} \times 56 = 176 \text{ cm}$$

$$\therefore \text{Perimeter of square}$$

$$= 272 - 176 = 96 \text{ cm}$$

$$\therefore \text{Side of square} = \frac{96}{4} = 24 \text{ cm}$$

$$\therefore \text{Area of square} = 24 \times 24$$

$$= 576 \text{ sq.cm.}$$

$$\text{Area of circle} = \pi r^2$$

$$= \frac{22}{7} \times 28 \times 28 = 2464 \text{ sq.cm.}$$

$$\therefore \text{Required sum}$$

$$= (576 + 2464) \text{ sq. cm.}$$

$$= 3040 \text{ sq. cm.}$$

3. (2) Base =  $4x$  cm.

$$\text{Height} = 5x \text{ cm.}$$

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$\Rightarrow 80 = \frac{1}{2} \times 4x \times 5x$$

$$\Rightarrow 10x^2 = 80$$

$$\Rightarrow x^2 = \frac{80}{10} = 8$$

$$\Rightarrow x = \sqrt{8} = 2\sqrt{2} \text{ cm}$$

4. (1)

Di

=

$\therefore A =$

Area

$$= \frac{\pi d^2}{4}$$

Accor

$$d^2 =$$

$$\Rightarrow d^2 =$$

$$\Rightarrow d^2 =$$

$$= \frac{168 \times 28}{28}$$

$$= \frac{168 \times 28}{28}$$

$$\therefore d =$$

$$\therefore \text{Rad}$$

$$= \frac{28}{2} =$$

5. (4) Curv

Total su

$$= 2\pi rh +$$

Accordin

$$\frac{2\pi rh}{2\pi r(h + r)}$$



$$= \frac{140}{100} \times 4\pi a^2$$

$$= 5.6\pi a^2 \text{ sq. units}$$

$$\text{Excess area} = 1.6\pi a^2$$

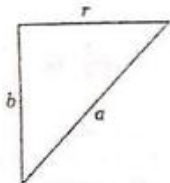
Two extra circles are formed.

Area of each circle

$$= \frac{1.6\pi a^2}{2} = 0.8\pi a^2$$

$$\therefore \pi r^2 = 0.8\pi a^2$$

$$\Rightarrow r^2 = 0.8a^2$$



$$\therefore b^2 + r^2 = a^2$$

$$\Rightarrow b^2 = a^2 - 0.8a^2 = 0.2a^2$$

$$= \frac{2}{10} a^2$$

$$\Rightarrow b^2 = \frac{a^2}{5} \Rightarrow b = \frac{a}{\sqrt{5}} \text{ units}$$

12. (1) Let, smaller parallel side

=  $x$  cm.

$\therefore$  Larger parallel side

=  $(x + 4)$  cm.

Area of trapezium =  $\frac{1}{2}$  (sum of parallel sides)  
 $\times$  perpendicular distance

$$\Rightarrow \frac{1}{2} (x + x + 4) \times 7 = 77$$

$$\Rightarrow 2x + 4 = \frac{77 \times 2}{7} = 22$$

$$\Rightarrow 2x = 22 - 4 = 18$$

$$\Rightarrow x = 9 \text{ cm.}$$

$$\therefore \text{Larger side} = x + 4 = 9 + 4 = 13 \text{ cm.}$$

13. (4) Side of square =  $\sqrt{9}$

= 3 cm. = radius of cone

$$\text{Height of cone} = 2 \times \frac{3}{2}$$

$$= 3 \text{ cm.}$$

Let number of cones be  $n$ .

$\therefore n \times \text{volume of a cone} = \text{Volume of sphere}$

$$\Rightarrow n \times \frac{1}{3} \pi \times 3 \times 3 \times 3$$

$$= \frac{4}{3}$$

$$\Rightarrow n$$

14. (3) R

Requ

$$= \frac{4}{36}$$

$$= \frac{77}{4}$$

15. (2) A

$$= \frac{7}{6}$$

$$= \frac{7}{6}$$

Sid

$$= \sqrt{1}$$

$$= 1$$

$$\therefore 1$$

$$= \frac{7}{6}$$

16. (3) A

$$\Rightarrow$$

$$\Rightarrow$$

$$\Rightarrow$$

$$\therefore$$

$$\therefore$$

$$\Rightarrow$$

$$\Rightarrow$$

$$\Rightarrow$$

$$\Rightarrow$$

$$\Rightarrow r$$

$$\therefore \text{Are}$$

$$= \frac{22}{7}$$

$$= \frac{22}{7}$$

17. (1) Are

$$= (80$$

$$= 560$$

$$\text{Area of}$$

$$\begin{aligned}\text{Total} &= (202.5 + 60 + 16.5) \text{ sq. cm.} \\ &= 279 \text{ sq. cm.} \\ \text{So, total area of key board} \\ &= (21 + 279) \text{ sq. cm.} \\ &= 300 \text{ sq. cm.}\end{aligned}$$

$$\therefore \text{Required breadth} = \frac{300}{12}$$

$$= 25 \text{ cm.}$$

24. (3) By taking Pythagoras theorem  
Diagonal of Rectangle X

$$= \sqrt{5^2 + 12^2} = 13 \text{ cm.}$$

Diagonal of rectangle Y

$$= \sqrt{5^2 + b^2} = \sqrt{89} \text{ cm.}$$

$$\Rightarrow 25 + b^2 = 89$$

$$\Rightarrow b^2 = 89 - 25 = 64$$

$$\Rightarrow b = \sqrt{64} = 8 \text{ cm.}$$

Required ratio = 13 : 8

25. (2) Radius =  $\frac{350}{2} = 175 \text{ cm.}$

Volume of tank =  $\pi r^2 h$

$$= \left(\frac{22}{7}\right) \times 175 \times 175 \times 240$$

Volume of each tin

$$= 25 \times 22 \times 35$$

Required number of tin

$$= \left(\frac{22}{7}\right) \times \frac{175 \times 175 \times 240}{25 \times 22 \times 35} = 1200$$

26. (1) Let the width of rectangle be  $x$  metre.

$$\therefore \text{Length} = 2x \text{ metre}$$

Let side of square =  $y$  metre

According to the question,

$$2x + y = 30 \dots (i)$$

$$x + y = 24 \dots (ii)$$

By equation (i) - (ii),

$$x = 6 \text{ metre}$$

From equation (ii)

$$6 + y = 24$$

$$\Rightarrow y = 24 - 6 = 18 \text{ metre}$$

$\therefore$  Area of square : Area of rectangle

$$= (18 \times 18) : (6 \times 12) = 9 : 2$$

27. (1) According to question,

$$l = \sqrt{15^2 + 8^2} = 17 \text{ cm}$$

$$\Rightarrow \frac{AC}{DC} = \frac{DC}{BC}$$

$$\frac{2}{6} = \frac{6}{BC} \Rightarrow BC = 18$$

$$\Rightarrow AB = 2 + 18 = 20$$

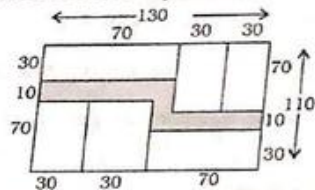
Radius of semicircle =  $\frac{20}{2} = 10$  and area

$$= \frac{\pi(10)^2}{2} = 50\pi \text{ cm}^2$$

7. (3) Number of tiles =  $\frac{\text{Area of floor}}{\text{Area of one tile}}$

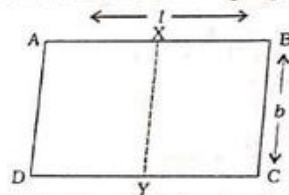
$$= \frac{110 \times 130}{70 \times 30} = \frac{11 \times 13}{7 \times 3} = 6.8$$

Number of tiles is equal to 6.



The shaded area is the uncovered area.

8. (2) **Method-1** : Check through options



Let ABCD be the original rectangle with  $AD = BC = 2$

When it is folded along XY [X is the midpoint of AB & Y is the midpoint of DC], we get small rectangle XYBC.

Option (1) If area of XYBC =  $4\sqrt{2}$ , where  $BC = 2$

$$\Rightarrow XB = \frac{4\sqrt{2}}{2} = 2\sqrt{2}$$

$$\text{Then } XB : BC = 2\sqrt{2} : 2 = \sqrt{2} : 1$$

Ratio of sides of bigger rectangle =  $AB : BC$

$$= (2\sqrt{2} + 2\sqrt{2}) : 2$$

$$= 4\sqrt{2} : 2 = 2\sqrt{2} : 1$$

$$XB : BC \neq AB : BC$$

(2) If area of smaller rectangle =  $2\sqrt{2}$

$$XB = \frac{2\sqrt{2}}{2} = \sqrt{2}$$

# MENSURATION : TWO AND THREE DIMENSIONS

416 m<sup>2</sup> > area of half the lawn which is equal to 400m<sup>2</sup>.  
 ⇒ No. of rounds are more than 3 but slightly less than 4.

If no. of rounds = 3.8

Area to be mowed after third round

$$= 400 - 324 = 76\text{m}^2$$

Area mowed in the 4th round alone

$$= 416 - 324 = 92\text{m}^2$$

92m<sup>2</sup> is mowed in 1 round [4th round alone]

76m<sup>2</sup> is mowed in  $\left(\frac{1}{92} \times 76\right)$  rounds = 0.82 rounds

Total no. of rounds = 3 + 0.82 = 3.8 rounds.

16. (3) Let a, b and c be sides of the triangle

$$a + b + c = 14$$

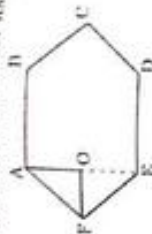
[Sum of any two sides of a triangle must be greater than the third side]

If a = 7 ⇒ b + c = 7 → not possible

This means that the length of each side of triangle

volume = 0, same is so if X = 6, so X must be between 1 and 5. You can now put different value of X. Also X cannot be greater than 6 because then X + X = 2 × X = 2 × 7 = 14 which is more than the side of the sheet itself.

18. (1)



Method-1 :

$$\angle AOF = 90^\circ$$

In a regular hexagon, all angles are equal to 120°.

$$\Rightarrow \angle AFE = 120^\circ$$

Join OE ΔAFO ≅ ΔEFO

$$AF = EF.$$

$$OF = OF$$

$$\angle AOF = \angle FOE = 90^\circ$$

## MENSURATION : TWO AND THREE DIMENSIONS

50. Required ratio

$$= \frac{1}{2} \text{ in } \Delta AFG \text{ or } \Delta AFG = \frac{1}{2} \times 6 \times 1.12$$

19. (4) Let radius of sphere A and B be r and R respectively.

$$4\pi r^2 = 4\pi R^2 + 3(4\pi r^2) = 16\pi r^2 \Rightarrow r^2 = 4r^2 \text{ or } r = 2r$$

$$\text{Let } r = 2 \Rightarrow R = 4$$

$$\text{Volume of A} = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(2)^3 = \frac{32}{3}\pi(8)$$

$$\text{Volume of B} = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi(4)^3 = \frac{256}{3}\pi(64)$$

$$\text{Required \%} = \frac{V_A - V_B}{V_B} \times 100 = \left[ \frac{\frac{32}{3}\pi(8) - \frac{256}{3}\pi(64)}{\frac{256}{3}\pi(64)} \right] \times 100$$

$$= \frac{4}{3}\pi(64 - 8)$$

$$= \frac{1}{2} \times 6 \times 1.12 = 3.36$$

$$\text{Also } BE^2 = AD \times DC$$

$$\Rightarrow BE^2 = \frac{18}{5} \times \left(10 - \frac{18}{5}\right)$$

$$\Rightarrow BE^2 = \frac{18}{5} \times \frac{32}{5} \Rightarrow BE = \frac{24}{5}$$

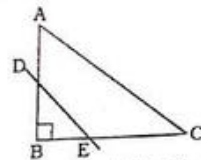
Still another method could find ar ΔABC by using Heron's formula then equate it to  $\frac{1}{2} AC \times BD$ . you will get BD. then proceed further.

21. (2)

# MENSURATION

Since  $\angle OBC = \angle OQC = \angle BCQ = 90^\circ$   
 this means  $\angle BOQ = 90^\circ$   
 Also  $OB = OQ = \text{radius of circle} =$   
 square and  $BC \parallel OQ$  and  $QC \parallel OB$   
 Now  $PA \perp LR$  (angle of rectangle)  
 $BC \parallel QO \Rightarrow PA \perp OQ \Rightarrow QS = 20$ .  
 Similarly  $LA \perp AP$  (angle in rectangle)  
 $PA$  while  $CL \parallel BR \Rightarrow LA \perp BR \Rightarrow BR$   
 Now in  $\triangle ASO$   
 $OA^2 = AS^2 + SO^2 \Rightarrow r^2 = (PS - PA)^2 +$   
 $r^2 = (r - 10)^2 + (r - 20)^2$   
 $r^2 = r^2 + 100 - 20r + r^2 + 400 - 40r$   
 $60r = 500 + r^2 \Rightarrow r^2 - 60r + 500 = 0$   
 $r(r - 50) - 10(r - 50) = 0 \Rightarrow r = 10$ ,  
 $r$  cannot be 10  $\Rightarrow r = 50$

23. (4)



Let  $ABC$  be the original triangle, with  
 square inches.  
 When it is cut along  $DE$  which is para  
 (hypotenuse), then  $DE = AC - 35\%$  of  
 of  $AC$ .

$\triangle DBE \sim \triangle ABC$

$\Rightarrow \frac{\text{ar}(\triangle DBE)}{\text{ar}(\triangle ABC)} = \text{ratio of square of corre}$   
 sides

$$\Rightarrow \frac{\text{ar} \triangle DBE}{34} = \frac{DE^2}{AC^2} = \left[ \frac{65}{100} AC \right]^2$$

$$\text{ar}(\triangle DBE) = \frac{65 \times 65 \times 34}{100 \times 100} = 14.365 \text{ sq. in.}$$

24. (5) Let the length, breadth and height of  
 be

$6x$ ,  $4x$  and  $2x$ .

Area of four walls  $= 2(l + b)h = 2(6x + 4x)$   
 $= 40x^2$

When breadth & height are halved while le  
 is doubled,

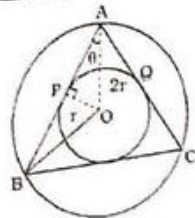
$$2(l + b)h = 2(12x + 2x)x = 28x^2$$

$\therefore$  Area decreases by  $\left( \frac{40x^2 - 28x^2}{40x^2} \right) 100\%$

$$= \frac{12}{40} \times 100 = 30\%$$



29. (3)



Area of outer circle = 4 times area of inner circle.

$$\pi R^2 = 4\pi r^2$$

$$R^2 = (2r)^2 \Rightarrow R = 2r$$

Radius of outer circle is equal to radius of inner circle.

i.e. if  $OP = r$ ,  $OA = 2r$

Let  $\angle PAO = \theta$ , then  $\sin \theta = \frac{OP}{OA} = \frac{r}{2r}$

In  $\triangle APO$ ,  $\angle OPA = 90^\circ$  (radius is perpendicular to the tangent at point of tangency)

$$\angle PAO = 30^\circ \Rightarrow \angle AOP = 60^\circ$$

$$\text{Also, } \angle OAQ = 30^\circ$$

$$\Rightarrow \angle PAC = \angle BAC = 30^\circ + 30^\circ = 60^\circ$$

In  $\triangle ABC$ ,  $\angle BAC = 60^\circ \Rightarrow \angle ABC + \angle ACB$

$$\angle ABC = \angle ACB = 60^\circ [\because AB = AC - \text{how?}]$$

$$\triangle BPO,$$

$$OA = OB = 2r, OP = OP, \angle OPA = \angle OPB =$$

$$AP = PB$$

Similarly  $AQ = QC \Rightarrow AB = 2AP$  and  $AC$  but

$$AP = AQ \Rightarrow AB = AC]$$

Hence  $\triangle ABC$  is an equilateral triangle

$$\text{Area of } \triangle ABC = \frac{\sqrt{3}}{4} (\text{side})^2 \dots\dots (i)$$

$$\text{We are given } \pi (2r)^2 = 12 \Rightarrow 4\pi r^2 = 12$$

$$r^2 = \frac{3}{\pi}$$

$$r = \sqrt{\frac{3}{\pi}}$$

$$\text{In } \triangle APO, AP^2 = 4r^2 - r^2 = 3r^2$$

$$AP = \sqrt{3}r = \sqrt{3} \left( \sqrt{\frac{3}{\pi}} \right) = \frac{3}{\sqrt{\pi}}$$

$$AB = 2AP = \frac{2 \times 3}{\sqrt{\pi}} = \frac{6}{\sqrt{\pi}}$$

Put in (i)

$$\text{Area of } \triangle ABC = \frac{\sqrt{3}}{4} \left( \frac{6}{\sqrt{\pi}} \right)^2 = \frac{\sqrt{3}}{4} \cdot \frac{36}{\pi} = \frac{9\sqrt{3}}{\pi}$$

(Note : If the circumradius is twice inradius, then the triangle that lies between the circumcircle and the in circle is equilateral triangle).

## DATA SUFFICIENCY EXPLAN.

1. (3) Let us check the data in statement

Length of diagonal =  $4\sqrt{2}$  cmWe know diagonal of square =  $\sqrt{2}a$   
the side of square.

$$\therefore \sqrt{2}a = 4\sqrt{2} \Rightarrow a = 4$$

We can find 'a', i.e. side of square  
area of the square which is equal  
 $\Rightarrow a^2 = 4 \times 4 = 16$  sq. cm.We can say that data in statement  
sufficient to answer the question.

Let us check statement II now

Perimeter of square = 16 cm  $\Rightarrow 4a$ 

$$\Rightarrow a = \frac{16}{4} \Rightarrow a = 4 \text{ cm.}$$

Again we can find the side and hence  
of the square.Thus, statement II alone is also sufficient  
to answer the question. Hence our  
choice is (3).

2. (1) From statement I,

$$4\pi r^2 = 5544$$

From here you can find the radius of  
and then the volume.

$$V = \frac{4}{3}\pi r^3$$

Statement II is redundant

3. (4) (From statement I) Volume of ball

$$= \frac{4}{3}\pi r^3 = \frac{500\pi}{3}$$

Volume of ball = Volume of water displaced

$$\pi r^2 h = \frac{500}{3}\pi$$

$$\Rightarrow r^2 \left(\frac{5}{3}\right) = \frac{500}{3} \Rightarrow r = 10$$

Radius of cylindrical water tank = 10 cm

Original Volume of water + Volume of ball  
= New Volume of water in tank.

$$\Rightarrow \pi(10^2)h + \frac{500}{3}\pi = \pi(10^2)\left(h + \frac{5}{3}\right)$$

$$\Rightarrow 100h + \frac{500}{3} = 100\left(h + \frac{5}{3}\right)$$

**Give answer (2)** if the data in statement I are sufficient to answer the question.

Give answer (3) if alone or in statement answer the question

**Give answer (4) if**  
**ments I and II together**  
**answer the question.**

**Give answer (5) 1**  
ments I and II together  
the question.

As we can see the "efficiency" and their so-  
 ferent patterns as  
 mathe-matical prob-  
 lems presented in parts and  
 Further, the answer  
 the right choice from  
 tailored in the "Direct  
 ally solving the pro-  
 answer. This schem  
 and choice of interr  
 student into a confu  
 while attempting to  
 tion, one should pro  
 with clarity. Elimina  
 sure is wrong or no

By looking at the prudent first to try data provided in the

There are two poss

(i) the data provided would be sufficient to

(ii) the data provided would not be sufficient

If the answer to (i) is  
or Option (3) is the a

**Directions (1-5) :** For each question, which of the given state determines the answer.

- I. A shopkeeper mixes A and B and sells it at 45 per kg. What is
- I. Cost price of rice A
- II. Cost price of rice B
- III. Cost price of rice B that of rice A.

11. What is the capacity of

- I. Radius of the base is
- II. Area of the base is
- III. Height of the cylinder

- (1) I and II Only
- (3) I and III Only
- (5) Any two of the three

12. What is the staff strength

- I. Male and female employees in the ratio of 2 : 3 respectively
- II. Of the officer employees
- III. Total number of officers

- (1) I and III only
- (2) II and either III or I
- (3) All I, II and III
- (4) Any two of the three
- (5) Question cannot be answered with the information in

13. How many articles were sold

- I. Total profit earned
- II. Cost-price per article
- III. Selling price per article

- (1) II and III only
- (3) All I, II and III
- (4) Any two of the three
- (5) Question cannot be answered with the information in all

14. What is the average age of

- A, B, C, D, E and F in the family
- I. Total age of D and E
- II. Average age of A, B, C
- III. Average age of A, B, C, D

- (1) Only I and II
- (3) Only II and III
- (5) None of these

15. What is the area of the triangle

- I. Base of the triangle is
- II. Height of the triangle is
- III. Hypotenuse of the triangle is

- (1) Only I and II
- (3) Only II and III
- (5) None of these

**Directions (16-20) :** In each of the following questions, a question is followed by three statements. You have to choose the correct answer from the options given in which one or more of the statements are necessary to answer the question.

16. What are the marks scored in English ?

B. One-fifth of the number.

27. In how many days will a piece of work be done?

A. If 16 children can do the work in 48 days.

B. If 9 men can do the work in 27 days.

28. How many women can do the work in 15 days?

A. If 12 women can do the work in 20 days.

B. If 10 men can do the work in 12 days.

29. What is the rate of simple interest on an amount of Rs. 12,000 for 3 years?

A. The difference between simple and compound interest.

B. The simple interest is Rs. 2,880.

30. What is the age of C, if the average age of A, B, and E whose average age is 30 years?

A. Average of the ages of A and B.

B. Average of the ages of B and E.

**Directions (31-35) :** In each of the following questions, a question followed by two statements numbered I and II are given. You have to decide whether the statements and then

the statements and then the question can be answered by using the data given in the statements.

**Give answer (1)** if the data in statement I alone are sufficient to answer the question, whereas the data given in statement II are not sufficient to answer the question.

**Give answer (2)** if the data in statement II alone are sufficient to answer the question, whereas the data given in statement I are not sufficient to answer the question.

**Give answer (3)** if the data in statement I alone or in statement II alone are sufficient to answer the question.

**Give answer (4)** if the data in both statements I and II are not sufficient to answer the question.

**Give answer (5)** if the data in both statements I and II are necessary to answer the question.

31. How many marks did Nigam get in English and Biology?

I. Nigam got 42 marks in English, which is half the marks he got in Biology.

II. Nigam's marks in Biology were 20% more than the total marks he got in English and Biology together.



50. What is the profit earned for ₹ 3,000?

- A. The cost price of 6 s to selling price of 5
- B. 20% profit is earned printer.

### LEVEL-II

**Directions (1-5) :** Each of en below consists of a questi ments numbered I and II given to decide whether the data pro ments is sufficient to answer both the statements.

**Give answer (1)** if the da alone is sufficient to answer the data in statement II alone answer the question.

**Give answer (2)** if the da alone is sufficient to answer the data in statement I alone answer the question.

**Give answer (3)** if the d alone or in statement II alone swer the question.

**Give answer (4)** if the dat ments I and II is not sufficient tion.

**Give answer (5)** if the dat ments I and II together is nece question.

1. What distance (in km) d downstream in 24 minut
  - I. The distance travelled by in 54 minutes is equal t elled by the boat downstr
  - II. The respective ratio betw boat in still water and the rent is 5 : 1.
2. What is the marked price
  - I. The marked price of the the cost price of the pen.
  - II. When a discount of 25 marked price of the pen. 5%. The cost price of the
3. In how many days can A finish a piece of work?
  - I. A and B together can fir of work in  $8\frac{8}{9}$  days. B finish the same piece of v

II. The sum becomes double at a per annum rate of simple interest.

III. The compound interest on the principal amount is Rs. 4500 at a rate of 8% compounded annually.

- (1) Only I
- (2) All I, II and III together
- (3) II and III together
- (4) Only I or III
- (5) I or II or III

17. What is the speed of the boat in still water?

**Statements :**

- I. The boat can cover 12 km distance in 2 hours.
- II. Speed of the stream is 3 km/h of boat in still water.
- III. The boat can cover 12 km distance in 4 hours.

- (1) I and III together
- (3) I and II together
- (5) Any of the two statements

18. A person borrowed some money at a rate of interest for 2 years. What was the amount required to return after 2 years?

**Statements :**

- I. If the amount was borrowed at a rate of 6% per annum, then after 5 years Rs. 10000 to pay as simple interest.
- II. The rate of interest is 6% per annum.
- III. The sum of money borrowed is Rs. 10000.

- (1) II and I or III
- (2) II and III together
- (3) I and III together
- (4) All statements are not sufficient to answer the question
- (5) All I, II and III

19. Shopkeeper gained how much profit on the sale of products in November 2016?

**Statements :**

- I. He earned 40% more profit in November 2016 as compared to October 2016.
- II. In December 2016 he earned 10% more profit than in November 2016.
- III. The total profit earned in November 2016 was Rs. 55,000.

- (1) Only I and III
- (2) Only II and III
- (3) Only I and II
- (4) All statements are not sufficient to answer the question
- (5) All I, II and III

20. In how many days 6 men can complete a work working together?

**Statements :**

- I. The ratio of efficiency of man A to man B is 3:2.
- II. Man A can complete the work in 10 days.
- III. Man B can complete the work in 15 days.

the data in Statement I and II to answer the question.

**Give answer (3)** if the data in Statements I and II together are sufficient to answer the question.

**Give answer (4)** if the data in Statement I alone or in Statement II alone are sufficient to answer the question.

**Give answer (5)** if the data in Statements I and II together are not sufficient to answer the question.

1. What will be the length of the train?

I. The train crosses a platform in 50 seconds.

II. The train crosses a pole in 30 seconds.

2. There are 450 employees in the human resources and information technology departments of an organization. How many employees are there in the human resources department?

I. 18% of all employees work in the human resources department and 24% work in the information technology department. 12% of all employees work in both information technology and human resources departments.

II. 12% of all employees work in the human resources department and 24% work in the information technology department. 12% of all employees work in both information technology and human resources departments.

3. In how many days will A and B together complete the piece of work?

I. A is half as good a workman as B.

II. B alone can do the same work in 24 days.

4. What is the two-digit number?

I. The sum of the digits of the number is 10.

II. The number obtained by interchanging the digits of the number is 72 more than the original number.

5. What will be the total cost of development of a garden in the shape of a circular field if the cost of development is Rs. 285 per square metre?

I. The respective ratio of the radius and the circumference of the field is 166 : 165.

II. The perimeter of the field is 166 metres.

**Directions (6 - 10) :** Each question below consists of a question followed by two statements numbered I and II. You have to decide whether the data given in the statements are sufficient to answer the question.

**Give answer (1)** if the data in Statement I alone are sufficient to answer the question.

**Give answer (2)** if the data in Statement II alone are sufficient to answer the question.

**Give answer (3)** if the data in Statements I and II together are sufficient to answer the question.

**Give answer (4)** if the data in Statement I alone or in Statement II alone are sufficient to answer the question.

**Give answer (5)** if the data in Statements I and II together are not sufficient to answer the question.

**Directions (6 - 10) :** Each question below consists of a question followed by two statements numbered I and II. You have to decide whether the data given in the statements are sufficient to answer the question.

**Give answer (1)** if the data in Statement I alone are sufficient to answer the question.

**Give answer (2)** if the data in Statement II alone are sufficient to answer the question.

**Give answer (3)** if the data in Statements I and II together are sufficient to answer the question.

**Give answer (4)** if the data in Statement I alone or in Statement II alone are sufficient to answer the question.

**Give answer (5)** if the data in Statements I and II together are not sufficient to answer the question.

**Directions (21 - 25):**

Each question consists of a statement followed by three information. You have to study the question and select the statement or statements which are correct. Mark the statement at question.

**21.** The ratio of the cost price of H to that of P is 6 : 5.

I. H incurred a loss of 2%.

II. H incurred a loss of 15%.

III. H and P sold their items at the same price.

(1) Only I and II (2) Only II and III

(3) Either I and III (or) II and III

(4) Either I and II (or) II and III

(5) All the three are required

**22.** The quantity of water after evaporation is 12 litres?

I. If 22 litres of the mixture is evaporated, the ratio of alcohol and water is 13 : 11.

II. When 20 litres of the mixture is evaporated, the ratio of alcohol and water is 8 : 11.

III. In a mixture in a tank, the ratio of alcohol to the water is 6 : 5.

(1) Only I and II (2) Only II and III

(3) Only I and III

(4) Either I and II (or) II and III

(5) All the three are required

**23.** The lower rate of interest is 16.67%.

I. A sum of money at a certain rate of interest becomes three times of itself in 14 years.

II. A sum of money at a certain rate of interest becomes three times of itself in 12 years.

III. A sum of money at a certain rate of simple interest doubles in 10 years.

(1) Only I (2) Only II (3) Only III

(4) Data inadequate

(5) All the three are required

**24.** Y alone can do a work in 40 days.

I. X, Y and Z together can do a work in 12 days in which Y takes three times as long as X and Z together take to do the work.

II. X takes twice as long as Y and Z together to do the work.

III. Z takes twice as long as X and Y together to do the work.

(1) Only I (2) Only II

(3) Only III (4) Only I and II

(5) All the three are required

$$\text{II \& III } 45\% = 27000$$

$$\therefore \text{ Salary} = 100\% = 27000 \times \frac{100}{45} =$$

Savings

$$= 60000 - (15000 + 27000)$$

$$= ₹ 18000$$

So, any two of the three are sufficient to determine the answer.

6. (4) We need equivalence between work of man and woman.

From statement I, we can get 1 man's work.

From statement II or III, we can get 1 day's work.

Hence, we can establish relation between man's and woman's work and get the answer.

7. (5) When a train crosses a pole, the

$$\text{Speed of train} = \frac{\text{Length of train}}{\text{Time taken}}$$

When a train crosses a platform, the

$$\text{Speed of train} = \frac{\text{Length of platform + length of train}}{\text{Time}}$$

Clearly, statement II and either I or III supplement the required data to determine the speed of train.

8. (5) Area of the square = (side)<sup>2</sup>

$$= \frac{1}{2} (\text{diagonal})^2$$

Again, Perimeter = 4 × side

Clearly, from any one of the three statements, we can determine area of the square.

9. (3) Let the number be  $10x + y$ .

From statement I,

$$(10y + x) - (10x + y) = 18$$

$$\Rightarrow 9(y - x) = 18$$

$$\Rightarrow y - x = 2 \quad \dots(i)$$

From statement II,

$$x + y = 14 \quad \dots(ii)$$

From statement III,

$$y - x = 2 \quad \dots(iii)$$

Clearly, statement II and either I or III supplement the data to know  $x$  and  $y$ , hence the number.

10. (2) From statement I, we do not know the principal.

From statement II, data are incomplete as principal is unknown.

From statement III, we get the required data, i.e.



28. (1) From statement 'A'

Days      Women

20  $\uparrow$       12  $\downarrow$

15  $\downarrow$        $x \uparrow$

$\therefore 15 : 20 = 12 : x$

$\Rightarrow 15 \times x = 12 \times 20$

$\Rightarrow x = \frac{12 \times 20}{15} = 16$  women

Statement B is superfluous.

29. (2) From statement B.

SI =  $\frac{\text{Principal} \times \text{time} \times \text{Rate}}{100}$

$\Rightarrow 2880 = \frac{12000 \times 2 \times \text{Rate}}{100}$

$\Rightarrow \text{Rate} = \frac{2880 \times 100}{12000 \times 2}$

$= 12\%$  per annum

30. (3) From both the statements.

$A+B+C+D+E = 5 \times 45$

$= 225$  years .....(i)

$A+B = 106$  years .....(ii)

$D+E = 94$  years .....(iii)

By Equation (i) - (ii) - (iii), the age of C can be determined.

31. (1) From statement I

Nigam's marks in Biology =  $2 \times 42 = 84$

32. (5) From statements I and II.

Side of the square =  $\sqrt{196} = 14$  cm.

$\therefore$  Circumference of the semi-circle =  $(\pi + 2) \times$

radius =  $7(\pi + 2)$  cm.

33. (5) From statement I.

$x \times \frac{40}{100} = \frac{50 \times 20}{100}$

$\Rightarrow x = 25$

From statement II.

$y \times \frac{30}{100} = \frac{72 \times 25}{100}$

$\Rightarrow y = 60$

$\therefore x : y = 25 : 60 = 5 : 12$

34. (5) From statement II.

If the present age of Shyam be  $x$  years then

Ram's present age =  $(x + 7)$  years

From statement I.

$\frac{x+7}{x} = \frac{4}{3}$

$\Rightarrow 4x = 3x + 21 \Rightarrow x = 21$

$\therefore$  Shyam's age after 6 years

$= 21 + 6 = 27$  years

35. (4) We do not have the average salary of D and E.

$$(B + C)'s\ 1\ day's\ work = \frac{3}{40} \dots (iii)$$

From statement II,

If time taken by C alone be  $x$  days, time taken by A alone =  $(x - 24)$  days

By equation (i) - (iii),

[A - C]'s 1 day's work

$$= \frac{9}{80} - \frac{3}{40}$$

$$= \frac{9-6}{80} = \frac{3}{80}$$

$$= \frac{1}{x-24} - \frac{1}{x} = \frac{3}{80}$$

$$= \frac{x-x-24}{x(x-24)} = \frac{3}{80}$$

$$= x(x-24) = 640$$

$$= 40(40-24)$$

$$\Rightarrow x = 40$$

$\therefore$  Time taken by A

$$= 40 - 24 = 16\ \text{days}$$

$\therefore$  From statement (i),

$$\frac{1}{16} + \frac{1}{y} = \frac{9}{80}$$

$$\Rightarrow \frac{1}{y} = \frac{9}{80} - \frac{1}{16} = \frac{9-5}{80}$$

$$= \frac{4}{80} = \frac{1}{20}$$

$\therefore$  Time taken by B = 20 days

$\therefore$  (A + B + C)'s 1 day's work

$$= \frac{1}{16} + \frac{1}{20} + \frac{1}{40}$$

$$= \frac{5+4+2}{80} = \frac{11}{80}$$

$$\therefore \text{Required time} = \frac{80}{11} = 7\frac{3}{11}\ \text{days}$$

4. (5) From both statements,

$$(a + b) = 13$$

$$a - b = 0$$

$$\therefore a^2 - b^2 = 0$$

5. (1) From statement I,

$$\text{Area of triangle} = \frac{1}{2}bh$$

$$\text{Area of rectangle} = bh$$

$$= 768\ \text{sq. metre}$$

$\therefore$  Area

$$= 384$$

6. (2) Qu

$$= \frac{6a}{a+b}$$

$$= \frac{6a}{a+b}$$

$$= \frac{a^2}{a^2+b^2}$$

$$= \frac{1}{2}$$

Quantity

$$= \frac{2ab}{a^2+b^2}$$

$$7. (2) \frac{b+c}{2}$$

$$\Rightarrow b+c$$

$$\Rightarrow c-a$$

8. (3) Radi

$\therefore$  Side

Area of

$$= \pi r^2$$

$$= \frac{5}{6}\pi r^2$$

$$\therefore \frac{5\pi r^2}{6}$$

$$\Rightarrow 5 \times \frac{2}{3}$$

$$\Rightarrow 5 \times \frac{2}{3}$$

$$\Rightarrow x^2 =$$

$$\Rightarrow x =$$

Quantity

$\therefore$  Area of

= Area of

$$(B + C)'s \text{ 1 day's work} = \frac{3}{40} \dots (ii)$$

From statement II,

If time taken by C alone be  $x$  days, time taken by A alone =  $(x - 24)$  days

By equation (i) - (ii),

$(A - C)'s \text{ 1 day's work}$

$$= \frac{9}{80} - \frac{3}{40}$$

$$= \frac{9-6}{80} = \frac{3}{80}$$

$$\Rightarrow \frac{1}{x-24} - \frac{1}{x} = \frac{3}{80}$$

$$\Rightarrow \frac{x-x+24}{x(x-24)} = \frac{3}{80}$$

$$\Rightarrow x(x-24) = 640$$

$$= 40(40-24)$$

$$\Rightarrow x = 40$$

$\therefore$  Time taken by A

$$= 40 - 24 = 16 \text{ days}$$

$\therefore$  From statement (i),

$$\frac{1}{16} + \frac{1}{y} = \frac{9}{80}$$

$$\Rightarrow \frac{1}{y} = \frac{9}{80} - \frac{1}{16} = \frac{9-5}{80}$$

$$= \frac{4}{80} = \frac{1}{20}$$

$\therefore$  Time taken by B = 20 days

$\therefore (A + B + C)'s \text{ 1 day's work}$

$$= \frac{1}{16} + \frac{1}{20} + \frac{1}{40}$$

$$= \frac{5+4+2}{80} = \frac{11}{80}$$

$$\therefore \text{Required time} = \frac{80}{11} = 7\frac{3}{11} \text{ days}$$

4. (5) From both statements,

$$(a + b) = 13$$

$$a - b = 0$$

$$\therefore a^2 - b^2 = 0$$

5. (1) From statement I,

$$\text{Area of triangle} = \frac{1}{2}bh$$

$$\text{Area of rectangle} = bh$$

$$= 768 \text{ sq. metre}$$

$\therefore$  Area

$$= 384$$

6. (2) Quantity

$$= \frac{(a+b)}{(a+b)}$$

$$= \frac{a^2 + b^2}{a^2 + b^2}$$

$$= \frac{1}{2}$$

Quantity

$$= \frac{1}{2ab(b^2)}$$

$$7. (2) \frac{b+c}{2}$$

$$\Rightarrow b + c$$

$$\Rightarrow c - a$$

8. (3) Radius

$\therefore$  Side of

Area of

$$= \pi r^2 - \left( \frac{1}{2} \right)$$

$$= \frac{5}{6} \pi r^2$$

$$\therefore \frac{5\pi x^2}{6}$$

$$\Rightarrow 5 \times \frac{22}{7}$$

$$\Rightarrow 5 \times \frac{22}{7}$$

$$\Rightarrow x^2 = \frac{38}{7}$$

$$\Rightarrow x = \sqrt{\frac{38}{7}}$$

Quantity

$\therefore$  Area of

= Area of

$$= \left( \frac{\sqrt{3}}{4} \right) \times 14$$

# DATA SUFFICIENCY

$$= \frac{5}{6} \times \frac{22}{7} \times r^2 = \frac{1540}{3}$$

$$= r^2 = \frac{1540}{3} \times \frac{6 \times 7}{5 \times 22} = 196$$

$$\Rightarrow r = \sqrt{196} = 14 \text{ cm.}$$

$$\therefore \text{Side of triangle} = 28 \text{ cm.}$$

Area of triangle

$$= \frac{\sqrt{3}}{4} \times 28 \times 28$$

$$= 196\sqrt{3} \text{ sq. cm.}$$

$$= 333.2 \text{ sq. cm.}$$

Area of all three sectors

$$= \frac{180}{360} \times \frac{22}{7} \times 14 \times 14$$

$$= 308 \text{ sq. cm.}$$

Area of shaded region-2

$$= 333.2 - 308$$

$$= 25.2 \text{ sq. cm.}$$

Clearly, quantity I > quantity II.

14. (4) C.P. of article A = Rs.  $x$   
Marked price of article A

$$= \text{Rs.} \left( \frac{x \times 140}{100} \right)$$

$$= \text{Rs.} \frac{7x}{5}$$

Marked price of article B

$$= x \times \frac{125}{100} = \text{Rs.} \frac{5x}{4}$$

$$\text{S.P. of article A} = \frac{7x}{5} \times \frac{80}{100}$$

$$= \text{Rs.} \frac{28x}{25}$$

S.P. of article B

$$= \frac{5x}{4} \times \frac{80}{100} = \text{Rs.} x$$

$$\therefore \frac{28x}{25} + x = 636$$

$$\Rightarrow \frac{53x}{25} = 636$$

$$\Rightarrow 53x = 636 \times 25$$

$$\Rightarrow x = \frac{636 \times 25}{53} = \text{Rs.} 300$$

Quantity

C.P. of ar

$$\therefore y \times \frac{12}{10}$$

$$\Rightarrow \frac{6y}{5} = 6$$

$$\Rightarrow \frac{6y}{5} = 6$$

$$\Rightarrow \frac{y}{5} = 60$$

$$\Rightarrow y = 300$$

$\therefore$  Quanti

15. (2) Quanti  
Expressio

$$\frac{ab(a^2 +$$

$$= \frac{-4ab}{a^2 + b^2}$$

$$\frac{ab(a +$$

$$= \frac{-4ab}{a^2 + b^2}$$

Quantity

$$\frac{1}{a^3b + 2ab}$$

16. (4) From s

$$\frac{PR^2}{10000} = 4$$

$$\Rightarrow \frac{P \times 64}{10000}$$

$$\Rightarrow P = \frac{45000}{64}$$

$$= \text{Rs.} 703$$

From stat

$$\text{C.I.} = P \left[ \left( 1 + \frac{r}{100} \right)^n - 1 \right]$$

$$\Rightarrow 4540 =$$

$$\Rightarrow 4540 =$$

$$\Rightarrow 4540 =$$

$\therefore$  1 boy will do  $\frac{18}{25}$  work in 48 days.

Thus, we get number of days for man.

21. (4) Gopi's present age

= Lalitha's age + 2

=  $6 + 2 = 8$  years

Atul's present age =  $8 \times 4 = 32$  years.

As per statement - 1, (Banu's age - 4)  $\times$  2

Atul's age - 4

$$\frac{(32 - 4)}{2} = (\text{Banu's age} - 4)$$

$14 + 4 = 18 = \text{Banu's age}$

Required sum = (Atul + Banu)'s age =  $32 + 18$   
= 50 years

22. (1) By taking statements I and II.

Ratio of shares next year

=  $(3 \times 110\%) : (5 \times 88\%) = 3 : 4$

As A's profit = Rs. 3450

$$\therefore \text{Total profit} = 3450 \times \left(\frac{7}{3}\right)$$

= Rs. 8050

23. (2) If B takes  $x$  days to complete the work

the  $\frac{1}{6}$ th of the work is done by B in  $\frac{x}{6}$  days

If A completes the  $\frac{1}{2}$  of the work as B does

So, A takes twice more days

$$\frac{2x}{6} = \frac{x}{3} \text{ days.}$$

$\therefore$  Time taken by A in doing 1 work =  $2x$  days

$\therefore$  (A + B)'s 1 day's work

$$= \frac{1}{x} + \frac{1}{2x} = \frac{3}{2x}$$

$$\therefore \frac{3}{2x} = \frac{1}{20} \Rightarrow x = 30 \text{ days.}$$

24. (4) From statements - I and II

P = Rs. 1000 and A = Rs. 1331 Time = 3 years

From Statement - III

The rate of interest was compounded annually because the simple interest of one year is equal to the compound interest of the first year.

Thus, we get P = 1000.

A = Rs. 1331 and Time = 3 years.

$$\therefore A = P \left(1 + \frac{R}{100}\right)^T$$



$$= 261 \text{ square metre}$$

$$\therefore \text{Total cost} = 261 \times 285$$

$$= \text{Rs. } 74385$$

6. (5) There are three unknown quantities and only two equations. Hence value of  $z$  cannot be determined.

7. (3) From both statements.

Number of students participating in elocution

$$= \frac{150 \times 100}{250} = 60$$

8. (2) From statement II.

$$\text{Speed of train} = \frac{\text{length of platform and train}}{\text{Time taken in crossing}}$$

$$= \frac{750 + 350}{55} = \frac{1100}{55}$$

$$= 20 \text{ m/second}$$

9. (3) From both statements.

Circumference of wheel

$$= \pi \times \text{diameter}$$

$$= \frac{22}{7} \times 140 = 440 \text{ cm}$$

Number of rotations

$$= \frac{880 \times 100}{440} = 200$$

10. (3) From both statements.

Speed of second person =  $x$  kmph

$$\therefore \frac{320}{x} - \frac{240}{6} = 3$$

$$\Rightarrow \frac{320}{x} - 40 = 3$$

$$\Rightarrow \frac{320}{x} = 43$$

$$\Rightarrow x = \frac{320}{43} = 7.4 \text{ kmph}$$

11. (3) It is the calculation of average speed

$$= \frac{(2 \times 40 \times 80)}{120} = \frac{160}{3}$$

$$\left( \frac{d}{40} \right) - \left( \frac{3d}{160} \right) = 1$$

$$\frac{(4d - 3d)}{160} = 1$$

$$\frac{d}{160} = 1$$

$$\text{So, } d = 160 \text{ km.}$$

21. (2) H's cost price

$$= \left( \frac{5457}{85} \right) \times 100 = \text{Rs } 6420$$

$$\text{P's cost price} = \left( \frac{5457}{102} \right) \times 100 = \text{Rs. } 350$$

$$\begin{aligned} \text{Required ratio} &= 6420 : 350 \\ &= 6 : 5 \end{aligned}$$

22. (3) If alcohol is 6x litres and water is 5x litres, then

$$\left[ 6x - \left( 22 \times \left( \frac{6}{11} \right) \right) \right] : \left[ 5x - \left( 22 \times \left( \frac{5}{11} \right) + 22 \right) \right]$$

$$= 9 : 13$$

$$\frac{(6x - 12)}{(5x + 12)} = \frac{9}{13}$$

$$\Rightarrow 78x - 156 = 45x + 108$$

$$\Rightarrow 78x - 45x = 156 + 108$$

$$\Rightarrow 33x = 264$$

$$\Rightarrow x = \frac{264}{33} = 8$$

So, water after replacement

$$= (5 \times 8) - 10 + 22 = 52 \text{ litres}$$

23. (2) If principal is P, then SI = Rs. 2P

$$\therefore \text{Rate} = \frac{(2P \times 100)}{(P \times 12)}$$

$$= \frac{50}{3} = 16.666\%$$

24. (1) (X + Z) in D days  $\rightarrow$  so Y completes in 3D days

$$\text{then, } \left( \frac{1}{D} \right) + \left( \frac{1}{3D} \right) = \frac{1}{10}$$

$$\Rightarrow \frac{4}{3D} = \frac{1}{10} \Rightarrow D = \frac{40}{3}$$

$$\text{So, Y will do in } 3D = 3 \times \left( \frac{40}{3} \right) = 40 \text{ days}$$

25. (5) X's work in % =  $\frac{100}{10} = 10\%$ 

$$\text{Y's work in \%} = \frac{100}{20} = 5\%$$

$$\text{Z's work in \%} = \frac{100}{40} = 2.5\%$$

26.

27.

28. (1)

29. (5) F

30. (3) F

We  
equ

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 $\Rightarrow A$ 

Also

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From

be c

Side

Area

 $\Rightarrow l$ 

From

 $\Rightarrow l$  $\Rightarrow l$  $\therefore R$ 

3x +

 $\Rightarrow 1$ 

Large

= 5x

 $\therefore S$ 

From

x +

 $\Rightarrow 2$  $\Rightarrow x$

section are based on Data Interpretation examinations, usually time allocation is in comparison to the number of questions. The purpose is to check the overall aptitude of the candidate. In this age of stiff competition, Data Interpretation is very important. The questions based on Data Interpretation will comparatively take less time and will provide opportunity for scoring marks.

In SSC Prelim exam, there are 10 questions of Data Interpretation out of 100 questions in Arithmetic Test. The question is not to be ignored and the score cannot be ignored.

Increasing importance of Statistical Data in Management is also evident from the questions asked in almost all competitive examinations. About 50 percent questions are based on Data Interpretation.

These questions require student to solve a given problem, willing to take a challenge and an ability to reason logically.

### PRESENTATION

Statistics are a set of numerical data. Numerical data constitute Statistical Data. The data collected are voluminous and are organised in a comprehensive manner. The data pertaining to any situation can be presented in one or more of the following ways:

- I. Numerical Data Tables
- II. Bar Diagrams
- III. Line Graphs
- IV. Pie Diagrams
- V. Other Diagrammatic Representations

Now we will discuss each way in detail.

### NUMERICAL DATA

A table is a systematic presentation of data in horizontal rows and columns. According to some salient features, tables are important devices for the presentation of data in a condensed and comprehensive manner. It provides the gateway for further statistical interpretation.

#### Parts of A Table

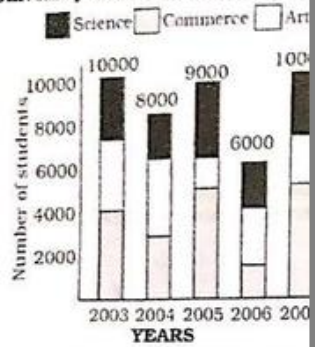
Before we go further in our study, it is imperative to know the parts of a table. Various parts of a table depend on the nature of the data and purpose of the investigation. The following features generally form the parts of a table. While embarking on attempting questions, you must pay attention to these parts.

- (i) **Table Number** : The table number is given at the top of the table.

shades, designs etc are used to distinguish various components. The students must pay attention to the colours and index provided. Otherwise their ignorance will lead to wrong answers.

Let us consider the following Bar Diagram.

**Faculty wise Distribution of Students in a University over the Years 2003-2007**

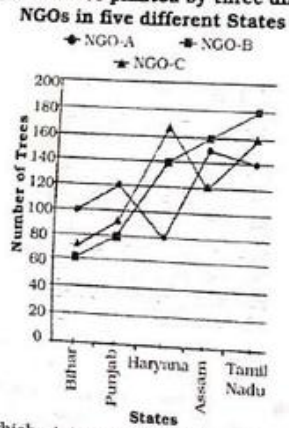


**MOD**

### LEVEL-I

**Directions (1-5) :** Study the following line graph carefully to answer the questions that follow.

**Number of trees planted by three different NGOs in five different States**



1. In which state was the total number of trees planted by NGO-A and NGO-B together seen to be the lowest?

- (1) Bihar
- (2) Punjab
- (3) Haryana
- (4) Assam
- (5) Tamil Nadu

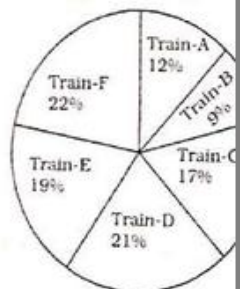
try in the previous year. If the  
from country E constituted 60%  
(males + females), in the year 2  
foreign students from country  
year were females?

- (1) 1400 (2) 1480 (3) 128  
(5) 1260

**Directions (16-20) :** Study the  
chart carefully to answer these ques

**Total number of Passengers in  
trains = 4800**

**Percentage wise distribution of**



16. What was the average number  
travelling in Train-A, Train-C and  
together?  
(1) 816 (2) 826 (3) 824  
(5) None of these
17. If cost of one ticket is ₹ 124, what  
amount paid by passengers of Train  
assuming all the passengers purchase  
cost of each ticket is equal)  
(1) ₹ 53,658 (2) ₹ 53,568  
(3) ₹ 53,558 (4) ₹ 53,468  
(5) None of these
18. Number of passengers in Train-E is  
mately what percentage of the total  
passengers in Train B and Train-D  
(1) 63 (2) 69 (3) 75  
(5) 79
19. What is the difference between the  
passengers in Train-C and number  
ger in Train-A?  
(1) 280 (2) 250 (3) 230  
(5) None of these
20. What is the total number of passengers  
D, Train-E and Train-F together?  
(1) 2796 (2) 3225 (3) 2976  
(5) None of these



35. What is the respective ratio of Company B (in tonnes) in the production of Company B (in tonnes) in 2008?

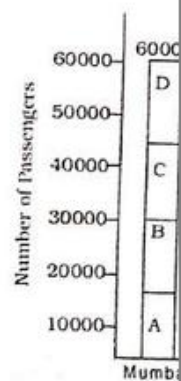
(1) 2 : 5 (2) 4 : 5 (3) 3 : 4  
(5) 1 : 4

**Directions (36–40) :** In the following table, the number of employees working in five companies and the corresponding ratio of male and female employees have been given. You are required to study the table carefully and answer the questions :

| Company | Number of Employees |
|---------|---------------------|
| L       | 400                 |
| M       | 600                 |
| N       | 800                 |
| O       | 1000                |
| P       | 1200                |

| Company | Male | Female |
|---------|------|--------|
| L       | 12   |        |
| M       | 15   |        |
| N       | 4    |        |
| O       | 13   |        |
| P       | 7    |        |

**Directions (41–45) :** In the following bar chart, the number of passengers travelling to four cities in first quarter of a year by four airlines A, B, C and D is shown. Study the chart carefully and answer the questions.



41. What is the difference between the number of passengers travelling to all cities by airline A and C ?  
(1) 5000 (2) 6000 (3) 5500  
(5) 15000
42. The number of passengers travelling to Mumbai and Kolkata by airline C is what per cent of the number of all passengers travelling to all cities by airline C ?

1. What is the respective ratio of total number of children from village O to the number of children attending school from the same village?

(1) 204 : 145  
 (2) 179 : 131  
 (3) 167 : 111  
 (4) 266 : 137  
 (5) None of these

2. What is the number of children attending school from village N?

(1) 145  
 (2) 159  
 (3) 170  
 (4) 164  
 (5) None of these

**Directions (6-10) :** Study the following table.

(Bank of Baroda PO Exam. 14.08.2014)

**Description of students studying in**

| School | Total Students | English | Maths |
|--------|----------------|---------|-------|
| A      | 480            | 78      | 68    |
| B      | 384            | 57      | 63    |
| C      | 435            | 64      | 73    |
| D      | 465            | 82      | 89    |
| E      | 389            | 52      | 77    |
| F      | 447            | 57      | 73    |

6. By **approximately** what percent is the number of students studying English less than that studying mathematics, in all schools taken together?

(1) 12  
 (2) 14  
 (3) 15  
 (4) 16  
 (5) 19

7. The respective ratio between the number of students studying Maths in schools A, C and D and that studying Social Science in schools B, D and F is

(1) 105 : 104  
 (2) 115 : 104  
 (3) 115 : 108  
 (4) 11 : 13  
 (5) None of these

8. By what percent is the number of students enrolled in Computer Science more than that enrolled in Hindi, taking all schools together?

16. What is the average annual salary of an employee in finance department from 2010 to 2012?

- (1) 3.6 (2) 3.8 (3) 4.0 (4) 4.2 (5) None of these

17. An employee who had worked in the finance department from 2010 to 2012, what was the sum of Rs. (in lakh)

- (1) 13.75 (2) 14.75 (3) 15.75 (4) 16.75 (5) None of these

18. If there were 50 employees in 2010 in the finance department, what amount was spent on their salaries in these years?

- (1) 430 (2) 420 (3) 410 (4) 400 (5) None of these

19. In the given years, there were 190 employees in production department. What amount did the company spend on their salaries?

- (1) 470 (2) 485 (3) 495 (4) 505 (5) 515

20. By what percent approximately did the annual salary of an employee in the Human Resources department increase from 2010 to the year 2012?

- (1) 16 (2) 12 (3) 10 (4) 8 (5) 23

**Directions (21-25) :** Study the following table carefully and answer the questions.

**The number of students who appeared for the examination in the year 2001.**

**Directions (26-30) :** Study the following table carefully and answer the questions.

**Number of students who appeared for the examination in the year 2001.**

| Years | 2001           |                      |
|-------|----------------|----------------------|
|       | Total Students | Percentage of female |
| A     | 1200           | 40%                  |
| B     | 2600           | 45%                  |

26. In the year 2003, what is the ratio of the number of female students who appeared for the examination in College A to that in College B?

- (1) 13 : 27 (2) 15 : 25 (3) 17 : 27 (4) 19 : 29 (5) 13 : 11

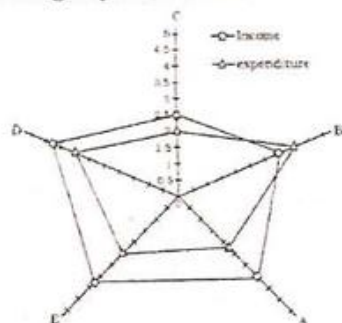
27. What is the difference between the number of female students who appeared for the examination in College A in the year 2003 and the average number of female students in the same year?

- (1) 214 (2) 172 (3) 130 (4) 98 (5) 198

8. By what percent is the sum total of Engg. graduates in B, C and F taken together greater than the sum total of Arts graduates in D and E taken together?
- (1) 56% (2) 68% (3) 65% (4) 72% (5) None of these
9. The number of Science graduates in A and B states combined is what percent of the number of graduates of all other streams taken together?
- (1) 28.5% (2) 27.9% (3) 29.7% (4) 30.1% (5) None of these
10. In which states is the difference between the number of Engg. and IT graduates higher than the difference between the number of Science and Arts graduates?
- (1) A, C and E (2) C, F and B  
(3) F, B and D (4) C, D and E  
(5) None of these

**Directions (11-15):** Study the given graph carefully to answer the following questions.

The income and expenditure of different companies during an year (in ₹ crore).



**Directions (16-20):** Study the following bar graph and answer the questions.

**Number of candidates appearing in the tests during 2005 and 2006**

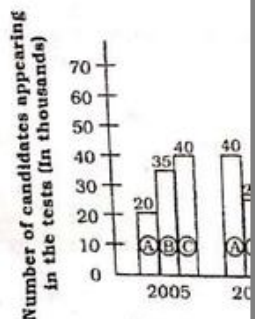
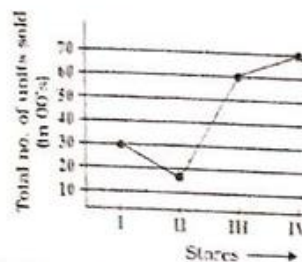


Table II

| Category → | Cordless Phone | Hair Dryer |
|------------|----------------|------------|
| Store I    | 2725           | 2585       |
| Store II   | 3190           | 3195       |
| Store III  | 3775           | 3815       |
| Store IV   | 4460           | 4480       |
| Store V    | 4905           | 5110       |

## LINE GRAPH



26. The no. of units of cordless phone sold by the five stores together is less than the units sold by all the five stores for hair dryers by —  
 (1) 5621 (2) 2956 (3) 2695  
 (4) cannot be determined  
 (5) None of these
27. By how much percent is the average monthly sales revenue earned per unit for hair dryers more than the average monthly sales revenue earned per unit for cordless phones?  
 (1) 80% (2) 75% (3) 68%  
 (4) cannot be determined  
 (5) None of these
28. Find the average monthly sales revenue earned per unit for store III, for all the units sold together.  
 (1) ₹ 3705 (2) ₹ 3759 (3) ₹ 5110  
 (4) cannot be determined  
 (5) None of these
29. In case of store IV, the annual sales revenue earned for cordless phones is less than the annual sales revenue earned for hair dryers by —  
 (1) ₹ 6.2 lakhs (2) ₹ 6.67 lakhs  
 (3) ₹ 66.67 lakhs (4) None of these
30. Number of units sold, in case of which category is highest for  
 (1) Store II (2) Store I  
 (3) Store I (4) Store II  
 (5) None of these



23. (5) Required speed of vehicle of C

$$= \left( \frac{810}{18} \right) \text{ kmph}$$

$$= \left( \frac{810}{18} \times \frac{5}{18} \right) \text{ m/sec.}$$

$$= 12.5 \text{ m/sec.}$$

24. (5) Required percentage

$$= \frac{636}{703} \times 100 = \frac{640}{700} \times 100 = 90$$

25. (2) Required ratio

$$= \frac{765}{15} : \frac{546}{14} = 51 : 39 = 17 : 13$$

26. (3) Total cost of wooden flooring

$$= \text{Rs. } (23 \times 29 + 13 \times 17) \times 170$$

$$= \text{Rs. } (667 + 221) \times 170$$

$$= \text{Rs. } 150960$$

Total cost of marble flooring

$$= \text{Rs. } (14 \times 13 + 21 \times 13 + 29 \times 21) \times$$

$$= \text{Rs. } (182 + 273 + 609) \times 190$$

$$= \text{Rs. } 202160$$

$\therefore$  Required ratio

$$= 150960 : 202160$$

$$= 1887 : 2527$$

27. (5) Area of ceiling and four walls of manager's room

$$= 13 \times 17 + 2 \times 12 (13 + 17)$$

$$= 221 + 720 = 941 \text{ sq. metre}$$

$$\therefore \text{Total cost} = 941 \times 190 + 13 \times 17 \times$$

$$= \text{Rs. } (178790 + 37570)$$

$$= \text{Rs. } 216360$$

28. (5) Area of wooden floors

$$= 667 + 221$$

$$= 888 \text{ sq. metre}$$

Area of marble floors

$$= 182 + 273 + 609$$

$$= 1064 \text{ sq. metre}$$

$$\text{Total area} = 1064 + 888$$

$$= 1952 \text{ sq. metre}$$

$$\text{Remaining area} = 2000 - 1952 = 48 \text{ sq.}$$

$$\therefore \text{Required cost} = 48 \times 110 = \text{Rs. } 5280$$

$$29. (2) \text{ Required percentage} = \frac{48}{2000} \times 100 =$$

$$30. (1) \text{ Required cost} = \text{Rs. } (23 \times 29 \times 170 +$$

$$21 \times 190)$$

$$= \text{Rs. } (113390 + 115710)$$

$$= \text{Rs. } 2,29,100$$

31. (4) Required percentage increase

$$= \frac{700 - 550}{550} \times 100 = 27$$

5. (5) Number of children in village L

$$= \frac{2040 \times 15}{100} = 306$$

Number of children attending school from vil-

$$\text{lage L} = \frac{1450 \times 14}{100} = 203$$

$$\text{Required percentage} = \frac{203}{306} \times 100 = 66\%$$

6. (1) Students who study English =  $78 + 57 + 64 + 82 + 52 + 57 = 390$

$$\begin{aligned} \text{Students who study maths} \\ &= 68 + 63 + 73 + 89 + 77 + 73 \\ &= 443 \end{aligned}$$

$$\text{Required percent} = \left( \frac{443 - 390}{443} \right) \times 100 = 12\%$$

7. (2) Required ratio

$$\begin{aligned} &= (68 + 73 + 89) : (46 + 93 + 69) \\ &= 230 : 208 = 115 : 104 \end{aligned}$$

8. (3) Students who study Computer Science = 524

$$\text{Students who study Hindi} = 415$$

$$\text{Required percent} = \frac{524 - 415}{415} \times 100 = 26$$

9. (4) Students of English + Maths + Science in school C

$$= 64 + 73 + 67 = 204$$

Students of English + Maths + Science in school D

$$= 82 + 89 + 62 = 233$$

Required percent

$$= \left( \frac{233 - 204}{233} \right) \times 100 = 12.45$$

10. (3) Required average

$$= \frac{426 + 402}{12} = \frac{828}{12} = 69$$

11. (1) Investment by Gaurang in schemes A and C

$$= \text{Rs. } (740 + 780) \text{ hundred}$$

$$= \text{Rs. } 152000$$

Investment by Varun in schemes B and E

$$= \text{Rs. } (560 + 380) \text{ hundred}$$

$$= \text{Rs. } 94000$$

$\therefore$  Required percentage

$$= \left( \frac{152000 - 94000}{94000} \right) \times 100 = 62\%$$

12. (2) Required average

$$= \text{Rs. } \frac{1}{6} (400 + 560 + 520 + 580 + 380 + 620) \text{ hundred}$$

2. (2) Volume of sales of brand C  
of 47000 = 6580 units  
To make this volume 7000,  
increment should be  
 $7000 - 6580 = 420$  units.  
So, required percentage

$$\frac{420}{47000} \times 100 = 0.9\%$$

3. (3) Global sales of brand C in  
= 112% of 47000 = 52640  
Volume of sales in USA in 2011  
= 18% of 47000 = 8460  
Volume of sales in USA in 2012  
= 134% of 8460 = 11336.4  
Volume of sales in China in 2011  
= 21% of 47000 = 9870  
Volume of sales in China in 2012  
= 122% of 9870 = 12041.4  
Volume of sales in rest of the world  
2011  
=  $47000 - (8460 + 9870)$   
=  $47000 - 18330$   
= 28670 units

Volume of sales of rest of the world  
=  $52640 - (11336.4 + 12041.4)$   
=  $52640 - 23377.8$   
= 29262.2 units

Therefore, required percentage sales  
in rest of the countries

$$= \frac{29262.2 - 28670}{28670} \times 100$$

$$= \frac{592.20}{28670} \times 100 = 2\%$$

4. (2) In 2012, combined sales of A and B  
 $32 \times 1000$   
= 64000 units  
and total sales of all the brands in 2012  
=  $(32 + 32 + 47 + 54 + 28 + 12) \times 1000$   
units

∴ Required percentage

$$= \frac{64000}{205000} \times 100$$

$$= 31.2\%$$

5. (3) Global sales of all brands in 2011  
of  
 $(21 + 32 + 47 + 54 + 28 + 12)$  thousand units  
= 133% of 194 thousand units  
= 258.02 thousand units

∴ Required volume

20. (3) Number of candidates appearing during all these years

$$\text{Test A} \Rightarrow (20 + 40 + 25 + 25 + 45 + 60) \text{ thousand} \\ = 215 \text{ thousand}$$

$$\text{Test B} \Rightarrow (35 + 25 + 35 + 30 + 30 + 40) \text{ thousand}$$

$$= 195 \text{ thousand}$$

$$\therefore \text{Required ratio} = 215 : 195$$

$$= 43 : 39$$

21. (5) Successful candidates from state

$$A = \frac{2400 \times 20}{100} = 480$$

$$\therefore \text{Successful female candidates} = 480 - 264 \\ = 216$$

$$\therefore \text{Required percent}$$

$$= \frac{216}{480} \times 100 = 45\%$$

22. (5) Candidates who appeared in the exam :

$$\text{State C} \Rightarrow \frac{3600 \times 12}{100} = 432$$

$$\text{State D} \Rightarrow \frac{3600 \times 28}{100} = 1008$$

$$\text{State E} \Rightarrow \frac{3600 \times 17}{100} = 612$$

Successful candidates :

$$\text{State C} \Rightarrow \frac{2400 \times 14}{100} = 336$$

$$\text{State D} \Rightarrow \frac{2400 \times 30}{100} = 720$$

$$\text{State E} \Rightarrow \frac{2400 \times 8}{100} = 192$$

$\therefore$  Unsuccessful candidates

$$= (432 - 336) + (1008 - 720) + (612 - 192) \\ = 96 + 288 + 420 = 804$$

$$\therefore \text{Required average} = \frac{804}{3}$$

$$= 268$$

23. (5)  $\therefore 100\% = 360^\circ$

$$\therefore 8\% = \frac{360}{100} \times 8 = 28.8^\circ$$

24. (4) Successful candidates from states C and D

$$= 2400 \times \frac{44}{100} = 1056$$

Appearing candidates in the exam :

$$\text{State C} \Rightarrow 432$$

$$\text{State D} \Rightarrow 1008$$

$\therefore$  Unsuccessful candidates :

(iii) If the rise is sharp initially but is decreasing, the series is likely to be formed by adding squared or cubed numbers.

(iv) The series showing irregular increase or decrease may have been formed in either of two ways :

(a) Intermixing of two different series or

(b) Two different operations being performed on successive terms.

**Example :** 1 4 27 256 3125 46656 ?

(1) 117649 (2) 279936 (3) 705894

(4) 16807 (5) 823543

**Ans. (5)** The given series is based on the following pattern :

$$1 = 1^1$$

$$4 = 2^2$$

## MODEL

### LEVEL-I

**Directions (1-10) :** What will come in place of question mark (?) in the given number series.

1. 4 11 26 57 120 ?

(1) 247 (2) 251 (3) 187 (4) 236  
(5) 215

2. 50 81 121 172 236 ?

(1) 392 (2) 309 (3) 361 (4) 379  
(5) 315

3. 84 82.4 85.6 80.8 ? 79.2

(1) 72.8 (2) 86.4 (3) 88.8 (4) 87.2  
(5) 85.7

4. 900 448 220 104 44 ?

(1) 8 (2) 16 (3) 24 (4) 42  
(5) 12

5. 60 ? 120 80 40 16

(1) 90 (2) 110 (3) 120 (4) 70  
(5) 100

6. 2 3 7 29 ? 3729

(1) 233 (2) 225 (3) 227 (4) 331  
(5) 191

7. 14 8 7 11.5 22 ?

(1) 54 (2) 64 (3) 62 (4) 58  
(5) 56

8. 8 14 25 46 82 ?

(1) 132 (2) 130 (3) 138 (4) 168  
(5) 172

9. 13 14 30 93 ? 1885

(1) 358 (2) 336 (3) 364 (4) 386  
(5) 376



42. 1 2 8 33 148 740 4626

2 (a) (b) (c) (d) (e) (f)

What will come in place of (e)

(1) 156 (2) 164 (3) 16

(5) None of these

43. 2 4.5 11 30 93 312

1 (a) (b) (c) (d) (e)

What will come in place of (b)

(1) 6 (2) 81 (3) 16

(5) None of these

44. 2 14 18 46 82 176 3

4 (a) (b) (c) (d) (e) (f)

What will come in place of (e)

(1) 235 (2) 338 (3) 21

(5) None of these

45. 1 3 7 11 21 43 85

4 (a) (b) (c) (d) (e) (f)

What will come in place of (f)

(1) 275 (2) 279 (3) 27

(5) None of these

**Directions (46-50) :** What will  
of the question mark (?) in the fo  
series which has only one number  
gin of + 1 or - 1 ? The first and las  
series are correct?

46. 93 95 99 ? 110 121 134

(1) 104 (2) 96 (3) 82

(5) None of these

47. 8 12 18 26 40.5 60.75 ? 136.68

(1) 104.125 (2) 12

(3) 96.125 (4) 83.

(5) None of these

48. 4 7 11 18 28

(1) 59 (2) 38 (3) 46

(5) None of these

49. 3 10 ? 172 886 5346 37471 2

(1) 39 (2) 27 (3) 24

(5) None of these

50. 15 22 57 183 ? 748 3755

(1) 709 (2) 698 (3) 74

(5) None of these

## LEVEL-II

**Directions (1-5) :** Find the wrong  
the given number series.

1. 212 242 278 324 392 482 60

(1) 212 (2) 324 (3) 392

(5) 278

7. Consider the following three numbers

I. 27, 35, 47, 63,  $x$

II. 6, 6, 18,  $y$ , 630

III. 156, 145, 123,  $z$ , 46

(1)  $x = y = z$  (2)  $x < y$

(3)  $x = y > z$  (4)  $x < y$

(5)  $x > y = z$

8. Consider the following three numbers

I. 3, 8, 27,  $x$ , 565

II. 625,  $y$ , 50, 30, 24

III. 56, 92,  $z$ , 133, 142

(1)  $x > y = z$  (2)  $x = y$

(3)  $x > y > z$  (4)  $x < y < z$

(5)  $x > y = z$

9. Consider the following three numbers

I. 17, 45, 87,  $x$ , 213

II. 54, 76, 109,  $y$ , 208

III. 4, 10, 33,  $z$ , 685

(1)  $x > y = z$  (2)  $x = y = z$

(3)  $x > y > z$  (4)  $x < y < z$

(5)  $x < y > z$

10. Consider the following three numbers

I. 31, 47,  $x$ , 58, 9

II.  $y$ , 127, 346, 733, 1336

III. 11,  $z$ , 87, 364, 1845

(1)  $x = y > z$  (2)  $x = y = z$

(3)  $x > y > z$  (4)  $x < y < z$

(5)  $x < y > z$

11. Find the sum of the series

$$\frac{7}{16} + \frac{77}{(16)^2} + \frac{777}{(16)^3} + \frac{7777}{(16)^4} + \dots \infty$$

(1)  $\frac{47}{48}$  (2)  $\frac{87}{48}$  (3)  $\frac{56}{45}$

(5)  $\frac{49}{149}$

12. Two sequences are such that one is in A.P. while other is in G.P. The first term of the A.P. is one less than the common difference of the given A.P. while the second term of the A.P. is the common ratio of the G.P. Also the sum of the first two terms of the G.P. is 10 and the sum of the first seven terms of the A.P. is 70. Which of the following can be the G.P. as per the above constraints?

(1)  $\frac{5}{2}, \frac{15}{2}, \frac{45}{2}, \frac{135}{2}, \dots$  (2) 1, 9, 81,  $\dots$

(3) 2, 8, 32, 128,  $\dots$  (4) Both (1) and (2)

(5) None of these

28. The 288th term of the sequence  $d, d, d, e, e, e, e, f, f, f, f, f, f, g, g, g, g, g, g, h, h, h, h, h, h, i, i, i, i, i, i, j, j, j, j, j, j, k, k, k, k, k, k, l, l, l, l, l, l, m, m, m, m, m, m, n, n, n, n, n, n, o, o, o, o, o, o, p, p, p, p, p, p, q, q, q, q, q, q, r, r, r, r, r, r, s, s, s, s, s, s, t, t, t, t, t, t, u, u, u, u, u, u, v, v, v, v, v, v, w, w, w, w, w, w, x, x, x, x, x, x, y, y, y, y, y, y, z, z, z, z, z, z$  is
- (1)  $u$                       (2)  $v$   
(3)  $p$
29. The sum of 3rd and 15th terms of an Arithmetic Progression is equal to the sum of 11th and 13th elements. Then which of the following elements should necessarily be equal to zero?
- (1) 1st  
(3) 12th  
(5) Data is insufficient
30. All the page numbers from 1 to 1000 were written beginning at page 1. However, the number 1000 was mistakenly added twice. How many times was the number 1000 added?
- (1) 44                      (2) 45  
(5) 20

## • SHORT ANSWERS

## LEVEL.

|         |         |         |         |
|---------|---------|---------|---------|
| 1. (1)  | 2. (5)  | 3. (4)  | 4.      |
| 7. (5)  | 8. (3)  | 9. (5)  | 10.     |
| 13. (1) | 14. (3) | 15. (1) | 16.     |
| 19. (4) | 20. (4) | 21. (1) | 22.     |
| 25. (2) | 26. (3) | 27. (1) | 28.     |
| 31. (1) | 32. (1) | 33. (3) | 34.     |
| 37. (5) | 38. (2) | 39. (4) | 40.     |
| 43. (5) | 44. (1) | 45. (3) | 46. (4) |
| 49. (1) | 50. (3) |         |         |

**LEVEL-II**

|         |         |         |         |
|---------|---------|---------|---------|
| 1. (2)  | 2. (5)  | 3. (1)  | 4. (4)  |
| 7. (4)  | 8. (3)  | 9. (4)  | 10. (2) |
| 13. (3) | 14. (1) | 15. (2) | 16. (1) |
| 19. (4) | 20. (2) | 21. (1) | 22. (2) |
| 25. (1) | 26. (4) | 27. (2) | 28. (5) |

**LEVEL-III**

|         |         |         |         |
|---------|---------|---------|---------|
| 1. (1)  | 2. (2)  | 3. (3)  | 4. (4)  |
| 7. (2)  | 8. (4)  | 9. (5)  | 10. (5) |
| 13. (2) | 14. (2) | 15. (1) | 16. (4) |
| 19. (1) | 20. (2) | 21. (4) | 22. (1) |
| 25. (2) | 26. (4) | 27. (3) | 28. (4) |

$$\begin{aligned}
 18. (3) & 963 - 1 \times 36 = 963 - 36 = 927 \\
 & 927 - 2 \times 36 = 927 - 72 = 855 \\
 & 855 - 3 \times 36 = 855 - 108 = 747 \\
 & 747 - 4 \times 36 = 747 - 144 = 603 \\
 & 603 - 5 \times 36 = 603 - 180 = 423 \\
 & \therefore ? = 423 - 6 \times 36 = 423 - 216 = 207
 \end{aligned}$$

$$\begin{aligned}
 19. (4) & 18 \times 1 + 2 = 18 + 2 = 20 \\
 & 20 \times 2 + 4 = 40 + 4 = 44 \\
 & 44 \times 3 + 6 = 132 + 6 = 138 \\
 & 138 \times 4 + 8 = 552 + 8 = 560 \\
 & 560 \times 5 + 10 = 2800 + 10 = 2810 \\
 & \therefore ? = 2810 \times 6 + 12 = 16860 + 12 = 16872
 \end{aligned}$$

$$\begin{aligned}
 20. (4) & \text{The pattern of the number series is } 3 \times 7 + 1 = 21 + 1 = 22
 \end{aligned}$$

$$22 \times 6 + 2 = 132 + 2 = 134$$

$$134 \times 5 + 3 = 670 + 3 = 673$$

$$673 \times 4 + 4 = 2692 + 4 = 2696$$

$$2696 \times 3 + 5 = 8088 + 5 = 8093$$

$$21. (1) \text{The pattern of the number series is } 6 \times 1 + 1 \times 7 = 6 + 7 = 13$$

$$13 \times 2 + 2 \times 6 = 26 + 12 = 38$$

$$38 \times 3 + 3 \times 5 = 114 + 15 = 129$$

$$129 \times 4 + 4 \times 4 = 516 + 16 = 532$$

$$532 \times 5 + 5 \times 3 = 2660 + 15 = 2675$$

$$22. (3) \text{The pattern of the number series is } 17 \times 0.5 + 0.5 = 9$$

$$9 \times 1 + 1 = 10$$

$$10 \times 1.5 + 1.5 = 16.5$$

$$16.5 \times 2 + 2 = 35$$

$$35 \times 2.5 + 2.5 = 87.5 + 2.5 = 90$$

$$23. (5) \text{The pattern of the number series is } \frac{286}{2} - 1 = 143 - 1 = 142$$

$$\frac{142}{2} - 1 = 71 - 1 = 70$$

$$\frac{70}{2} - 1 = 35 - 1 = 34$$

$$\frac{34}{2} - 1 = 17 - 1 = 16$$

$$\frac{16}{2} - 1 = 8 - 1 = 7$$

$$24. (3) \text{The pattern is : } 5 \times 0.5 + 0.5 = 2.5 + 0.5 = 3$$

$$3 \times 1.5 + 1.5 = 4.5 + 1.5 = 6$$

$$6 \times 2.5 + 2.5 = 15 + 2.5 = 17.5$$

$$17.5 \times 3.5 + 3.5 = 61.25 + 3.5 = 64.75$$

$$3 \times 1 + (1)^2 = 4 \dots\dots(a)$$

$$4 \times 2 + (2)^2 = 4 \dots\dots(b)$$

$$4 \times 3 + (3)^2 = \boxed{21} \dots\dots(c)$$

$$21 \times 4 + (4)^2 = 68 \dots\dots(d)$$

Therefore, the 21 will come in place of (c).

42. (5) The series is based on following pattern

$$1 \times 1 + (1)^2 = 2$$

$$2 \times 2 + (2)^2 = 8$$

$$8 \times 3 + (3)^2 = 33$$

$$33 \times 4 + (4)^2 = 148$$

$$148 \times 5 + (5)^2 = 765$$

$$765 \times 6 + (6)^2 = 4626$$

Similarly,

$$2 \times 1 + (1)^2 = 3 \dots\dots(a)$$

$$3 \times 2 + (2)^2 = 10 \dots\dots(b)$$

$$10 \times 3 + (3)^2 = 39 \dots\dots(c)$$

$$39 \times 4 + (4)^2 = \boxed{172} \dots\dots(d)$$

Therefore, the number 172 will come in place of (d).

43. (5) The series is based on following pattern

$$2 \times 2 + 0.5 = 4.5$$

$$4.5 \times 2 + (0.5) \times 4 = 11$$

$$11 \times 2 + 2 \times 4 = 30$$

$$30 \times 2 + 8 \times 4 = 92$$

$$92 \times 2 + 32 \times 4 = 312$$

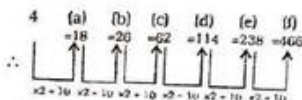
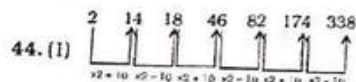
$$312 \times 2 + 128 \times 4 = 1136$$

Similarly,

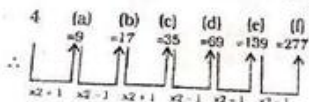
$$1 \times 2 + 0.5 = 2.5 \dots\dots(a)$$

$$2.5 \times 2 + (0.5) \times 4 = \boxed{7} \dots\dots(b)$$

Therefore, the number 7 will come in place of (b).



In the given series 176 should be replace by 174. 238 will come in place of (e).





13. (3) The pattern is :

$$84 - 19 = 65$$

$$134 - 84 = 50 (= 65 - 15)$$

$$171 - 134 = 37 (= 50 - 13)$$

$$197 - 171 = 26 (= 37 - 11)$$

$$\therefore ? = 197 + 17 (= 26 - 9)$$

$$= \boxed{214}$$

14. (1) The pattern is :

$$51 + 2 = 53$$

$$53 + 6 (= 2 + 4) = 59$$

$$59 + 10 (= 6 + 4) = 69$$

$$69 + 14 (= 10 + 4) = 83$$

$$83 + 18 (= 14 + 4) = \boxed{101}$$

15. (2) The pattern is :

$$13 + 1 \times 14 = 13 + 14 = 27$$

$$27 + 3 \times 14 = 27 + 42 = 69$$

$$69 + 5 \times 14 = 69 + 70 = 139$$

$$139 + 7 \times 14 = 139 + 98 = 237$$

$$237 + 9 \times 14 = 237 + 126$$

$$= \boxed{363}$$

16. (1) The pattern is :

$$48 \times \frac{1}{2} = 24$$

$$24 \times \frac{3}{2} = 36$$

$$36 \times \frac{5}{2} = 90$$

$$90 \times \frac{7}{2} = 315$$

Similarly,

$$a = 20 \times \frac{1}{2} = 10$$

$$b = 10 \times \frac{3}{2} = 15$$

$$c = 15 \times \frac{5}{2} = \frac{75}{2}$$

$$d = \frac{75}{2} \times \frac{7}{2} = \frac{525}{4} = \boxed{131.25}$$

17. (1) The pattern is :

$$2 \times 5 + 1^3 = 10 + 1 = 11$$

$$11 \times 4 + 2^3 = 44 + 8 = 52$$

$$52 \times 3 + 3^3 = 156 + 27 = 183$$

$$182 \times 2 + 4^3 = 366 + 64 = 430$$

$$430 \times 1 + 5^3 = 430 + 125 = 555$$

a :

b =

18. (1)

7 ×

9 ×

21 :

67 :

273

Sim

a =

b =

19. (4)

9 ×

8 ×

12 ×

27 ×

Sim

a =

b =

c =

20. (2) T

8 ×

9 ×

21 ×

68 ×

279

Simi

a = 5

b = 6

c = 1

d = 5

21. (1) T

5 ×

3 ×

4 ×

7.5 :

17 ×

= 45

45 ×

22. (2) T

9 +

16 +

25 +

38 +

57 +

$$99 \times 4 - 4^2 = 396 - 16 = 380$$

$$380 \times 5 - 5^2 = 1900 - 25 = 1875$$

6. (3) Look at the pattern :

I.  $10 + 15 \times 0.5 = 17.5$

$$17.5 + 15 \times 1 = 32.5$$

$$32.5 + 15 \times 1.5 = 55$$

$$55 + 15 \times 2 = 85$$

$$\therefore x = 32.5$$

Look at the pattern :

II.  $7 + (2 \times 2 + 5) = 16$

$$16 + (3 \times 3 + 5) = 30$$

$$30 + (4 \times 4 + 5) = 51$$

$$51 + (5 \times 5 + 5) = 81$$

$$\therefore y = 30$$

Look at the pattern :

III.  $2^2 - 2 = 6$

$$3^2 - 3 = 24$$

$$4^2 - 4 = 60$$

$$5^2 - 5 = 120$$

$$\therefore z = 24$$

7. (2) Look at the pattern :

I.  $27 + 8 = 35$

$$35 + 12 = 47$$

$$47 + 16 = 63$$

$$63 + 20 = 83$$

$$\therefore x = 83$$

Look at the pattern :

II.  $6 \times 1 = 6$

$$6 \times 3 = 18$$

$$18 \times 5 = 90$$

$$90 \times 7 = 630$$

$$\therefore y = 90$$

Look at the pattern :

III.  $156 - 11 = 145$

$$145 - 22 = 123$$

$$123 - 33 = 90$$

$$90 - 44 = 46$$

$$\therefore z = 90$$

8. (4) Look at the pattern :

I.  $(3 \times 2) + 2 = 8$

$$(8 \times 3) + 3 = 27$$

$$(27 \times 4) + 4 = 112$$

$$(112 \times 5) + 5 = 565$$

$$\therefore x = 112$$

Look at the pattern :

II.  $625 \times \left(\frac{1}{5}\right) = 125$

$$125 \times \left(\frac{2}{5}\right) = 50$$

5

3

$\therefore y$

Look

III. 5

9

1

13

$\therefore z$

9. (5) Lo

I. 17

45

87

14

$\therefore x =$

Look

II. 54

76

10

153

$\therefore y =$

Look

III. (4

(10

(33

(13

$\therefore z =$

10. (5) Loo

I. 31

47

22

58

Look at

II.  $3^3 +$

$5^3 +$

$7^3 +$

$9^3 +$

$11^3 +$

Look at

III. (11

(26

(87

(364

$$S_{\infty} = \pi r^2 + \pi \left(\frac{r}{2}\right)^2 + \pi \left(\frac{r}{4}\right)^2 + \pi \left(\frac{r}{8}\right)^2 + \dots$$

$$= \pi r^2 + \frac{\pi r^2}{4} + \frac{\pi r^2}{16} + \frac{\pi r^2}{64} + \dots$$

$$= \pi r^2 \left(1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots\right)$$

$$= \pi r^2 \left(\frac{1}{1 - \frac{1}{4}}\right) = \pi r^2 \left(1 \times \frac{4}{3}\right) = \frac{4}{3} \pi r^2$$

$$15. (1) S_n = \log x^2 + \log \left(\frac{x^4}{y}\right) + \log \left(\frac{x^6}{y^2}\right) + \log \left(\frac{x^8}{y^3}\right) + \dots + \log \left(\frac{x^{2n}}{y^{n-1}}\right)$$

$$= \log \left(x^2 \cdot \frac{x^4}{y} \cdot \frac{x^6}{y^2} \cdot \frac{x^8}{y^3} \dots \frac{x^{2n}}{y^{n-1}}\right)$$

$$= \log \left[\frac{x^{2+4+6+\dots+2n}}{y^{1+2+3+\dots+(n-1)}}\right] = \log \left[\frac{x^{n(n+1)}}{y^{\frac{(n-1)n}{2}}}\right]$$

$$= \log \left[\frac{x^{2(n+1)}}{y^{n-1}}\right]^{\frac{n}{2}}$$

16. (4) Let no. of hours be 'n'.

Tom covers 20, 18, 16, 14, ..., n.

$$S_n = \frac{n}{2} [2 \times 20 + (n-1)(-2)]$$

$$= \frac{n}{2} [40 - 2n + 2] = \frac{n}{2} [42 - 2n] = n(21 - n)$$

Jerry covers 5, 6, 7, 8, .....

$$S_n = \frac{n}{2} [10 + (n-1)] = \frac{n}{2} [9 + n]$$

$$\Rightarrow n(21 - n) + \frac{n}{2} (9 + n) = 94$$

$$\Rightarrow 21n - n^2 + \frac{9}{2}n + \frac{n^2}{2} = 94$$

$$\Rightarrow \frac{51}{2}n - \frac{n^2}{2} = 94$$

$$\Rightarrow 51n - n^2 = 188$$

$$\Rightarrow n^2 - 51n - 188 = 0$$

$$\Rightarrow n^2 - 4n - 47n - 188 = 0$$

$$\Rightarrow n(n-4) - 47(n-4) = 0$$

$$\Rightarrow n = 47, 4$$

n = 4

would

17. (1) D

81 ×

36 ×

i.e. 8

16 ×

= 81

= 81

= 81

18. (3) Let

10

Let n

$S_n = 1$

$S_2 = 1$

$S_3 = S_2$

Check

(1)  $2a$

(2)  $n +$

(3)  $2n^2$

19. (1) At t

At time

[1 old c

At time

reprode

At time

so on.

Series b

utes.

Since t

only, he

will die.

utes = 3

=  $3^6$  ( $3^6$ )

20. (2) A

S

D

Let ABCD

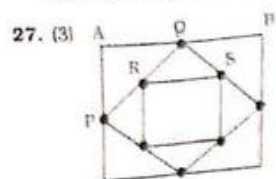
units PQRS

joining midp

$\Delta APS$  is a r

enuse.

Now when  $d = 999$  then we cannot have 3 elements in S. hence  $d = 999$  is ruled out.  
Possible cases are 7 in all.



Given  $AB = a$   
In  $\Delta APQ$  = (right triangle).  $PQ^2 = AP^2 + AQ^2$

$$PQ = \sqrt{\left(\frac{a}{2}\right)^2 + \left(\frac{a}{2}\right)^2} = \sqrt{\frac{2a^2}{4}} = \frac{a\sqrt{2}}{2}$$

$$RS = \sqrt{\left(\frac{a\sqrt{2}}{4}\right)^2 + \left(\frac{a\sqrt{2}}{4}\right)^2} = \sqrt{\frac{4a^2}{16}} = \frac{a}{2}$$

AB, PQ, RS are sides of  $S_1, S_2, S_3$  respectively.

Hence,  $P_1 + P_2 + P_3 + \dots = 4a + 4$

$$\left(\frac{a}{2}\sqrt{2}\right) + 4\left(\frac{a}{2}\right) + \dots$$

$$= 4\left[a + \frac{\sqrt{2}}{2}a + \frac{a}{2} + \dots\right]$$

$$= 4\left[\frac{a}{1 - \frac{\sqrt{2}}{2}}\right] = \frac{4a \times 2}{2 - \sqrt{2}} = \frac{8a}{2 - \sqrt{2}}$$

$$\text{Also, } A_1 + A_2 + A_3 + \dots = a^2 + \left(\frac{a}{2}\sqrt{2}\right)^2 + \left(\frac{a}{2}\right)^2 + \dots$$

$$= a^2 + \frac{a^2}{2} + \frac{a^2}{4} + \dots$$

$$= \frac{a^2}{1 - \frac{1}{2}} = 2a^2$$

$$\therefore \text{Required ratio} = \frac{\frac{8a}{2 - \sqrt{2}}}{2a^2} = \frac{8a}{(2 - \sqrt{2})2a^2}$$

$$= \frac{4}{a(2 - \sqrt{2})} = \frac{4(2 + \sqrt{2})}{a(4 - 2)} = \frac{4(2 + \sqrt{2})}{2a}$$

$$= \frac{2(2 + \sqrt{2})}{a}$$

28. (4) W

14 -

No

4 +

= 26

=

have

be

Hence

error

to 2

'w'.

would

be

[Till  $t_{27}$

will have

[Other

26 z's 2

$t_{320} - t_{55}$

$t_{391} - t_{42}$

$t_{277} - t_{30}$

29. (3)  $t_3 + t_5$

$\Rightarrow 2a +$

$a + 11d$

30. (3) We a

$+ 4 +$

twice, h

than 10

i.e.  $\frac{n(n+1)}{2}$

Making

If  $n = 24$

If  $n = 34$

If  $n = 44$

1000.

Hence, a

ber that